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Working correlation selection in generalized estimating equations

Mi Jin Jang

University of Iowa

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WORKING CORRELATION SELECTION IN GENERALIZED ESTIMATING
EQUATIONS

by
Mi Jin Jang

An Abstract

Of a thesis submitted in partial fulfillment
of the requirements for the Doctor of
Philosophy degree in Biostatistics
in the Graduate College of
The University of Iowa

December 2011

Thesis Supervisor: Professor Jane F. Pendergast

ABSTRACT

Longitudinal data analysis is common in biomedical research area. Generalized estimating equations (GEE) approach is widely used for longitudinal marginal models. The GEE method is known to provide consistent regression parameter estimates regardless of the choice of working correlation structure, provided $\sqrt{n}$ consistent nuisance parameters are used. However, it is important to use the appropriate working correlation structure in small samples, since it improves the statistical efficiency of $\hat{\beta}$. Several working correlation selection criteria have been proposed (Rotnitzky and Jewell, 1990; Pan, 2001; Hin and Wang, 2009; Shults et. al, 2009). However, these selection criteria have the same limitation in that they perform poorly when over-parameterized structures are considered as candidates.

In this dissertation, new working correlation selection criteria are developed based on generalized eigenvalues. A set of generalized eigenvalues is used to measure the disparity between the bias-corrected sandwich variance estimator under the hypothesized working correlation matrix and the model-based variance estimator under a working independence assumption. A summary measure based on the set of the generalized eigenvalues provides an indication of the disparity between the true correlation structure and the misspecified working correlation structure. Motivated by the test statistics in MANOVA, three working correlation selection criteria are proposed: PT (Pillai's trace type criterion), WR (Wilks' ratio type criterion) and RMR (Roy's Maximum Root type criterion). The relationship between these generalized eigenvalues and the CIC measure is revealed.

In addition, this dissertation proposes a method to penalize for the over-parameterized working correlation structures. The over-parameterized structure converges to the true correlation structure, using extra parameters. Thus, the true correlation structure and the over-parameterized structure tend to provide similar $\text{cov}(\hat{\beta})$,

and similar working correlation selection criterion values. However, the over-parameterized structure is more likely to be chosen as the best working correlation structure by “the smaller the better” rule for criterion values. This is because the over-parameterization leads to the negatively biased sandwich variance estimator, hence smaller selection criterion value. In this dissertation, the over-parameterized structure is penalized through cluster detection and an optimization function. In order to find the group (“cluster”) of the working correlation structures that are similar to each other, a cluster detection method is developed, based on spacings of the order statistics of the selection criterion measures. Once a cluster is found, the optimization function considering the trade-off between bias and variability provides the choice of the “best” approximating working correlation structure.

The performance of our proposed criterion measures relative to other relevant criteria (QIC, RJ and CIC) is examined in a series of simulation studies.

Abstract Approved: _____
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Graduate College
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CERTIFICATE OF APPROVAL

PH.D. THESIS

This is to certify that the Ph.D. thesis of

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has been approved by the Examining Committee
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To My Family

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Longitudinal data analysis is common in biomedical research area. Generalized estimating equations (GEE) approach is widely used for longitudinal marginal models. The GEE method is known to provide consistent regression parameter estimates regardless of the choice of working correlation structure, provided $\sqrt{n}$ consistent nuisance parameters are used. However, it is important to use the appropriate working correlation structure in small samples, since it improves the statistical efficiency of $\hat{\beta}$. Several working correlation selection criteria have been proposed (Rotnitzky and Jewell, 1990; Pan, 2001; Hin and Wang, 2009; Shults et. al, 2009). However, these selection criteria have the same limitation in that they perform poorly when over-parameterized structures are considered as candidates.

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CHAPTER 1

INTRODUCTION

1.1 Modeling Approaches for Longitudinal Data

Prospective longitudinal study designs, as well as retrospective, observational longitudinal data, are common in biomedical research area. Multiple measurements on the same participant are inherently correlated, since several characteristics (e.g., genetic, environmental, social factors) influence all of the repeated measures on that individual. For instance, in a weight loss study, participants are instructed to eat the same diet and their weights are measured on multiple occasions over the period of study. Genetic factors and lifestyle behaviors that stay consistent over the time of the study and have an influence or impact on weight would be a source of within-person correlation over time.

Marginal models and random effects models are often used for the analysis of longitudinal data. In this context, the term “marginal model” is usually used to describe a model in which the mean response, “marginalized” or averaged over individuals at each time point, is specified in terms of fixed effect covariates, and within-person correlation or association over time is incorporated into the estimation methodology. Marginal models are useful when the inference is intended to be population-based, since the means being modeled are the population averages of participants with the same covariate values. In contrast, models containing individual-level random effects target the individual by describing his/her own personal mean profile over time, and are referred to as subject-specific models. Thus, marginal models are often referred to as population-averaged models and random effects models are referred to as subject-specific models. This distinction has been blurred by the introduction of marginal models specified using random effects (Heagerty, 1999) but for the purpose of this dissertation, only marginal models with fixed effect covariates will be considered.

1.2 Introduction to Generalized Estimating Equations

Generalized estimating equations (GEE) approach is based on quasi-likelihood estimation and is a widely used estimation method for longitudinal marginal models (Liang and Zeger, 1986). The GEE method does not require full specification of the multivariate distribution of the repeated responses, but requires only specification of the first two moments of the outcome vectors. The within-subject associations or correlations among the repeated measures are taken into account by defining a so-called “working” correlation structure and incorporating that structure into the estimation. This working correlation matrix is often denoted $R(\alpha)$, where α is a vector of parameters that characterize the structure. When the sample size is large enough to estimate α parameters reasonably well, the closer $R(\alpha)$ is to the true underlying structure, the greater the gain in efficiency in estimation of the mean parameters.

To establish notation, consider independent observations from n individuals. For each individual i , a response y_{it} and a $p \times 1$ covariate vector $x_{it} = (x_{it1}, x_{it2}, \dots, x_{itp})'$ are obtained at times $t = 1, 2, \dots, m_i$. Let $Y_i = (y_{i1}, y_{i2}, \dots, y_{im_i})'$ be the $m_i \times 1$ vector of responses for the i^{th} individual and $X_i = \{x'_{i1}, x'_{i2}, \dots, x'_{im_i}\}'$ be the $m_i \times p$ corresponding covariate matrix. Responses within the same individual may be correlated to each other, but responses among different individuals are assumed to be independent. Covariates may be either time-varying or time-stationary.

The GEE approach requires the specification of the marginal mean model, the mean-variance relationship of the distribution of y_{it} , and the working correlation structure of Y_i . First, the marginal expectation of the response, $E(y_{it} | x_{it}) = \mu_{it}$, is related to a linear predictor through a known link function $g(\mu_{it}) = \eta_{it} = x_{it}'\beta$, where $\beta = (\beta_1, \beta_2, \dots, \beta_p)'$ is a $p \times 1$ vector of regression parameters and x_{it}' is the i^{th} row of X_i . Second, the variance of each y_{it} given the covariates, may depend on the mean, $\text{var}(y_{it}) = \phi v(\mu_{it})$, where ϕ is a scale parameter and $v(\cdot)$ is a known variance function. Lastly, the $m_i \times m_i$ working correlation matrix for each y_{it} , $R_i(\alpha)$, is assumed to be fully

specified by $q \times 1$ vector of unknown parameters, α , which is the same for all individuals. Note that the i subscript is used to denote potentially different dimensions of this matrix across subjects, but the structure within the matrix is assumed to be the same for all subjects. This implies that the working covariance matrix for y_{it} is $V_i = \phi A_i^{1/2} R_i(\alpha) A_i^{1/2}$, where $A_i = \text{diag}(v(\mu_{it}))$; $t = 1, \dots, m_i$. The working correlation structures considered in this dissertation are given in Table 1.1.

The GEE estimate of β , $\hat{\beta}$, is obtained by solving the following estimating equation:

$$S(\hat{\beta}, R_i, \Delta) = \sum_{i=1}^n D_i' V_i(\hat{\alpha})^{-1} (Y_i - \mu_i) = 0, \quad (1.1)$$

where $D_i = \partial \mu_i(\beta) / \partial \beta$ and $\Delta = \{(Y_1, X_1), \dots, (Y_n, X_n)\}$ indicates the data at hand. Since the GEE depend on both β and correlation parameters α and have no closed-form solution, iterative two-stage estimation procedure of β and the nuisance parameters (α and ϕ) is required.

Provided that the mean model is correctly specified and we have $n^{1/2}$ consistent estimators $\hat{\alpha}$ and $\hat{\phi}$ under mild regularity conditions, Liang and Zeger (1986) showed that $n^{1/2}(\hat{\beta} - \beta)$ is asymptotically normal with mean zero and covariance matrix V_G given by

$$\begin{aligned} V_G &= \lim_{n \rightarrow \infty} n \left[\sum_{i=1}^n D_i' V_i^{-1} D_i \right]^{-1} \left[\sum_{i=1}^n D_i' V_i^{-1} \text{cov}(Y_i) V_i^{-1} D_i \right] \left[\sum_{i=1}^n D_i' V_i^{-1} D_i \right]^{-1} \quad (1.2) \\ &= \lim_{n \rightarrow \infty} n M_0^{-1} M_1 M_0^{-1}. \end{aligned}$$

Liang and Zeger suggested that if you estimated $\text{cov}(\hat{\beta})$ using the empirical (“sandwich”) estimator:

$$\hat{\Sigma}_S = \hat{M}_0^{-1} \hat{M}_1 \hat{M}_0^{-1}, \quad (1.3)$$

where

$$\hat{M}_0 = \sum_{i=1}^n \hat{D}_i' V_i(\hat{\alpha}, \hat{\phi})^{-1} \hat{D}_i,$$

$$\hat{M}_1 = \left[\sum_{i=1}^n \hat{D}_i' V_i(\hat{\alpha}, \hat{\phi})^{-1} (Y_i - \mu_i(\hat{\beta})) (Y_i - \mu_i(\hat{\beta}))' V_i(\hat{\alpha}, \hat{\phi})^{-1} \hat{D}_i \right], \text{ and}$$

$\hat{\alpha}$ and $\hat{\phi}$ are any $n^{1/2}$ consistent estimators of α and ϕ , respectively, then

$$n\hat{\Sigma}_S - V_G \xrightarrow{p} 0.$$

The model-based (“naïve”) estimator of $\text{cov}(\hat{\beta})$ is given by

$$\hat{\Sigma}_M = \hat{M}_0^{-1}. \quad (1.4)$$

Given that the mean model and variance function $v(\mu_{it})$ relating the mean to the variance are correctly specified, the $\hat{M}_1$, in the sandwich estimator will eventually override a poor choice of the working correlation structure and still yield consistent estimates of the regression parameters with large samples of individuals. Thus, if the working correlation structure in GEE is incorrectly specified, but the mean model is correct and the α , ϕ estimates are $\sqrt{n}$ -consistent, the sandwich estimator can provide consistent estimates of regression parameters, whereas the model-based estimator will not. However, if V_i is correctly modeled, $V_i = \text{cov}(Y_i)$, $\hat{M}_1 - M_0 \xrightarrow{p} 0$, $\hat{M}_0 - M_0 \xrightarrow{p} 0$, and $M_1 M_0^{-1} \rightarrow I_p$ as $n \rightarrow \infty$, so that

$$\lim_{n \rightarrow \infty} n M_0^{-1} M_1 M_0^{-1} = \lim_{n \rightarrow \infty} n M_0^{-1} = V_G.$$

According to the extended Gauss-Markov theorem, the generalized estimating equation given by (1.1) is optimal for β in the class of linear functions of Y_i (Wang and Carey, 2003).

Table 1.1 Working Correlation Structures Considered

Working Correlation Structure Estimator		Example: 4×4 matrix
Independent (R_{IN})	$Corr(Y_{ij}, Y_{ik}) = \begin{cases} 1 & j = k \\ 0 & j \neq k \end{cases}$	$R_{IN} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Exchangeable (R_{EX})	$Corr(Y_{ij}, Y_{ik}) = \begin{cases} 1 & j = k \\ \alpha & j \neq k \end{cases}$	$R_{EX} = \begin{bmatrix} 1 & \alpha & \alpha & \alpha \\ \alpha & 1 & \alpha & \alpha \\ \alpha & \alpha & 1 & \alpha \\ \alpha & \alpha & \alpha & 1 \end{bmatrix}$
AR-1 (R_{AR-1})	$Corr(Y_{ij}, Y_{i,j+t}) = \alpha^t \text{ for } t = 0, 1, \dots, n_i - j$	$R_{AR-1} = \begin{bmatrix} 1 & \alpha & \alpha^2 & \alpha^3 \\ \alpha & 1 & \alpha & \alpha^2 \\ \alpha^2 & \alpha & 1 & \alpha \\ \alpha^3 & \alpha^2 & \alpha & 1 \end{bmatrix}$
Toeplitz (R_{TOEP})	$Corr(Y_{ij}, Y_{i,j+t}) = \begin{cases} 1 & j = k \\ \alpha_t & j = 1, 2, \dots, n_i - t \end{cases}$	$R_{TOEP} = \begin{bmatrix} 1 & \alpha_1 & \alpha_2 & \alpha_3 \\ \alpha_1 & 1 & \alpha_1 & \alpha_2 \\ \alpha_2 & \alpha_1 & 1 & \alpha_1 \\ \alpha_3 & \alpha_2 & \alpha_1 & 1 \end{bmatrix}$
Unstructured (R_{UN})	$Corr(Y_{ij}, Y_{ik}) = \begin{cases} 1 & j = k \\ \alpha_{jk} & j \neq k \end{cases}$	$R_{UN} = \begin{bmatrix} 1 & \alpha_{12} & \alpha_{13} & \alpha_{14} \\ \alpha_{12} & 1 & \alpha_{23} & \alpha_{24} \\ \alpha_{13} & \alpha_{23} & 1 & \alpha_{34} \\ \alpha_{14} & \alpha_{24} & \alpha_{34} & 1 \end{bmatrix}$

1.3 Statistical Efficiency Considerations in GEE

The robustness property of the sandwich variance estimator might tempt the analyst to care little about correctly specifying the within-subject association among repeated measures because consistent β estimates are obtained. However, there are three

reasons why an appropriate choice of working correlation matrix is important, especially in terms of statistical efficiency.

First, the robustness property of the sandwich variance estimator to misspecification of the form of working correlation matrix is an asymptotic property, and cannot be assumed to hold in all situations. Use of the sandwich variance estimator implicitly relies on there being independent replications of the vector of responses with enough data so that

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\mu}_i)(Y_i - \hat{\mu}_i)' \rightarrow \text{cov}(Y_i) \text{ as } n \rightarrow \infty.$$

So, when there are too few replications, highly irregularly spaced observations, or too many missing components of Y_i (even if missing completely at random) to estimate the true underlying covariance matrix well, the sandwich variance estimator is less appealing. If the number of individuals (n) is small but the number of repeated measures (m) on each individual is large, sandwich variance estimator is not recommended (Liang and Zeger, 1986; Fitzmaurice et al., 2004).

Second, a working correlation structure which closely approximates the true correlation matrix yields greater efficiency of $\hat{\beta}$ (Albert and McShane, 1995; Wang and Carey, 2003; Wang and Lin, 2005). Wang and Carey (2003) showed that the asymptotic relative efficiency depended on the three features: 1) disparity between the working correlation structure and the true underlying correlation structure, 2) the estimation method of the correlation parameters, and 3) the design (e.g., the structures of the predictor matrices and the sizes of clusters). Even when conditions 2) and 3) are not problematic, misspecification of the working correlation structure leads to a loss of efficiency and inflated variance estimates.

Third, misspecified working correlation structure sometimes yields inconsistent $\hat{\alpha}$ estimates. Crowder (1995) addressed some issues regarding inconsistency of $\hat{\alpha}$ under a misspecified working correlation structure. If, for example, the true correlation matrix is

AR-1, he showed that $\hat{\alpha}$ of the exchangeable working correlation structure does not exist or does not have unique solution in certain cases. For occasions where $\hat{\alpha}$ is not a $\sqrt{n}$ -consistent estimate, that is, it is a violation of an assumption of the Liang and Zeger's theorem, the consistency property of the regression parameter estimates (along with their impact on the variance estimates), is no longer assured.

Even though $\hat{\beta}$ is consistent, misspecified working correlation structure leads the sandwich variance estimate to be inefficient. The most efficient statistical inference in GEE would be to use the model-based variance estimator under the correctly specified working correlation structure. But, of course, one does not know a priori which correlation structure is correct. An appropriate choice of the working correlation matrix can result in large efficiency gains and bias reduction or elimination. Therefore, it is important to try and choose a working correlation matrix that is close to the true correlation matrix, and to do so, the analyst needs a decision tool that performs well.

1.4 Overall Goal of the Dissertation

The goal of this dissertation is to develop new selection criteria for the working correlation structure in GEE. The new working correlation selection criteria will be developed based on generalized eigenvalues, which measure discrepancy between two matrices. The new working correlation selection criteria are expected to effectively distinguish working correlation structures which closely approximate the true correlation matrix from other candidates.

In addition, penalization for the over-parameterized working correlation structure will be considered. Even though over-parameterized structure tends to closely estimate the correlation structure of the observed data, over-fitting yields high variability in the model estimates, especially in small samples. Nevertheless, the existing working correlation selection criterion measures do not consider penalizing over-parameterized structures or do not effectively penalize them. The new working correlation selection

criteria are expected to perform better than those in the current statistical literature by penalizing over-parameterized structure using a combination of a cluster detection method to restrict attention to those candidate structures that fit the data well and a function to penalize over-parameterized structures, thus providing a decision tool to choose among the candidate models..

The performance of the new working correlation selection criteria will be evaluated in various simulation settings in GEE suitable for modeling correlated Gaussian, binary and Poisson response data.

1.5 Outline

The remainder of this dissertation is organized as follows:

Chapter 2 includes an overview of working correlation structures. Estimation and asymptotic properties of the working correlation matrix are presented. The issues of the estimation of the correlation parameters proposed by Liang and Zeger (1986) are also discussed.

In chapter 3, we discuss the importance of the working correlation structure. The effect of the misspecified working correlation structure on the asymptotic relative efficiency is presented. Several bias-corrected sandwich variance estimators are reviewed. The impacts of the misspecified working correlation structure on the bias-corrected sandwich variance estimators are assessed through simulation study.

Chapter 4 consists of the literature review on the selection criteria for the working correlation structure (Pan, 2001; Hin and Wang, 2009; Shults and Chaganty, 1998; and Rotnitzky and Jewell, 1990).

In chapter 5, we present the concept of the generalized eigenvalues and their use in the multivariate test statistics and their application to the working correlation structure selection problem.

In chapter 6, we derive the generalized eigenvalue-based selection criteria for the working correlation structure. The performance of the proposed selection criteria are compared with the existing selection criterion measures, using a simulation study based on a factorial experimental design.

In chapter 7, we propose a new penalization method for over-parameterized working correlation structures. The proposed penalization method consists of cluster detection based on the spacings of the order statistics of the selection criterion measures and an optimization function (or loss function) considering the trade-off between bias and variability. The performance of the penalization method is evaluated using a simulation study.

In chapter 8, we present another penalization method using the bootstrapped data. Using ANOVA and Tukey's HSD grouping, the group of working correlation structures which are not statistically significant is found. Then the most parsimonious working correlation structure is selected within the group. Simulation study is performed to evaluate whether this approach effectively penalize the over-parameterized structures.

In chapter 9, we discuss the working correlation selection criteria presented in chapters 6 through 8 and describe future directions suggested by this work.

CHAPTER 2

AN OVERVIEW OF WORKING CORRELATION STRUCTURES

This chapter addresses the estimation of the working correlation structures. A general method of estimating correlation parameters and the asymptotic properties of the working correlation structures are discussed.

2.1 Estimation of the Working Correlation Structure

Some characteristics of the several working correlation structures are introduced. Then, a general method for estimating correlation parameters for each working correlation structure is presented, along with issue that arise in the estimation.

2.1.1 Characteristics of Working Correlation Structures

A working correlation structure $R(\alpha)$ is a $m \times m$ correlation matrix for repeated or clustered measurements from each individual $Y_i = (y_{i1}, y_{i2}, \dots, y_{im})'$ and its structure is specified by an analyst. A working independent structure (R_{IN}) is an identity matrix, so no pairwise correlation coefficients are estimated in this case. It is used when the multiple measurements on the same sampling unit (e.g., person) are assumed to be uncorrelated to each other.

The exchangeable working correlation structure (R_{EX}) assumes that the correlation between any pair of measurements on the same individual is the same α . It is often assumed in experiments using a split-plot design, where a within-plot factor is randomly allocated to subplots within main plots. The R_{EX} correlation structure is, however, somewhat unappealing for the longitudinal studies because randomization across time points is not possible, and often correlations are expected to decline as the time interval between the measurements increases. Nevertheless, R_{EX} is often used as a practical choice in small samples, since R_{EX} is very parsimonious with only one parameter.

A characteristic of the autoregressive of order one correlation structure (R_{AR-1}) is that the magnitude of the (positive) correlations quickly decrease over time as the separation between pairs of repeated measures increases ($\alpha_1 = \rho, \alpha_2 = \rho^2, \dots, \alpha_m = \rho^m$). It is also a parsimonious structure with one parameter and widely used in the longitudinal data analysis. However, it should be noted that in many real longitudinal situations, the AR-1 decline in correlations is too aggressive. The R_{AR-1} is appropriate when the measurements are made at approximately equal intervals of time, but the general concept of modeling the decline in correlations as a function of the time between measurement points is captured in a class of structures known as spatial correlation structures. These, like the AR-1 structure also are parsimonious, often with only one parameter.

The Toeplitz working correlation structure (R_{TOEP}) is a more flexible structure (relying on more parameters), and R_{EX} and R_{AR-1} are special cases of R_{TOEP} . The R_{TOEP} structure assumes that any pair of responses that are equally separated in time (or other ordering) have the same correlation. Like R_{AR-1} , it is appropriate when measurements are obtained at approximately equal intervals. The $m \times m$ R_{TOEP} matrix has $m-1$ parameters, one for each off-diagonal of the matrix. The number of parameters linearly increases as the number of multiple measurements on the same individual or within the cluster increases.

An unstructured working correlation matrix (R_{UN}) has no explicit pattern, but rather every correlation coefficient is allowed to be different. Any patterned matrix is nested within the R_{UN} structure. All $0.5m(m-1)$ elements of $m \times m$ matrix R_{UN} are estimated. So, even though R_{UN} is the most flexible structure, the number of parameters grows rapidly with the number of measurement occasions, m . The use of R_{UN} is appealing only when the sample size (n) is large enough to estimate $0.5m(m-1)$ parameters in it with some reasonable level of accuracy. Otherwise, the estimation of R_{UN} can be unstable. When the longitudinal data are severely unbalanced or sample sizes are small, it is desirable (or even necessary) to use more parsimonious structures.

In this dissertation, working correlation structures are categorized into three groups. The working correlation matrix that has the same structure as the true correlation is called *correctly specified*. A working correlation structure that is more complex than the true correlation matrix is called *over-parameterized* or *over-specified*. For instance, if the true correlation structure is AR-1, unstructured and Toeplitz working correlation structures are called over-parameterized. A working correlation structure that is neither equivalent nor over-parameterized (e.g., does not have enough parameters or the right structure to fit to the true structure) is called *misspecified*. In what follows, the subscript T denotes “under the truth”.

2.1.2 Estimation of correlation parameters α

The GEE estimator of the regression parameter β is obtained by solving the estimating equation, which is a non-linear system of equations for β , $R(\alpha)$ and ϕ . Therefore, a two-stage iterative fitting procedure for β and the nuisance parameters is required. Given the current value of β , the correlation parameter vector $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_q)'$ which fully characterizes $R(\alpha)$ and the scale parameter ϕ are estimated from the residuals in the iterative fitting process.

Consider

$$e_{ij} = (y_{ij} - \mu_{ij}) / \sqrt{\text{var}(\mu_{ij}) / \phi}, \quad i = 1, 2, \dots, n; \quad j = 1, 2, \dots, m \quad (2.1)$$

where $E(e_{ij}) = 0$, $E(e_{ij}^2) = \phi$ and $E(e_{ij}e_{ik}) = \phi \text{corr}(y_{ij}, y_{ik}) = \phi \rho_{jk}$. Let $\tilde{e}_{ij}$ denote an unbiased estimator of e_{ij} satisfying $E(\tilde{e}_{ij}^2) = \phi$ and $E(\tilde{e}_{ij}\tilde{e}_{ik}) = \phi \rho_{jk}$. The estimator of e_{ij} suggested by Liang and Zeger and its related issue are discussed in section 2.3. In this section, a general way for estimating α and ϕ using any unbiased estimator $\tilde{e}_{ij}$ will be discussed.

Let r_{jk} be (j, k) -component of a $m \times m$ working correlation matrix. For the exchangeable structure, one scalar $\alpha = \alpha_1$ fully characterizes $R_{EX}(\alpha)$, regardless of the size of the matrix. In this case, $r_{(EX)jk} = \alpha_1$ for all $j \neq k$. Similarly, one scalar $\alpha = \alpha_1$ fully

characterizes $R_{AR-1}(\alpha)$ but $r_{(AR-1)jk} = \alpha_1^{|j-k|}$. For the $m \times m$ Toeplitz working correlation structure, $\alpha = (\alpha_1, \dots, \alpha_{m-1})'$ characterizes $R_{TOEP}(\alpha)$ and $r_{(TOEP)j, (j+t)} = \alpha_t$ ($t = 1, \dots, m-1$). For an unstructured matrix, each $r_{(UN)jk}$ ($j < k$) has different value α . Using the moment estimators, the scale parameter can be estimated by

$$\tilde{\phi} = \frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^m \tilde{e}_{ij}^2. \quad (2.2)$$

The estimation method of α varies by the structure of $R(\alpha)$ chosen by an analyst. For unstructured correlations, $\tilde{r}_{(UN)jk}$, which would estimate the (j,k) component of $R_{UN}(\alpha)$ matrix, can be estimated by

$$\tilde{r}_{(UN)jk} = \frac{1}{n\tilde{\phi}} \sum_{i=1}^n \tilde{e}_{ij} \tilde{e}_{ik}, \text{ where } j \neq k. \quad (2.3)$$

Elements of the unstructured matrix are used to estimate other correlation parameters of the Toeplitz, exchangeable and AR-1 structures – or any others. For the Toeplitz structure, $\tilde{\alpha}_t = \tilde{r}_{(TOEP)j, j+t}$ ($t = 1, \dots, m-1$) is obtained by averaging t -th diagonal components of the estimated unstructured matrix:

$$\tilde{r}_{(TOEP)j, j+t} = \tilde{\alpha}_{(TOEP)t} = \frac{1}{n(m-t)\tilde{\phi}} \sum_{i=1}^n \sum_{j=1}^{m-t} \tilde{e}_{ij} \tilde{e}_{i(j+t)}, \text{ where } j \neq k. \quad (2.4)$$

For the exchangeable structure, $\tilde{\alpha}_{EX}$ is obtained by averaging $\tilde{r}_{(UN)jk}$ over j and k :

$$\tilde{r}_{(EX)jk} = \tilde{\alpha}_{EX} = \frac{1}{0.5nm(m-1)\tilde{\phi}} \sum_{i=1}^n \sum_{j < k} \tilde{e}_{ij} \tilde{e}_{ik}, \text{ for all } j \neq k. \quad (2.5)$$

That is, $\tilde{\alpha}$ of the exchangeable structure can be regarded as the average of the upper (or lower) off-diagonal components of the estimated unstructured matrix. For AR-1, $\tilde{\alpha}_{AR-1}$ is obtained from the following estimating equation (Liang and Zeger, 1986):

$$\sum_{i=1}^n \sum_{j < k} (\tilde{e}_{ij} \tilde{e}_{ik} - \phi \alpha_{AR-1}^{|j-k|}) = 0. \quad (2.6)$$

Other estimators are also used. The SAS package estimates $\tilde{\alpha}_{AR-1}$ using the correlations between the two adjacent points $\tilde{e}_{ij} \tilde{e}_{i(j+1)}$:

$$\tilde{r}_{(AR-1)j,(j+1)} = \tilde{\alpha}_{AR-1} = \frac{1}{n(m-1)\tilde{\phi}} \sum_{i=1}^n \sum_{j \leq m-1} \tilde{e}_{ij} \tilde{e}_{i(j+1)}, \text{ where } j \neq k. \quad (2.7)$$

In this dissertation, equation (2.7) is used to estimate $\tilde{\alpha}_{AR-1}$ in the simulation studies.

2.2 Asymptotic Properties of the Working Correlation

Structure

As described in the section 2.1.2, the unstructured working correlation matrix, $R_{UN}(\alpha)$, is estimated from the $\tilde{e}_{ij}$ values. One or more elements of $R_{UN}(\tilde{\alpha})$ are combined to estimate α of other working correlation matrices. Recall that $\tilde{e}_{ij}$ is an estimator satisfying $E(\tilde{e}_{ij}^2) = \phi$ and $E(\tilde{e}_{ij} \tilde{e}_{ik}) = \phi \rho_{jk}$. In this section, the limiting values of the estimated working correlation structures are presented.

2.2.1 Unstructured Working Correlation Matrix

As the sample size increases, each (j, k) -component of $R_{UN}(\tilde{\alpha})$ converges to the (j, k) -component of the true correlation matrix, $R_T(\rho)$:

$$\tilde{r}_{(UN)jk} = \frac{1}{n\tilde{\phi}} \sum_{i=1}^n \tilde{e}_{ij} \tilde{e}_{ik} = \frac{\frac{1}{n} \sum_{i=1}^n \tilde{e}_{ij} \tilde{e}_{ik}}{\frac{1}{nm} \sum_{i=1}^n \sum_{j=1}^m \tilde{e}_{ij}^2} \xrightarrow{P} \frac{\phi \rho_{jk}}{\phi} = \rho_{jk} \text{ as } n \rightarrow \infty. \quad (2.8)$$

Therefore, the estimated unstructured working correlation matrix will converge to any type of the true correlation matrix R_T .

2.2.2 Toeplitz Working Correlation Matrix

For the Toeplitz working correlation structure, $\tilde{\alpha}_t$ converges to the average of the t -th diagonal components of $R_T(\rho)$. As $n \rightarrow \infty$,

$$\tilde{r}_{(TOEP)j(j+t)} = \tilde{\alpha}_{(TOEP)t} = \frac{1}{n(m-t)\tilde{\phi}} \sum_{i=1}^n \sum_{j=1}^{m-t} \tilde{e}_{ij} \tilde{e}_{i(j+t)} \xrightarrow{P} \frac{1}{m-t} \sum_{j=1}^{m-t} \rho_{j(j+t)}, \quad (2.9)$$

where $1 \leq t \leq (m-1)$.

If the true correlation structure is Toeplitz, $R_{TOEP}(\tilde{\alpha})$ converges to $R_T(\rho)$. Since AR-1 and exchangeable structures are special cases of the Toeplitz matrix, if the true

correlation is AR-1 or exchangeable, $R_{TOEP}(\tilde{\alpha})$ also converges to $R_T(\rho)$. However, if the true correlation structure is more complicated than the Toeplitz structure, $R_{TOEP}(\tilde{\alpha})$ would not converge to $R_T(\rho)$.

2.2.3 Exchangeable Working Correlation Matrix

The $\tilde{\alpha}$ estimated under the exchangeable working correlation structure converges to the average of the upper off-diagonal components of the true correlation matrix $R_T(\rho)$: as $n \rightarrow \infty$,

$$\tilde{r}_{(EX)jk} = \tilde{\alpha}_{(EX)} = \frac{1}{0.5nm(m-1)\tilde{\phi}} \sum_{i=1}^n \sum_{j < k} \tilde{e}_{ij} \tilde{e}_{ik} \xrightarrow{P} \frac{1}{0.5m(m-1)} \sum_{j < k} \rho_{jk}. \quad (2.10)$$

If $R_T(\rho)$ is exchangeable, the average of all the upper off-diagonal components is

$\rho = \rho_1$. Therefore, $\tilde{\alpha}_{EX} \xrightarrow{P} \rho_1$ and $R_{EX}(\tilde{\alpha})$ converges to $R_T(\rho)$. If $R_T(\rho_1)$ is AR-1 and $|\rho_1| < 1$ ($\rho_1 \neq 0$), the average of the all upper off-diagonal components is

$$\varpi \equiv \frac{1}{0.5m(m-1)} \sum_{j < k} \rho_{jk} = \frac{1}{0.5m(m-1)} \left(\frac{1-\rho_1}{\rho_1} \right) \left(m - \frac{1-\rho_1^m}{1-\rho_1} \right),$$

which depends on the size of the matrix, m , as well as $\rho = \rho_1$. As the sample size increases, $\tilde{\alpha}_{EX} \rightarrow \varpi$. However, $\varpi \neq \rho_1$ since $|\rho_1| < 1$. Therefore, if the true correlation is AR-1, the use of the exchangeable working correlation structure yields inconsistent $\sqrt{n}$ α estimate and $R_{EX}(\tilde{\alpha})$ does not converge to the true correlation matrix.

2.2.4 AR-1 Working Correlation Structure

We have shown that $\tilde{\alpha}$ estimated under the $R_{AR-1}(\alpha)$ working correlation assumption could be estimated using either (2.6) or (2.7). If $\tilde{\alpha}$ is estimated by (2.7) under the AR-1 assumption, $\tilde{\alpha}$ converges to the average of the first off-diagonal components of $R_T(\rho)$: as $n \rightarrow \infty$,

$$\tilde{r}_{(AR-1)jk} = \tilde{\alpha}_{(AR-1)} = \frac{1}{n(m-1)\tilde{\phi}} \sum_{i=1}^n \sum_{j=1}^{m-1} \tilde{e}_{ij} \tilde{e}_{i(j+1)} \xrightarrow{P} \frac{1}{(m-1)} \sum_{j=1}^{m-1} \rho_{j(j+1)}. \quad (2.11)$$

If the true correlation structure is exchangeable with ρ_1 parameter, then $\tilde{\alpha}_{(AR-1)} \xrightarrow{P} \rho_1$ as $n \rightarrow \infty$. For the r -th off-diagonal components of $R_{AR-1}(\tilde{\alpha})$, $\tilde{\alpha}^r \xrightarrow{P} \rho_1^r$ as $n \rightarrow \infty$. On the other hand, all off-diagonal components of $R_T(\rho_1)$ are ρ_1 . Since we consider $|\rho_1| < 1$ ($\rho_1 \neq 0$), $\rho_1 \neq \rho_1^r$ for $r > 1$. Therefore, even though $\tilde{\alpha}_{(AR-1)}$ is a $\sqrt{n}$ consistent estimate for ρ_1 , the sandwich variance estimate under $R_{AR-1}(\tilde{\alpha})$ does not converge to the asymptotic variance of $\hat{\beta}$ since $R_{AR-1}(\tilde{\alpha})$ does not converge to the true correlation matrix.

Now, suppose $\tilde{\alpha}$ of $R_{AR-1}(\alpha)$ is estimated by (2.6). Crowder (1995) indicated that there is no solution for $\tilde{\alpha}$ in some cases if the true correlation matrix ($R_T(\rho)$) is exchangeable and AR-1 working correlation structure is used. The limiting values of the two terms in the estimating equation (2.6) are as follows:

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \sum_{j < k} \tilde{e}_{ij} \tilde{e}_{ik} &\xrightarrow{P} 0.5m(m-1)\phi\rho, \\ \frac{1}{n} \sum_{i=1}^n \sum_{j < k} \phi \alpha^{|j-k|} &\xrightarrow{P} \phi \left(\frac{\alpha}{1-\alpha} \right) \left(m - \frac{1-\alpha^m}{1-\alpha} \right), \quad n \rightarrow \infty. \end{aligned} \quad (2.12)$$

Now, consider the function $q(\alpha)$ which is the difference of those two terms:

$$q(\alpha) = 0.5m(m-1)\phi\rho - \phi \left(\frac{\alpha}{1-\alpha} \right) \left(m - \frac{1-\alpha^m}{1-\alpha} \right). \quad (2.13)$$

It can be shown that $dq(\alpha)/d\alpha < 0$ for $-1 < \alpha < 1$. For $-1 < \rho < 1$ and fixed n ,

$$\lim_{\alpha \rightarrow 1} q(\alpha) = 0.5\phi m(m-1)(\rho-1) < 0 \quad \text{and} \quad \lim_{\alpha \rightarrow -1} q(\alpha) = 0.5\phi m(m-1)(\rho + \alpha_m^{-1}),$$

where $\alpha_m = m-1$ for m even and $\alpha_m = m$ for m odd. Therefore, the unique solution exists if $\rho + \alpha_m^{-1} \geq 0$. Otherwise, no solution for $\tilde{\alpha}$ exists.

2.3 Issues of the estimation of correlation parameters

proposed by Liang and Zeger (1986)

In the previous sections, we assumed $\tilde{e}_{ij}$ is an unbiased estimator of e_{ij} satisfying $E(\tilde{e}_{ij}^2) = \phi$ and $E(\tilde{e}_{ij}\tilde{e}'_{ik}) = \phi\rho_{jk}$. Liang and Zeger (1986) suggest simply using the moment estimators based on Pearson residuals defined by

$$\hat{e}_{jk} = (y_{jk} - \hat{\mu}_{jk}) / \sqrt{\text{var}(\hat{\mu}_{jk}) / \hat{\phi}}. \quad (2.14)$$

Statistical software such as SAS and R estimate α and ϕ based on $\hat{e}_{jk}$. This method poses the question of whether $\hat{e}_{jk}$ satisfies $E(\hat{e}_{ij}^2) = \phi$ and $E(\hat{e}_{ij}\hat{e}_{ik}) = \phi\rho_{jk}$.

First, consider the diagnostics for the multiple linear regression model. For example, Pearson residuals are used to detect extreme observations. The estimated variance-covariance matrix of Pearson residuals is biased since the residuals are correlated to each other. That is, $\text{var}(\hat{e}) = \sigma^2(I - \Lambda)$, where $\Lambda = \{h_{ij}\} = X(X'X)^{-1}X'$ is called the hat matrix. The studentized residuals are scaled, using the corresponding diagonal element of $I - \Lambda = (r_i = \hat{e}_i / \sqrt{MSE(1 - h_{ii})})$ to remove the dependency of $\text{var}(\hat{e}_i)$ on the hat matrix, yielding a constant variance of 1 for $r_i, i=1, 2, \dots, n$. For this reason, the studentized residuals are often preferred over the usual Pearson residuals for the model diagnostics.

Likewise, the use of $\hat{e}_{jk}$ on the working correlation matrix estimation in GEE yields biased estimates of the variance. When $\hat{e}_{jk}$ is used to estimate $R(\alpha)$ and $\text{cov}(Y_i)$, the sandwich variance estimator is negatively biased. Thus, several bias-corrected sandwich variance estimators have been proposed (Mancl and DeRouen, 2001; Kauermann and Carroll, 2001; Pan, 2001; Wong and Long, 2010) to improve the estimation of $\text{cov}(Y_i)$. Their definitions and properties are described in Section 3.2.

On the other hand, the biasedness of $\hat{\alpha}$ has not received much attention. Recently, Lu et al. (2007) proposed a bias-corrected estimator for α . They provided bias-corrected covariance estimates that extend those of Mancl and DeRouen (2001) to encompass the correlation parameters as well as the mean parameters. Their proposed estimating equation for α is

$$\sum_{i=1}^n S_i' W_i^{-1} (\tilde{R}_i - \rho_i(\alpha)) = 0, \quad (2.15)$$

where $\tilde{R}_i = (\tilde{\zeta}_{i12}, \tilde{\zeta}_{i13}, \dots, \tilde{\zeta}_{i(m-1)m})'$ is a $[0.5m(m-1)] \times 1$ vector with elements

$\tilde{\zeta}_{ijk} = g_{ij}' s_{ik} \ (j < k)$. The row vector g_{ij}' corresponds to the j th row of the symmetric

matrix $G_i = (I - V_i^{-1/2} H_i V_i^{-1/2})^{-1}$ and s_{ik} is the k th column vector of $S_i = (Y_i - \hat{\mu}_i)(Y_i - \hat{\mu}_i)'$. The matrix $H_i = D_i \left(\sum_{i=1}^n D_i' V_i^{-1} D_i \right)^{-1} D_i' V_i$ is called the cluster leverage by Preisser and Qaqish (1996). A weighting matrix $W_i = \text{diag}(w_{i12}, w_{i13}, \dots, w_{i(m-1)m})$ is a diagonal matrix with elements w_{ijk} , which is the variance of $\hat{e}_{ij}\hat{e}_{ik}$ (Prentice, 1988):

$$w_{ijk} = 1 + \frac{(1 - 2\hat{\mu}_{ij})(1 - 2\hat{\mu}_{ik})\rho_{ijk}}{(v_{ij}v_{ik})^{1/2}} - \rho_{ijk}^2. \quad (2.16)$$

The estimating equations approach given by equation (2.15) to estimate the α parameters in the vector $\rho_i(\alpha)$ for the binary data case is implemented in the SAS/IML software (<http://www.bios.unc.edu/~jpreisse>).

The performance of the bias-corrected α estimation by Lu et al. (2007) has not been sufficiently investigated. Since G_i includes $R(\alpha)$, derivation of the quantity of bias correction needed for α estimation is not straightforward. Moreover, their method is based on Mancl and DeRouen's bias correction method, which has a tendency to overcorrect the bias of the sandwich variance estimator. Therefore, it is not clear how well their bias-corrected $\hat{\alpha}$ performs.

This dissertation is more focused on the selection of the working correlation matrix structure rather than accuracy of the estimation of individual components working correlation matrix. Therefore, Liang and Zeger's approach for estimating α based on Pearson residuals will be used throughout this dissertation. However, future work includes an assessment of whether implementing a bias-corrected estimator of the α vector has any impact on the ability to select an appropriate working correlation structure.

CHAPTER 3

THE IMPORTANCE OF THE WORKING CORRELATION STRUCTURE SELECTION

In the GEE method, if the working covariance matrix is correctly specified, along with correct model specification, the β estimate becomes optimal within this asymptotic framework and V_G , the limiting covariance matrix of the regression β parameters, is reduced to V_{opt} :

$$V_{opt} = \lim_{n \rightarrow \infty} n \left(\sum_{i=1}^n D_i V_i^{-1} D_i \right)^{-1}. \quad (3.1)$$

For any working covariance matrix, $V_G - V_{opt}$ is a non-negative definite matrix (Wang and Lin, 2005). Therefore, a misspecified working covariance matrix results in a loss of efficiency.

Nevertheless, the modeling of $R(\alpha)$ has not received much attention. In small samples, a parsimonious structure with no parameters (independence) or one parameter has been preferred by many researchers, since extra nuisance parameters can cause less accurate estimates with high variability. Recently, several bias-corrected sandwich variance estimators have been developed in order to improve the statistical efficiency regardless of the choice of the working correlation structure in small samples.

In this section, the effect of misspecification of the working correlation structure on the statistical efficiency is presented. First, the literature review on the asymptotic relative efficiency of the variance estimator under misspecified working correlation structure is presented. Later, several bias-corrected sandwich variance estimators are introduced. Then, the sensitivity of the bias-corrected sandwich variance estimator to the misspecified working correlation structure is evaluated using a simulation study.

3.1 Asymptotic Relative Efficiency of the Variance Estimator under the Misspecified Working Correlation Structure

Asymptotic relative efficiency (ARE) for a regression parameter estimator is defined as the ratio of the appropriate diagonal element in the “optimal” estimation variance and the corresponding “working” estimation variance. That is, the numerator of ARE is the diagonal element in V_{opt} and the denominator is the diagonal element in V_G under a misspecified working correlation structure corresponding to a single β parameter (Wang and Lin, 2005). Some simulation studies use V_G under the independence assumption to compute the ARE.

Since the analytical expression for asymptotic relative efficiency is not available in general, many simulation studies reported in the literature were carried out to evaluate the amount of the efficiency loss observed. Among those simulation studies, however, there have been some controversies regarding the efficiency loss resulting from the misspecified working covariance matrix. For instance, Zhao and Prentice (1990) and Liang et al. (1990) showed in their simulation studies that the variance estimators under R_{IN} (independence working correlation) were similar to those under the true correlation structure, although there was a little efficiency loss. On the other hand, Liang and Zeger (1986), Zhao et al. (1992), Fitzmaurice (1995) and Wang and Carey (2003) showed that the loss of efficiency due to the misspecification of the working correlation matrix depended on both the strength of the correlation between the responses and the covariate design matrix. If the true correlation was small to moderate, the sandwich variance estimators were similar, regardless of the choice of working correlation structure. If the true correlation was high, the substantial improvement in statistical efficiency could be obtained by correctly specifying the working correlation structure. If the covariate design includes at least one within-cluster covariate, the independence assumption can lead to a substantial loss of efficiency. Therefore, Fitzmaurice et al. (2004) and other authors

recommended some attempt should be made to model within-subject association correctly, even though those parameters are regarded as a nuisance parameters and the true underlying correlation structure is unknown.

Nevertheless, there are some practical concerns regarding the working correlation selection in small samples. McDonald (1993) described a concern that if the sample sizes were small, the extra nuisance parameters could cause convergence difficulties and inflate the variances, even if the specification of the nuisance parameters was correct. For these reasons, parsimonious working correlation structures with one or zero parameters have been favored by many researchers.

3.2 Bias-Corrected Sandwich Variance Estimators

Recently, new attempts to estimate the asymptotic variance of β more precisely by removing inherent bias has been made to improve the efficiency, regardless of the choice of the working correlation structure in small samples. In this section, several bias-corrected sandwich variance estimators are introduced.

It has been widely-known that the sandwich variance estimator proposed by Liang and Zeger (1986) is biased downward and generally has a larger variability than the model-based variance estimator (Paik, 1988; Sherman and le Cessie, 1997; Kauermann and Carroll, 2001; Mancl and DeRouen, 2001; Lu et al., 2007; Wong and Long, 2010). Use of the negatively-biased variance estimator can result in the hypothesis tests of the regression coefficients that are too liberal and confidence intervals on the β s that are too narrow, especially in smaller samples. So, various alternative estimators to the GEE sandwich estimator have been proposed (Paik, 1988; Lipsitz et al., 1990; Qu et al., 1994; Sherman and le Cessie, 1997; Kauermann and Carroll, 2001; Mancl and DeRouen, 2001; Pan, 2001; Wong and Long, 2010). Among them, four recent approaches will be described.

Kauermann and Carroll (2001) showed that the sandwich variance estimator used by Liang and Zeger (1986) under the correctly specified working covariance is negatively biased with order $1/n$ and proposed a bias-corrected sandwich variance estimator assuming a correctly specified variance structure. Their estimators for $\text{cov}(Y_i)$ and the bias-corrected sandwich variance estimator (KC) are presented in equations (3.2) and (3.3).

$$\hat{\text{cov}}(Y_i)_{KC} = (I - \hat{H}_i)^{-1/2} (Y_i - \hat{\mu}_i)(Y_i - \hat{\mu}_i)' (I - \hat{H}_i)^{-1/2}, \quad (3.2)$$

where $H_i = D_i \left(\sum_{i=1}^n D_i' V_i^{-1} D_i \right)^{-1} D_i' V_i^{-1}$ is called the cluster leverage (Preisser and Qaqish, 1996).

$$KC = \left(\sum_{i=1}^n \hat{D}_i' \hat{V}_i^{-1} \hat{D}_i \right)^{-1} \left[\sum_{i=1}^n \hat{D}_i' \hat{V}_i^{-1} \hat{\text{cov}}(Y_i)_{KC} \hat{V}_i^{-1} \hat{D}_i \right] \left(\sum_{i=1}^n \hat{D}_i' \hat{V}_i^{-1} \hat{D}_i \right)^{-1}. \quad (3.3)$$

Mancl and DeRouen (2001) proposed another bias-corrected sandwich variance estimator. Unlike Kauermann and Carroll (2001), they did not assume a correctly specified working covariance and ignored one term from its first order expansion resulting in overcorrection. Thus, the proposed estimator of Mancl and DeRouen tends to overcorrect the bias. Mancl and DeRouen's bias-corrected $\hat{\text{cov}}(Y_i)$ and their sandwich variance estimator (MD) are presented in the equations (3.4) and (3.5).

$$\hat{\text{cov}}(Y_i)_{MD} = (I - \hat{H}_i)^{-1} (Y_i - \hat{\mu}_i)(Y_i - \hat{\mu}_i)' (I - \hat{H}_i)^{-1}. \quad (3.4)$$

$$MD = \left(\sum_{i=1}^n \hat{D}_i' \hat{V}_i^{-1} \hat{D}_i \right)^{-1} \left[\sum_{i=1}^n \hat{D}_i' \hat{V}_i^{-1} \hat{\text{cov}}(Y_i)_{MD} \hat{V}_i^{-1} \hat{D}_i \right] \left(\sum_{i=1}^n \hat{D}_i' \hat{V}_i^{-1} \hat{D}_i \right)^{-1}. \quad (3.5)$$

Note the difference between these two estimators. Kauermann and Carroll included the term $(I - \hat{H}_i)^{-1/2}$ in their expression for $\hat{\text{cov}}(Y_i)$, whereas Mancl and DeRouen used $(I - \hat{H}_i)^{-1}$ instead.

Lu et al. (2007) compared the small sample performance of these two bias-corrected sandwich variance estimators. The marginal mean model and the within-cluster correlations were correctly specified in their simulation study. They considered the

correlated binary data setting with the several balanced cluster sizes ($m=4, 6, 10, 40, 80$) and sample sizes ($n=10, 14, 20, 40, 80$). The marginal mean model they used was

$$\text{logit}(\mu_{ij}) = \beta_0 + \beta_1 X_{1ij} + \beta_2 X_{2ij} + \beta_3 X_{1ij} X_{2ij},$$

where $X_{1ij}=1$ for an intervention community and 0 for a control community; $X_{2ij}=1$ for posttest (year1), and 0 for baseline (pretest) and $\beta = (\beta_0, \beta_1, \beta_2, \beta_3)' = (0.25, 0, -0.1, -0.3)'$.

The true correlation structure was defined by $\rho_{ijj'} = z'_{ijj'} \alpha = 0.10 z_{0ijj'} + 0.05 z_{1ijj'}$ where $z_{ijj'} = (1, 0)'$ if $X_{2ij} = X_{2ij'}$ and $z_{ijj'} = (0, 1)'$ if $X_{2ij} \neq X_{2ij'}$. To evaluate the performance of the variance estimators, the marginal mean model was fixed. An evaluation of $\hat{\text{var}}(\hat{\beta}_1)$, $\hat{\text{var}}(\hat{\beta}_2)$ and $\hat{\text{var}}(\hat{\beta}_3)$ comparing the two bias-correction approaches to each other and to the original Liang and Zeger sandwich estimator was presented. When the sample size was less than 40, the Mancl and DeRouen (MD) estimator consistently overestimated the three variances and was more variable than the Kauermann and Carroll (KC) and the Liang and Zeger (LZ) sandwich estimators. With respect to the bias reduction in the estimation of $\text{cov}(\hat{\beta})$, the KC estimator performed better than the MD estimator. However, the MD estimator had better coverage probabilities than the KC estimator. This is because the overcorrection of the negative bias is offset by the greater chance of undercoverage resulting from the high variability of the sandwich estimator.

Pan (2001) suggested a different approach for estimating the $\text{cov}(Y_i)$ term in the sandwich variance estimator. Recall that the original Liang and Zeger (1986) proposal was to estimate $\text{cov}(Y_i)$ by $S_i S_i'$ where $S_i = (Y_i - \hat{\mu}_i)(Y_i - \hat{\mu}_i)'$. Mancl and DeRouen (2001) and Kauermann and Carroll (2001) estimate $\text{cov}(Y_i)$ by $(I - H_i)^{-1} S_i S_i' (I - H_i)^{-1}$ and $(I - H_i)^{-1/2} S_i S_i' (I - H_i)^{-1/2}$, respectively. One common characteristic of these estimators is that the estimate of $\text{cov}(Y_i)$ is based only on the estimated deviations $S_i = (Y_i - \hat{\mu}_i)(Y_i - \hat{\mu}_i)'$ from subject i , rather than pooling across subjects, as proposed by Pan (2001) and others

Pan (2001) suggested pooling observations across different subjects by assuming there is a common correlation structure R_c across all subjects and the variance function is correctly modeled. For the modification to work, he assumed two conditions:

- (A1) The conditional variance of Y_{ij} given X_{ij} is correctly specified;
- (A2) A common correlation structure, R_c , exists across all subjects.

He proposed that $\text{cov}(Y_i)$ can be estimated by

$$\hat{\text{cov}}(Y_i)_{PAN} = \phi A_i^{-1/2} R_{UN} A_i^{-1/2} = \phi A_i^{-1/2} \left(\frac{1}{n} \sum_{i=1}^n A_i^{-1/2} S_i S_i' A_i^{-1/2} \right) A_i^{-1/2}, \quad (3.6)$$

where R_{UN} indicates an unstructured working correlation matrix. If R_{UN} is estimated by $\sum_{i=1}^n A_i^{-1/2} S_i S_i' A_i^{-1/2} / n$, the diagonal elements of the estimated R_{UN} are not necessarily one.

So he recommended using the non-diagonal elements of R_{UN} and stipulating all diagonal elements of R_{UN} to be 1's. The middle factor of his sandwich variance estimator is

$M_{1(PAN)} = \sum_{i=1}^n D_i' V_i^{-1} \text{cov}(Y_i)_{PAN} V_i^{-1} D_i$. Pan compared $M_{1(PAN)}$ with $M_{1(LZ)}$, the middle term of the sandwich variance estimator used by Liang and Zeger. He proved that his estimator is more efficient than Liang and Zeger's estimator in that the variability of $M_{1(PAN)}$ is smaller than that of $M_{1(LZ)}$ asymptotically. That is, as the sample size increases ($n \rightarrow \infty$), $\text{cov}\{\text{vec}(M_{1(LZ)})\} - \text{cov}\{\text{vec}(M_{1(PAN)})\}$ converges to a matrix M that is non-negative definite with probability 1.

Wong and Long (2010) noted that $\hat{\text{cov}}(Y_i)_{PAN}$ still includes $S_i S_i'$. Recall that Mancl and DeRoun (2001) and Kauermann and Carroll (2001) provided bias-correction methods for $S_i S_i'$. Using the work of Mancl and DeRouen (2001), Wong and Long (2010) provided a modification of the Pan (2001) estimator that incorporated their bias-correction:

$$\text{cov}(Y_i)_{WL} = A_i^{-1/2} \left(\frac{1}{n} \sum_{i=1}^n A_i^{-1/2} (I - H_i)^{-1} S_i S_i' (I - H_i)^{-1} A_i^{-1/2} \right) A_i^{-1/2}. \quad (3.7)$$

The middle term of the Wong and Long sandwich variance estimator is

$M_{1(WL)} = \sum_{i=1}^n D_i' V_i^{-1} \text{cov}(Y_i)_{WL} V_i^{-1} D_i$, while the two outside terms remain unchanged. Under

the usual regulatory conditions for GEE and assumptions (A1) and (A2), they showed that $\text{cov}(\text{vec}(M_{1(LZ)})) - \text{cov}(\text{vec}(M_{1(WL)}))$ and $\text{cov}(\text{vec}(M_{1(MD)})) - \text{cov}(\text{vec}(M_{1(WL)}))$ are both non-negative definite matrices with probability 1, while $\text{cov}(\text{vec}(M_{1(PAN)})) - \text{cov}(\text{vec}(M_{1(WL)})) \rightarrow 0$ with probability 1 as $n \rightarrow \infty$. This implies that Wong and Long's estimator and Pan's estimator are asymptotically equivalent in terms of the variability of their estimators. However, Wong and Long's estimator is asymptotically less variable than those proposed by Liang and Zeger and Mancl and DeRouen.

3.3 Simulation Study I: The Performance of the Bias-Corrected Sandwich Variance Estimators under Misspecified Working Correlation Structures

The performance of the bias-corrected sandwich variance estimators described in Section 3.2 has been evaluated in the literature under somewhat limited conditions. When the true correlations among responses were low, the bias-corrected sandwich variance estimators were not sensitive to the misspecification of the working correlation structure. However, in most simulations, only very parsimonious working correlation structures with zero or one parameter were considered. Moreover, the impact of the strength of the true correlations among responses on the performances of the bias-corrected sandwich variance estimators was not assessed.

Simulation Study I is aimed at evaluating the performance of the bias-corrected sandwich variance estimators in various settings. Five working correlation structures (independence, exchangeable, AR-1, Toeplitz and unstructured), different sample sizes, cluster sizes, and true correlation structures are considered. As the correct specification of the mean model is not the focus of this dissertation, all simulation studies will use the same, simple, and correctly-specified mean model.

3.3.1 Simulation Setting

This simulation study addresses the performance of the six variance estimators described more fully above when the sample size is relatively small:

1. The model-based variance estimator of Liang and Zeger (MB)
2. The sandwich variance estimator by Liang and Zeger (LZ)
3. The bias-corrected sandwich variance estimator of Kauermann and Carroll (KC)
4. The bias-corrected sandwich variance estimator by Mancl and DeRouen (MD)
5. The bias-corrected sandwich variance estimator by Pan (PAN)
6. The bias-corrected sandwich estimator of Wang and Long (WL).

The simulation design has a factorial design, with the factor levels as described in Table 3.1. For each cell of the study, 1000 random samples are generated, and all estimators are computed using the same data from each sample drawn. The marginal mean model is correctly modeled.

Since AR-1 and EX correlation matrices only depend on one α parameter, they are simple to define. On the other hand, TOEP and UN correlation structures require $(m-1)$ and $m(m-1)/2$ parameters, respectively. Note that the AR-1 and exchangeable structures are nested within the Toeplitz structure, and all patterned matrices are nested within the unstructured matrix form. The form of the underlying, true Toeplitz and unstructured correlation structures were chosen to be distinct from other candidate forms, and the structure for the $m \times m$ matrix was chosen to be nested within that of the $(m+1) \times (m+1)$ matrix, $m=4, 6$, as shown in Table 3.2(a) and Table 3.2 (b).

Table 3.1 Simulation Study I Design Parameters

Factor	Levels
Distribution (D)	Multivariate Bernoulli responses: $\text{logit}(\mu_{it}) = \beta_1 x_{1t} + \beta_2 x_{2t}$, where x_{1t} and x_{2t} are independently generated from $U[0.5, 1]$, $\beta_1 = \beta_2 = 0.3$.
Response vector dimension (m)	$m = 4, 6$
True Correlation Structure $C(\alpha)$	Exchangeable: EX(α), $\alpha=0.3, 0.7$ Autoregressive of order 1: AR-1(α), $\alpha=0.3, 0.7$ Toeplitz: TOEP(m), : α values in Table 3.2(a) Unstructured: UN(m), : α values in Table 3.2(b)
Working Correlation structure $R(\alpha)$	Independence (IN), Exchangeable (EX), Autoregressive (AR-1), Toeplitz (TOEP), and unstructured (UN)
Variance Estimator for $\hat{\beta}_2$	MB, LZ, KC, MD, PAN, WL
Sample Sizes (n)	$n = 20, 40, 50$

Table 3.2 True Correlations in Toeplitz and Unstructured Correlation Matrices

(a) True TOEP correlation ($m=4$ to $m=6$)

$m=4$				$m=5$	$m=6$
1.00	0.25	0.25	0.70	0.25	0.30
0.25	1.00	0.25	0.25	0.70	0.25
0.25	0.25	1.00	0.25	0.25	0.70
0.70	0.25	0.25	1.00	0.25	0.25
0.25	0.70	0.25	0.25	1.00	0.25
0.30	0.25	0.70	0.25	0.25	1.00

(b) True UN correlation ($m=4$ to $m=6$)

$m=4$				$m=5$	$m=6$
1.00	0.80	0.60	0.14	0.10	0.23
0.80	1.00	0.70	0.18	0.17	0.18
0.60	0.70	1.00	0.25	0.24	0.22
0.14	0.18	0.25	1.00	0.45	0.22
0.10	0.17	0.24	0.45	1.00	0.16
0.23	0.18	0.22	0.22	0.16	1.00

We focus our attention on $\hat{\text{var}}(\hat{\beta}_2)$, and for ease of notation, drop the “2” subscript (i.e., $\beta \equiv \beta_2$). The variance of a single mean β parameter, $\text{var}(\hat{\beta})$ is estimated by six different approaches described in Section 3.2: MB, LZ, MD, KC, PAN and WL. The small-sample performance of the six variance estimators will be evaluated using (i) the relative difference between sandwich variance estimators and the Monte Carlo variance of $\hat{\beta}(R_T)$, (ii) the Monte Carlo variance of the variance estimators, (iii) the coverage probability and (iv) the width of the confidence interval.

(1) Relative difference of the variance estimator with respect to the Monte Carlo variance of $\hat{\beta}(R_T)$

The percent relative difference of a variance estimator is defined as

$$\left[\frac{1}{1000} \left\{ \sum_{s=1}^{1000} \text{var}(\hat{\beta}(R))_s \right\} - \text{var}_{MC}(\hat{\beta}(R_T)) \right] \times 100\%, \quad (3.8)$$

where $\text{var}_{MC}(\hat{\beta}(R_T))$ is an estimate of the Monte-Carlo (or simulation) variance of $\hat{\beta}(R_T)$

:

$$\text{var}_{MC}(\hat{\beta}(R_T)) = \frac{1}{999} \sum_{s=1}^{1000} \left(\hat{\beta}(R_T)_s - \frac{1}{1000} \sum_{s=1}^{1000} \hat{\beta}(R_T)_s \right)^2. \quad (3.9)$$

(2) Monte Carlo variance of the variance estimator

The Monte Carlo variance of the variance estimator is calculated by

$$\text{var}_{MC}[\text{var}(\hat{\beta}(R))] = \frac{1}{999} \sum_{s=1}^{1000} \left\{ \text{var}(\hat{\beta}(R))_s - \frac{1}{1000} \sum_{s=1}^{1000} \text{var}(\hat{\beta}(R))_s \right\}^2. \quad (3.10)$$

where $\text{var}(\hat{\beta}(R))_s$ is the estimated variance of $\hat{\beta}$ under the working correlation structure R from the s^{th} simulation.

(3) Coverage probability and relative error of the coverage probability

Coverage probability is defined as the probability that a 95% Wald-type confidence intervals for β , using the working correlation structure R , from 1000 simulations will contain the true parameter:

$$CP \equiv \hat{P}[\hat{\beta}(R) - 1.96\sqrt{\text{var}(\hat{\beta}(R))} < \beta < \hat{\beta} + 1.96\sqrt{\text{var}(\hat{\beta}(R))}]. \quad (3.11)$$

The estimated coverage probability is the proportion of 95% confidence intervals computed as $\hat{\beta}(R) \pm 1.96\sqrt{\text{cov}(\hat{\beta}(R))}$ that contain the true parameter value β . The relative error of the coverage probability is defined as

$$(CP - 0.95) / 0.95 \times 100\%. \quad (3.12)$$

(4) Width of the confidence interval

The width of the 95% Wald confidence interval is defined as

$$2 \times 1.96 \times \sqrt{\text{var}(\hat{\beta}(R))}. \quad (3.13)$$

For each variance estimator, the average of the widths of the confidence intervals from 1000 independent simulations is computed.

3.3.2 Simulation Results

Table 3.3 through Table 3.6 provide the relative difference and coverage probability of all six variance estimators when the data are binomial and the true correlation matrix is AR-1 ($\alpha = 0.3, 0.7$ and $m = 4, 6$). Corresponding simulation results for cases where the true correlation matrix is independence (IN), exchangeable (EX), Toeplitz, or Unstructured are shown in Appendix A (Tables A.1-A.8), but are discussed below. Figure 3.1 through Figure 3.12 present the scatter plots of the relative difference against the relative error of the coverage probability of the two sandwich variance estimators (LZ and WL) under the five working correlation structures (IN, AR-1, EX, Toeplitz, and Unstructured). Other bias-corrected sandwich variance estimators are omitted from the figures because they show patterns that are very similar to the WL estimator, and would be difficult to distinguish on the plots. The first four figures (Figure 3.1 – Figure 3.4) are from simulations in which AR-1 is the correct model. Figure 3.5 through Figure 3.8 correspond to cases where exchangeable is the correct underlying model. Since the α parameters for the Toeplitz and unstructured correlation matrices when $m = 4$ are embedded in that for $m = 6$, only two figures are needed for cases where the underlying true correlation is Toeplitz (Figure 3.9, Figure 3.10) or unstructured (Figure 3.11, Figure 3.12). In each scatter plot, the closer the point is to the origin, the closer the variance estimator approximates the asymptotic variance of β , since both the relative difference and the relative error of the coverage probability are close to zero.

3.3.2.1 Relative Difference of the Variance Estimator

Overall, the direction and the magnitude of the relative difference of the variance estimator depend on the true underlying correlation structure $C(\alpha)$, the sample size (n), the cluster size (m), and the choice of the working correlation structure.

If the working correlation structure is correctly specified or over-parameterized, the amount of the relative difference of the six variance estimators decreases as n

increases. However, under the misspecified working correlation structure, the relative difference of the variance estimators does not necessarily decrease as n increases. For instance, we see that the working independence matrix leads to a highly inflated $\text{var}(\hat{\beta})$, especially when the true α parameter is a high positive value (Table 3.4, Table 3.6, Figure 3.2, Figure 3.4, Figure 3.6, Figure 3.8).

Generally speaking, the three bias-corrected sandwich variance estimators (KC, PAN, and WL) tend to have smaller relative difference as compared to the sandwich variance estimator by Liang and Zeger (LZ). The MD estimator tends to overcorrect the bias in comparison to other variance estimators in small samples, which is consistent with the simulation results of Lu et al. (2007), and has more variability in its bias estimates. The model-based estimator of Liang and Zeger under model misspecification has similar relative difference as their sandwich estimator (LZ) when the dimension of the covariance matrix is $m=4$, but performs worse when the dimension of the matrix is larger ($m=6$).

The simulation results clearly indicate that the bias-corrected sandwich variance estimators are not robust to the misspecification of the working correlation structure. If the true pairwise correlations between the responses are low and the number of multiple measurements is small, the variance estimates under the different misspecified working correlation structures are similar (Figure 3.1, Figure 3.5). However, if the true pairwise correlations between the responses are high, the misspecified working correlation structures cause inflated variance estimates. For example, if the true correlation structure is AR-1, the exchangeable and the independence working correlation structure do not approximate the true correlation matrix, even in large samples. This results in a positive bias, so the corresponding sandwich variance estimators tend to have inflated values. The observed relative difference of the variance estimators under independence assumption went up to 180%.

The unstructured working correlation matrix yields more negatively-biased variance estimates as compared to other working correlation matrices. The use of the bias-corrected variance estimators does not fully overcome this negative bias when the structure has a relatively large number of parameters which can be used to match any particular data set. In particular, when the true correlations between the responses are weak and the responses of the same individual are measured on many occasions, the variance estimates under R_{UN} tend to underestimate $\text{var}(\hat{\beta})$ more severely than other working correlation structures (Table 3.5).

Lastly, the relative difference of the model-based variance estimator using the correctly specified working correlation structure is not always closest to zero, as one might expect, due to variability in small samples. Even though the working correlation structure is correctly specified, if the true correlation structure is complicated, the parameter estimation can be inefficient in the smaller samples. Figure 3.11 and Figure 3.12 show that the variance estimates under R_{UN} , which is the correctly specified structure, are more negatively biased than the variance estimates under more parsimonious structure such as R_{TOEP} , for sample sizes of $n=20$ and $n=40$, but although still negative, come closer to 0 for $n=50$ than the other under-parameterized structures.

3.3.2.2 Coverage Probability

Coverage probability appears to be less sensitive to the misspecification of the working correlation structure than relative difference of the regression parameter variance estimator. In this simulation study, the mean model is correctly specified, so the GEE estimated regression parameter ($\hat{\beta}$) is close to the true value β_T . Unless the estimated variance is too small (due primarily to the negative bias) or highly variable (due to small sample sizes), the true parameter value is within the confidence interval. The misspecified, but parsimonious working correlation structure sometimes results in the

coverage probability close to the nominal value. For example, when $C(\alpha)$ is AR-1(0.3), $m=4$, $n=40$ and the KC estimator is used (Table 3.3), the coverage probabilities under R_{IN} , R_{EX} and R_{AR-1} are 95%, 94.2% and 93.4%, respectively. On the other hand, overparameterized structures, especially R_{UN} in cases where the number of repeated measurements is large ($m=6$), result in more serious undercoverage, since $\text{var}(\hat{\beta})$ is more severely underestimated (Table 3.5 and Table 3.6).

Table 3.3 Relative difference and coverage probability of each six variance estimator (MB, LZ, MD, KC, PAN, WL) when the true correlation structure for the binary responses is AR-1(0.3) and $m=4$.

	Relative difference (%)					Coverage probability (%)				
	IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
<i>n=20</i>										
MB	-0.7	-8.1	-13.5	-18.8	-27.3	95.0	94.2	94.4	93.0	90.4
LZ	-7.8	-14.8	-20.1	-24.5	-30.5	93.1	92.8	91.1	90.5	87.9
MD	10.0	0	-6.2	-10.8	-16.9	94.7	94.2	94.1	92.9	90.9
KC	0.6	-7.8	-13.5	-18	-24.5	93.9	93.5	92.6	91.6	89.7
PAN	-2.7	-10.2	-15.4	-20	-27.1	94.7	94.0	94.1	92.5	90.4
WL	4.4	-5.3	-10.8	-15.3	-22.3	95.2	94.3	94.5	93.3	91.7
<i>n=40</i>										
MB	13.2	5.4	-0.7	-3.6	-8.1	95.3	94.7	94.7	93.9	92.8
LZ	10.0	1.6	-4.2	-7	-10.7	94.7	94.0	93.4	92.8	91.4
MD	20.0	9.8	3.7	0.8	-3.1	95.5	94.6	94.5	93.7	92.9
KC	14.9	5.6	-0.3	-3.2	-7	95.0	94.2	93.9	93.4	91.8
PAN	12.5	4.1	-1.7	-4.4	-8.6	95.2	94.8	94.4	93.5	92.9
WL	16.4	6.7	0.8	-2	-6.2	95.5	95.0	94.7	93.9	93.4
<i>n=50</i>										
MB	9.0	1.8	-4.2	-6.6	-10.0	94.8	94.4	94.9	93.9	93.7
LZ	6.5	-1.3	-6.7	-8.8	-12.2	94.5	94.2	94.2	94.0	93.4
MD	14.2	5	-0.7	-2.8	-6.4	95.2	94.9	95.1	94.8	94.6
KC	10.2	1.8	-3.8	-5.8	-9.3	94.6	94.7	94.6	94.4	94.1
PAN	8.5	0.5	-5.1	-7.3	-10.7	94.1	93	93.7	93.3	92.9
WL	11.5	2.4	-3.3	-5.4	-8.8	94.6	93.4	94.1	93.4	93.4

Table 3.4 Relative difference and coverage probability of each six variance estimator (MB, LZ, MD, KC, PAN, WL) when the true correlation structure for the binary response is AR-1(0.7) and $m=4$.

	Relative difference (%)					Coverage probability (%)				
	IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
<i>n=20</i>										
MB	107.9	22.9	-3.1	-11.8	-25.8	95.0	94.2	94.7	91.7	88.3
LZ	92.3	13.2	-11.6	-12.8	-18.9	92.7	93.5	92.4	91.1	90.4
MD	135.1	31.9	4.3	4.0	-0.1	95.3	95.1	95.1	93.6	94.2
KC	112.5	22.1	-4.1	-4.9	-13.7	93.8	94.8	93.8	92.3	92.0
PAN	104.3	19.8	-7.5	-8.9	-15.4	94.5	93.7	94.2	92.4	89.9
WL	125.5	25.8	-2.1	-3.0	-8.7	96.3	95.0	95.2	93.9	92.2
<i>n=40</i>										
MB	128.8	32.4	3.4	-2.1	-13.5	94.9	96.3	95.7	94.1	92.2
LZ	125.1	28.9	-2.7	-3.6	-13.8	94.4	94.4	94.2	92.4	92.8
MD	148.8	39.1	5.5	4.7	-5.9	95.0	95.2	95.1	93.9	93.5
KC	136.6	33.9	1.2	0.4	-9.9	94.9	94.8	94.8	93.5	93.1
PAN	130.4	30.7	1.2	0.2	-8.5	94.2	94.0	94.0	92.8	92.4
WL	141.8	33.7	3.9	3.1	-5.6	94.6	94.2	94.5	93.3	92.8
<i>n=50</i>										
MB	143.7	40.6	9.2	5.2	-5.0	94.8	96.1	96.1	94.6	94.0
LZ	138.3	33.7	3.1	2.3	-5.3	93.4	95.7	95.3	94.5	92.9
MD	158.1	41.9	9.9	9.1	1.4	94.1	95.8	95.7	95.1	93.9
KC	148.0	37.7	6.4	5.6	-2.4	93.7	95.7	95.6	94.9	93.5
PAN	146.6	39.6	7.1	6.2	-3.2	93.5	95.0	94.3	94.0	93.0
WL	156.3	42.1	9.3	8.6	-0.9	94.6	95.0	94.5	94.8	93.1

Table 3.5 Relative difference and coverage probability of each six variance estimator (MB, LZ, MD, KC, PAN, WL) when the true correlation structure for the binary response is AR-1(0.3) and $m=6$.

	Relative difference (%)					Coverage probability (%)				
	IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
<i>n=20</i>										
MB	18.9	13.0	3.0	-5.1	-27.2	96.9	96.6	96.5	95.3	87.0
LZ	11.5	5.5	-3.9	-10.0	-27.7	94.2	94.7	93.8	93.4	86.2
MD	30.0	21.2	10.6	4.1	-14.7	95.1	95.7	95.0	94.6	89.6
KC	20.3	13.0	3.1	-3.3	-21.6	94.8	95.0	94.4	94.1	87.7
PAN	17.5	10.9	1.2	-5.5	-24.6	96.6	96.7	96.4	95.3	88.2
WL	23.7	15.1	5.0	-1.6	-20.6	96.9	96.7	96.9	95.7	89.4
<i>n=40</i>										
MB	10.3	5.1	-4.1	-7.7	-17.1	95.0	94.9	95.0	93.8	92.2
LZ	8.2	2.0	-6.5	-9.9	-18.1	94.5	94.2	94.0	93.5	90.6
MD	16.7	9.2	0.2	-3.3	-11.9	95.3	95.2	95.0	94.5	92.0
KC	12.3	5.6	-3.2	-6.7	-15.0	94.9	94.5	94.5	93.8	91.5
PAN	9.6	3.7	-5.3	-8.4	-17.0	94.4	94.4	94.8	93.3	93.0
WL	12.3	5.4	-3.7	-6.8	-15.3	94.7	94.4	94.8	93.7	93.3
<i>n=50</i>										
MB	15.1	9.7	0.0	-2.7	-10.5	95.4	95.0	94.5	94.3	92.1
LZ	13.7	7.5	-1.8	-4.5	-12.2	93.9	94.5	94.1	92.6	91.0
MD	20.8	13.4	3.7	1.0	-7.0	94.7	94.8	94.3	93.6	91.9
KC	17.2	10.4	0.9	-1.8	-9.6	94.3	94.7	94.2	93.2	91.7
PAN	15.9	9.6	-0.1	-3.0	-10.2	95.1	94.8	94.6	93.8	92.5
WL	18.1	11.0	1.2	-1.6	-8.9	95.9	94.8	94.7	93.8	92.5

Table 3.6 Relative difference and coverage probability of each six variance estimator (MB, LZ, MD, KC, PAN, WL) when the true correlation structure for the binary response is AR-1(0.7) and $m=6$.

	Relative difference (%)					Coverage probability (%)				
	IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
<i>n=20</i>										
MB	151.2	67.5	12.0	-1.4	-24.9	96.2	96.7	95.2	92.8	87.1
LZ	141.3	56.7	1.3	0.0	2.0	94.4	94.1	94.0	90.9	87.8
MD	188.3	79.1	16.5	16.3	36.1	95.8	95.9	95.2	93.3	90.8
KC	163.6	67.5	8.5	7.7	-3.8	95.2	95.2	94.6	92.2	89.2
PAN	152.8	65.2	9.1	7.7	6.8	95.7	94.1	94.1	92.1	87.3
WL	173.8	71.1	13.8	13.1	18.0	96.7	94.5	95.2	92.8	88.5
<i>n=40</i>										
MB	136.2	54.9	2.9	-3.8	-21.3	94.1	94.5	95.7	93.9	88.6
LZ	135.2	48.1	-3.1	-4.7	-8.7	93.9	93.6	94.8	94.0	90.2
MD	157.1	58.1	3.9	2.4	-0.2	95.1	94.4	95.3	94.6	92.0
KC	145.9	53.0	0.3	-1.2	-13.0	94.7	94.0	94.9	94.5	90.9
PAN	140.0	52.4	-0.1	-1.7	30.0	94.4	93.8	95.0	94.2	90.0
WL	149.6	54.9	1.9	0.4	39.6	94.5	93.9	95.4	94.5	90.3
<i>n=50</i>										
MB	128.4	50.1	-0.7	-6.2	-20.6	95.1	95.1	95.2	94.1	91.5
LZ	130.7	46.4	-5.1	-6.2	-18.2	95.0	94.9	94.3	93.9	92.1
MD	147.7	54.3	0.4	-0.6	-13.1	95.7	95.3	94.7	94.1	93.0
KC	139.1	50.3	-2.4	-3.5	-15.7	95.6	95.0	94.7	94.0	92.4
PAN	133.4	48.5	-3.0	-4.3	-11.5	94.8	94.2	94.0	93.8	91.2
WL	140.7	50.4	-1.5	-2.7	-9.8	95.3	94.2	94.3	94.0	91.6

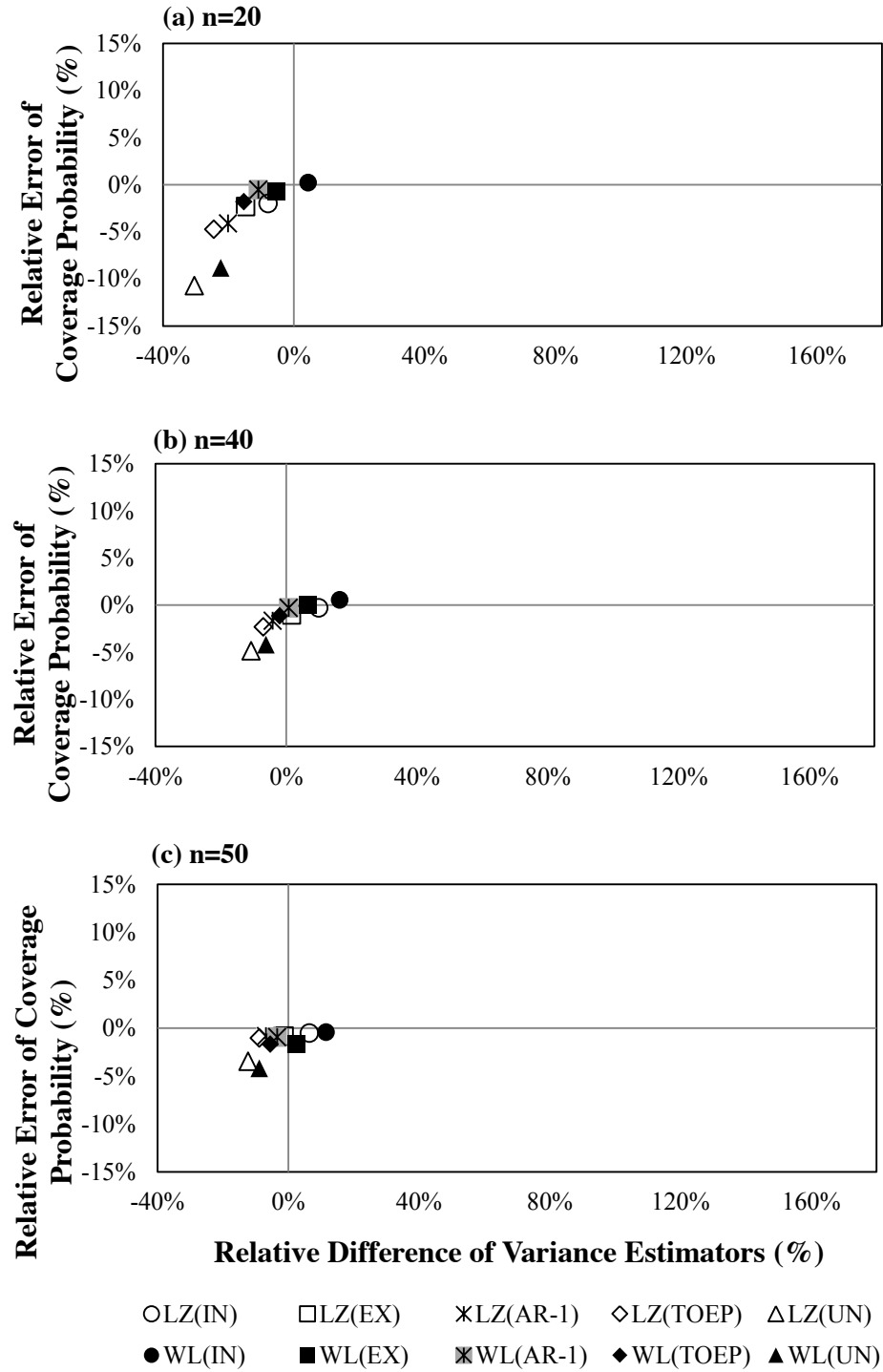


Figure 3.1 Relative error of coverage probability (y-axis) versus relative difference of variance estimator (x-axis) when the true correlation structure for the binary responses is AR-1(0.3) and $m=4$.

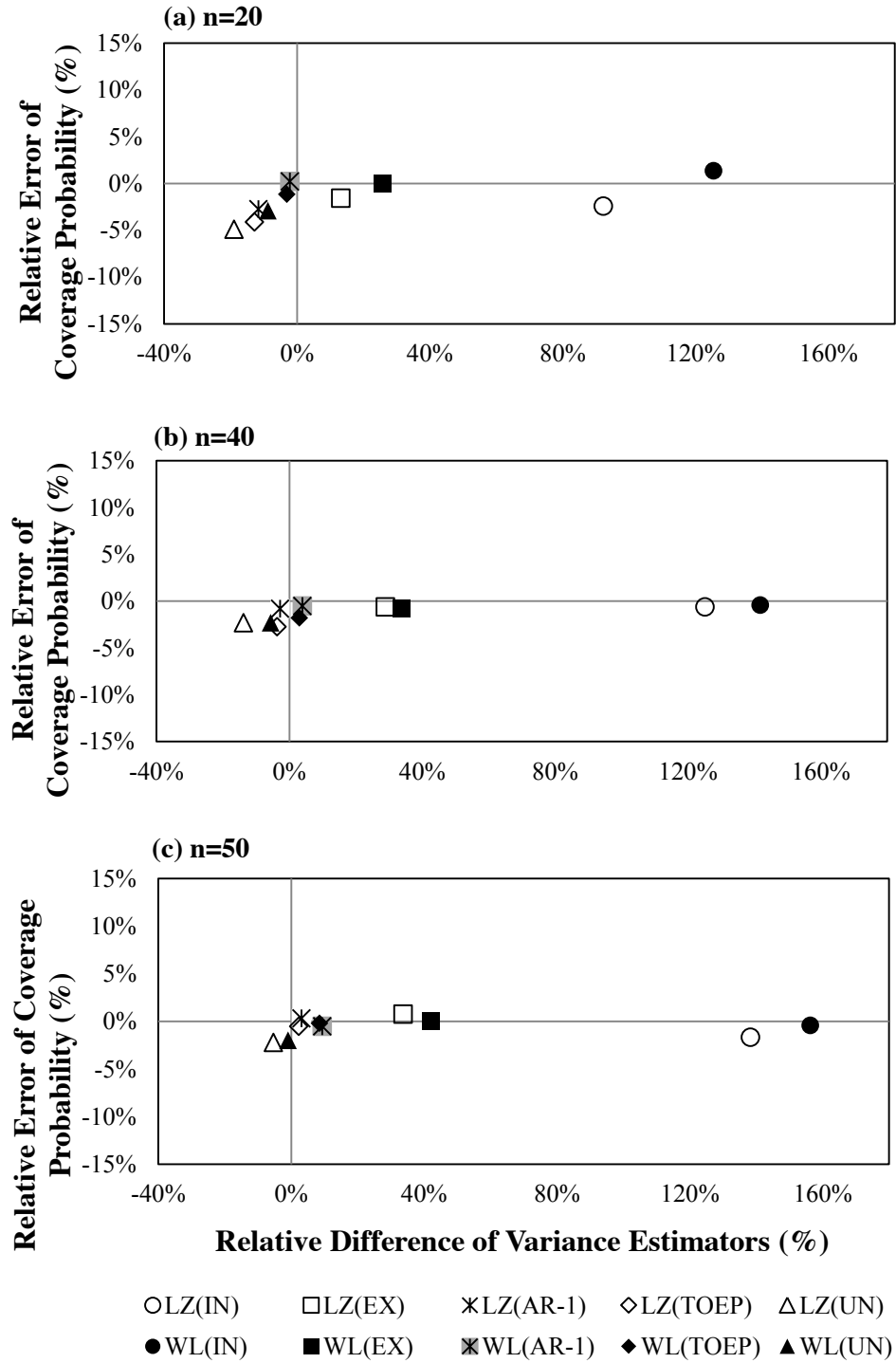


Figure 3.2 Relative error of coverage probability (y-axis) versus relative difference of variance estimator (x-axis) when the true correlation structure for the binary responses is AR-1(0.7) and $m=4$.

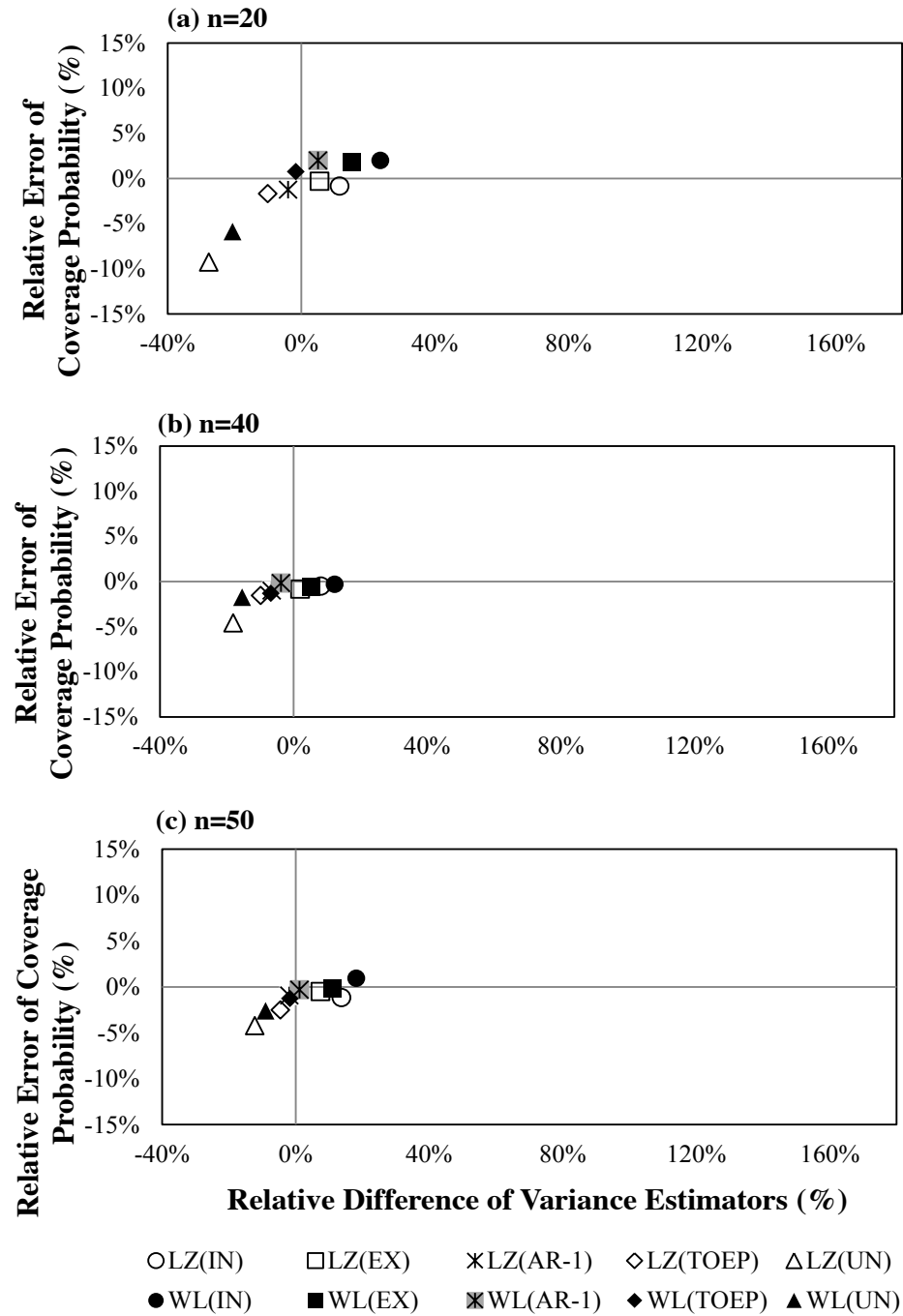


Figure 3.3 Relative error of coverage probability (y-axis) versus relative difference of variance estimator (x-axis) when the true correlation structure for the binary responses is AR-1(0.3) and $m=6$.

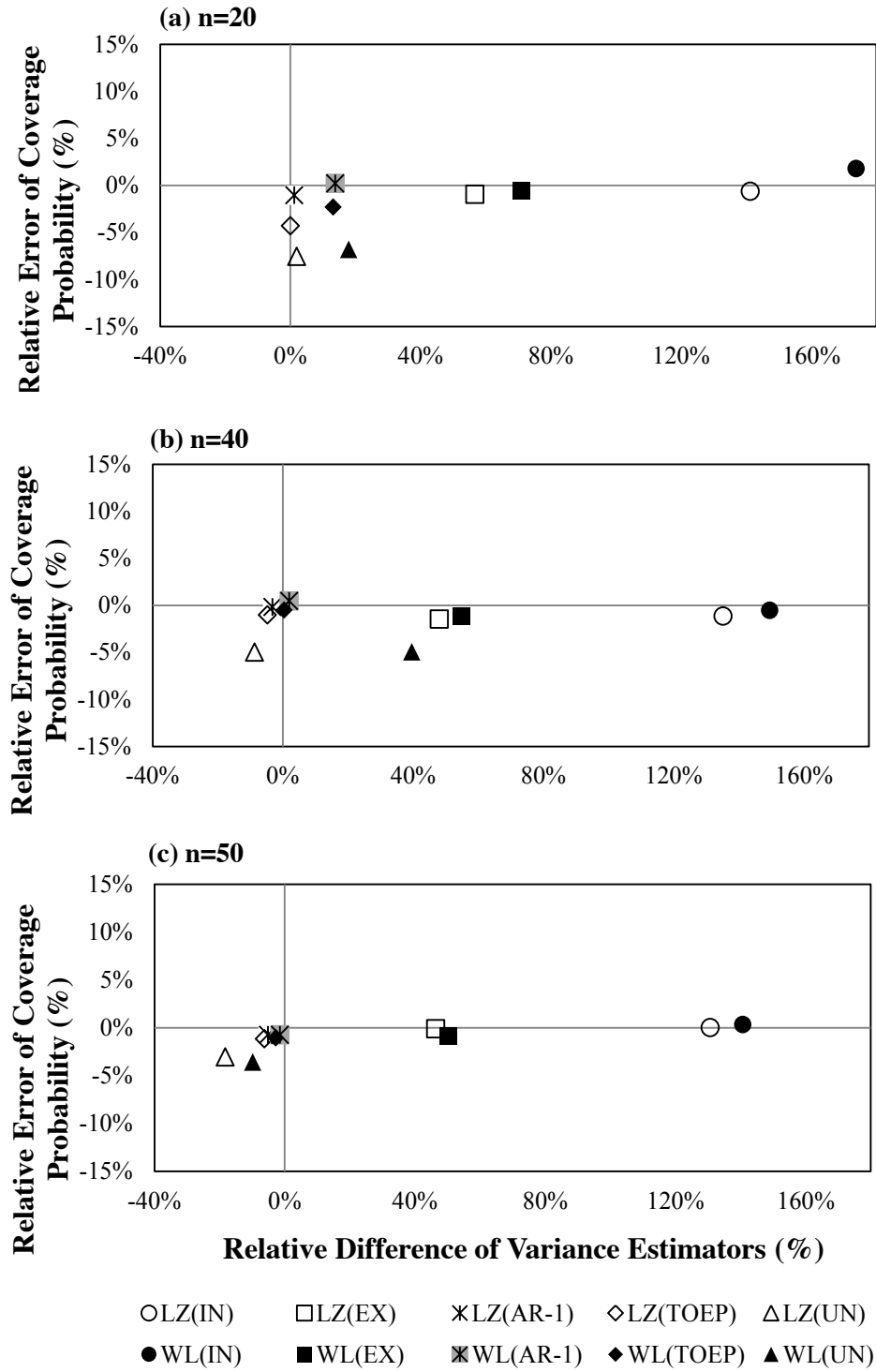


Figure 3.4 Relative error of coverage probability (y-axis) versus relative difference of variance estimator (x-axis) when the true correlation structure for the binary responses is AR-1(0.7) and $m=6$.

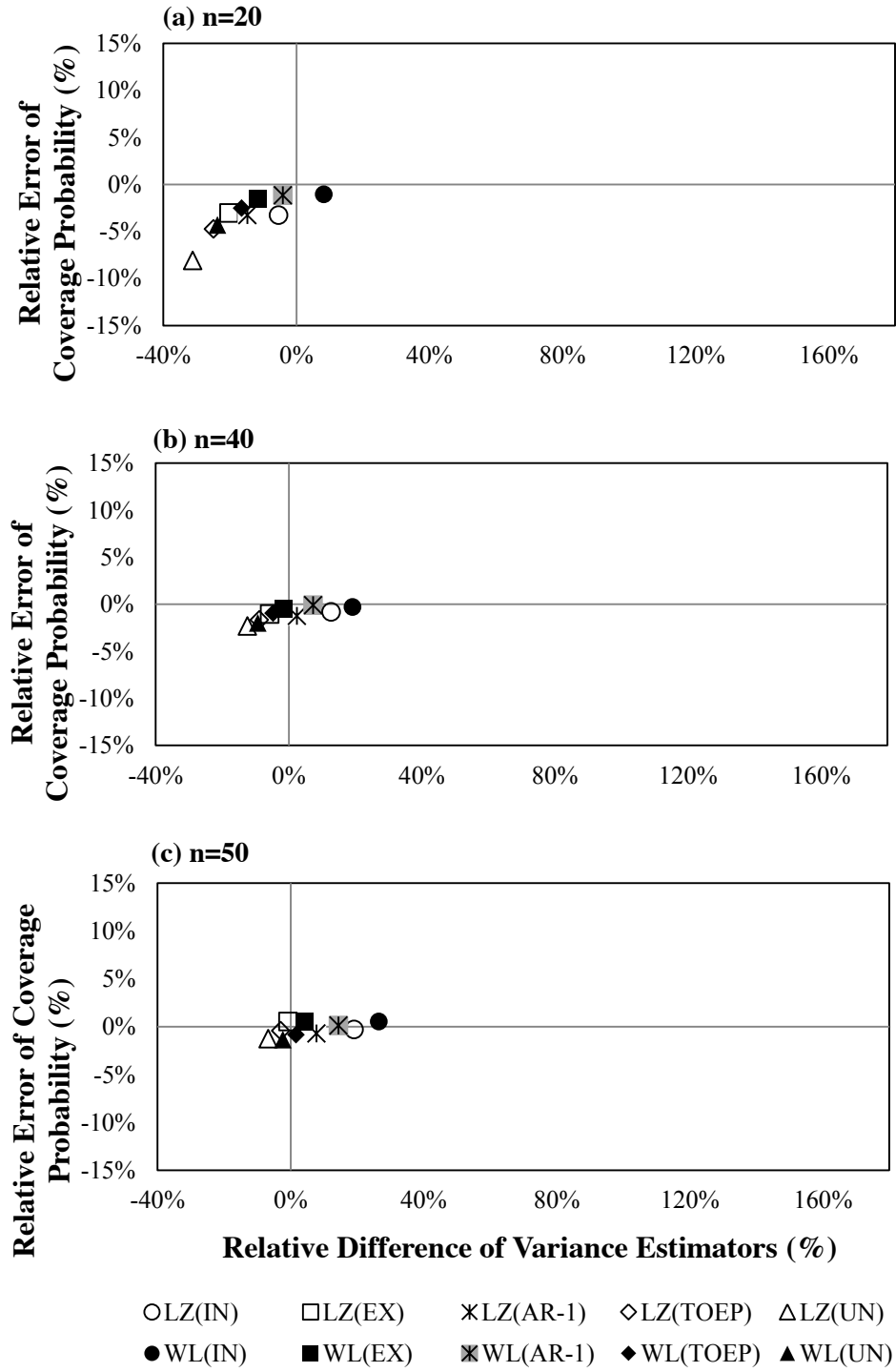


Figure 3.5 Relative error of coverage probability (y-axis) versus relative difference of variance estimator (x-axis) when the true correlation structure for the binary responses is EX (0.3) and $m=4$.

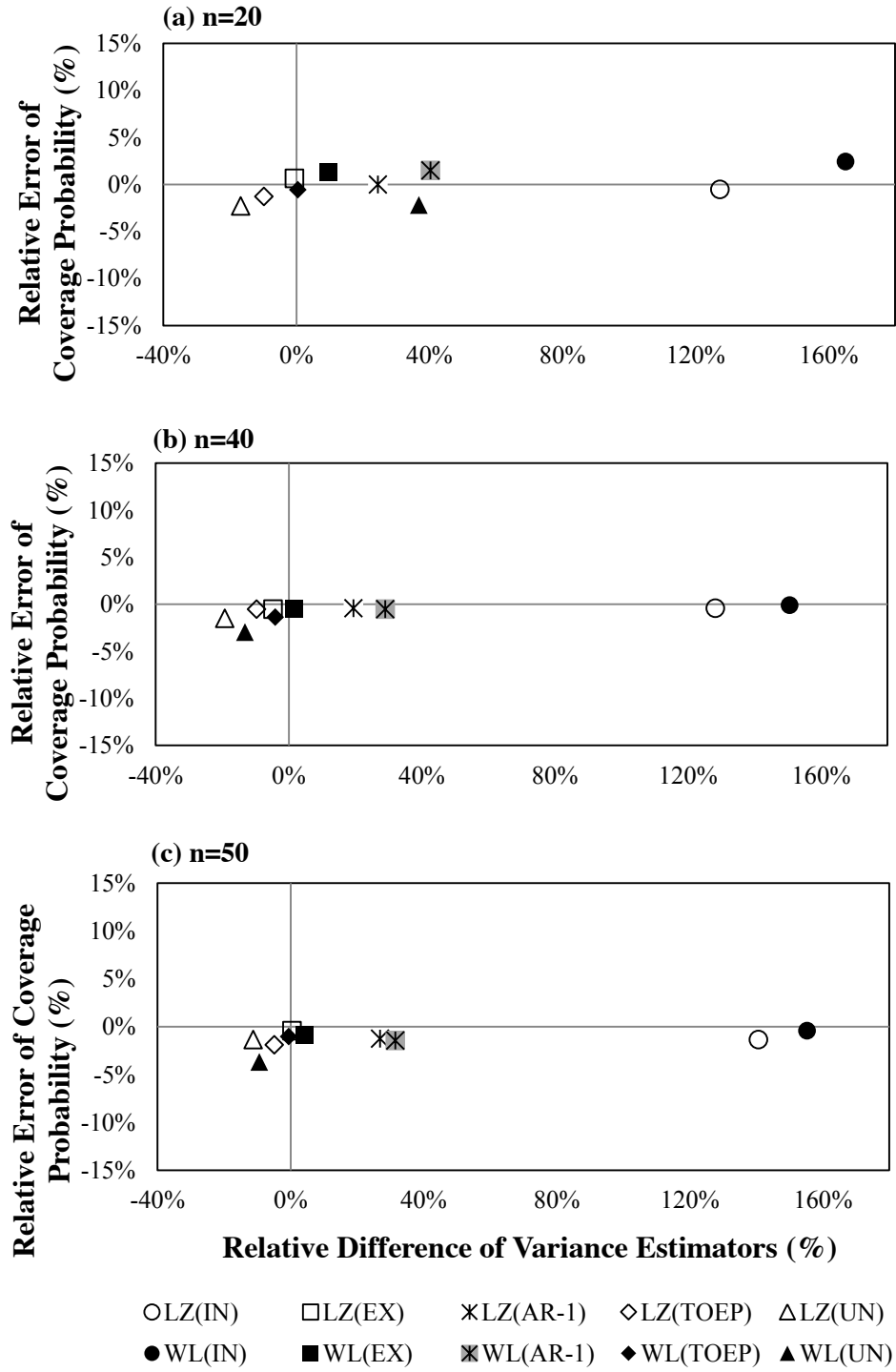


Figure 3.6 Relative error of coverage probability (y-axis) versus relative difference of variance estimator (x-axis) when the true correlation structure for the binary responses is EX (0.7) and $m=4$.

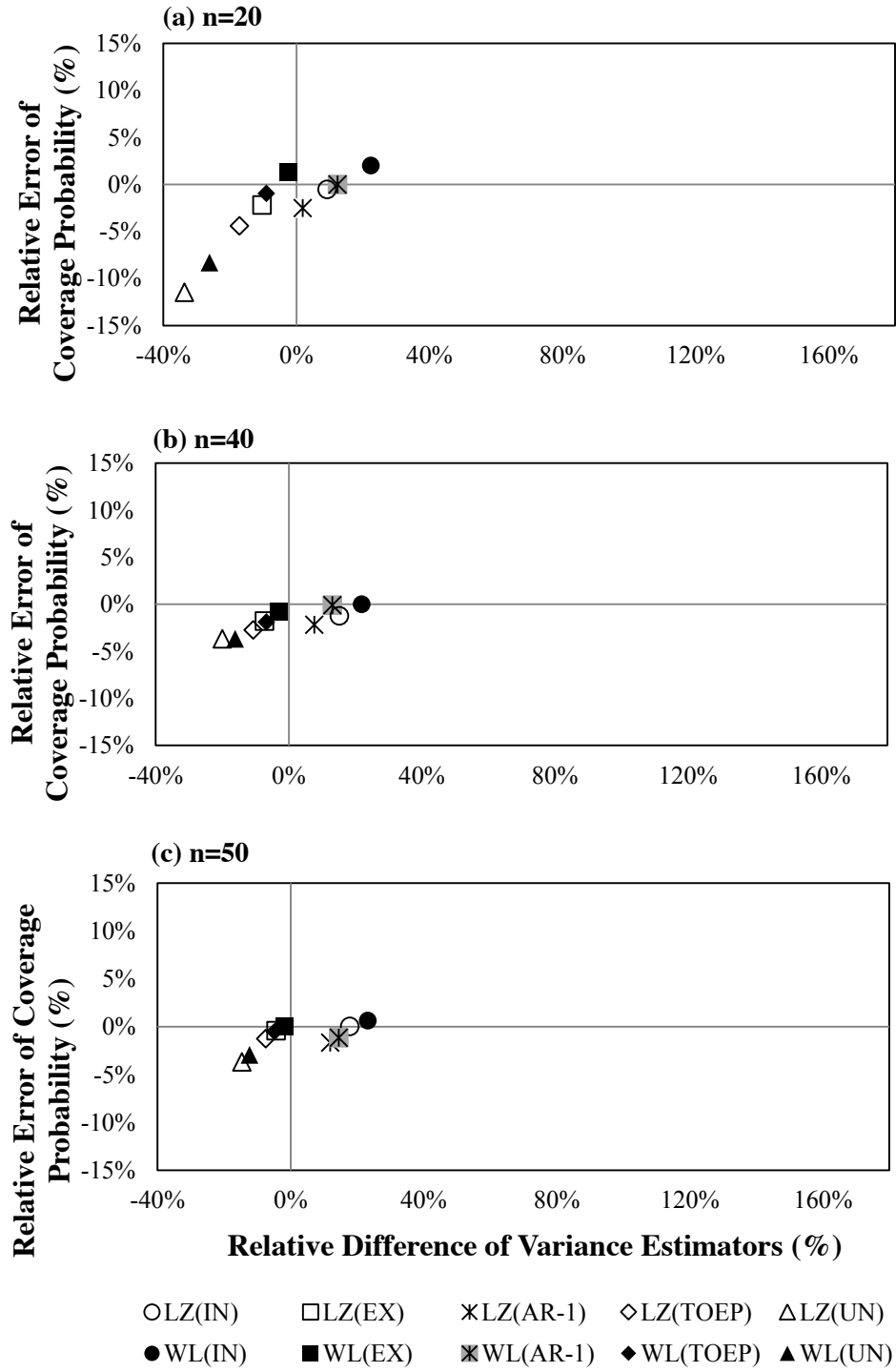


Figure 3.7 Relative error of coverage probability (y-axis) versus relative difference of variance estimator (x-axis) when the true correlation structure for the binary responses is EX (0.3) and $m=6$.

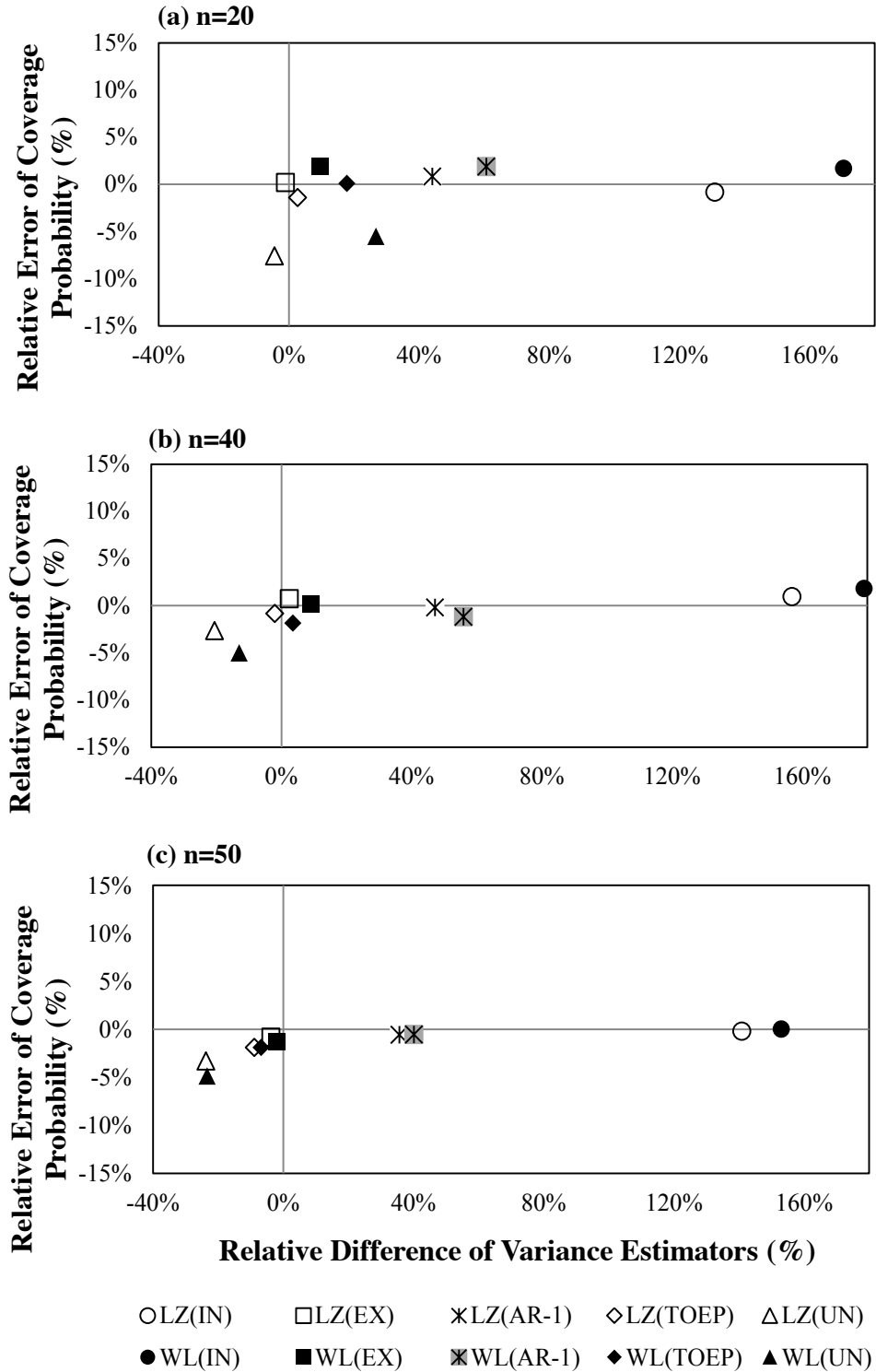


Figure 3.8 Relative error of coverage probability (y-axis) versus relative difference of variance estimator (x-axis) when the true correlation structure for the binary responses is EX (0.7) and $m=6$.

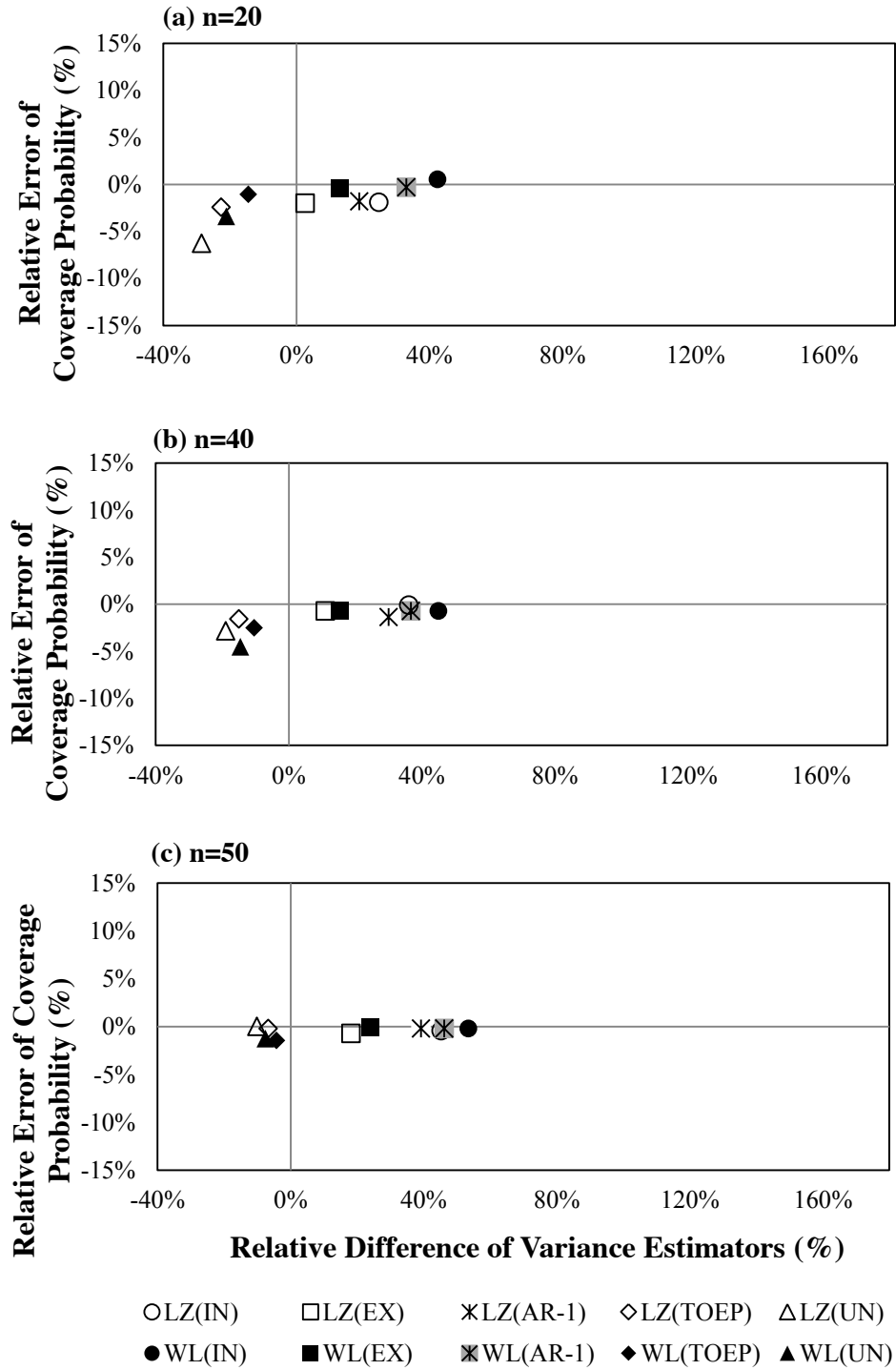


Figure 3.9 Relative error of coverage probability (y-axis) versus relative difference of variance estimator (x-axis) when the true correlation structure for the binary responses is TOEP and $m=4$.

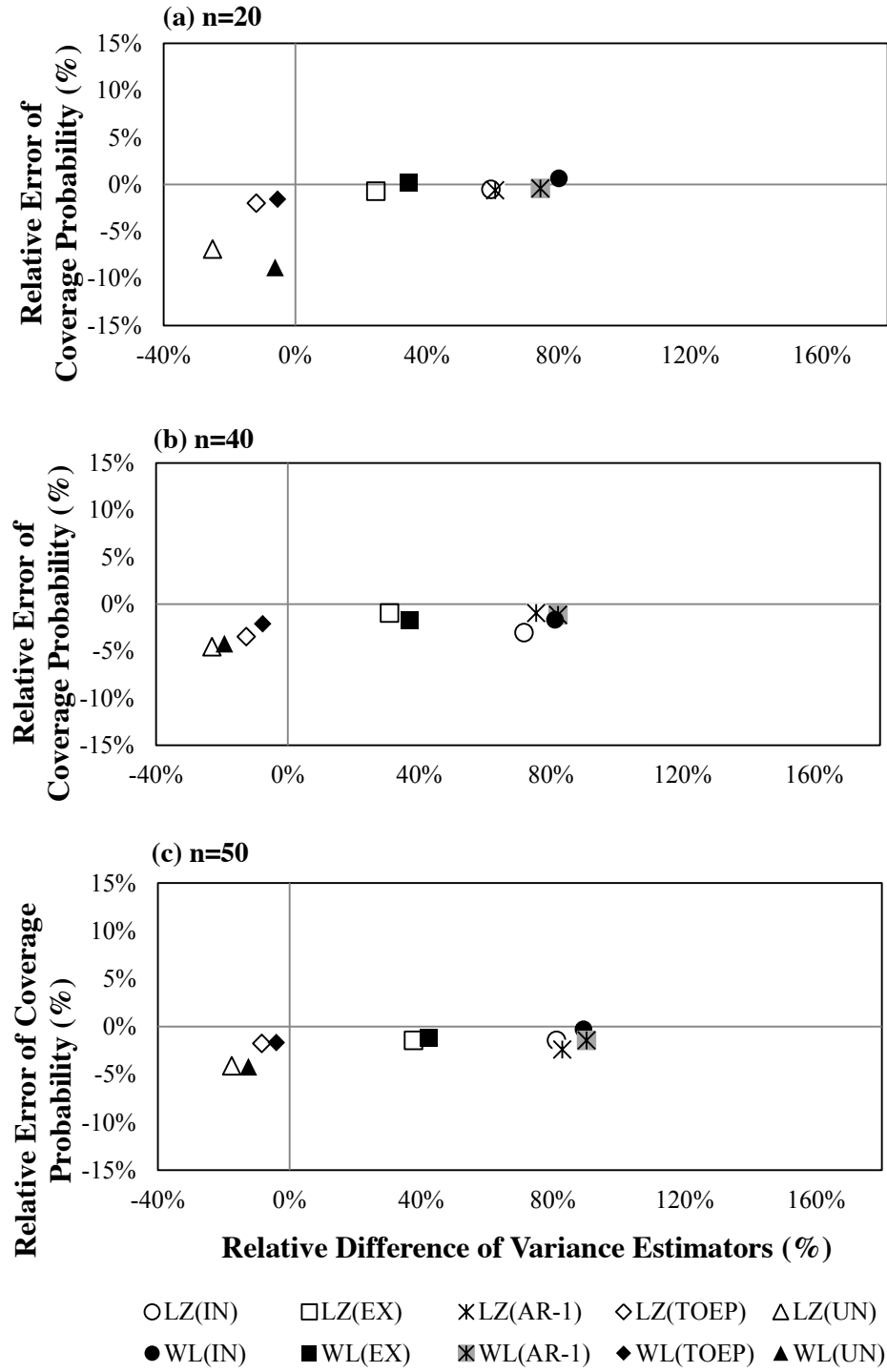


Figure 3.10 Relative error of coverage probability (y-axis) versus relative difference of variance estimator (x-axis) when the true correlation structure for the binary responses is TOEP and $m=6$.

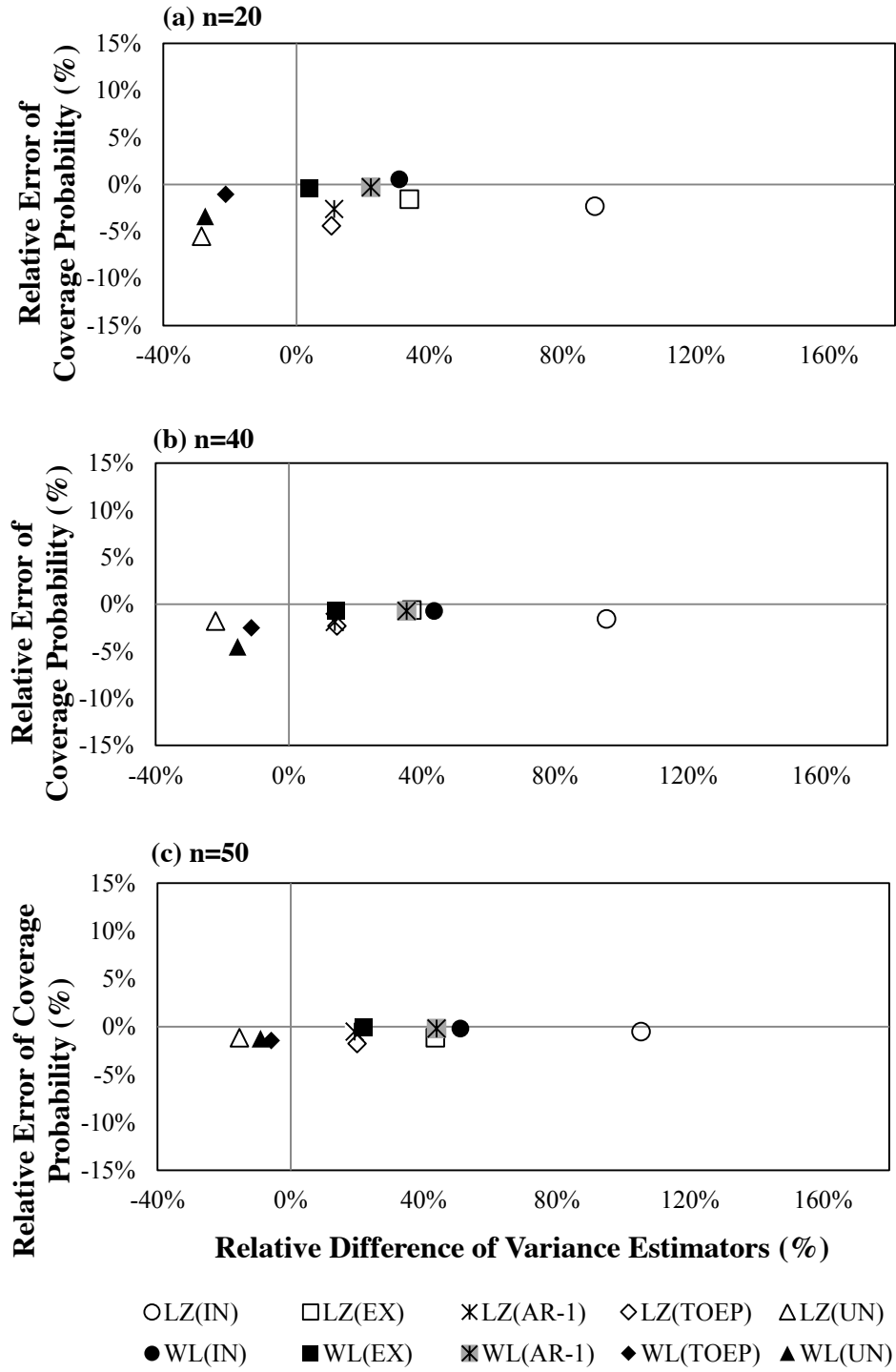


Figure 3.11 Relative error of coverage probability (y-axis) versus relative difference of variance estimator (x-axis) when the true correlation structure for the binary responses is UN and $m=4$.

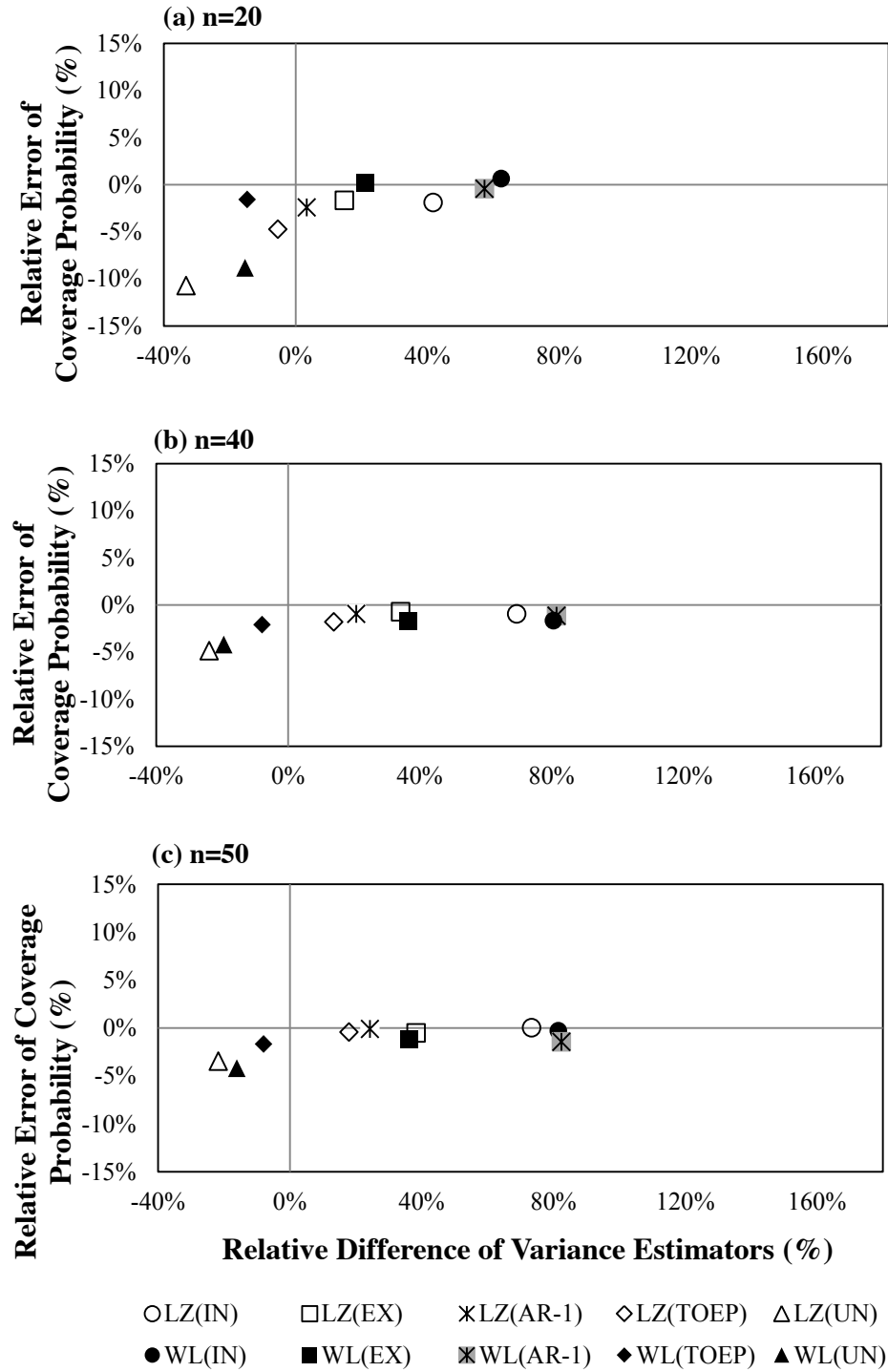


Figure 3.12 Relative error of coverage probability (y-axis) versus relative difference of variance estimator (x-axis) when the true correlation structure for the binary responses is UN and $m=6$.

3.3.2.3 The Monte Carlo Variance of the Variance

Estimator and Its Relationship to the Coverage Probability

The sandwich variance estimators generally have a higher variability than the corresponding model-based (“naïve”) variance estimators, due to the fact that the “meat” of the sandwich is an unstructured matrix with a relatively large number of parameters that must be estimated. Among the sandwich variance estimators, the MD, LZ, and KC estimators tend to have higher variability than the PAN and WL estimators (e.g., Figure 3.13 where the true structure is AR-1($\alpha = 0.3$), $m=6$, $n=20$; others presented in Appendix A, Tables A.9-A.20). It is consistent with the simulation results of Lu et al. (2007). As discussed in the section 2.2, the PAN and WL estimator calculate $\text{cov}(Y_i)$ by pooling (averaging) observations across different subjects, while other sandwich variance estimators calculate $\text{cov}(Y_i)$ for each subject i . This pooling or averaging has the impact of shrinking the overall variability of the final estimator.

Since even the sandwich variance estimator using the true correlation structure has a high variability in small samples, it is worth considering the tradeoff between the smaller variability of the naïve variance estimator and the robustness of the sandwich estimator – especially if more guidance can be given to the analyst to select a reasonable working correlation structure.

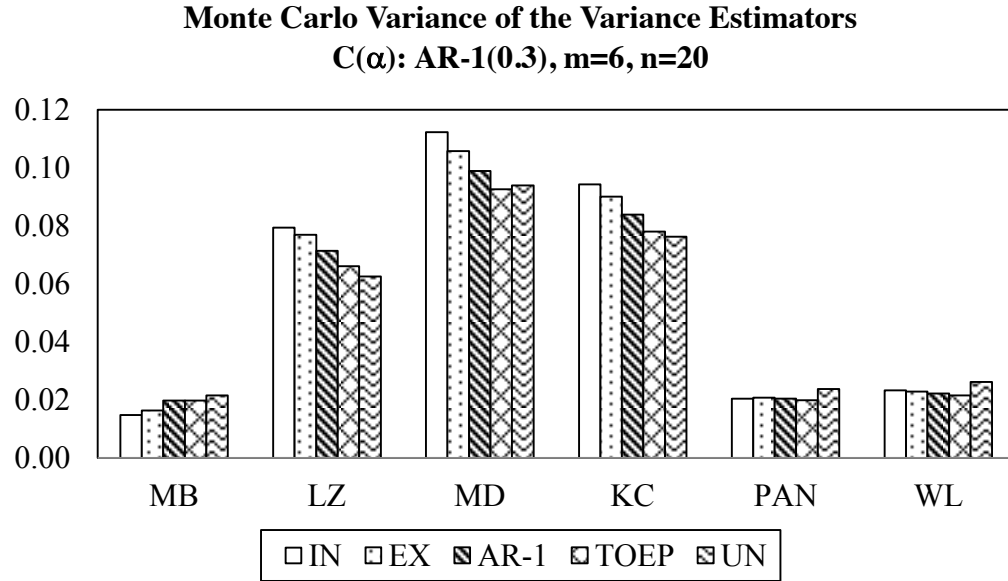


Figure 3.13 Monte Carlo variance of the six variance estimators when the true correlation structure for the correlated binary responses is AR-1(0.3), $m=6$ and $n=20$.

3.3.2.4 The Width of the Confidence Intervals

In general, the confidence level width depends on the sample size, variability of the estimator and the confidence level. Table 3.7 through Table 3.18 provide the widths of 95% confidence interval for all six variance estimators. As the sample size increases, the confidence interval width tends to decrease. The confidence interval width also depends on the choice of the working correlation structure and the variance estimator.

Overall, over-parameterized working correlation structure results in narrower confidence interval width. Misspecified working correlation structure results in wider confidence interval width. For example, when the true correlation structure is AR-1($\alpha=0.3$) and $m=4$ (Table 3.7), the confidence interval width under independence or exchangeable working correlation structure is wider than that under the true correlation structure, AR-1. On the other hand, the confidence interval width under Toeplitz or

unstructured matrix is narrower. When the true correlation structure is UN, all misspecified working correlation structures lead to wider confidence intervals.

Among the variance estimators, the LZ estimator tends to lead to narrower confidence intervals due to the negative bias for the given sample size and working correlation structure. The MD estimator causes wider confidence intervals than other variance estimators. It is because that the MD estimator tends to overcorrect the bias.

Table 3.7 The width of the 95% confidence intervals of each six variance estimator when the true correlation structure for the binary response is AR-1(0.3) and $m=4$.

	IN	EX	AR-1	TOEP	UN
<i>n=20</i>					
MB	4.708	4.526	4.386	4.244	4.003
LZ	4.483	4.299	4.154	4.035	3.849
MD	4.893	4.651	4.496	4.380	4.198
KC	4.682	4.470	4.320	4.202	4.011
PAN	4.656	4.470	4.334	4.213	4.009
WL	4.823	4.589	4.449	4.334	4.136
<i>n=40</i>					
MB	3.228	3.113	3.019	2.974	2.903
LZ	3.164	3.037	2.947	2.902	2.841
MD	3.305	3.157	3.065	3.020	2.959
KC	3.233	3.096	3.005	2.960	2.899
PAN	3.216	3.093	3.004	2.962	2.894
WL	3.271	3.131	3.041	2.999	2.932
<i>n=50</i>					
MB	2.875	2.777	2.692	2.658	2.609
LZ	2.830	2.723	2.645	2.615	2.564
MD	2.930	2.808	2.728	2.699	2.647
KC	2.880	2.765	2.686	2.656	2.605
PAN	2.868	2.759	2.680	2.649	2.599
WL	2.907	2.785	2.706	2.676	2.626

Table 3.8 The width of the 95% confidence intervals of each six variance estimator when the true correlation structure for the binary response is AR-1(0.7) and $m=4$.

	IN	EX	AR-1	TOEP	UN
<i>n=20</i>					
MB	4.750	3.616	3.193	3.041	2.756
LZ	4.517	3.393	2.984	2.970	2.760
MD	4.991	3.655	3.232	3.233	3.031
KC	4.747	3.520	3.104	3.097	2.872
PAN	4.694	3.566	3.125	3.102	2.874
WL	4.929	3.653	3.214	3.200	2.984
<i>n=40</i>					
MB	3.241	2.454	2.162	2.104	1.971
LZ	3.198	2.398	2.076	2.068	1.940
MD	3.362	2.489	2.160	2.154	2.023
KC	3.279	2.443	2.118	2.110	1.982
PAN	3.248	2.437	2.142	2.132	2.027
WL	3.327	2.465	2.171	2.162	2.059
<i>n=50</i>					
MB	2.879	2.178	1.916	1.880	1.783
LZ	2.835	2.105	1.846	1.840	1.761
MD	2.950	2.168	1.905	1.900	1.821
KC	2.892	2.136	1.875	1.869	1.788
PAN	2.891	2.168	1.897	1.890	1.798
WL	2.947	2.188	1.917	1.911	1.819

Table 3.9 The width of the 95% confidence intervals of each six variance estimator when the true correlation structure for the binary response is AR-1 (0.3) and $m=6$.

	IN	EX	AR-1	TOEP	UN
<i>n=20</i>					
MB	3.758	3.661	3.491	3.348	2.920
LZ	3.598	3.495	3.331	3.221	2.865
MD	3.882	3.745	3.571	3.462	3.108
KC	3.737	3.617	3.448	3.338	2.983
PAN	3.731	3.624	3.459	3.340	2.972
WL	3.829	3.692	3.524	3.409	3.052
<i>n=40</i>					
MB	2.609	2.546	2.431	2.385	2.259
LZ	2.569	2.494	2.385	2.342	2.229
MD	2.668	2.579	2.469	2.424	2.312
KC	2.618	2.536	2.427	2.382	2.270
PAN	2.601	2.529	2.415	2.375	2.260
WL	2.632	2.550	2.436	2.396	2.282
<i>n=50</i>					
MB	2.321	2.265	2.162	2.132	2.044
LZ	2.297	2.232	2.132	2.102	2.014
MD	2.367	2.293	2.191	2.161	2.072
KC	2.332	2.262	2.161	2.131	2.043
PAN	2.328	2.264	2.160	2.129	2.047
WL	2.351	2.278	2.175	2.144	2.062

Table 3.10 The width of the 95% confidence intervals of each six variance estimator when the true correlation structure for the binary response is AR-1 (0.7) and $m=6$.

	IN	EX	AR-1	TOEP	UN
<i>n=20</i>					
MB	3.774	3.068	2.496	2.339	1.982
LZ	3.653	2.912	2.335	2.318	2.102
MD	3.990	3.110	2.502	2.496	2.313
KC	3.817	3.009	2.416	2.405	2.121
PAN	3.776	3.040	2.464	2.444	2.205
WL	3.929	3.094	2.517	2.504	2.289
<i>n=40</i>					
MB	2.624	2.120	1.723	1.667	1.501
LZ	2.603	2.053	1.657	1.644	1.550
MD	2.720	2.121	1.715	1.704	1.612
KC	2.661	2.086	1.686	1.674	1.558
PAN	2.643	2.102	1.701	1.687	1.597
WL	2.694	2.119	1.718	1.705	1.621
<i>n=50</i>					
MB	2.332	1.887	1.531	1.489	1.366
LZ	2.333	1.850	1.486	1.478	1.373
MD	2.417	1.899	1.528	1.520	1.415
KC	2.374	1.875	1.507	1.499	1.393
PAN	2.355	1.876	1.515	1.505	1.426
WL	2.391	1.887	1.527	1.517	1.440

Table 3.11 The width of the 95% confidence intervals of each six variance estimator when the true correlation structure for the binary response is EX(0.3) and $m=4$.

	IN	EX	AR-1	TOEP	UN
<i>n=20</i>					
MB	4.723	4.328	4.176	4.164	3.924
LZ	4.496	4.105	4.240	3.972	3.780
MD	4.924	4.433	4.598	4.297	4.108
KC	4.703	4.264	4.414	4.130	3.938
PAN	4.672	4.273	4.442	4.148	3.934
WL	4.862	4.382	4.567	4.258	4.052
<i>n=40</i>					
MB	3.231	2.967	2.856	2.917	2.847
LZ	3.173	2.894	3.017	2.844	2.786
MD	3.320	3.007	3.141	2.955	2.897
KC	3.245	2.950	3.078	2.899	2.841
PAN	3.216	2.942	3.069	2.894	2.820
WL	3.278	2.975	3.109	2.928	2.855
<i>n=50</i>					
MB	2.872	2.631	2.525	2.594	2.539
LZ	2.831	2.577	2.684	2.547	2.496
MD	2.936	2.656	2.772	2.627	2.575
KC	2.883	2.616	2.727	2.587	2.535
PAN	2.883	2.632	2.754	2.597	2.544
WL	2.928	2.656	2.782	2.621	2.569

Table 3.12 The width of the 95% confidence intervals of each six variance estimator when the true correlation structure for the binary response is EX(0.7) and $m=4$.

	IN	EX	AR-1	TOEP	UN
<i>n=20</i>					
MB	4.705	3.094	2.609	2.845	2.573
LZ	4.505	2.895	3.193	2.732	2.531
MD	5.002	3.116	3.474	2.957	2.762
KC	4.746	3.002	3.329	2.831	2.600
PAN	4.645	3.047	3.402	2.887	2.674
WL	4.902	3.124	3.508	2.973	2.829
<i>n=40</i>					
MB	3.246	2.105	1.735	2.009	1.882
LZ	3.189	2.024	2.252	1.962	1.844
MD	3.359	2.099	2.349	2.039	1.919
KC	3.273	2.061	2.300	1.998	1.881
PAN	3.266	2.100	2.349	2.032	1.916
WL	3.353	2.124	2.384	2.058	1.944
<i>n=50</i>					
MB	2.339	1.510	1.099	1.444	1.289
LZ	2.364	1.479	1.747	1.436	1.306
MD	2.457	1.517	1.800	1.475	1.344
KC	2.410	1.498	1.773	1.455	1.325
PAN	2.383	1.498	1.783	1.458	1.312
WL	2.429	1.508	1.799	1.469	1.324

Table 3.13 The width of the 95% confidence intervals of each six variance estimator when the true correlation structure for the binary response is EX(0.3) and $m=6$.

	IN	EX	AR-1	TOEP	UN
<i>n=20</i>					
MB	3.777	3.436	3.191	3.264	2.835
LZ	3.634	3.278	3.493	3.145	2.784
MD	3.943	3.505	3.754	3.371	3.009
KC	3.784	3.389	3.620	3.255	2.893
PAN	3.761	3.403	3.643	3.278	2.912
WL	3.884	3.463	3.718	3.340	2.984
<i>n=40</i>					
MB	2.615	2.373	2.188	2.317	2.187
LZ	2.583	2.312	2.491	2.269	2.142
MD	2.691	2.390	2.582	2.346	2.218
KC	2.636	2.350	2.536	2.307	2.179
PAN	2.630	2.363	2.547	2.315	2.192
WL	2.672	2.382	2.571	2.334	2.212
<i>n=50</i>					
MB	2.327	2.111	1.937	2.074	1.983
LZ	2.315	2.083	2.251	2.048	1.966
MD	2.392	2.138	2.317	2.103	2.021
KC	2.353	2.110	2.283	2.075	1.994
PAN	2.349	2.110	2.275	2.076	1.990
WL	2.378	2.123	2.292	2.089	2.003

Table 3.14 The width of the 95% confidence intervals of each six variance estimator when the true correlation structure for the binary response is EX(0.7) and $m=6$.

	IN	EX	AR-1	TOEP	UN
<i>n=20</i>					
MB	3.786	2.498	1.903	2.281	1.869
LZ	3.652	2.337	2.790	2.309	2.035
MD	4.021	2.490	3.002	2.494	2.245
KC	3.831	2.412	2.893	2.361	2.076
PAN	3.797	2.471	2.966	2.426	2.141
WL	3.984	2.517	3.035	2.495	2.249
<i>n=40</i>					
MB	2.625	1.684	1.229	1.587	1.378
LZ	2.615	1.634	1.944	1.593	1.411
MD	2.744	1.686	2.016	1.647	1.465
KC	2.679	1.660	1.979	1.620	1.429
PAN	2.671	1.689	2.011	1.640	1.459
WL	2.735	1.704	2.034	1.657	1.479
<i>n=50</i>					
MB	2.339	1.510	1.099	1.444	1.289
LZ	2.364	1.479	1.747	1.436	1.306
MD	2.457	1.517	1.800	1.475	1.344
KC	2.410	1.498	1.773	1.455	1.325
PAN	2.383	1.498	1.783	1.458	1.312
WL	2.429	1.508	1.799	1.469	1.324

Table 3.15 The width of the 95% confidence intervals of each six variance estimator when the true correlation structure for the binary response is TOEP and $m=4$.

	IN	EX	AR-1	TOEP	UN
<i>n=20</i>					
MB	4.714	4.268	4.180	3.546	3.409
LZ	4.498	4.056	4.367	3.437	3.291
MD	4.930	4.379	4.743	3.750	3.605
KC	4.707	4.213	4.549	3.602	3.452
PAN	4.658	4.205	4.550	3.589	3.428
WL	4.851	4.312	4.687	3.707	3.549
<i>n=40</i>					
MB	3.228	2.932	2.873	2.513	2.466
LZ	3.175	2.859	3.096	2.470	2.412
MD	3.324	2.970	3.226	2.576	2.517
KC	3.249	2.914	3.160	2.523	2.465
PAN	3.228	2.900	3.153	2.537	2.476
WL	3.293	2.933	3.196	2.573	2.512
<i>n=50</i>					
MB	2.873	2.602	2.544	2.267	2.232
LZ	2.827	2.542	2.761	2.235	2.195
MD	2.932	2.620	2.853	2.312	2.270
KC	2.879	2.581	2.807	2.273	2.233
PAN	2.871	2.593	2.815	2.267	2.227
WL	2.916	2.617	2.845	2.293	2.253

Table 3.16 The width of the 95% confidence intervals of each six variance estimator when the true correlation structure for the binary response is TOEP and $m=6$.

	IN	EX	AR-1	TOEP	UN
<i>n=20</i>					
MB	3.765	3.340	3.077	2.574	2.225
LZ	3.624	3.185	3.627	2.639	2.334
MD	3.938	3.403	3.899	2.851	2.550
KC	3.777	3.291	3.760	2.727	2.379
PAN	3.755	3.295	3.744	2.731	2.459
WL	3.887	3.353	3.825	2.797	2.551
<i>n=40</i>					
MB	2.614	2.303	2.066	1.823	1.674
LZ	2.593	2.258	2.613	1.833	1.699
MD	2.704	2.333	2.708	1.901	1.765
KC	2.648	2.295	2.660	1.867	1.727
PAN	2.631	2.306	2.656	1.886	1.742
WL	2.675	2.324	2.681	1.905	1.763
<i>n=50</i>					
MB	2.335	2.064	1.863	1.648	1.540
LZ	2.320	2.018	2.326	1.638	1.544
MD	2.399	2.071	2.394	1.685	1.592
KC	2.359	2.044	2.360	1.661	1.564
PAN	2.349	2.049	2.369	1.679	1.583
WL	2.381	2.062	2.387	1.692	1.597

Table 3.17 The width of the 95% confidence intervals of each six variance estimator when the true correlation structure for the binary response is UN and $m=4$.

	IN	EX	AR-1	TOEP	UN
<i>n=20</i>					
MB	4.705	3.967	3.591	3.461	2.763
LZ	4.485	3.731	3.388	3.382	2.569
MD	4.939	4.025	3.667	3.676	2.814
KC	4.705	3.874	3.523	3.524	2.677
PAN	4.658	4.205	4.550	3.589	3.428
WL	4.851	4.312	4.687	3.707	3.549
<i>n=40</i>					
MB	3.241	2.737	2.481	2.436	2.002
LZ	3.190	2.656	2.417	2.426	1.958
MD	3.347	2.758	2.514	2.527	2.046
KC	3.267	2.706	2.464	2.475	2.003
PAN	3.228	2.900	3.153	2.537	2.476
WL	3.293	2.933	3.196	2.573	2.512
<i>n=50</i>					
MB	2.879	2.418	2.187	2.165	1.784
LZ	2.846	2.369	2.154	2.162	1.777
MD	2.957	2.441	2.224	2.233	1.841
KC	2.901	2.405	2.189	2.197	1.806
PAN	2.871	2.593	2.815	2.267	2.227
WL	2.916	2.617	2.845	2.293	2.253

Table 3.18 The width of the 95% confidence intervals of each six variance estimator when the true correlation structure for the binary response is UN and $m=6$.

	IN	EX	AR-1	TOEP	UN
<i>n=20</i>					
MB	3.770	3.413	3.027	3.051	2.281
LZ	3.650	3.273	3.094	2.963	2.277
MD	3.962	3.499	3.320	3.179	2.487
KC	3.802	3.384	3.204	3.068	2.325
PAN	3.755	3.295	3.744	2.731	2.459
WL	3.887	3.353	3.825	2.797	2.551
<i>n=40</i>					
MB	2.620	2.363	2.082	2.154	1.665
LZ	2.596	2.305	2.183	2.122	1.687
MD	2.705	2.382	2.262	2.197	1.757
KC	2.650	2.343	2.222	2.159	1.709
PAN	2.631	2.306	2.656	1.886	1.742
WL	2.675	2.324	2.681	1.905	1.763
<i>n=50</i>					
MB	2.330	2.105	1.858	1.927	1.528
LZ	2.313	2.062	1.952	1.903	1.527
MD	2.390	2.117	2.008	1.955	1.576
KC	2.351	2.089	1.980	1.929	1.548
PAN	2.349	2.049	2.369	1.679	1.583
WL	2.381	2.062	2.387	1.692	1.597

3.3.2.5 Summary

In summary, a misspecified working correlation structure, in a small to moderate sample size setting, can be expected to impact the inferences made on regression parameters. Even though the bias-corrected sandwich variance estimators perform better than the sandwich variance estimator by Liang and Zeger, they are still influenced by the misspecified working correlation structures, in a manner that cannot be overcome by larger sample sizes.

An under-specified structure leads to poorly estimated and inflated variance estimate. An over-parameterized structure results in negatively-biased variance estimators, which in turn result in under-coverage of confidence intervals and liberal hypothesis tests on the regression parameters. Use of the correct correlation structure is the best choice in terms of the bias and the coverage probability, provided the true correlation structure is not too complicated for the available sample size. If the true correlation structure is quite complicated and the sample size is small, the analyst may need to compromise by picking a reasonable, but more parsimonious working correlation structure in order to achieve better bias and coverage probability properties. If the analyst has the ability to make an informed choice on the structure of the working correlation in a small sample setting – using either statistical tools or prior knowledge – one can expect the Liang and Zeger’s model-based variance estimator to work better than corresponding sandwich estimators. This is because the sandwich variance estimators are much more variable than the model-based variance estimators when the number of independent clusters is small and that variability affects inferences on the regression parameters.

CHAPTER 4

AN OVERVIEW OF WORKING CORRELATION STRUCTURE SELECTION CRITERIA

In this chapter, a literature review on the selection criteria for the working correlation structure is presented.

4.1 QIC Criterion

Model selection is aimed at finding the model that is nearest to the true model from a set of potential candidate models. For many model selection criterion measures, the best fitting candidate model is chosen based on an expected discrepancy measure that gauges the separation between the true model and a candidate model. The Akaike Information Criterion (AIC), perhaps the most well-known model selection criterion, is derived as an estimator of the expected Kullback-Leibler discrepancy between the true model and a candidate model. The Bayesian Information Criterion (BIC) provides a large-sample estimator of a transformation of the Bayesian posterior probability associated with the approximating model. Traditional model selection criteria such as AIC or BIC, however, cannot be directly applied for correlation structure selection in GEE because they are likelihood-based and full multivariate likelihoods are not expressed or used in GEE estimation. Instead, the estimation is based (in part) on quasi-likelihood. While defining the full multivariate likelihood is tractable under the assumption of independent responses and in some particular multivariate data settings, but is often not so in the general multivariate, correlated data setting.

Pan (2001) proposed a modification of AIC, called ‘quasi-log-likelihood under the independence model information criterion’ (QIC). The QIC was constructed by replacing the likelihood in the Kullback-Leibler information with the quasi-likelihood under the working independence assumption. The quasi-likelihood approach is natural in this setting because the quasi-likelihood estimating equations have the same form as the

maximum likelihood estimating equations in the GLM-type models when the canonical link is used. If each observation is independent, the quasi-likelihood is the sum of quasi-score function for each observation. Due to the lack of a general, tractable, multivariate analogue of the quasi-likelihood for correlated response data under any general working correlation structure, the quasi-likelihood in QIC was constructed under the independence working correlation assumption.

To derive the QIC, and in recognition of the lack of a full joint likelihood in the GEE setting, the Kullback-Leibler information based on the quasi-likelihood under the working independence model was used. Let M_C be a candidate model and M_T be the true model. The Kullback-Leibler discrepancy between M_C and M_T using the quasi-likelihood under the working independence model is

$$\Delta(\hat{\beta}, \beta_T, I) = E_{M_T} [-2Q(\beta; I, \Delta)], \quad (4.1)$$

where $Q(\beta; I, \Delta)$ is the quasi-likelihood under independence assumption of the dataset Δ and the expectation E_{M_T} is taken under the true model M_T . The underlying parameter vectors of M_C and M_T are β and β_T , respectively. The estimate, $\hat{\beta} = \hat{\beta}(R)$ is the estimated regression parameter vector under the working correlation matrix R of the candidate model M_C . The expectation of this Kullback-Leibler discrepancy can be approximated using Taylor's expansion as

$$\begin{aligned} E_{M_T} [\Delta(\hat{\beta}, \beta_T, I)] \approx \\ -2E_{M_T} [Q(\hat{\beta}; I, \Delta)] + 2E_{M_T} [(\hat{\beta} - \beta_T)' S(\hat{\beta}; I, \Delta)] + 2tr [\Sigma_{M(IN)}^{-1} \Sigma_{S(R)}] \end{aligned} \quad (4.2)$$

where $S(\hat{\beta}; I, \Delta) = \partial Q(\beta; I, \Delta) / \partial \beta|_{\beta=\hat{\beta}}$ is a quasi-likelihood score equation under independence assumption. By ignoring the second term, which would be difficult to obtain, the QIC for GEE is defined as

$$\text{QIC}(R) = -2Q[\hat{\beta}(R); I, \Delta] + 2tr [\hat{\Sigma}_{M(IN)}^{-1} \hat{\Sigma}_{S(R)}], \quad (4.3)$$

where $\hat{\Sigma}_{M(IN)}$ is the model-based variance estimator under independence working correlation structure and $\hat{\Sigma}_{S(R)}$ is the sandwich variance estimator under the working correlation structure R . However, the second term, $2E_{M_T}[(\hat{\beta} - \beta_T)'S(\hat{\beta}; I, \Delta)]$, is not necessarily negligible. It disappears only if $\hat{\beta} = \hat{\beta}(I)$, since in that case, $S(\hat{\beta}(I); I, \Delta) = 0$. Pan (2001) showed in his somewhat limited simulation studies that ignoring the second term does influence the performance of QIC, but not dramatically, within his restricted simulation settings.

Hardin and Hilbe (2003) have suggested a slight modification of QIC. They noted that the parameter estimates $\hat{\beta} = \hat{\beta}(R)$ and the nuisance parameters $\hat{\phi} = \hat{\phi}(R)$ in QIC are calculated under the hypothesized working correlation structure R , while the quasi-likelihood is constructed under an assumed independence structure. They suggested the use of the estimates of parameters under independence assumption (i.e., $\hat{\beta}(I)$ and $\hat{\phi}(I)$ instead of $\hat{\beta}(R)$ and $\hat{\phi}(R)$) in QIC, which provides more stability, and can be justified since an incorrectly-specified working correlation structure tends to have little impact on the β parameter estimates.

$$\text{QIC}(R)_{HH} = -2Q[\hat{\beta}(I); I, \Delta] + 2tr\left[\hat{\Sigma}_{M(IN)}^{-1}\hat{\Sigma}_{S(R)}\right] \quad (4.4)$$

It should be noted that under certain conditions, QIC and QIC_{HH} have the same values for independence (IN) and exchangeable (EX) structures; hence, they cannot distinguish IN from EX. Barnett et al. (2010) pointed out that if the covariate matrix X does not contain at least one covariate that is time-dependent and one that is cluster-specific, then the $\hat{\Sigma}_{S(IN)}$ is identical to $\hat{\Sigma}_{S(EX)}$. This is because cancellation of the terms involving $\partial\hat{\mu}_i / \partial\hat{\beta}$ in the equation of the sandwich variance estimator leads to both covariance structures forming the same regression parameter estimates.

Either version of the QIC can be used to compare models with different mean and/or covariance structures, in the same spirit as AIC. The QIC measures tend to be more sensitive to changes in the mean structure than changes in the covariance structure,

for reasons that are described in the next section on the another selection measure called CIC.

4.2 CIC Criterion

Hin and Wang (2009) proposed using half of the second term in QIC alone for the selection of the working correlation structure selection in GEE. This statistic is called the CIC (Correlation Information Criterion).

$$\text{CIC} = tr \left[\hat{\Sigma}_{S(R)} \hat{\Sigma}_{M(IN)}^{-1} \right] \quad (4.5)$$

The first term in QIC, which is based on the quasi-likelihood, is free from both the working correlation structure as well as the true correlation structure, so it would not be informative in the selection of the covariance structure. Moreover, the form of quasi-likelihood is constructed under the assumption of the independent observations, although the parameters are estimated under the hypothesized working correlation structure. On the other hand, the second term in QIC contains information about the hypothesized correlation structure via the sandwich variance estimator. Even though the second term plays a role as a penalty term for mean model variable selection, the QIC is more heavily impacted by the first term. Hence, QIC is not a particularly sensitive measure to use for working correlation structure selection.

Hin and Wang (2009) and Wang and Hin (2010) showed the performance of CIC to be much better than the QIC in selecting the correct correlation structure. However, it should be noted that the scale parameter ϕ was fixed in their simulation settings. The scale parameter is usually estimated by

$$\hat{\phi} = \frac{1}{N - p} \sum_{i=1}^n \sum_{j=1}^{m_i} (y_{ij} - \hat{y}_{ij})^2 \quad (4.6)$$

where $N = \sum_{i=1}^n m_i$ is the total number of measurements and p is the number of regression parameters. In the case of Gaussian responses, if the scale parameter is

estimated differently in each candidate model, QIC becomes $(N-p)+2CIC$. In that case, both CIC and QIC will select the same working correlation structure.

Still, CIC has one primary limitation. Hin and Wang (2009) recognized that comparison of two working correlation matrices may not be advisable when the numbers of correlation parameters in each are quite different. The CIC does not include a penalty term that increases with the number of correlation parameters, and thus cannot penalize for over-parameterization.

4.3 SC Criterion

Shults and Chaganty (1998) proposed a criterion (denoted SC) that chooses the working correlation structure that minimizes the generalized error sum of squares. The SC criterion minimizes the weighted error sums of squares, where the weight is the inverse covariance matrix. Generalized error sum of square is

$$\begin{aligned}\Theta(\alpha, \beta) &= \sum_{i=1}^n [Y_i - E(Y_i)]' \text{cov}(Y_i)^{-1} [Y_i - E(Y_i)] \\ &= \sum_{i=1}^n Z_i'(\beta) R_i^{-1}(\alpha) Z_i(\beta),\end{aligned}\tag{4.7}$$

where $Z_i(\beta) = A_i^{1/2}[Y_i - E(Y_i)]$ and $E(Y_i)$ depends on β . The SC criterion is defined as the estimated adjusted/averaged residual generalized sum of squares

$$SC = \Theta(\hat{\alpha}, \hat{\beta}) / (N - p - q)\tag{4.8}$$

where $N = \sum_{i=1}^n m_i$ is the total number of measurements, p is the number of regression parameters and q is the number of parameters in the working correlation structure.

4.4 RJ Criterion

If both the mean and covariance models in GEE are specified correctly, one could expect $\hat{\Sigma}_{M(R)}$ and $\hat{\Sigma}_{S(R)}$ to be almost identical with reasonably large sample sizes.

Rotnitzky and Jewell (1990) used this as motivation and suggested so-called RJ criterion to be used for working correlation structure. Let $\Omega = \hat{\Sigma}_{S(R)} \hat{\Sigma}_{M(R)}^{-1}$, where $\hat{\Sigma}_{M(R)}$ and $\hat{\Sigma}_{S(R)}$ are estimated $p \times p$ covariance matrices. If the working correlation structure is

correctly specified, Ω should be close to an identity matrix, and hence their proposed measures RJ1 and RJ2 should be close to one, where

$$RJ1 = \text{tr}(\hat{\Sigma}_{S(R)} \hat{\Sigma}_{M(R)}^{-1}) / p = \text{tr}(\Omega) / p \quad (4.9)$$

and

$$RJ2 = \text{tr}(\Omega^2) / p. \quad (4.10)$$

Rotnitzky and Jewell provided another measure using RJ1 and RJ2:

$$RJ3 = \sqrt{(\text{RJ1} - 1)^2 + (\text{RJ2} - 1)^2}. \quad (4.11)$$

All three criterion measures, RJ1, RJ2 and RJ3, can be used for working correlation structure selection. In this dissertation, what is termed the “RJ criterion” is that described in equation (4.9).

4.5 Comparisons among QIC, CIC, SC, and RJ Criteria

Working correlation structure selection criteria in GEE introduced so far have been compared to each other in several papers. To investigate the performance of the proposed criteria as compared to others, simulation studies were conducted under different settings and restrictions, which are summarized below.

4.5.1 AIC vs. QIC

Pan (2001) compared the performance of QIC with AIC in selecting the working correlation structure in a marginal logistic regression under the true exchangeable correlation structure. Only independence (IN), exchangeable (EX), and autoregressive of order 1 (AR-1) structures were considered as candidate models in a simulation study with 1000 replications. The percent correctly identified correlations structures were 67.8% versus 83.6% (QIC versus AIC) when $n=50$, and 72.1% versus 94.6% (QIC vs. AIC) when $n=100$. Two reasons why AIC was more efficient than QIC for working correlation selection were discussed. First, the MLE of β is more efficient than the GEE

estimator. Second, information on the true correlation structure is embedded in the likelihood function in AIC but not directly in the quasi-likelihood in QIC. Perhaps it is obvious that the maximum likelihood approach would be more efficient than GEE, but it is useful to see how much accuracy is lost when quasi-likelihood is used.

Barnett et al. (2010) compared the performance of AIC and QIC for multivariate Gaussian responses. They considered a relatively small sample size ($n=30$) and a relatively large cluster size ($m=8$). For exchangeable correlation, two different within-cluster correlation levels were considered: $\alpha = 0.2$ and $\alpha = 0.5$. For the first-order autoregressive correlation, different α values were used: $\alpha = 0.3$ and $\alpha = 0.7$. Direct comparison of AIC and QIC is limited in their simulation study, since different models with different types of estimation were used to obtain the AIC and QIC. Mixed models using the maximum likelihood estimation were fitted to obtain AIC. Marginal models and GEE estimation were used for QIC. Nevertheless, it is worth commenting on their results because they included the unstructured correlation structure (R_{UN}) in the candidate models along with IN, AR-1 and EX, which Pan (2001) and others did not.

Their simulation results showed that QIC did particularly poorly when the true covariance structure was independence or had a weak exchangeable ($\alpha=0.2$) or autoregressive ($\alpha=0.3$) structure (0~14% correct). The QIC was strongly biased towards selecting R_{UN} . This highlights the fact that the QIC does not penalize for the added complexity of the $m(m-1)/2$ parameters required for R_{UN} , so of course a more flexible correlation structure could fit the data better. If we have large enough sample size to estimate the parameters in UN working correlation structure as well as mean parameters precisely, over-parameterization would not be a computational problem, although it would still lead to a negative bias of variance parameters. However, with small sample size and large cluster size, over-parameterization can lead to a problem fitting the model.

4.5.2 RJ vs. QIC

Hin et al. (2007) compared RJ with QIC to identify the true correlation structure via simulations with Gaussian and binary response data, covariates varying at the cluster or observation level, and using only the EX or AR-1 intraclass correlation structures. They evaluated the performance of these two criteria based on the frequency of selecting the correct structure among IN, EX and AR-1 out of 10,000 simulations. Each independent replication contained 100 balanced clusters of size 5. Their simulation studies show that RJ has high sensitivity in terms of identifying EX, but low specificity. When the true correlation structure is EX, the correct identification rates for RJ range from 84% to 100%. However, when AR-1 is the true correlation, RJ still selects EX 50-71% of the time. When compared to RJ, QIC appears to be a more robust criterion in this setting. When the true correlation is EX, the correct identification rates for QIC range from 65% to 77%. When the true correlation is AR-1, the correct identification rates for QIC range from 64% to 81%. Hin et al. (2007) concluded that neither QIC nor RJ can be regarded as a dominant criterion for working correlation structure selection in GEE. We note that these comparisons were again only made among candidate correlation structures with none or only one α correlation parameter.

4.5.3 CIC vs. QIC

In a simulation study of Hin and Wang (2009), CIC (and similarly, CIC_{HH}) showed remarkable improvement over QIC in selecting the correct correlation structure. They considered $m=5$ observations per subject and simulated data for $n=30$ and $n=100$ subjects. Through 1000 independent simulations, the performance of QIC and CIC were assessed. Two true correlation structures, EX and AR-1, were considered, with moderate pairwise correlation $\alpha = 0.4$. The IN, EX, and AR-1 were the candidate structures. The dispersion parameter ϕ was held fixed (hence not estimated by (2.2)) for all models.

When the true correlation was EX, the correct identification rates for QIC and CIC were 62-72% and 81-96%, respectively. When the true correlation was AR-1, the correct identification rates for QIC and CIC were 57-73% and 82-97%, respectively. The QIC_{HH} criterion performance was similar to QIC.

The improvement in working correlation structure of the CIC over the QIC can be easily explained. Regardless of the working correlation structure, one can expect the GEE estimates of the β parameters to be similar. The first term of the QIC, based on the quasi-likelihood under independence and those similar β estimates, cannot help distinguish between competing correlation models, but rather only makes the QIC more variable than the CIC.

Wang and Hin (2010) also compared CIC to QIC under different simulation settings. Gaussian responses with fixed dispersion parameter and lognormal responses were considered. Each simulation dataset contained 100 balanced clusters ($n=100$) of size $m=5$, and 1000 independent replications. When EX was the true correlation, the correct identification rates were approximately 62-76% for QIC and 81-96% for CIC. When AR-1 was the true correlation, the correct identification rates were 64-70% for QIC and 87-99.9% for CIC. The set of candidate models contained IN, EX, and AR-1 structures only, again avoiding the issue of comparing correlation structures among candidate models with varying numbers of correlation parameters.

4.5.4 SC vs. RJ

Shults et al. (2009) compared the performance of SC criterion with $|RJ1-1|$ (equation (4.9)), $|RJ2-1|$ (equation (4.10)), and $DBAR=RJ1-2RJ1+1$ for binary responses. They considered the multivariate binary distribution with different sample sizes ($n=20, 40, 60, 80, 160$) and a larger cluster size ($m=8$) than other simulation studies. As candidate models, AR-1, EX, and tri-diagonal structure (3-dependent structure) were considered. They also provided two rule-out criteria to exclude any structure that

violated standard constraints for binary data. Overall, the performances of |RJ1-1|, |RJ2-1| and DBAR were better than that of the SC criterion. The detection rate of SC was at most 40% and the overall performance of SC was much weaker when compared to other three criteria. In their discussion section, they mentioned that it is perhaps not surprising that the SC model performed poorly. The residual is a function of $\hat{\beta}$ and we know that $\hat{\beta}$ is robust to the choice of the working correlation pattern, in the sense that $\hat{\beta}$ will be typically be estimated consistently, even when the working correlation structure is misspecified. Thus, any discriminating power for $\text{cov}(\hat{\beta})$ was, in a sense, ‘washed out’ by the lack of discriminating power in $\hat{\beta}$ and therefore in the residual.

4.5.5 Summary

Several existing criteria for identification of a working correlation structure in GEE were described in the previous sections. The RJ criterion has high sensitivity in identifying exchangeable correlation structure but it has low specificity, which limits its usefulness, even in comparisons of structures with only one parameter (e.g., exchangeable and AR-1). The SC criterion performs even worse than the RJ criterion in ability to correctly identify the underlying true covariance structure. As discussed in Shults et al. (2009), it is function of $\hat{\beta}$, which is robust to the misspecified working correlation structure. This robustness property tends to make the SC criterion insensitive to the choice of working correlation structure.

The QIC criterion generally performs better than the SC and RJ criterion in identifying the working correlation structure. However, similar to the SC criterion, the quasi-likelihood under independence assumption ($Q(\hat{\beta}(R); I, \Phi)$), which is the first term in QIC, dilutes the impact of different working correlation structures on this measure, while retaining sensitivity to differences in the mean structures of competing models. It was shown by Hin and Wang (2009) that excluding the first term in QIC improved the correct identification rates of the covariance structure. However, most simulation studies

presented in the literature (including Hin and Wang, 2009) avoided overparameterized working correlation structures as candidate models. Barnett et al. (2010) showed the QIC criterion performed very poor when an unstructured correlation structure was included in a set of candidate models. No studies to date have examined the performance of the CIC when overparameterized structures are included among the candidate covariance models, but without some sort of penalization for over-parameterization, it would not be expected to perform well.

CHAPTER 5

GENERALIZED EIGENVALUES AND APPLICATIONS

In this dissertation, new selection criteria for the working correlation structure are developed based on generalized eigenvalues. In this chapter, the concept and properties of generalized eigenvalues are presented. Generalized eigenvalues have been used to define hypothesis test statistics in the classical MANOVA setting, which is discussed. We also introduce how they can be used to define working correlation structure selection criteria.

5.1 Generalized Eigenvalues and Eigenvectors

Generalized eigenvalues and eigenvectors are an extended version of the (ordinary) eigenvalues and eigenvectors. After the definition and properties of the ordinary eigenvalues and eigenvectors are reviewed, the generalized eigenvalues and eigenvectors are introduced.

5.1.1 Ordinary Eigenvalues and Eigenvectors

Measures that capture different aspects of a matrix can be very useful when comparing one matrix to another. In this section, common measures of matrix properties are described and placed in the context of this research.

5.1.1.1 Ordinary Eigenvalues and Eigenvectors

Eigenvalues and eigenvectors are properties of a matrix that describe orientation and spread within a multidimensional space. Let A be a $p \times p$ matrix and let x be a $p \times 1$ non-null vector such that $Ax = \lambda x$. Then the scalar λ is said to be an eigenvalue (or characteristic value) of the matrix A and x is its corresponding eigenvector (or characteristic vector). One can think of the matrix A as a way to change the magnitude and direction of the vector x , resulting in the transformed vector $y = Ax$. When the vector x is an eigenvector of the matrix A , the expression $Ax = \lambda x$ implies that the direction of x is

either unchanged or switched to the opposite direction (rotated 180°). The eigenvalue λ is the amount of stretch or shrink of the eigenvector (x) when transformed by the matrix A . In general, the eigenvalues of real matrix A can be real, complex, or both. But when the matrix A is a covariance matrix, it must be a symmetric positive definite matrix. In that case, the number of non-zero eigenvalues will be p , the rank of the matrix A , and all p eigenvalues must be positive real numbers.

5.1.1.2 Trace of a Matrix

In the context of this research, we are interested in the $p \times p$ covariance matrix Σ of $\hat{\beta} = (\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_p)'$, which is a symmetric positive-definite matrix whose $(i, j)^{\text{th}}$ element is the covariance between $\hat{\beta}_i$ and $\hat{\beta}_j$. Eigenvalues and eigenvectors of the covariance matrix give important information regarding the variability of $\hat{\beta}$. Denote the eigenvalue-eigenvector pairs of Σ as $(\lambda_1, x_1), (\lambda_2, x_2), \dots, (\lambda_p, x_p)$, ordered so that $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p > 0$. The sum of eigenvalues of Σ , is called the trace of Σ (denoted $tr(\Sigma)$). Each eigenvalue explains a portion of the total variance, so in that sense, the sum of the eigenvalues can be considered a measure of total variability or spread in multidimensional space. As the largest eigenvalue, λ_1 captures the highest proportion of the total variance, λ_2 has the second largest proportion, and λ_p has the minimum proportion of the total variance.

5.1.1.3 Determinant of a Matrix

The product of the eigenvalues of Σ , by definition, is the determinant of Σ , denoted $|\Sigma|$. The determinant is a measure of spread in a covariance matrix that is sometimes called the *generalized variance* — a term used in the context of normal distributions by Wilks (1934). Geometrically, the determinant of 2×2 covariance matrix is the squared area of a parallelogram whose lengths of the sides are the standard deviations for the variables.

Consider 2×2 sample covariance matrix

$$\hat{\Sigma} = \begin{pmatrix} s_1^2 & s_{12} \\ s_{12} & s_2^2 \end{pmatrix}.$$

s_1^2 and s_2^2 are the sample variance of variable 1 and variable 2, respectively and s_{12} is the sample covariance for the two variables. Then, the sample correlation coefficient is $r_{12} = s_{12}/(s_1 \cdot s_2)$. The determinant of the matrix $|\hat{\Sigma}| = s_1^2 s_2^2 - s_{12}^2$. The cosine of the angle between the sides of the parallelogram is equal to the correlation between the variables (Gentle, 2007: p.37): that is, $\cos \theta = r_{12}$. The area of the parallelogram is

$$Area = s_2 \cdot h = s_2 \cdot s_1 \sqrt{1 - \cos^2 \theta} = s_1 \cdot s_2 \sqrt{1 - r_{12}^2} = \sqrt{s_1^2 s_2^2 - s_{12}^2} = |\hat{\Sigma}|^{1/2}.$$

If two variables are uncorrelated, the angle between two sides of the parallelogram is 90° and the area is maximized. That is, we have the largest determinant (generalized variance). As the covariance between two variables increases, the angle goes down from 90° and the area gets smaller. In other words, the generalized variance gets smaller as two variables are highly correlated. Similarly, the determinant in a higher dimension of covariance matrix explains the extent of the spread of the variables.

5.1.2 Generalized Eigenvalues and Eigenvectors

In Section 5.1.1 above, eigenvalues and eigenvectors were defined to describe a single matrix, which will now be extended to the comparison of two covariance matrices. As the eigenvalues and eigenvectors yield important information for one variance matrix, generalized eigenvalues (or *relative* eigenvalues) and generalized eigenvectors (or *relative* eigenvectors) provide important information on the comparison of two variance matrices (Bock, 1975; Schott, 2005; Gentle, 2007; Seber, 2007).

Definition:

Suppose A and B are $p \times p$ matrices, u is a $p \times 1$ vector, and λ is a scalar such that

$$Au = \lambda Bu.$$

Then u is a generalized eigenvector of A *with respect to* B (eigenvector of A *in the metric of* B or eigenvector of A *relative to* B) with a generalized eigenvalue λ (eigenvalue of A *in the metric of* B or eigenvalue of A *relative to* B).

Simply, if B is the identity matrix, the generalized eigenvalues of A with respect to B are equal to the ordinary eigenvalues of A . Here, we consider only the case where both A and B are positive definite, as we are concerned with comparing covariance matrices. Several useful properties regarding generalized eigenvalues are presented below. If λ and u are a generalized eigenvalue and generalized eigenvector of A relative to B , then

- (a) λ and u are an ordinary eigenvalue and eigenvector of $B^{-1}A$;
- (b) $(1/\lambda)$ and u are an ordinary eigenvalue and eigenvector of $A^{-1}B$;
- (c) λ and $x \equiv Bu$ are an ordinary eigenvalue and eigenvector of AB^{-1} ;
- (d) $1+\lambda$ is a generalized eigenvalue of $A+B$ with respect to B , with a generalized eigenvector $x \equiv Bu$;
- (e) $\lambda/(1+\lambda)$ is a generalized eigenvalue of A with respect to $A+B$ with corresponding generalized eigenvector u ; and
- (f) $1/(1+\lambda)$ is a generalized eigenvalue of B with respect to $A+B$ with corresponding generalized eigenvector $x \equiv Bu$.

To summarize, let A and B be $p \times p$ covariance matrices of $p \times 1$ random vectors X and Y , respectively. A set of ordinary eigenvalues of one covariance matrix contains information about the variability of the components of those random vectors. The sum of the eigenvalues is a measure of the total variance. The product of the eigenvalue is called the generalized variance and it is a measure of spread. Likewise, a set of generalized eigenvalues of A with respect to B captures multidimensional spread of the points X relative to the variability of the points Y in p -dimensions (Schott, 2005). If two matrices

are comparable to each other, all generalized eigenvalues of A with respect to B are close to 1.

If A is a covariance matrix, the ordinary eigenvalues of A can be thought of as a measure of ‘size’. Likewise, if both A and B are covariance matrices, the generalized eigenvalues measure the ‘size’ of A relative to B (Thisted, 1988). Note that A could have a smaller variance in one direction, but a large variance in another. So, ‘size’ should be interpreted as a measure of spread which captures all dimensions simultaneously.

5.1.3 Use of Generalized Eigenvalues in Multivariate

Analysis of Variance

Many problems of classical multivariate statistics involve generalized eigenvalues. In multivariate analysis of variance (MANOVA), test statistics such as Pillai’s trace, Hotelling-Lawley trace, Wilks’ lambda, and Roy’s maximum root are based on the generalized eigenvalues of two matrices. Following traditional notation in normal-theory based multivariate analysis of variance, let Y_{hij} denote the response at time j from the i th subject in group h , for $h=1,2,\dots,g$; $i=1,2,\dots,n_h$; and $j=1,\dots,m$. The observed data vectors $Y_{hi} = (Y_{hi1}, Y_{hi2}, \dots, Y_{him})'$ are assumed to be independent and normally distributed with mean $\mu_h = (\mu_{h1}, \dots, \mu_{hm})' = \mu + \tau_h$ and common covariance matrix Σ , $Y_{hi} \sim N_i(\mu + \tau_h, \Sigma)$. Here, μ is an overall mean vector and τ_h represents the h^{th} group effect – that is the difference between the mean vector in the h^{th} group and the grand mean μ --with $\sum_{h=1}^g n_h \tau_h = 0$. The MANOVA model for comparing g population mean vectors is specified as (Johnson and Wichern, 2007):

$$Y_{hi} = \mu + \tau_h + e_{hi} \quad i = 1, 2, \dots, n_h \quad \text{and} \quad h = 1, 2, \dots, g \quad (5.1)$$

where e_{hi} are independent $N_m(0, \Sigma)$.

A vector of observations can be decomposed as $y_{hi} = \bar{y} + (\bar{y}_h - \bar{y}) + (y_{hi} - \bar{y}_h)$, where $\bar{y}$ is the overall sample average vector and $\bar{y}_h$ is the sample average within the h^{th}

group. This decomposition leads to the multivariate analog of the univariate sum of squares decomposition as follows:

$$\sum_{h=1}^g \sum_{i=1}^{n_h} (y_{hi} - \bar{y})(y_{hi} - \bar{y})' = \sum_{h=1}^g n_h (\bar{y}_h - \bar{y})(\bar{y}_h - \bar{y})' + \sum_{h=1}^g \sum_{i=1}^{n_h} (y_{hi} - \bar{y}_h)(y_{hi} - \bar{y}_h)' \quad (5.2)$$

$$\left(\begin{array}{l} \text{total sum of squares and} \\ \text{cross products} \end{array} \right) \quad \left(\begin{array}{l} \text{Hypothesis (Between)} \\ \text{sum of squares and} \\ \text{cross products} \equiv H \end{array} \right) \quad \left(\begin{array}{l} \text{Error (Within)} \\ \text{sum of squares and} \\ \text{cross products} \equiv E \end{array} \right)$$

The hypothesis of no group effects, $H_0 : \tau_1 = \tau_2 = \dots = \tau_g = 0$ is tested by comparing relative sizes of the hypothesis (H) and error (E) sum of squares and cross products. Let λ_j be the generalized eigenvalue of H with respect to E . Both H and E are $t \times t$ symmetric matrices, which can be shown to be independent with Wishart distributions. Then, $\theta_j = \lambda_j / (1 + \lambda_j)$ is the generalized eigenvalue of H with respect to $(H+E)$ and $1 - \theta_j = 1 / (1 + \lambda_j)$ is the generalized eigenvalue of E with respect to $(H+E)$, for $j=1, 2, \dots, m$, where m is the number of repeated measurements on each individual i . In this setting, there is no uniformly most powerful test for comparing response means. The four test statistics below, based on the generalized eigenvalues of H with respect to E and of H with respect to $H+E$, are often cited in this context.

(a) Hotelling-Lawley trace: $tr[HE^{-1}] = \sum_{j=1}^t \lambda_j = \sum_{j=1}^t \theta_j / (1 - \theta_j)$

(b) Pillai's trace: $tr[H(H+E)^{-1}] = \sum_{j=1}^t \theta_j$

(c) Wilks' Λ : $|E|/|H+E| = \prod_{j=1}^t (1 - \theta_j)$

(d) Roy's maximum root: $\max\{\theta_j\}$

All these statistics can be considered as overall measures of the size of H relative to E (or relative to $(H+E)$). Large generalized eigenvalues of H with respect to E provide evidence against the null hypothesis of no group differences. So, the null hypothesis of no group differences is rejected for small values of Wilks' Λ and large values for other three test statistics. Under the normality assumption, the test statistics

can be transformed to have either an exact (under certain conditions) or approximate central F distributions under the null hypothesis of equal mean vectors (with common variance matrix) and non-central F under the alternative of non-equal mean vectors (Seber, 1984).

5.2 The Existing Working Correlation Selection Criteria Based on Generalized Eigenvalues

5.2.1 Motivation for Using Generalized Eigenvalues on the Working Correlation Selection

When choosing both a mean and variance structure in a multivariate setting, it has been suggested that an analyst find a well-fitting covariance structure first using a fully-parameterized or highly parameterized mean model and then working to find a well-fitting, more parsimonious mean model. The logic behind this suggested modeling process is that the estimated mean parameters are relatively insensitive to the working correlation structure. However, if the mean model is underspecified, the fit of the covariance matrix will be impacted by any resulting bias in the regression parameter estimates. By starting with a fully-specified (perhaps over-specified) mean model, one can focus first on choosing a correlation structure with presumably unbiased β estimates. So the first step in the process would be to concentrate only on correlation structure while holding the full mean model fixed. If we are only interested in choosing working correlation structure while holding the mean model is fixed, the working correlation selection problem can be viewed as comparison of covariance matrices under different working correlation structure assumptions. The concept of generalized eigenvalues is useful to measure the discrepancy between two matrices.

Although Rotnitzky and Jewell (1990) and Hin and Wang (2009) did not mention the generalized eigenvalues in their derivations, in fact, the RJ1 $\left(\text{tr}(\hat{\Sigma}_{S(R)} \hat{\Sigma}_{M(R)}^{-1}) / p \right)$, and CIC $\left(\text{tr}(\hat{\Sigma}_{S(R)} \hat{\Sigma}_{M(IN)}^{-1}) \right)$ compare two covariance matrices based on generalized

eigenvalues. In the next sections, the RJ and CIC criterion measures are reviewed from a perspective of the comparison of the matrices.

5.2.2 RJ Criterion

Rotnitzky and Jewell (1990) provided a modified “working” Wald statistic T_w^* using the model-based variance estimator:

$$T_w^* = (\hat{\beta}(R) - \beta_0)' \hat{\Sigma}_{M(R)}^{-1} (\hat{\beta}(R) - \beta_0). \quad (5.3)$$

Under mild regularity conditions and provided the marginal mean model is correctly specified, they showed that $T_w^* = \sum_{j=1}^p c_j \chi_j^2 + o_p(1)$, where χ_j^2 ($j=1, \dots, p$) are independent $\chi^2(1)$ random variables, $c_1 \geq c_1 \geq \dots \geq c_p$ are the eigenvalues of $Q = M_0^{-1} M_1 + o^{p \times p}(1)$.

Since $M_0^{-1} M_1 = M_0^{-1} M_1 M_0^{-1} (M_0^{-1})^{-1} = \Sigma_{S(R)} \Sigma_{M(R)}^{-1}$, the weights c_j ($j=1, \dots, p$) can be regarded as the generalized eigenvalues of the sandwich variance estimator with respect to the model based variance estimator under a specified working correlation structure. If the working correlation structure is correctly specified, $c_j=1$ for all $j=1, \dots, p$. Rotnitzky and Jewell mentioned that the weights c_j ($j=1, \dots, p$) summarize the effect on the statistical inference about β of the particular choice of the working correlation structure. The c_j s may provide information on how close the working correlation structure is to the true correlation structure by measuring distance of the c_j s from 1. Therefore, Rotnitzky and Jewell proposed selecting an adequate working correlation structure using the average of the generalized eigenvalues c_j s,

$$RJ1 = \sum_{j=1}^p c_j / p = \text{tr}(\hat{\Sigma}_{S(R)} \hat{\Sigma}_{M(R)}^{-1}) / p. \quad (5.4)$$

However, as Hin et al. (2007) and Shults et al.(2009) showed, the RJ criterion tended to select the exchangeable structure for most cases considered, which were structures that had no more than one α parameter in the working structure $R(\alpha)$. It implies that $\hat{\Sigma}_{S(EX)}$ and $\hat{\Sigma}_{M(EX)}$ are almost always close to each other regardless of the

true correlation. It is not reported in any literature yet why $\hat{\Sigma}_{S(EX)}$ and $\hat{\Sigma}_{M(EX)}$ are close to each other as compared to $\hat{\Sigma}_{S(R)}$ and $\hat{\Sigma}_{M(R)}$ under other working correlation structures R .

5.2.3 CIC Criterion

The CIC criterion $\left(tr(\hat{\Sigma}_{S(R)} \hat{\Sigma}_{M(IN)}^{-1}) \right)$ is defined as the half of the second term of the QIC. In the derivation of the QIC, the CIC term is obtained from the following relationship (Pan, 2001):

$$\begin{aligned}
 E_{M_T} \left[(\hat{\beta} - \beta_T)' \Sigma_{M(IN)}^{-1} (\hat{\beta} - \beta_T) \right] &= E_{M_T} \left[tr \left\{ (\hat{\beta} - \beta_T)' \Sigma_{M(IN)}^{-1} (\hat{\beta} - \beta_T) \right\} \right] \\
 &= E_{M_T} \left[tr \left\{ (\hat{\beta} - \beta_T) (\hat{\beta} - \beta_T)' \Sigma_{M(IN)}^{-1} \right\} \right] \\
 &= tr \left[E_{M_T} \left\{ (\hat{\beta} - \beta_T) (\hat{\beta} - \beta_T)' \Sigma_{M(IN)}^{-1} \right\} \right] \\
 &= tr \left[cov(\hat{\beta}) \Sigma_{M(IN)}^{-1} \right].
 \end{aligned} \tag{5.5}$$

$cov(\hat{\beta})$ can be consistently estimated by the sandwich variance estimator, $\hat{\Sigma}_{S(R)}$. $\hat{\Sigma}_{M(IN)}^{-1} = -\partial^2 Q(\beta; I, \Phi) / \partial \beta \partial \beta'$, which is the inverse of the model based variance estimator under $R=I$.

The expectation of the CIC, $E_{M_T} \left[(\hat{\beta} - \beta_T)' \Sigma_{M(IN)}^{-1} (\hat{\beta} - \beta_T) \right]$, can be expressed by (5.6) by adding and subtracting $\Sigma_{M(R)}^{-1}$ and $\Sigma_{M(T)}^{-1}$:

$$\begin{aligned}
 E_{M_T} \left[(\hat{\beta}(R) - \beta_T)' \Sigma_{M(IN)}^{-1} (\hat{\beta}(R) - \beta_T) \right] &= E_{M_T} \left[(\hat{\beta}(R) - \beta_T)' \Sigma_{M(R)}^{-1} (\hat{\beta}(R) - \beta_T) \right] \\
 &\quad + E_{M_T} \left[(\hat{\beta}(R) - \beta_T)' (\Sigma_{M(T)}^{-1} - \Sigma_{M(R)}^{-1}) (\hat{\beta}(R) - \beta_T) \right] \\
 &\quad + E_{M_T} \left[(\hat{\beta}(R) - \beta_T)' (\Sigma_{M(IN)}^{-1} - \Sigma_{M(T)}^{-1}) (\hat{\beta}(R) - \beta_T) \right] \\
 &= E_{M_T} \left[\sum_{k=1}^p c_k \chi_k^2 \right] \\
 &\quad + E_{M_T} \left[(\hat{\beta}(R) - \beta_T)' (\Sigma_{M(T)}^{-1} - \Sigma_{M(R)}^{-1}) (\hat{\beta}(R) - \beta_T) \right] \\
 &\quad + E_{M_T} \left[(\hat{\beta}(R) - \beta_T)' (\Sigma_{M(IN)}^{-1} - \Sigma_{M(T)}^{-1}) (\hat{\beta}(R) - \beta_T) \right].
 \end{aligned} \tag{5.6}$$

The first term $E_{M_T} \left[\sum_{k=1}^p c_k \chi_k^2 \right]$ is the expectation of the RJ criterion under the true model M_T . The third term has little impact on the working correlation structure selection since $\hat{\beta}(R)$ is robust to the misspecification to R and $\Sigma_{M(IN)}^{-1} - \Sigma_{M(T)}^{-1}$ does not depend on the

choice of R . The second term has information on how close the working correlation structure is to the true correlation structure.

If the working correlation structure is correctly specified or over-parameterized, the first term $E_{M_T} [\sum_{k=1}^p c_k \chi_k^2] = p$ since $c_k = 1$ for all $k=1, \dots, p$ and χ_k^2 ($k=1, \dots, p$) are independent $\chi^2(1)$ random variables (Rotnitzky and Jewell, 1990). The second term disappears since $\Sigma_{M(T)}^{-1} = \Sigma_{M(R)}^{-1}$. The third term becomes

$E_{M_T} [(\hat{\beta}(R_T) - \beta_T)'(\Sigma_{M(IN)}^{-1} - \Sigma_{M(T)}^{-1})(\hat{\beta}(R_T) - \beta_T)]$. Thus, the expectation of CIC, which is the left-hand side of equation (5.6), becomes

$$p + E_{M_T} [(\hat{\beta}(R_T) - \beta_T)'(\Sigma_{M(IN)}^{-1} - \Sigma_{M(T)}^{-1})(\hat{\beta}(R_T) - \beta_T)] = p + C. \quad (5.7)$$

$C \equiv E_{M_T} [(\hat{\beta}(R_T) - \beta_T)'(\Sigma_{M(IN)}^{-1} - \Sigma_{M(T)}^{-1})(\hat{\beta}(R_T) - \beta_T)]$ has little impact on working correlation selection. As the sample size increases, variances are shrinking down but $\Sigma_{M(IN)}^{-1} - \Sigma_{M(T)}^{-1}$ does not converge to a zero matrix since $I - R_T \rightarrow \mathbb{N} \neq O$ as $n \rightarrow \infty$ unless $R_T = I$.

If the working correlation structure is misspecified, $c_k \neq 1$ for some $k=1, \dots, p$. $\sum_{k=1}^p c_k$ is shown to be larger than p in the simulation study unless R is exchangeable. The second term $E_{M_T} [(\hat{\beta}(R) - \beta_T)'(\Sigma_{M(T)}^{-1} - \Sigma_{M(R)}^{-1})(\hat{\beta}(R) - \beta_T)]$ is non-zero if $R \neq R_T$ and has additional information on the disparity between R_T and misspecified R . The expectation of CIC becomes

$$\begin{aligned} & \sum_{k=1}^p c_k \\ & + E_{M_T} [(\hat{\beta}(R) - \beta_T)'(\Sigma_{M(T)}^{-1} - \Sigma_{M(R)}^{-1})(\hat{\beta}(R) - \beta_T)] \\ & + E_{M_T} [(\hat{\beta}(R) - \beta_T)'(\Sigma_{M(IN)}^{-1} - \Sigma_{M(T)}^{-1})(\hat{\beta}(R) - \beta_T)]. \end{aligned} \quad (5.8)$$

Since the regression parameter estimate is robust to the misspecification of the working correlation structure in large samples (i.e., $\hat{\beta}(R) \approx \hat{\beta}(R_T)$), the equation (5.8) can be approximated by

$$\sum_{k=1}^p c_k + E_{M_T} [(\hat{\beta}(R) - \beta_T)'(\Sigma_{M(T)}^{-1} - \Sigma_{M(R)}^{-1})(\hat{\beta}(R) - \beta_T)] + C. \quad (5.9)$$

Therefore, CIC is better able to discern which misspecified working correlation structure is closer to the truth than is the RJ criterion, because it contains the information in both the first two terms of (5.9), whereas the RJ criterion is using only information in the first term. However, when the working correlation model is over-parameterized, the middle term vanishes, and neither CIC nor RJ can detect the overfitting.

CIC, which is $tr(\hat{\Sigma}_{S(R)}\hat{\Sigma}_{M(IN)}^{-1})$, is interpreted as the sum of the generalized eigenvalues of the sandwich variance estimator under working correlation R with respect to the model-based variance estimator under the working independence assumption. These generalized eigenvalues have information on the disparity between the misspecified R and R_T . As shown in chapter 3, the independence working correlation assumption results in positive bias in diagonal elements of $\hat{\Sigma}_{M(IN)}$ when the data are positively correlated. As the generalized eigenvalues are comparing the magnitude of sandwich estimator $\hat{\Sigma}_{S(R)}$ to the magnitude of $\hat{\Sigma}_{M(IN)}$, the closer R is to R_T , the smaller the generalized eigenvalues would tend to be. Within the context of generalized eigenvalues, this explains why the best working correlation is chosen based on the minimum value of CIC among candidate models.

CHAPTER 6

PROPOSED SELECTION CRITERION MEASURES BASED ON GENERALIZED EIGENVALUES

In this chapter, the new working correlation structure selection criteria based on generalized eigenvalues are presented. As discussed in the previous section, generalized eigenvalues are useful to measure the discrepancy between two covariance matrices. A set of generalized eigenvalues gives the information on the disparity between one covariance matrix and the other covariance matrix, as it measures the relative “size” of one compared to the other.

Let $\lambda_1, \lambda_2, \dots, \lambda_p$ be generalized eigenvalues of $\hat{\Sigma}_{S(R)}$ with respect to $\hat{\Sigma}_{M(IN)}$, where the sandwich estimator, $\hat{\Sigma}_{S(R)}$, is fit under the working correlation structure R specified by user and the model-based estimator, $\hat{\Sigma}_{M(IN)}$ is fit under the working independence assumption. If the two matrices are comparable in size in the j^{th} orthogonal direction e_j , λ_j is close to 1. As shown in the Simulation Study I, $\hat{\Sigma}_{M(IN)}$ tends to have larger positive bias than any other sandwich variance estimators that are fit under the working correlation structure $R \neq I$. Without any correlation parameters to fit, $\hat{\Sigma}_{M(IN)}$ is regarded as a stable structure, but presumably the worst choice, as it cannot account for the within cluster association. In the presence of positive correlations among observations in the response vector, as is often expected in longitudinal data, the size of $\hat{\Sigma}_{S(R)}$ will usually be smaller than that of $\hat{\Sigma}_{M(IN)}$. Thus, small eigenvalue λ_j would indicate large disparity between the two matrices in the direction of the corresponding generalized eigenvector e_j .

A general form for the working correlation selection criteria based on the generalized eigenvalues is expressed by $f(\lambda_1, \lambda_2, \dots, \lambda_p, e_1, e_2, \dots, e_p)$. A function f can be a multidimensional function considering both the generalized eigenvalues and eigenvectors. More simply, one dimensional summary measure, such as the sum of the

generalized eigenvalues, can be considered. In particular, if $f(\lambda_j, e_j; j=1, \dots, p) = \sum_{j=1}^p \lambda_j$, it becomes CIC. An optimization criteria and a resulting optimal function f for the working correlation structure selection has yet to be developed. Nevertheless, multivariate test statistics in MANOVA lead us to devise some useful expressions for f in the working correlation structure selection problem.

The test statistics in MANOVA (Pillai's trace, Wilk's Lambda, Wilks' ratio, Hotelling-Lawley Trace, and Roy's Maximum Root) are related to the covariance selection problem in the sense that they each are a measure of similarity/difference between two matrices: H , the hypothesis matrix, and E , the error matrix. However, in that problem, the two matrices being compared are stochastically independent with known distribution, which is not the situation here. However, the forms of these test statistics are still useful in comparing the "size" of one matrix to another via generalized eigenvalues. If ξ_j is an eigenvalue of HE^{-1} , Hotelling-Lawley's test statistic is the sum of ξ_j . Other test statistics such as Pillai's trace, Wilks' ratio and Roy's maximum root test statistics use $\xi_j / (1 + \xi_j)$ instead of ξ_j . Pillai's trace is the sum of $\xi_j / (1 + \xi_j)$, Wilks' Λ is the product of $1 / (1 + \xi_j)$, Wilks' ratio is the product of $\xi_j / (1 + \xi_j)$ and Roy's maximum root is $\xi_1 / (1 + \xi_1)$, where ξ_1 is the largest eigenvalue. Wilks' ratio is not generally used in the MANOVA setting, but is nonetheless useful to consider in this setting. Even though none are always most powerful in the MANOVA setting, Pillai's trace is considered the most reliable and robust measure in small samples, and Wilk's Λ tends to perform in the middle across a wide variety of group mean configurations when the alternative hypothesis is true. If eigenvalues are similar in magnitude to each other, Pillai's trace is more powerful than others. However, if eigenvalues are substantially different, Hotelling-Lawley's statistic is known to be more powerful than others, unless only the largest is substantially different, in which case Roy's Maximum root test performs best. Each different statistic presented here for the MANOVA problem can be considered a different form of the function $f(\lambda_j, e_j; j=1, \dots, p)$.

The CIC and RJ measures can be regarded as the generalized eigenvalue-based criteria, which in fact, resemble the form of Hotelling-Lawley's test statistic. It is useful to explore whether the forms of other multivariate statistics would perform better or worse in the working correlation structure selection problem, which involves correlated matrices, especially in small to moderate samples.

6.1 Definition of the New Working Correlation Selection

Criterion Measures

Motivated by the MANOVA test statistics and CIC, three generalized eigenvalue based criteria for working correlation structure selection are proposed and investigated:

Measure 1: Pillai's Trace Type Criterion (PT)

$$PT = tr \left[\hat{\Sigma}_{S(R)} \left(\hat{\Sigma}_{S(R)} + \hat{\Sigma}_{M(IN)} \right)^{-1} \right] = \sum_{j=1}^p \frac{\lambda_j}{1 + \lambda_j} \quad (6.1)$$

Measure 2: Wilks' Ratio Type Criterion (WR)

$$WR = \det \left[\hat{\Sigma}_{S(R)} \left(\hat{\Sigma}_{S(R)} + \hat{\Sigma}_{M(IN)} \right)^{-1} \right] = \prod_{j=1}^p \frac{\lambda_j}{1 + \lambda_j} \quad (6.2)$$

Measure 3: Roy's Maximum Root Type Criterion (RMR)

$$RMR = \max \left\{ \frac{\lambda_j}{1 + \lambda_j}; j = 1, \dots, p \right\} = \frac{\lambda_1}{1 + \lambda_1} \quad (6.3)$$

where $\hat{\Sigma}_{S(R)}$ is the sandwich variance estimator of $\hat{\beta}$ under the hypothesized working correlation structure, and $\hat{\Sigma}_{M(IN)}$ is the model based variance estimator of $\hat{\beta}$ under the independent assumption. The number of mean parameters, p , is the length of the vector $\hat{\beta}$. The generalized eigenvalues of $\hat{\Sigma}_{S(R)}$ with respect to $\hat{\Sigma}_{M(IN)}$ are denoted as $\lambda_1 \geq \lambda_2 \geq \dots \lambda_p > 0$.

These measures compare the size of $\hat{\Sigma}_{S(R)}$ to that of the fixed reference matrix $\hat{\Sigma}_{M(IN)}$. Since the "size" of a well-fitting, $\hat{\Sigma}_{S(R)}$ should be small relative to that of a

poorly-fitting, positively-biased, $\hat{\Sigma}_{M(IN)}$, we are looking for small values of λ_j . This implies better-fitting models will have smaller values of PT, WR, and RMR.

One can see from the above that Wilks' Λ and Wilks' ratio are related statistics. Wilks' ratio in the MANOVA setting (Wilks, 1932), is defined as $\det(H(H + E)^{-1})$, whereas Wilks' Λ is $\det[E(H + E)^{-1}]$. Wilks' Λ , rather than Wilks' ratio, is commonly seen in statistical software packages for MANOVA, and is derived under likelihood ratio principles. We note that when defined in this context, the form of Wilks' Λ , $\prod_{j=1}^p 1/(\lambda_j + 1)$, is not as sensitive as others for the working correlation selection as discovered in simulation studies not presented here. Therefore, attention is focused on the forms of Pillai's trace, Wilks' ratio and Roy's maximum root as working correlation structure selection criteria are developed and assessed.

6.2 Simulation Study II

In the Simulation Study II, the new selection criteria for working correlation structure (PT, WR and RMR) are compared with the existing selection criteria QIC, CIC, and RJ in two different settings.

1. For each criterion, the best approximating working correlation structure is chosen among independence, exchangeable and AR-1 only.
2. For each criterion, the best approximating working correlation structure is chosen among independence, exchangeable, AR-1, Toeplitz and unstructured working correlation structures.

The first setting allows a direct comparison to the comparable studies presented in the literature, in which the competing working correlation structures have none or one α parameter. The second reflects a more realistic setting in which overparameterized candidate structures are included.

6.2.1 Simulation Setting

In all simulation studies, correlated Gaussian, binary, and Poisson correlated responses are considered. We consider different sample sizes ($n=20, 30, 50, 100, 200$), different sizes of the balanced clusters ($m_i = m = 4, 6$), and the true correlation structures (AR-1, EX, TOEP, UN). In situations where the true correlation structure is AR-1 or EX, different true α parameters ($\alpha = 0.3, 0.6$) are considered. The simulation design is a factorial with the following factor levels listed in Table 6.1.

Table 6.1 Simulation Study II Design Parameters

Factor	Levels
Distribution (D)	Multivariate Gaussian Responses Multivariate Bernoulli Responses Multivariate Poisson Responses
Response vector dimension (m)	$m = 4, 6$
True Correlation Structure $C(\alpha)$	Exchangeable: EX(α), $\alpha = 0.3, 0.6$ Autoregressive of order 1: AR-1(α), $\alpha = 0.3, 0.6$ Toeplitz: TOEP(m), α values in Table 3.2(a) Unstructured: UN(m), α values in Table 3.2(b)
Working Correlation structure $R(\alpha)$	Independence (IN), Exchangeable (EX), Autoregressive (AR-1), Toeplitz (TOEP), and unstructured (UN).
Sample Sizes (n)	$n = 20, 30, 50, 100, 200$

The marginal mean of the t -th component of the correlated Gaussian response vector y_i is $E(y_{it}) = \mu_{it} = \beta_1 x_{1t} + \beta_2 x_{2t}$, where $\beta_1 = \beta_2 = 0.3$, the true scale parameter $\phi_T = 1$, x_{1t} and x_{2t} are independently generated from $U[0,1]$. This is the same marginal mean model used by Hin and Wang (2009). For the correlated binary responses, the components of the mean vector are modeled as $\text{logit}(\mu_{it}) = \beta_1 x_{1t} + \beta_2 x_{2t}$, where x_{1t} and x_{2t} are independently generated from $U[0.5,1]$ and $\beta_1 = \beta_2 = 0.3$. For correlated Poisson responses, the components of the mean vector are modeled as $\log(\mu_{it}) = \beta_1 x_{1t} + \beta_2 x_{2t}$, where x_{1t} and x_{2t} are independently generated from $U[0.5,1]$ and $\beta_1 = \beta_2 = 0.3$. For all simulations, x_{1t} and x_{2t} are observational-level covariates.

All correlation/covariance matrices used in the simulation are positive definite, and the same covariance structures are used for the m -dimensional Gaussian, multivariate Bernoulli, and multivariate Poisson random vectors. For the correlated binary responses, the marginal means limit the ranges of the correlations (Qaqish, 2003). The correlation coefficients (i.e., α parameters) of the true Toeplitz and unstructured matrices presented in Table 3.2 were chosen so that they satisfied the correlation bounds for the binary responses, as well as being proper pairwise correlations for the Poisson and Gaussian distributions.

A sample of n independent simulated observations with response vector of size m , true correlation structure $C(\alpha)$, and distribution D are generated under the marginal mean and true correlation structure. The model is fit under the correct marginal mean model and the correct variance function, but assuming each of five different working correlation structures ($R(\alpha)=\text{IN, EX, AR-1, TOEP, UN}$) and using that one data set. This is replicated $r = 1000$ times.

Working correlation selection criterion measures are computed using the bias-corrected sandwich variance estimator by Wong and Long (WL), which was not the original proposed structure of QIC or CIC. As shown in Simulation Study I, the WL variance estimator effectively corrects the negative bias of the Liang and Zeger's

sandwich variance estimator and tends to have smaller variability in small samples than other sandwich variance estimators. Thus, it appeared to be the best choice for this investigation, so as not to confound issues of bias with issues of working correlation selection.

Each selection criterion selects the best approximating working correlation structure within the listed set of five candidate structures. Through 1000 independent simulation replications, the performance of each criterion is determined based on the frequency of choosing the correct structure out of 1000.

All simulations are performed using *R* version 2.11.1. Correlated Gaussian random variables are generated using the MASS library, and correlated binary data are generated using the binarySimCLF library (Qaqish, 2003). For every mean vector and correlation matrix based on correlated binary responses, conditional linear family (CLF) compatibility is checked. As defined in Qaqish (2003), the conditional linear family is a subset of all multivariate binary distributions, and used in this algorithm for efficiently generating large multivariate vectors with specified marginal probabilities and correlation matrices. If a single observation (Y_i, X_i) simulated data does not meet the CLF compatibility criterion, it is discarded and a new independent correlated binary random vector is generated. Correlated Poisson random variables are generated using the method proposed by Yahav and Shmueli (2011). Generating random vectors in this manner is repeated until 1000 samples of size n are obtained. The GEE fitting within *R* is performed using the `geese()` function in the `geepack` package.

6.2.2 Simulation Results: Correlated Binary Response

Figure 6.1 through Figure 6.4 present the correct identification rates of the six criteria (PT, WR, RMR, CIC, QIC, RJ) for the correlated binary responses under the different true correlation structures. In particular, the frequencies of the working correlation structure identified by the six criteria when $n=50$ and $m=4$ are tabulated in

Table 6.2. Simulation results for the correlated Gaussian and Poisson responses also present similar results, so their results are not shown here, but are included in Appendix B.

The correct identification rate for each criterion is defined as the percent of selecting the true correlation structure out of 1000 independent simulations. For a given simulated data set (i.e., one replication of the simulation), the best working correlation structure is selected from two sets of the candidate structures $Wc1 = \{IN, EX, AR-1\}$ and $Wc2 = \{IN, EX, AR-1, TOEP, UN\}$ separately.

If the true correlation structure $C(\alpha)$ is included in $Wc1 = \{IN, EX, AR-1\}$ and TOEP and UN are not considered as candidates, PT, WR, and CIC select the correctly-specified working correlation structure with high rates. RMR performs a little bit worse. This is probably because RMR uses only the largest generalized eigenvalue whereas PT, WR, CIC utilize all generalized eigenvalues to measure the discrepancy between two covariance matrices. If AR-1 is the true correlation structure, QIC and RJ are poor at identifying the correct structure as compared to PT, WR and CIC (Figure 6.1(a)(c) and Figure 6.2(a)(c)). On the contrary, if EX is the true correlation structure, RJ performs very similarly to PT, WR, and CIC in selecting the correct structure (Figure 6.3(a)(c) and Figure 6.4(a)(c)).

If $Wc2$ is $\{IN, EX, AR-1, TOEP, UN\}$ and the true correlation structure is any patterned matrix, all six criteria perform poorly. Overall, PT, WR, CIC, and RMR prefer the unstructured working correlation structure, reflecting the ability of that structure to fit to any dataset (Table 6.2). They rarely choose the incorrect under-parameterized structures (i.e., misspecified structures). For instance, if the true correlation structure is AR-1, they rarely select IN or EX. On the other hand, QIC and RJ select the misspecified structures approximately 30% of the time. Independent working correlation structure is sometimes chosen by QIC, whereas other criterion measures hardly ever select it. Exchangeable working correlation is preferred by RJ for most cases. The RJ criterion, as

designed, chooses the working correlation structure where the sandwich variance estimate is nearest to the model-based variance estimate. The simulation results show that RJ tends to select the exchangeable structure regardless of the true underlying correlation structure, which is also shown in Hin et al. (2007). RJ seems to perform better than other criterion measures when $Wc2 = \{IN, EX, AR-1, TOEP, UN\}$, but its correct identification rate is still below 50%.

In summary, the proposed selection criteria, as well as CIC, perform similarly well when the true correlation structure is parsimonious and a set of candidate structures does not contain any over-parameterized structures. However, they tend to select an over-parameterized working correlation structure much of the time – with QIC doing so less often because it is choosing independence structure. The unstructured, and to a lesser extent, the Toeplitz working correlation structures are more likely to be selected by those criterion measures. Since the unstructured matrix closely approximates the true correlation structure in large samples, the loss of statistical efficiency of $\hat{\beta}$ would be trivial in that case. However, in small to moderate sample sizes, the use of the unstructured matrix causes the hypothesis testing to be liberal, as shown in chapter 2.

6.2.3 Simulation Results: Correlated Gaussian Response

When the response vector is the correlated Gaussian response, the similar patterns are identified (Table B.13 ~ Table B.24 in Appendix B). Note that both QIC and CIC select the same working correlation structure all the time for the correlated normal responses if the scale parameter is estimated for each different model. When the scale parameter is estimated for each different model, QIC becomes $(N - p) + 2CIC$ for the correlated normal responses where $N = n \times m$ and p is the dimension of $\hat{\beta}$. Since $(N - p)$ is the constant, QIC and CIC select the same working correlation structure. Hin and Wang (2009) fixed scale parameter in their simulations and showed the superiority of

CIC over QIC for working correlation selection. It is worth to evaluate the effect of fixed scale parameter on the performance of the criterion measures. Except QIC, the performance of other criterion measures are found to be hardly influenced by the fixed scale parameter as compared to simulation results when the scale parameter is estimated. On the other hand, QIC performs worse than PT, WR, and CIC when the scale parameter is fixed (Tables C.1 ~ C.4 in Appendix C). In practice, the true scale parameter is unknown. Regression parameter estimates and covariance estimates based on the estimated scale parameter are used for inference. Moreover, it was shown that other criterion measures which are superior to QIC were little impacted by whether or not scale parameter is fixed. Therefore, for the rest of this paper, simulation results when the scale parameter is estimated differently in each candidate models will be shown.

6.2.4 Simulation Results: Correlated Poisson Response

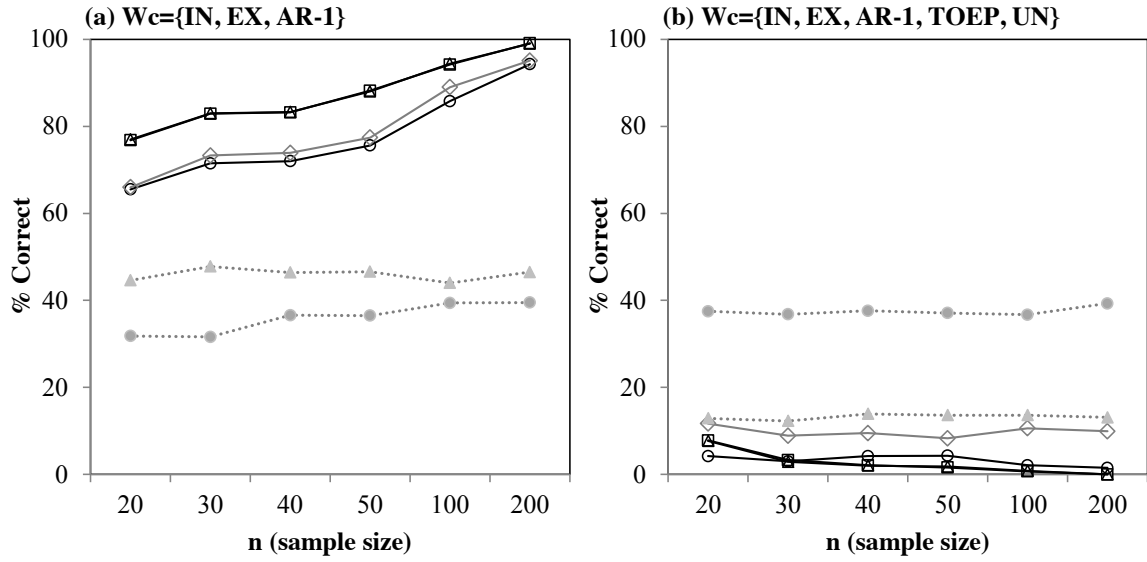
When the response vector is the correlated Poisson response, the similar patterns are identified (Table B.25 ~ Table B.36 in Appendix B). When the over-parameterized structure is included (i.e., $Wc2 = \{IN, EX, AR-1, TOEP, UN\}$) and the true underlying correlation structure is parsimonious such as AR-1 or exchangeable, all six criteria tend to select the over-parameterized structure.

Therefore, penalization for over-parameterized structure is required to improve the performance of the proposed selection criteria in small samples. In the next chapter, a new method for penalizing over-parameterized structure is introduced.

Table 6.2 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC, and RJ from 1000 independent replications. The true correlation structures for the correlated binary responses are AR-1 ($\alpha = 0.3, 0.6$) and EX($\alpha = 0.3, 0.6$), $m = 4$ and $n = 50$.

$C(\alpha)$	Criterion	$W_c = \{IN, EX, AR-1\}$			$W_c = \{IN, EX, AR-1, TOEP, UN\}$				
		IN	EX	AR	IN	EX	AR	TOEP	UN
AR-1(0.3) $m = 4$ $n = 50$	PT	0	118	882	0	1	16	75	908
	WR	0	120	880	0	0	18	75	907
	RMR	71	173	756	28	9	43	50	870
	CIC	39	187	774	19	46	83	130	722
	QIC	288	246	466	206	139	136	148	371
	RJ	146	489	365	1	268	371	126	234
AR-1 (0.6) $m = 4$ $n = 50$	PT	6	37	957	5	11	73	118	793
	WR	6	35	959	5	11	76	120	788
	RMR	14	83	903	7	10	68	136	779
	CIC	6	86	908	5	25	141	219	610
	QIC	292	258	450	244	194	114	167	281
	RJ	8	590	402	5	306	358	113	218
EX (0.3) $m = 4$ $n = 50$	PT	0	887	113	0	16	1	94	889
	WR	0	884	116	0	16	1	96	887
	RMR	116	729	155	36	43	17	69	835
	CIC	35	787	178	13	106	40	142	699
	QIC	329	301	370	218	72	255	96	359
	RJ	9	676	315	0	337	7	259	397
EX (0.6) $m = 4$ $n = 50$	PT	25	930	45	25	64	3	136	772
	WR	25	928	47	25	64	3	130	778
	RMR	32	917	51	18	84	3	121	774
	CIC	27	868	105	26	158	33	221	562
	QIC	348	349	303	277	122	242	101	258
	RJ	25	825	150	25	375	1	261	338

Correlated Binary Responses $C(\alpha)$: AR-1(0.3), $m = 4$



Correlated Binary Responses $C(\alpha)$: AR-1 (0.6), $m = 4$

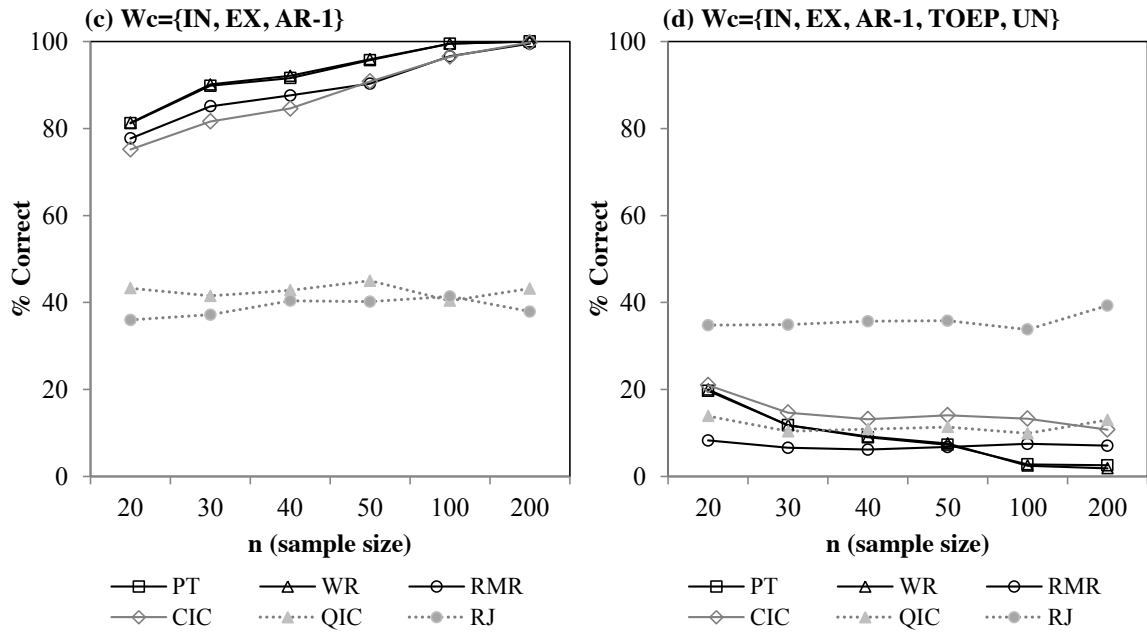


Figure 6.1 The correct identification rates of PT, WR, RMR, CIC, QIC, and RJ. The 4×4 true correlation structure for the correlated binary responses: AR-1(0.3) for (a) and (b); AR-1(0.6) for (c) and (d). (a) and (c) select the best working correlation structure from $\{IN, EX, AR-1\}$. (b) and (d) select the best working correlation structure from $\{IN, EX, AR-1, TOEP, UN\}$.

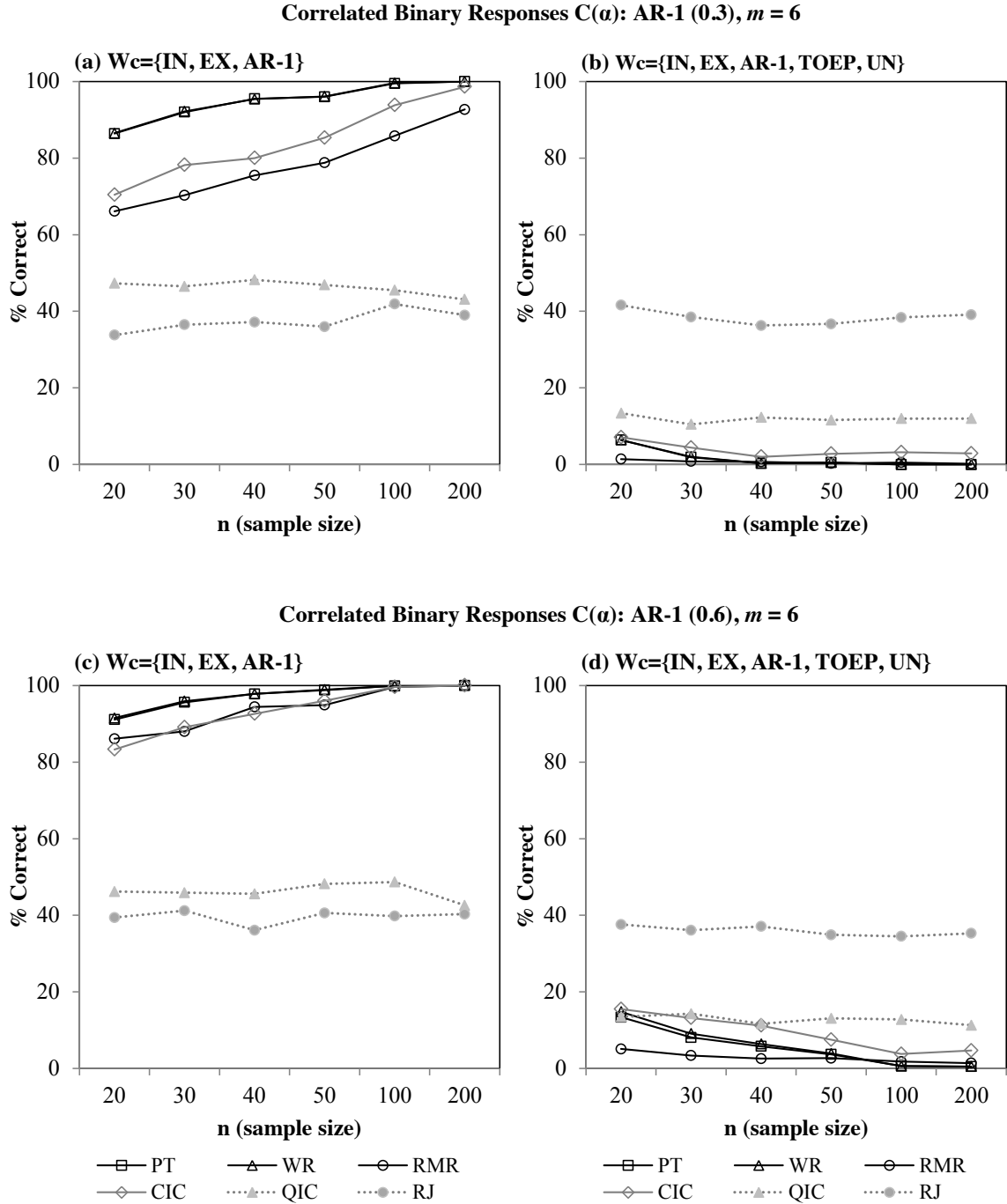


Figure 6.2 The correct identification rates of PT, WR, RMR, CIC, QIC, and RJ. The 6×6 true correlation structure for the correlated binary responses: AR-1(0.3) for (a) and (b); AR-1(0.6) for (c) and (d). (a) and (c) select the best working correlation structure from $\{IN, EX, AR-1\}$. (b) and (d) select the best working correlation structure from $\{IN, EX, AR-1, TOEP, UN\}$.

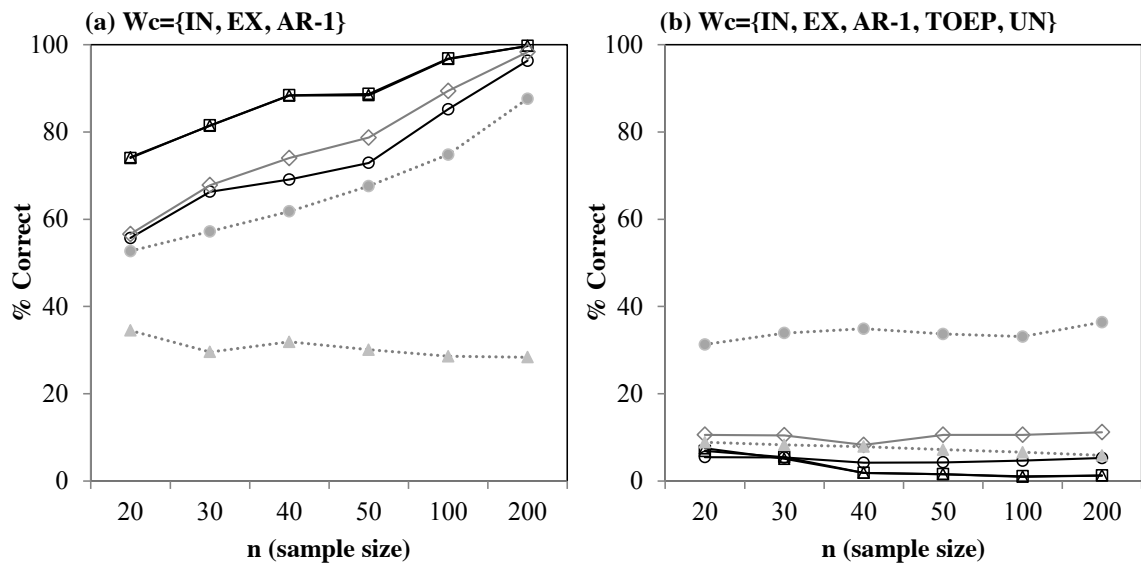
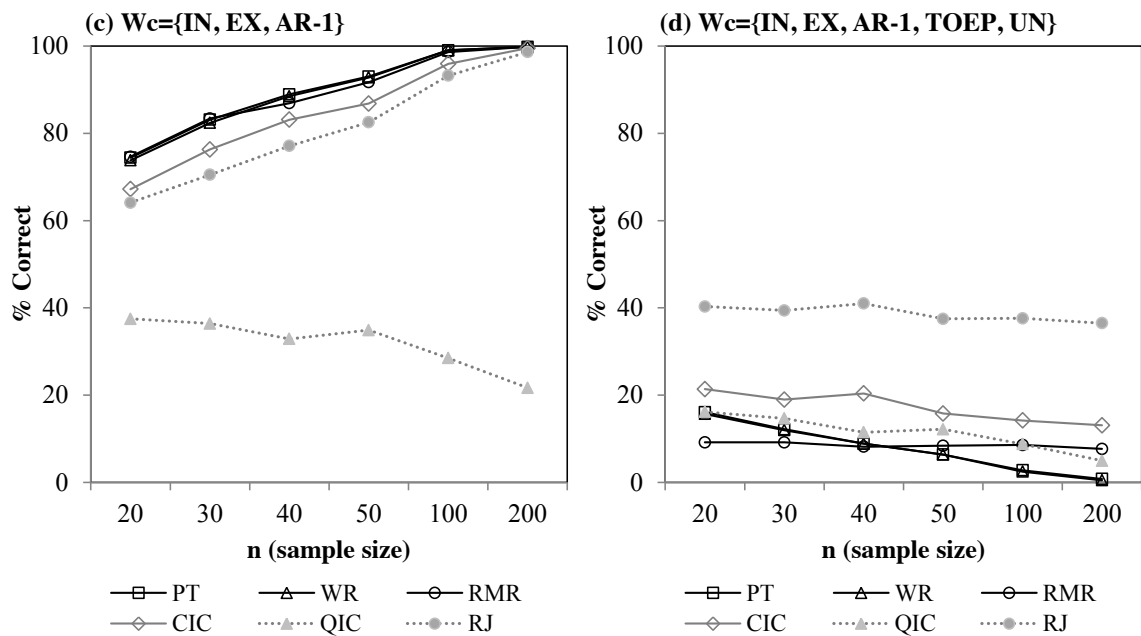
Correlated Binary Responses $C(\alpha)$: EX (0.3), $m = 4$ Correlated Binary Responses $C(\alpha)$: EX (0.6), $m = 4$ 

Figure 6.3 The correct identification rates of PT, WR, RMR, CIC, QIC, and RJ. The 4×4 true correlation structure for the correlated binary responses: EX (0.3) for (a) and (b); EX (0.6) for (c) and (d). (a) and (c) select the best working correlation structure from $\{IN, EX, AR-1\}$. (b) and (d) select the best working correlation structure from $\{IN, EX, AR-1, TOEP, UN\}$.

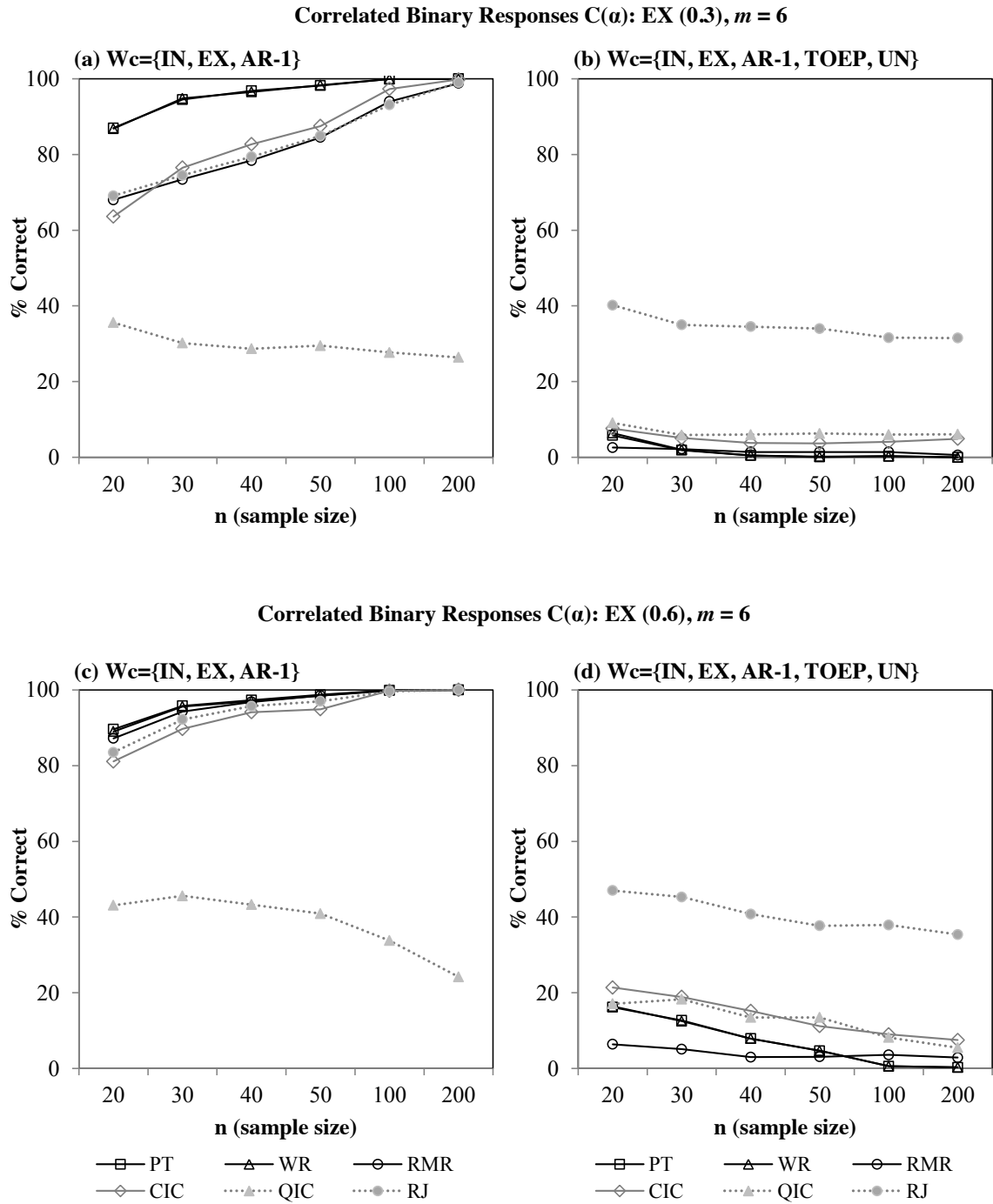


Figure 6.4 The correct identification rates of PT, WR, RMR, CIC, QIC, and RJ. The 6×6 true correlation structure for the correlated binary responses: EX (0.3) for (a) and (b); EX (0.6) for (c) and (d). (a) and (c) select the best working correlation structure from $\{IN, EX, AR-1\}$. (b) and (d) select the best working correlation structure from $\{IN, EX, AR-1, TOEP, UN\}$.

CHAPTER 7

PENALIZATION FOR OVER-PARAMETERIZED STRUCTURES

Traditional bias correction methods use the expected bias from the likelihood function or Kullback-Leibler discrepancies, in the univariate situation where all measurements are independent. For instance, AIC ($-2 \ln f(y | \hat{\theta}_k) + 2k$) is an asymptotically unbiased estimator of the expected discrepancy $E_{M_T} \{d(\hat{\theta}_k)\}$, where E_{M_T} denotes the expectation under the true distribution of Δ (i.e., under the generating model M_T), $f(y | \hat{\theta}_k)$ is the maximum likelihood for the estimated model and k is the number of parameters in the model, and $d(\hat{\theta}_k) = E_{M_T} \{-2 \ln f(y | \theta_k)\} |_{\theta_k = \hat{\theta}_k}$. The expected discrepancy reflects the overall separation between the generating model and a fitted model. The estimated discrepancy, which is $-2 \ln f(y | \hat{\theta}_k)$, reflects how well the fitted model predicts the data at hand. Thus, it is a negatively biased estimator of the expected discrepancy. Correcting for this bias leads to the penalty term of AIC:

$E_{M_T} \{d(\hat{\theta}_k)\} - E_{M_T} \{-2 \ln f(y | \hat{\theta}_k)\} = 2k + o(1)$. The penalty term of QIC, useful in a multivariate setting where observations are vectors of correlated measurements, is constructed in a similar way by using the quasi-likelihood under the working independence assumption. Note that the penalty term of QIC is $2 \times$ CIC and it plays a role as a penalty term for variable selection, but it does not penalize for over-parameterization of the working correlation structure. Therefore, it is difficult to find the penalty term for the CIC using the traditional bias correction methods because we are now in a multivariate setting, generally without an expression for the full multivariate likelihood.

If the true correlation structure (R_T) is nested within an over-parameterized structure (R_{ov}), $R_{ov} \rightarrow R_T$ as $n \rightarrow \infty$. For example, if R_T is AR-1, the estimated R_{TOEP} and R_{UN} converge to R_{AR-1} as $n \rightarrow \infty$. Thus, they are not biased estimators, but they estimate the true correlation structure with extra parameters. As discussed earlier, the criterion measures that can be used to select a working correlation structure do not effectively

penalize for over-parameterization. We do not have a full multivariate likelihood in this setting, which makes the use of generalized eigenvalues in our selection criteria attractive, as they do not require such. However, the challenge now is to capitalize on the behaviors of those selection criterion measures when the working correlation is over-, under- and correctly-specified. In this chapter, a new method for identifying which working correlation structures are reasonable fits to the data and choosing among them is presented.

7.1 Use of the Cluster Detection Method on the Penalization for Over-Parameterized Structures

The sandwich variance estimator under an over-parameterized working correlation R_{ov} and that under an assumption that matches the true correlation structure tend to have similar values of the selection measures based on generalized eigenvalues, since the component matrices being compared converge (as $n \rightarrow \infty$) to the same quantities whether over-parameterized or correctly specified. The corresponding working correlation selection criterion values are close in magnitude to each other. They could be considered “*clustered*,” when compared to the magnitudes of selection criterion measures that clearly do not fit the data. When the working correlation is under-specified, then there will be inherent bias in the correlation matrix, which will cause the eigenvalue-based criteria to reflect that bias and be different from those computed under correctly specified or over-specified structures.

Suppose the selection criterion values under all candidate working correlation structures, including the true structure, are ordered from smallest to largest. The selection criterion value under an over-specified structure R_{ov} , for example, R_{UN} , tends to have the minimum value, due to its ability to overfit the data. However, the underlying true structure – and any structure that is very similar to the true structure – will also have a relatively small selection criteria measure. As the fit becomes worse, the selection

criterion measure gets larger and moves away from the minimum value and those “close” to the minimum. We will refer to those measures that are similar in value and close to the minimum a “*cluster*.” On the other hand, working correlation structures that are neither the true correlation nor an over-parameterized structure, will tend to have relatively larger selection criterion values. If the fit is poor, they would be located outside the *cluster*.

Once a *cluster* of working correlation structures is found, any working correlation structure within it could be viewed as reasonable, for they all fit the data about as well or better than any of the other candidate working correlations considered that are outside the cluster. Some researchers might prefer the most parsimonious structure, even though its selection criterion value is not the minimum, and might represent a smoothing of parameter values in the true underlying correlation matrix. Others may prefer a more flexible structure with smaller selection criterion value, indicating a closer fit to the data, and accept the potential that it may be over-parameterized. An optimization function that compromises the parsimony of parameters and goodness of fit would be useful to find the “best” approximating working correlation structure from among those considered. Identifying working correlation structures within the cluster, which all represent similar fits to the data, as well as identifying the “best” approximating structure would be useful information for the analyst.

We begin by describing the relevant published literature related to this approach.

7.2 Cluster Detection Methodology

7.2.1 Literature Review on the Cluster Detection Methods

There is an extensive literature regarding cluster detection in spatial statistics. One approach to detect nonrandom clusters of points on a line is to use a test statistic based on a function of the spacings between observed points. Given a set of independent random variables, $Z_1, Z_2, \dots, Z_k$, let $Z_{(1)} \leq Z_{(2)} \leq \dots \leq Z_{(k)}$ denote the order statistics of these

variables. Then, *spacing* is defined as $D_i = Z_{(i+1)} - Z_{(i)}$, which is the difference between the i^{th} and $(i+1)^{th}$ successive order statistics of a sample. Thus, spacings of points in a cluster are smaller than those of points outside the cluster, relative to what would be the expected spacings. The expected distance between any two order statistics in the sample depends upon the underlying distribution of the original sample.

Molinari et al. (2001) proposed a multiple cluster detection method based on a piecewise constant regression model for spacings of points. Under the no-clustering hypothesis, random variables are assumed to be uniformly distributed $U(a, b)$. The transformed random variables, $Z_1, Z_2, \dots, Z_k$ are assumed to be random variables from uniform $U(0,1)$. Under H_0 , $Z_{(i)}$, the i^{th} order statistic, follows a beta distribution $\beta(1, k - i + 1)$ and the spacing D_i follows a $\beta(1, k)$ distribution. Mean spacings within a cluster would be smaller than the overall mean spacing. For a given number of clusters, the least squares estimation method is applied to compute the bounds of each potential cluster. Once the bounds are computed for each potential number of clusters, the “optimal” number of clusters are determined by AIC or BIC. In general, the best model is determined by the smallest criterion value. However, if AIC (or BIC) differences are very small, the acceptance of a single model with the minimum criterion based on one dataset at hand might lead to a false sense of confidence. Therefore, Molinari et al. (2001) computed criterion values based on 1000 bootstrapped samples and determined the significance of AIC (or BIC) differences using that bootstrapped estimate of variability.

Demattei and Molinari (2007) proposed a test statistic for the mean spacing on the predetermined portion of the number line. The same general idea behind the method of Molinari et al. (2001) was used, but the bootstrapping method was avoided. Under the no-clustering hypothesis, the transformed random variables, $Z_1, Z_2, \dots, Z_k$, are assumed to be random variables from uniform $U(0,1)$. Under H_0 , $Z_{(i)}$ follows a beta distribution $\beta(1, k - i + 1)$ and the spacing D_i follows $\beta(1, k)$. They proposed to test H_0 (no

clustering) against the alternative hypothesis H_1 that the mean D_i on the considered portion is less than the mean of a $\beta(1, k)$. Their approach is implemented as follows. For all k random variables, define

$$T = \frac{1}{k-1} \sum_{i=1}^{k-1} (k+1)D_i. \quad (7.1)$$

The p -value corresponding to the observed mean distance (or spacing) on a given portion of the support, u , is calculated as follows:

$$P(T \leq u) \leq p_u \quad \text{with } p_u = \exp\left(-\frac{(k-1)(1-u)^2}{2k / (k+2) + 2(1-u) / 3}\right). \quad (7.2)$$

This method is more flexible, in that multiple clusters can be found by repeatedly applying this procedure to each potential cluster on different sections of the support, whereas Molinari et al. (2001) tests the model in the whole study region. The flexibility comes with a price, of course, of inflated overall Type I error rates from the multiple testing.

7.2.2 Limitations on the Use of Existing Cluster Detection

Methods to Working Correlation Selection

Let Z be any working correlation selection criterion. We consider k different working correlation structures. Let $Z_{(1)} \leq Z_{(2)} \leq \dots \leq Z_{(k)}$ denote the ordered selection criterion values under k different working correlation structures. If the set of candidate models contains the true underlying correlation structure, the working correlation structure associated with $Z_{(1)}$ shows the tightest fit to the data – but might be a structure that over-fits the data. Through the cluster detection method, we aim to find the set of working correlation structures with selection criterion measures close to $Z_{(1)}$, say $\{Z_{(1)}, Z_{(2)}, \dots, Z_{(i)}\}$. We are not interested in finding clusters of poorly-fitting working correlations, so detecting only one cluster around the minimum is our only interest. Multiple cluster detection is not necessary.

It should be noted that both Molinari et al. (2001) and Demattei and Molinari (2007) used approaches based on the underlying standard uniform distribution of the random variables. Data transformation to $U(0,1)$ prior to the hypothesis testing is required, in order to measure distance D_i in a meaningful and comparable way. The true distribution for a working correlation selection criterion measure in GEE depends upon the underlying distribution of the data as well as the choice of the working correlation – which could either be correct, over-parameterized, under-parameterized, or simply incorrect in some other way. Thus, the set of criterion measures $Z_1, Z_2, \dots, Z_k$ will not have the same underlying distributions and individually are generally unknown. Also, they are not independent random variables, because all of them will be functions of the same set of n data points. However, when bias correction is used in the estimation of the GEE robust variance estimator, it is intuitively reasonable that all Z_i values associated with either the correct or over-parameterized working correlation structures are trying to estimate the same quantity, and would tend to “cluster” near that true value. Those associated with ill-fitting structures should be larger, and thus skew the empirical histogram of the set to the right. However, if all candidate models fit or over-fit the true working correlation, all values will tend to be within the same “cluster”. It is argued that this characterization of what is expected could be effective in identifying that cluster of similarly fitting criterion measures from among the set of k candidates. In order to use this approach, however, it is still necessary to scale the range of Z_i values onto a $[0, 1]$ range. The traditional data normalization method $f(t_i) = [t_i - \min(t)] / [\max(t) - \min(t)]$ can be used. However, it may be overly influenced by extreme values in this setting. For a highly skewed distribution, it may cause those measures at the low end to appear more similar than we would want. Additionally, the earlier work of Demattei and Molinari (2007) under the assumption of a uniform random sample showed that the p -value for cluster testing would tend to be high in the small sample size setting. Practically speaking, most analysts will not consider a large set of working correlation structures.

To base a cluster-detection method on their hypothesis testing approach in our small sample setting, even if their distributional assumptions were a reasonable approximation, would allow too many working correlation structures in the cluster. The proposed method of scaling is described in Section 7.3.

7.3 Proposed Cluster Detection Method for Working

Correlation Selection

The general idea of the work of Molinari et al. (2001) is that the mean spacing of a cluster is lower than the overall mean spacing. A spacing smaller than the overall mean spacing suggests that the two points are part of the same cluster. Spacing larger than the overall mean spacing suggests that the two points are not in the same cluster. The proposed cluster detection method is based on the idea of Molinari et al. (2001). However, rather than using the hypothesis testing, we rely on a comparison of a spacing to the mean spacing for a given portion to determine the *cluster*. This approach is defined more precisely below.

Step 1

Let Z be any working correlation selection criterion. We consider k different working correlation structures, including independence, which would be expected to fit poorly in this correlated data setting. For a set of k candidate working correlation structures, compute the working correlation selection criterion values for each candidate:

$Z_1, Z_2, \dots, Z_k$. Let Z_j indicate the criterion value under the j -th working correlation structure. Then, sort Z_j values from the smallest to the largest: $Z_{(1)} \leq Z_{(2)} \leq \dots \leq Z_{(k)}$.

Step 2

Obtain the spacings $D_i = Z_{(i+1)} - Z_{(i)}$, $i = 1, 2, \dots, k-1$ and the overall mean spacing

$\bar{D} = \sum_{i=1}^{k-1} D_i / (k-1)$. Calculate the mean spacing $\bar{D}_u$ for a potential cluster

$\{Z_{(1)}, Z_{(2)}, \dots, Z_{(u+1)}\}$:

$$\bar{D}_u = \frac{1}{u} \sum_{i=1}^u D_i \quad \text{where } u = 1, 2, \dots, k-1. \quad (7.3)$$

Step 3

The cluster $\{Z_{(1)}, Z_{(2)}, \dots, Z_{(i)}\}$ is found by finding i such that $\bar{D}_1, \bar{D}_2, \dots, \bar{D}_{i-1}$ are smaller than $\bar{D}$ but $\bar{D}_i$ is larger than $\bar{D}$. Because we are not statistically comparing the average distance to an expected average distance, we are not assuming any underlying distribution of the measures, nor are we using any estimate of variability of the measures (e.g., through bootstrapping). Once the cluster $\{Z_{(1)}, Z_{(2)}, \dots, Z_{(i)}\}$ is found, any working correlation structure within this cluster is considered a candidate for use.

We consider one example that illustrates the proposed cluster detection method. For the working correlation selection, PT criterion is used. Five working correlation structures {IN, EX, AR-1, TOEP, UN} are considered.

Table 7.1 Example illustrating the proposed clustering detection method

$Z_{(1)}$	$Z_{(2)}$	$Z_{(3)}$	$Z_{(4)}$	$Z_{(5)}$
PT(UN)	PT(TOEP)	PT(AR-1)	PT(EX)	PT(IN)
=0.943	=0.974	=0.998	=1.183	=1.222
	$\bar{D}_1 = 0.031$	$D_2 = 0.024$	$D_3 = 0.185$	$D_4 = 0.039$
	$\bar{D}_1 = 0.031$	$\bar{D}_2 = 0.0275$	$\bar{D}_3 = 0.080$	$\bar{D}_4 = 0.0698$
$\bar{D} = 0.0698$	$\bar{D}_1 < \bar{D}$	$\bar{D}_2 < \bar{D}$	$\bar{D}_3 > \bar{D}$	
Cluster: $\{Z_{(1)}, Z_{(2)}, Z_{(3)}\} \Rightarrow \{\text{UN, TOEP, AR-1}\}$				

Since PT(UN) is the minimum, if we do not consider penalization, the unstructured working correlation matrix would be chosen by PT. However, the proposed clustering detection method provides us with more useful information. The cluster includes {UN, TOEP, AR-1}, which indicates that PT(TOEP) and PT(AR-1) are relatively close the

minimum value $PT(UN)$. Therefore, adding the structure of Toeplitz and AR-1 did not change the estimated eigenvalues as much as when the exchangeable and independence more restrictive assumptions were made.

7.4 Optimization function

Some researchers would prefer the most parsimonious working correlation structure such as AR-1. Others might prefer a more flexible structure such as TOEP or UN within the cluster, depending on the goal of the analysis, the sample size and number of clustered measurements. An optimization function (or loss function) considering the trade-off between bias and variability would help us to find the “best” working correlation structure within the cluster.

Using an optimization function, the goal is to find a parsimonious structure that closely approximates the true correlation matrix, without adding unnecessary parameters. An appropriate optimization function should effectively penalize the working correlation structure with many parameters when the sample size is small or the responses are measured on many occasions. It should also be able to penalize the working correlation structure which is parsimonious, but has a worse-fit. Therefore, the sample size (n), the number of repeated measurements (m) and the number of parameters of the working correlation structure (q) could be considered when constructing an optimization function. Even though we expect the amount of penalization increases as m (number of clustered measurements) or q (number of parameters in the correlation matrix) increase and n decreases, a loss function must be chosen. It should be noted that the term “optimization” is only intended to mean optimal in the sense that it minimized a particular loss function, not a more general form of optimality.

Squared Euclidean distance is a commonly used loss function. If we consider k cost components, the squared weighted Euclidean distance of $\varphi = (\varphi_1, \varphi_2, \dots, \varphi_k)'$ from the origin is $\sum_{j=1}^k w_j \varphi_j^2$ where $\sum_{j=1}^k w_j = 1$. For penalizing the over-parameterized

structure, we consider two cost components: 1) the number of parameters (q) in the working correlation structure and 2) the selection criterion measure such as PT or WR. The q value is related to the parsimony and the selection criterion measure is related to the goodness-of-fit of the structure to the data. Even though m and n are not considered directly as the cost components, they are related to q and the criterion value. As m increases, the q of the over-parameterized structure, for example, perhaps R_{TOEP} and R_{UN} , increases. As n increases, the criterion values are expected to be smaller since the model fits the data better, and they will be closer to each other, since all correlation parameter estimates are converging to the same quantities.

Assume the *cluster* includes i different working correlation structures, $\{R_1, \dots, R_i\}$. Each R_j has the corresponding (Z_j, q_j) . Let $Z_{\max} = \max\{Z_j \mid j=1, \dots, i\}$ and $q_{\max} = \max\{q_j \mid j=1, \dots, i\}$. The weighted Euclidean distance, $[wZ_j^2 + (1-w)q_j^2]^{1/2}$ might be considered as a loss function, depending on whether the investigator puts more weight/emphasis on goodness of fit or on parsimony. However, we should note that the scales of Z and q are quite different. The number q is determined by m and the candidate structure, R . For a given m , q is $0.5m(m-1)$ for R_{UN} , and 1 for R_{EX} and 1 for R_{AR-1} . On the other hand, the proposed Z is not a function of m (then number of repeated measures on each individual) but a function of p (the number of regression parameters), which is the dimension of the covariance matrices being compared by the generalized eigenvalues. If $\hat{\Sigma}_{S(R)} \approx \hat{\Sigma}_{M(IN)}$, PT, WR and CIC are:

$$\begin{aligned} \text{PT} &= \text{tr}[\hat{\Sigma}_{S(R)}(\hat{\Sigma}_{S(R)} + \hat{\Sigma}_{M(IN)})^{-1}] \approx p/2, \\ \text{WR} &= \det[\hat{\Sigma}_{S(R)}(\hat{\Sigma}_{S(R)} + \hat{\Sigma}_{M(IN)})^{-1}] \approx (0.5)^p, \\ \text{CIC} &= \text{tr}[\hat{\Sigma}_{S(R)}\hat{\Sigma}_{M(IN)}^{-1}] \approx p. \end{aligned} \tag{7.4}$$

Since the distributions of these functions of $\hat{\Sigma}_{S(R)}$ and $\hat{\Sigma}_{M(IN)}$ are unknown, the range (or variance) of each criterion measure is generally unknown. However, the simulation results show that they are not too far away from the values in the equation (7.4) even

when $\hat{\Sigma}_{S(R)} \neq \hat{\Sigma}_{M(IN)}$. If the responses are measured at ten different time points ($m=10$) and we consider two regression parameters in the marginal mean model ($p=2$), q ranges from zero to 45, but PT values locate around at 1. Therefore, the normalization in (7.5) is suggested by dividing the Z and q from each candidate model by $Z_{\max}$ and $q_{\max}$, respectively.

A general expression of the optimization function for the j^{th} candidate model is defined as follows:

$$f(Z_j, q_j) = \left[(1-w) \left(Z_j / Z_{\max} \right)^2 + w \left(q_j / q_{\max} \right)^2 \right]^{1/2}, \quad (7.5)$$

where Z_j is the selection criterion measure under the j^{th} candidate working correlation structure R_j within the *cluster* and q_j is the number of parameters of R_j . w ($0 \leq w \leq 1$) is a common weighting factor. If $w=0$, the working correlation structure with the minimum criterion value is selected as the “best” approximating working correlation structure. If $w=1$, the most parsimonious structure is selected.

The proposed optimization loss function $f^*(Z, q)$ in this dissertation weighs both terms equally by using $w=0.5$ and multiplies $f(Z, q)$ by the constant $\sqrt{2}$:

$$\begin{aligned} f^*(Z, q) &= \sqrt{2} f(Z, q) \\ &= \sqrt{2} \left[\frac{1}{2} \left(Z / Z_{\max} \right)^2 + \frac{1}{2} \left(q / q_{\max} \right)^2 \right]^{1/2} \\ &= \left[\left(Z / Z_{\max} \right)^2 + \left(q / q_{\max} \right)^2 \right]^{1/2} \\ &= \left[(Z^*)^2 + (q^*)^2 \right]^{1/2}. \end{aligned} \quad (7.6)$$

The proposed optimization function can be regarded as the Euclidean distance from the origin in the graph of Figure 7.1 to the normalized point $(Z^*, q^*) = (Z / Z_{\max}, q / q_{\max})$. As the normalized point is further away from the origin, the corresponding the working correlation structure has more parameters and/or a larger selection criterion measure than other candidate models. On the other hand, the normalized point closer to the origin indicates that the corresponding working correlation structure has a smaller selection criterion measures and/or fewer parameters than other candidate models. Therefore,

using this particular loss function, the working correlation structure having the minimum $f^*(Z, q)$ is chosen as the best approximating working correlation structure.

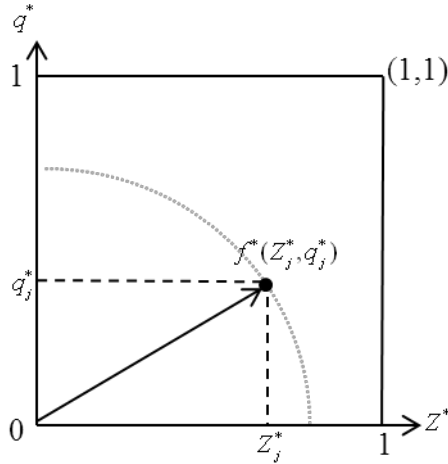


Figure 7.1 Optimization function: distance from the origin to the point (Z^*, q^*)

In summary, our proposed penalization approach for overparameterized working correlation structures consists of two steps: 1) the cluster detection of the working correlation structures and 2) the selection of the best approximating working correlation structure within the cluster using the optimization function. The proposed optimization function equally weights $Z^*(=Z/Z_{max})$ and $q^*(=q/q_{max})$ and it is more powerful to detect the true correlation structure when we exclude the working correlation structure with extremely large criterion values since it affects the scaling of Z . Therefore, unless we develop more sophisticated optimization function considering the optimal weights and cost components, the cluster detection is required to exclude those extreme values and narrow down the candidates. In the next section, the performance of the proposed penalization approach is examined through simulation study.

7.5 Simulation Study III

The goal of the Simulation Study III is to examine the effect of the method of penalization for the over-parameterized structure on the working correlation structure selection presented in Sections 7.3 and 7.4. The penalization method is applied to all six criterion measures (PT, WR, RMR, QIC, CIC, RJ).

The simulated datasets which are generated from the Simulation Study II are used in this simulation study. We consider correlated binary responses, correlated Poisson responses, and correlated Gaussian responses. The best approximating working correlation structure is selected from a set of candidate structures $W_{c2} = \{IN, EX, AR-1, TOEP, UN\}$. The correct identification rates of each criterion, with and without use of the penalization method, is used to assess performance of the criteria select the best working correlation structure based on their minimum criterion values.

7.5.1 Simulation III Results: Correlated Binary Responses

Results of this simulation study are used to assess the performance of the proposed method for penalizing over-parameterized working correlation structures relative to the same measures without using it. For the correlated binary responses, the frequencies of the working correlation structure out of 1000 independent replications are tabulated in Table 7.2 through Table 7.13. Corresponding simulation results for cases where the outcome vector is correlated Gaussian responses and the correlated Poisson random variables are shown in Appendix D, Tables D.1 ~ D.12 (Gaussian) and Tables D.13 ~ D.24 (Poisson).

7.5.1.1 The True Correlation Structure: AR-1

First, consider the simulation results for cases where the true correlation structure $C(\alpha)$ is AR-1 (Table 7.2 ~ Table 7.5). Without penalization for the over-parameterized structure, the correct identification rates (frequency/1000) of PT, WR, RMR, CIC and

QIC are in the range of 0% to 20% and tend to decrease as n increases. This is because the estimation of the over-parameterized structure R_{ov} as well as R_T becomes more precise as n increases. The unstructured matrix is more likely to be chosen as n increases, reflecting the small deviations from the truth as exhibited in that particular dataset, thus over-fitting to the data and producing a slightly smaller criterion measure. The RJ criterion selects the correct correlation structure with approximately 30 ~ 40 % but it also selects the exchangeable structure with approximately 30% even when $n = 200$.

With penalization for the over-parameterized structure, the correct identification rates of PT, WR and CIC increase as n increases, approaching 100% by $n = 200$. The RMR criterion performs worse since it uses less information of the disparity between two covariance matrices as compared to PT, WR and CIC. The RJ criterion also performs better as n increases, but its correct identification rates are only approximately 50% when $n = 200$. Unlike other criterion measures, QIC sometimes selects the working independence structure and its performance in working correlation selection is shown to be lower than other criterion measures considered in this simulation study.

In small sample sizes and binary data, the correct identification rates depend heavily on the strength of the correlation coefficients of R_T and the size of the matrix (m). When the true correlations are low, and the size of the matrix small (Table 7.2), the unstructured matrix is still preferred by PT, WR, RMR and CIC in the unpenalized setting at similar rates with AR-1, which is the true correlation structure in this case. Penalization improves the correct identification dramatically. If the number of repeated measurements is large and the true correlation coefficient is low (Table 7.4), the proposed penalization method cannot effectively penalize the unstructured matrix. Even though the bias-corrected sandwich variance estimator is used, the sandwich variance estimator under the unstructured matrix is still severely negatively biased in small samples. Thus, the proposed clustering detection method tends to capture the unstructured matrix only.

If the true α parameter is high (Tables 7.3 and 7.5), the true underlying correlation structure is more distinguishable from other structures. In this case, the disparity between the variance estimate under R_T and the variance estimate under the misspecified structure becomes large. Moreover, the negative bias of the sandwich variance estimator under the over-parameterized structure is reduced for the high true α parameter. Therefore, R_T is more likely to be included in the *cluster* by the proposed cluster detection method. Even if the criterion measure under the over-parameterized structure, especially R_{UN} , is the minimum, the optimization function selects the parsimonious R_T by penalizing the over-parameterized structures.

Although the percent correct selection using penalization is similar among PT, WR, and CIC, both PT and WR tend to outperform CIC in most cases. The measure based only on the largest generalized eigenvalue (RMR) generally performs better than the others when the true underlying structure is more complicated (TOEP and UN), and worse for the simple structures. The RJ criterion is clearly inferior to the others. We note that a variation of this method could be used without first identifying a cluster, but rather moving directly to the use of the optimization function using all candidate structures. This would be useful in settings where the correlations are all expected to be quite low and/or when all candidate structures would be expected to fit reasonably well.

Table 7.2 Frequencies of the working correlation structure identified using PT, WR, RMR, CIC, QIC and RJ with/without considering penalization for the over-parameterized structures from 1000 independent replications (Correlated binary response; $C(\alpha)$: AR-1(0.3), $m=4$).

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	15	117	457	411	15	83	346	88	468
	WR	0	17	123	463	397	14	85	349	95	457
	RMR	0	17	128	400	455	87	36	219	75	583
	CIC	10	68	194	318	410	16	77	346	87	474
	QIC	10	68	194	318	410	316	79	197	125	283
	RJ	0	306	338	209	147	29	406	560	1	4
30	PT	0	0	100	475	425	2	70	410	55	463
	WR	0	1	114	474	411	2	70	422	61	445
	RMR	0	0	102	421	477	70	36	270	62	562
	CIC	0	41	204	350	405	4	79	412	53	452
	QIC	0	41	204	350	405	326	85	221	92	276
	RJ	0	374	353	177	96	5	440	555	0	0
50	PT	0	0	93	490	417	0	69	569	16	346
	WR	0	1	102	484	413	0	74	578	17	331
	RMR	0	0	81	429	490	68	35	376	26	495
	CIC	0	28	181	369	422	0	65	563	16	356
	QIC	0	28	181	369	422	364	91	239	76	230
	RJ	0	330	386	192	92	1	454	545	0	0
100	PT	0	0	83	481	436	0	39	825	2	134
	WR	0	0	87	497	416	0	40	831	2	127
	RMR	0	0	65	391	544	30	35	563	14	358
	CIC	0	7	199	368	426	0	39	800	3	158
	QIC	0	7	199	368	426	422	122	249	37	170
	RJ	0	342	382	197	79	0	475	525	0	0
200	PT	0	0	83	455	462	0	9	973	0	18
	WR	0	0	96	460	444	0	10	973	0	17
	RMR	0	0	65	365	570	12	15	771	8	194
	CIC	0	1	218	328	453	0	14	961	0	25
	QIC	0	1	218	328	453	401	138	307	12	142
	RJ	0	357	375	188	80	0	471	529	0	0

Table 7.3 Frequencies of the working correlation structure identified using PT, WR, RMR, CIC, QIC and RJ with/without considering penalization for the over-parameterized structures from 1000 independent replications (Correlated binary response; $C(\alpha)$: AR-1(0.6), $m=4$).

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	35	55	197	209	504	36	111	685	10	158
	WR	35	56	200	203	506	35	110	686	26	143
	RMR	15	24	83	150	728	68	72	426	33	401
	CIC	42	96	210	224	428	39	121	679	14	147
	QIC	240	169	139	204	248	338	169	259	85	149
	RJ	35	278	348	129	210	37	441	521	0	1
30	PT	2	27	90	149	732	13	62	807	3	115
	WR	2	28	92	148	730	13	58	813	8	108
	RMR	7	11	62	131	789	45	51	563	17	324
	CIC	2	59	132	226	581	14	82	799	1	104
	QIC	264	196	109	158	273	376	172	253	64	135
	RJ	2	309	357	112	220	14	431	555	0	0
50	PT	5	11	73	118	793	5	27	932	0	36
	WR	5	11	76	120	788	5	25	940	0	30
	RMR	7	10	68	136	779	23	55	739	10	173
	CIC	5	25	141	219	610	5	49	912	0	34
	QIC	244	194	114	167	281	379	166	297	42	116
	RJ	5	306	358	113	218	5	474	521	0	0
100	PT	0	3	28	78	891	0	5	992	0	3
	WR	0	3	25	74	898	0	5	993	0	2
	RMR	1	1	75	139	784	2	18	925	4	51
	CIC	0	6	133	164	697	0	8	990	0	2
	QIC	283	225	99	130	263	413	191	283	12	101
	RJ	0	340	338	125	197	0	478	522	0	0
200	PT	0	0	26	62	912	0	0	1000	0	0
	WR	0	0	19	65	916	0	0	1000	0	0
	RMR	0	0	71	129	800	0	5	991	0	4
	CIC	0	0	108	193	699	0	0	1000	0	0
	QIC	287	228	130	117	238	422	198	315	6	59
	RJ	0	308	393	116	183	0	448	552	0	0

Table 7.4 Frequencies of the working correlation structure identified using PT, WR, RMR, CIC, QIC and RJ with/without considering penalization for the over-parameterized structures from 1000 independent replications (Correlated binary response; $C(\alpha)$: AR-1(0.3), $m=6$).

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	1	5	64	215	715	2	16	196	123	663
	WR	0	5	64	215	716	2	16	193	137	652
	RMR	4	8	14	38	936	29	6	59	37	869
	CIC	19	24	71	204	682	2	18	193	125	662
	QIC	150	101	134	249	366	307	48	157	184	304
	RJ	3	199	416	177	205	18	414	560	7	1
30	PT	0	1	19	77	903	0	6	170	41	783
	WR	0	1	20	76	903	0	7	175	52	766
	RMR	8	7	8	20	957	22	5	66	27	880
	CIC	11	12	44	109	824	0	5	154	54	787
	QIC	143	118	105	169	465	321	56	140	97	386
	RJ	0	202	385	158	255	2	438	559	1	0
50	PT	0	0	5	9	986	0	5	253	13	729
	WR	0	0	5	11	984	0	7	287	19	687
	RMR	4	4	3	6	983	7	11	105	14	863
	CIC	4	13	28	58	897	0	5	224	18	753
	QIC	193	108	116	135	448	355	63	182	59	341
	RJ	0	249	367	141	243	0	450	550	0	0
100	PT	0	0	0	2	998	0	2	584	4	410
	WR	0	0	0	2	998	0	1	632	2	365
	RMR	0	0	5	4	991	3	6	252	8	731
	CIC	0	7	32	47	914	0	2	558	4	436
	QIC	211	147	120	127	395	366	87	200	38	309
	RJ	0	265	384	137	214	0	447	553	0	0
200	PT	0	0	0	3	997	0	0	923	0	77
	WR	0	0	0	3	997	0	0	944	0	56
	RMR	0	0	2	3	995	1	1	455	4	539
	CIC	1	0	29	80	890	0	1	887	0	112
	QIC	243	176	120	123	338	410	105	250	21	214
	RJ	0	301	391	155	153	0	468	532	0	0

Table 7.5 Frequencies of the working correlation structure identified using PT, WR, RMR, CIC, QIC and RJ with/without considering penalization for the over-parameterized structures from 1000 independent replications (Correlated binary response; $C(\alpha)$: AR-1(0.6), $m=6$).

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	5	31	134	334	496	5	40	607	31	317
	WR	5	31	148	320	496	5	38	615	58	284
	RMR	2	9	51	102	836	20	14	243	42	681
	CIC	12	50	155	325	458	6	47	587	59	301
	QIC	226	160	135	243	236	341	129	263	106	161
	RJ	5	294	376	176	149	6	445	546	3	0
30	PT	1	19	81	242	657	1	23	728	7	241
	WR	1	19	91	227	662	1	22	705	71	201
	RMR	5	3	34	95	863	21	15	361	24	579
	CIC	2	32	132	249	585	1	34	665	56	244
	QIC	204	183	143	190	280	338	131	273	67	191
	RJ	1	304	361	154	180	1	464	530	5	0
50	PT	0	2	37	130	831	0	4	917	0	79
	WR	0	2	39	127	832	0	2	918	23	57
	RMR	1	1	27	65	906	3	10	581	9	397
	CIC	0	7	75	165	753	0	6	873	13	108
	QIC	205	171	131	183	310	317	140	290	47	206
	RJ	0	321	349	148	182	0	467	532	1	0
100	PT	0	0	7	26	967	0	0	999	0	1
	WR	0	0	6	25	969	0	0	999	0	1
	RMR	0	0	18	53	929	0	1	873	3	123
	CIC	0	0	38	107	855	0	0	992	1	7
	QIC	250	157	128	118	347	342	135	331	19	173
	RJ	0	301	345	141	213	0	499	501	0	0
200	PT	0	0	5	11	984	0	0	1000	0	0
	WR	0	0	5	14	981	0	0	1000	0	0
	RMR	0	0	14	38	948	0	0	991	0	9
	CIC	0	0	47	77	876	0	0	1000	0	0
	QIC	258	231	113	111	287	363	199	312	10	116
	RJ	0	329	353	137	181	0	472	528	0	0

7.5.1.2 The True Correlation Structure: Exchangeable

When the true correlation structure is exchangeable, similar patterns are identified except the RJ criterion. As discussed above, the RJ criterion is known to have a tendency to select the exchangeable structure regardless of the true correlation structure. If the true correlation structure is exchangeable, its performance is better than PT, WR and CIC. However, the true underlying correlation structure is generally unknown.

The improvement in correct identification seen when the true structure is exchangeable and the correlation is high is not as strong as that seen when it is an AR-1 structure with high correlation, unless the sample size is 200.

Table 7.6 Frequencies of the working correlation structure identified using PT, WR, RMR, CIC, QIC and RJ with/without considering penalization for the over-parameterized structures from 1000 independent replications (Correlated binary response; $C(\alpha)$: EX(0.3), $m=4$).

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	22	69	31	169	709	37	326	141	78	418
	WR	22	75	31	164	708	29	332	141	89	409
	RMR	44	55	30	77	794	184	130	81	35	570
	CIC	58	106	83	173	580	47	316	140	70	427
	QIC	189	89	173	156	393	323	111	197	80	289
	RJ	30	313	60	258	339	33	868	88	4	7
30	PT	2	55	9	113	821	11	450	110	42	387
	WR	2	51	10	116	821	7	466	115	45	367
	RMR	54	54	26	76	790	247	146	76	29	502
	CIC	33	105	63	144	655	18	439	109	34	400
	QIC	180	83	225	114	398	322	105	240	49	284
	RJ	4	339	29	253	375	5	950	43	0	2
50	PT	0	16	1	94	889	1	640	87	10	262
	WR	0	16	1	96	887	0	652	92	10	246
	RMR	36	43	17	69	835	293	181	67	17	442
	CIC	13	106	40	142	699	3	604	85	10	298
	QIC	218	72	255	96	359	370	109	272	35	214
	RJ	0	337	7	259	397	0	985	15	0	0
100	PT	0	11	0	71	918	0	887	29	1	83
	WR	0	10	0	71	919	0	895	31	2	72
	RMR	28	47	4	66	855	499	184	44	7	266
	CIC	2	106	11	150	731	0	863	30	0	107
	QIC	243	66	309	79	303	397	108	315	10	170
	RJ	0	331	0	278	391	0	1000	0	0	0
200	PT	0	13	0	68	919	0	990	3	0	7
	WR	0	13	0	70	917	0	992	3	0	5
	RMR	6	53	0	65	876	655	219	14	2	110
	CIC	0	112	2	168	718	0	986	1	0	13
	QIC	252	59	358	68	263	440	121	345	5	89
	RJ	0	364	0	260	376	0	1000	0	0	0

Table 7.7 Frequencies of the working correlation structure identified using PT, WR, RMR, CIC, QIC and RJ with/without considering penalization for the over-parameterized structures from 1000 independent replications (Correlated binary response; $C(\alpha)$: EX(0.6), $m=4$).

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	100	161	28	251	460	100	495	264	24	117
	WR	100	157	28	252	463	100	477	273	46	104
	RMR	38	92	20	151	699	168	328	75	31	398
	CIC	114	214	69	239	364	116	489	248	29	118
	QIC	311	162	162	166	199	388	208	268	49	87
	RJ	100	403	47	244	206	102	783	115	0	0
30	PT	72	122	14	172	620	72	608	219	8	93
	WR	72	120	14	172	622	72	600	226	25	77
	RMR	26	92	15	125	742	172	469	62	10	287
	CIC	76	190	57	197	480	76	619	198	11	96
	QIC	299	147	189	133	232	393	185	276	47	99
	RJ	72	394	14	225	295	72	846	81	1	0
50	PT	25	64	3	136	772	25	767	177	2	29
	WR	25	64	3	130	778	25	754	181	14	26
	RMR	18	84	3	121	774	221	580	42	3	154
	CIC	26	158	33	221	562	25	773	166	4	32
	QIC	277	122	242	101	258	395	173	304	28	100
	RJ	25	375	1	261	338	25	915	60	0	0
100	PT	2	28	0	83	887	2	912	82	0	4
	WR	2	25	0	77	896	2	912	82	1	3
	RMR	4	86	0	133	777	207	750	7	0	36
	CIC	2	142	10	185	661	2	919	75	0	4
	QIC	305	88	300	76	231	462	120	335	9	74
	RJ	2	376	0	234	388	2	962	36	0	0
200	PT	0	8	0	63	929	0	975	25	0	0
	WR	0	5	0	55	940	0	974	26	0	0
	RMR	0	77	0	116	807	224	774	2	0	0
	CIC	0	131	1	181	687	0	980	20	0	0
	QIC	336	50	369	64	181	449	107	391	3	50
	RJ	0	365	0	253	382	0	986	14	0	0

Table 7.8 Frequencies of the working correlation structure identified using PT, WR, RMR, CIC, QIC and RJ with/without considering penalization for the over-parameterized structures from 1000 independent replications (Correlated binary response; $C(\alpha)$: EX(0.3), $m=6$).

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	5	58	4	203	730	7	229	70	88	606
	WR	4	64	6	199	727	6	216	71	113	594
	RMR	22	26	15	45	892	50	96	22	29	803
	CIC	33	76	32	194	665	16	214	67	99	604
	QIC	158	91	161	164	426	275	105	185	98	337
	RJ	6	402	12	283	297	7	950	38	3	2
30	PT	1	19	0	100	880	2	235	47	41	675
	WR	1	20	0	99	880	2	223	46	87	642
	RMR	14	22	6	42	916	53	123	18	18	788
	CIC	8	51	26	117	798	3	216	43	50	688
	QIC	158	59	165	126	492	268	85	200	58	389
	RJ	1	350	2	280	367	1	980	17	1	1
50	PT	0	2	0	21	977	0	425	14	8	553
	WR	0	1	0	22	977	0	459	15	25	501
	RMR	5	14	4	19	958	41	218	22	13	706
	CIC	5	37	7	72	879	0	373	16	15	596
	QIC	159	63	242	73	463	288	100	246	35	331
	RJ	0	340	1	259	400	0	998	2	0	0
100	PT	0	3	0	9	988	0	827	3	0	170
	WR	0	3	0	11	986	0	866	3	1	130
	RMR	1	14	1	19	965	21	439	13	6	521
	CIC	0	41	2	69	888	0	728	2	0	270
	QIC	222	60	263	77	378	336	105	282	17	260
	RJ	0	316	0	256	428	0	1000	0	0	0
200	PT	0	0	0	5	995	0	989	0	0	11
	WR	0	0	0	5	995	0	992	0	0	8
	RMR	0	6	0	14	980	15	764	4	2	215
	CIC	0	49	0	73	878	0	965	0	0	35
	QIC	279	61	298	70	292	430	126	293	7	144
	RJ	0	315	0	278	407	0	1000	0	0	0

Table 7.9 Frequencies of the working correlation structure identified using PT, WR, RMR, CIC, QIC and RJ with/without considering penalization for the over-parameterized structures from 1000 independent replications (Correlated binary response; $C(\alpha)$: EX(0.6), $m=6$).

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	50	162	12	287	489	53	355	256	100	236
	WR	50	164	12	284	490	53	350	267	107	223
	RMR	14	64	14	118	790	63	263	25	24	625
	CIC	72	214	44	233	437	63	384	225	112	216
	QIC	275	171	133	188	233	361	193	252	80	114
	RJ	50	470	7	266	207	51	842	99	8	0
30	PT	22	127	2	272	577	22	434	313	38	193
	WR	22	125	1	273	579	22	396	312	103	167
	RMR	8	51	5	86	850	40	409	16	10	525
	CIC	25	189	21	262	503	25	451	277	62	185
	QIC	284	183	112	174	247	371	215	250	58	106
	RJ	22	453	0	277	248	22	855	123	0	0
50	PT	7	47	0	122	824	7	586	307	8	92
	WR	7	47	0	118	828	7	571	311	54	57
	RMR	3	31	0	61	905	18	740	2	4	236
	CIC	9	112	12	167	700	8	606	278	30	78
	QIC	254	135	178	140	293	349	180	289	66	116
	RJ	7	377	0	257	359	7	872	121	0	0
100	PT	0	6	0	24	970	0	797	202	0	1
	WR	0	6	0	24	970	0	789	203	8	0
	RMR	0	36	0	58	906	6	977	0	0	17
	CIC	0	90	1	135	774	0	815	179	1	5
	QIC	287	82	253	94	284	371	156	333	47	93
	RJ	0	379	0	235	386	0	922	78	0	0
200	PT	0	3	0	8	989	0	913	87	0	0
	WR	0	3	0	4	993	0	912	88	0	0
	RMR	0	29	0	56	915	0	1000	0	0	0
	CIC	0	75	0	143	782	0	915	85	0	0
	QIC	317	55	338	57	233	410	128	383	24	55
	RJ	0	354	0	255	391	0	967	33	0	0

7.5.1.3 The True Correlation Structure: Toeplitz

When the true correlation structure is Toeplitz (defined in Table 3.2) and the distribution is a multivariate binomial, PT, WR and CIC perform better by penalizing over-parameterized structures (Table 7.10 and Table 7.11). Note that both R_{TOEP} and R_{UN} are penalized. Since R_{UN} contains more parameters, the optimization function penalizes R_{UN} more than R_{TOEP} . In this simulation setting, the true 4×4 Toeplitz correlation structure has considerable similarity to the exchangeable structure, with most of its correlations being exactly .25. Therefore, R_{EX} is chosen at similar rates as R_{TOEP} , which is the true correlation structure. Whereas PT and WR with penalization perform better as n increases, the accuracy rates of CIC with penalization decrease for 4×4 matrix since R_{EX} is more likely to be chosen by CIC as n increases (Table 7.10). For the 6×6 Toeplitz structure, we do not observe this decrease in correct identification as n gets larger in CIC, since the underlying Toeplitz matrix now is more easily distinguished from exchangeable (Table 7.11).

The RMR and QIC do not perform well as much as PT, WR and CIC when the proposed penalization method is used. The RJ criterion still tends to select the exchangeable structure. Furthermore, since the true underlying Toeplitz structure (defined in is Table 3.2) has similarity to the exchangeable structure, it selects the exchangeable structure with 100% when $n = 100$ and 200.

Table 7.10 Frequencies of the working correlation structure identified using PT, WR, RMR, CIC, QIC and RJ with/without considering penalization for the over-parameterized structures from 1000 independent replications (Correlated binary response; $C(\alpha)$: TOEP, $m=4$).

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	15	277	11	132	565	24	428	42	310	196
	WR	13	278	11	133	565	21	419	44	342	174
	RMR	26	50	6	116	802	232	155	5	192	416
	CIC	59	266	27	153	495	67	417	37	275	204
	QIC	196	167	191	144	302	354	152	162	144	188
	RJ	15	524	5	211	245	20	950	19	9	2
30	PT	3	233	1	117	646	8	464	23	382	123
	WR	3	233	1	122	641	7	420	23	443	107
	RMR	17	33	0	105	845	343	109	0	213	335
	CIC	22	249	9	167	553	42	465	21	338	134
	QIC	178	170	231	146	275	370	165	186	131	148
	RJ	3	538	2	217	240	5	982	12	1	0
50	PT	1	143	0	64	792	2	473	7	475	43
	WR	1	143	0	64	792	1	401	7	563	28
	RMR	6	19	0	86	889	562	87	0	190	161
	CIC	7	171	1	170	651	22	540	6	377	55
	QIC	167	130	260	147	296	359	139	212	152	138
	RJ	1	492	1	219	287	1	997	2	0	0
100	PT	0	61	0	65	874	0	432	0	565	3
	WR	0	61	0	66	873	0	336	0	662	2
	RMR	2	3	0	77	918	850	11	0	107	32
	CIC	0	74	0	176	750	6	623	0	367	4
	QIC	191	80	297	155	277	384	103	244	160	109
	RJ	0	465	0	257	278	0	1000	0	0	0
200	PT	0	9	0	62	929	0	405	0	595	0
	WR	0	9	0	62	929	0	249	0	751	0
	RMR	0	0	0	80	920	950	1	0	49	0
	CIC	0	9	0	217	774	1	725	0	274	0
	QIC	161	54	321	173	291	367	108	270	175	80
	RJ	0	463	0	256	281	0	1000	0	0	0

Table 7.11 Frequencies of the working correlation structure identified using PT, WR, RMR, CIC, QIC and RJ with/without considering penalization for the over-parameterized structures from 1000 independent replications (Correlated binary response; $C(\alpha)$: TOEP, $m=6$).

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	3	153	1	382	461	13	207	76	426	278
	WR	3	152	1	384	460	10	193	82	464	251
	RMR	3	29	6	125	837	67	102	8	105	718
	CIC	29	175	14	369	413	25	227	75	385	288
	QIC	202	158	159	230	251	327	147	170	180	176
	RJ	3	546	1	267	183	8	942	35	13	2
30	PT	2	106	0	327	565	6	168	52	553	221
	WR	2	106	0	323	569	4	137	57	627	175
	RMR	7	15	5	118	855	63	162	8	116	651
	CIC	8	125	3	345	519	17	215	47	461	260
	QIC	187	139	170	237	267	313	136	192	181	178
	RJ	2	523	0	265	210	4	968	15	13	0
50	PT	0	44	0	250	706	0	87	29	770	114
	WR	0	44	0	253	703	0	53	29	839	79
	RMR	1	9	1	78	911	28	268	4	154	546
	CIC	1	55	0	303	641	4	137	29	670	160
	QIC	207	94	213	213	273	340	117	221	157	165
	RJ	0	473	0	281	246	1	987	11	1	0
100	PT	0	2	0	97	901	0	23	8	962	7
	WR	0	2	0	99	899	0	2	8	984	6
	RMR	0	6	0	50	944	19	513	0	210	258
	CIC	0	2	0	178	820	0	42	8	939	11
	QIC	203	74	259	151	313	348	88	267	139	158
	RJ	0	440	0	266	294	0	997	3	0	0
200	PT	0	0	0	37	963	0	3	0	997	0
	WR	0	0	0	40	960	0	0	0	1000	0
	RMR	0	0	0	33	967	8	800	0	157	35
	CIC	0	0	0	179	821	0	9	0	991	0
	QIC	238	66	264	127	305	372	87	282	118	141
	RJ	0	456	0	249	295	0	1000	0	0	0

7.5.1.4 The True Correlation Structure: Unstructured

Lastly, if the true correlation structure is unstructured (defined in Table 3.2), the proposed penalization method in the binary data setting decreases the correct identification rates of the criterion measures when the matrix size is small 4×4 (Table 7.12). The 4×4 true underlying structure in this simulation study has considerable similarity with AR-1. Therefore, AR-1 is more likely to be chosen in small samples since both R_{TOEP} and R_{UN} are penalized. However, when $m=6$, all criterion measures, whether penalized or not, perform effectively the same in the binary data setting when the underlying structure is unstructured.

7.5.2 Simulation Results III: Correlated Gaussian

Responses and Poisson Responses

The frequencies of the working correlation structure out of 1000 independent replications for the correlated Gaussian responses (Table D.1 ~ Table D.12) and for the correlated Poisson responses (Table D.13 ~ Table D.24) are presented in Appendix D. They present similar results to the simulation results for the correlated binary responses.

Table 7.12 Frequencies of the working correlation structure identified using PT, WR, RMR, CIC, QIC and RJ with/without considering penalization for the over-parameterized structures from 1000 independent replications (Correlated binary response; $C(\alpha)$: UN, $m=4$).

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	38	62	253	245	402	1	76	515	7	401
	WR	38	65	255	238	404	0	82	529	13	376
	RMR	5	8	75	104	808	11	96	510	22	361
	CIC	47	87	229	265	372	0	96	575	11	318
	QIC	260	196	135	238	171	115	124	413	23	325
	RJ	38	308	317	126	211	1	505	418	0	76
30	PT	7	30	195	225	543	0	34	503	3	460
	WR	7	30	211	209	543	0	37	537	4	422
	RMR	2	5	38	76	879	3	43	511	8	435
	CIC	10	49	200	253	488	0	44	606	5	345
	QIC	241	170	130	247	212	80	76	495	9	340
	RJ	7	283	366	117	227	2	498	442	0	58
50	PT	4	8	175	176	637	0	12	454	1	533
	WR	4	8	186	165	637	0	12	511	1	476
	RMR	0	1	24	39	936	0	15	506	0	479
	CIC	4	15	174	214	593	0	17	626	0	357
	QIC	249	192	133	226	200	41	45	545	1	368
	RJ	4	300	309	131	256	0	494	480	0	26
100	PT	0	1	107	104	788	0	0	371	0	629
	WR	0	1	113	98	788	0	0	451	0	549
	RMR	0	0	5	9	986	0	0	416	0	584
	CIC	0	3	95	129	773	0	1	627	0	372
	QIC	234	195	99	228	244	13	8	575	0	404
	RJ	0	282	305	75	338	0	542	454	0	4
200	PT	0	0	39	32	929	0	0	199	0	801
	WR	0	0	43	28	929	0	0	310	0	690
	RMR	0	0	0	0	1000	0	0	260	0	740
	CIC	0	0	27	45	928	0	0	549	0	451
	QIC	249	190	92	189	280	1	2	594	0	403
	RJ	0	303	256	90	351	0	543	457	0	0

Table 7.13 Frequencies of the working correlation structure identified using PT, WR, RMR, CIC, QIC and RJ with/without considering penalization for the over-parameterized structures from 1000 independent replications (Correlated binary response; $C(\alpha)$: UN, $m=6$).

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	5	39	68	467	421	0	103	340	55	502
	WR	5	35	79	459	422	0	79	318	121	482
	RMR	10	21	17	74	878	10	174	290	44	482
	CIC	25	102	108	355	410	0	125	312	106	457
	QIC	215	142	214	210	219	176	105	244	33	442
	RJ	5	396	147	272	180	0	716	187	11	86
30	PT	2	10	48	417	523	0	58	323	28	591
	WR	2	9	59	407	523	0	41	291	89	579
	RMR	3	16	5	61	915	3	116	295	27	559
	CIC	12	58	84	334	512	0	63	324	96	517
	QIC	212	149	220	207	212	140	94	250	32	484
	RJ	2	385	91	274	248	0	828	109	6	57
50	PT	0	1	14	338	647	0	29	243	8	720
	WR	0	1	16	336	647	0	15	231	55	699
	RMR	1	7	2	13	977	1	69	237	3	690
	CIC	1	34	45	279	641	0	42	282	49	627
	QIC	237	149	215	133	266	89	80	239	7	585
	RJ	0	386	46	267	301	0	903	66	0	31
100	PT	0	0	5	174	821	0	7	145	0	848
	WR	0	0	6	173	821	0	1	143	19	837
	RMR	0	1	0	3	996	0	39	140	0	821
	CIC	0	6	16	163	815	0	16	174	22	788
	QIC	241	138	264	67	290	44	43	187	1	725
	RJ	0	370	9	241	380	0	969	9	0	22
200	PT	0	0	0	66	934	0	0	70	0	930
	WR	0	0	1	65	934	0	0	72	3	925
	RMR	0	0	0	0	1000	0	8	78	0	914
	CIC	0	1	0	65	934	0	1	102	5	892
	QIC	213	130	250	53	354	17	9	153	1	820
	RJ	0	380	1	181	438	0	995	2	0	3

CHAPTER 8

ANOTHER PENALIZATION APPROACH USING BOOSTRAPPED DATA

The motivation behind the proposed penalization approach using the cluster detection is that the over-parameterized structure and the true correlation structure have similar measures, but the misspecified structure does not. To find the group (or cluster) of working correlation structures that are similar to each other, the cluster detection method based on the spacings of the order statistics of the criterion measures was proposed in chapter 7. In this chapter, we discuss another method we investigated to find the group of working correlation structures which are not statistically different.

8.1 Incorporating Measure Variability into the Selection

An over-parameterized model and the true model would tend to have similar measures, but in practice, there is no reliable rule to determine how small a deviation between the two measures implies “similar.” The bootstrap method can be used to construct an estimate of the standard deviation of the proposed selection measures. This information can be incorporated into the covariance structure selection process by using it to determine if two measures are statistically different from each other. The approach proposed is to fit the data using different working correlation structures, compute the selection criterion for each, find the bootstrap variance of that measure, and incorporate the information into a classical ANOVA analysis and follow-up comparisons to look for measures that differ significantly. This approach results in identification of groups of working correlation models with criterion measures that are “not” statistically different from each other. Then, we would propose to declare the most parsimonious structure within the (best) group as the best approximating structure.

8.2 Simulation Study IV

The effectiveness of this approach in the GEE covariance selection problem is assessed in the simulation study IV. Figure 8.1 shows a flow chart of a specific simulation process. The empirical bootstrap method will be used, in which B bootstrap samples from the observed (simulated) data are drawn using resampling with replacement. For this simulation experiment, B is defined as the half of the sample size (i.e., $B = 0.5n$). To get one bootstrap sample, a simple random sample of size n is taken, with replacement, from the n independent vectors of the observed data. This process is repeated until B bootstrap samples are generated by resampling with replacement and they are treated as a set of B independent real samples from population.

For each bootstrap sample, five GEE models under different working correlation structures (IN, EX, AR-1, TOEP, UN) are fit. Mean models are correctly specified so that effects of different working correlation assumptions are the only model fit component being assessed. For each GEE model fit, the PT, WR, RMR, CIC and RJ are calculated. That is, within one bootstrap sample, PT(IN), PT(EX), PT(AR-1), PT(TOEP), PT(UN) are obtained. This is repeated until we have obtained B samples of each PT(IN), PT(EX), PT(AR-1), PT(TOEP), and PT(UN) measure. Then, an applicable hypothesis testing procedure, such as ANOVA is performed to compare the mean of the B criterion measures under each of the five different working structures. Since small values of the selection criterion measures indicate strong fit to the data, the goal is to find the set of candidate working covariance models whose criterion measures are not significantly different from the minimum value for that criterion. Once those groups are found (e.g., through Tukey's HSD testing) the most parsimonious working correlation is chosen as the best working correlation structure. If two or more structures in that group have the same number of parameters (e.g., AR-1 and EX), we propose the working correlation structure with the lowest criterion value will be chosen as "best". We assess

the performance of the bootstrapping approach by 100 independent replicated simulations.

In the Simulation Study IV, we consider the correlated binary responses. The simulation design is a factorial with the factor levels described in Table 8.1. We denote PT, WR, RMR, CIC, QIC and RJ applying the penalization method using bootstrapped data by PT_b , WR_b , RMR_b , CIC_b , QIC_b and RJ_b .

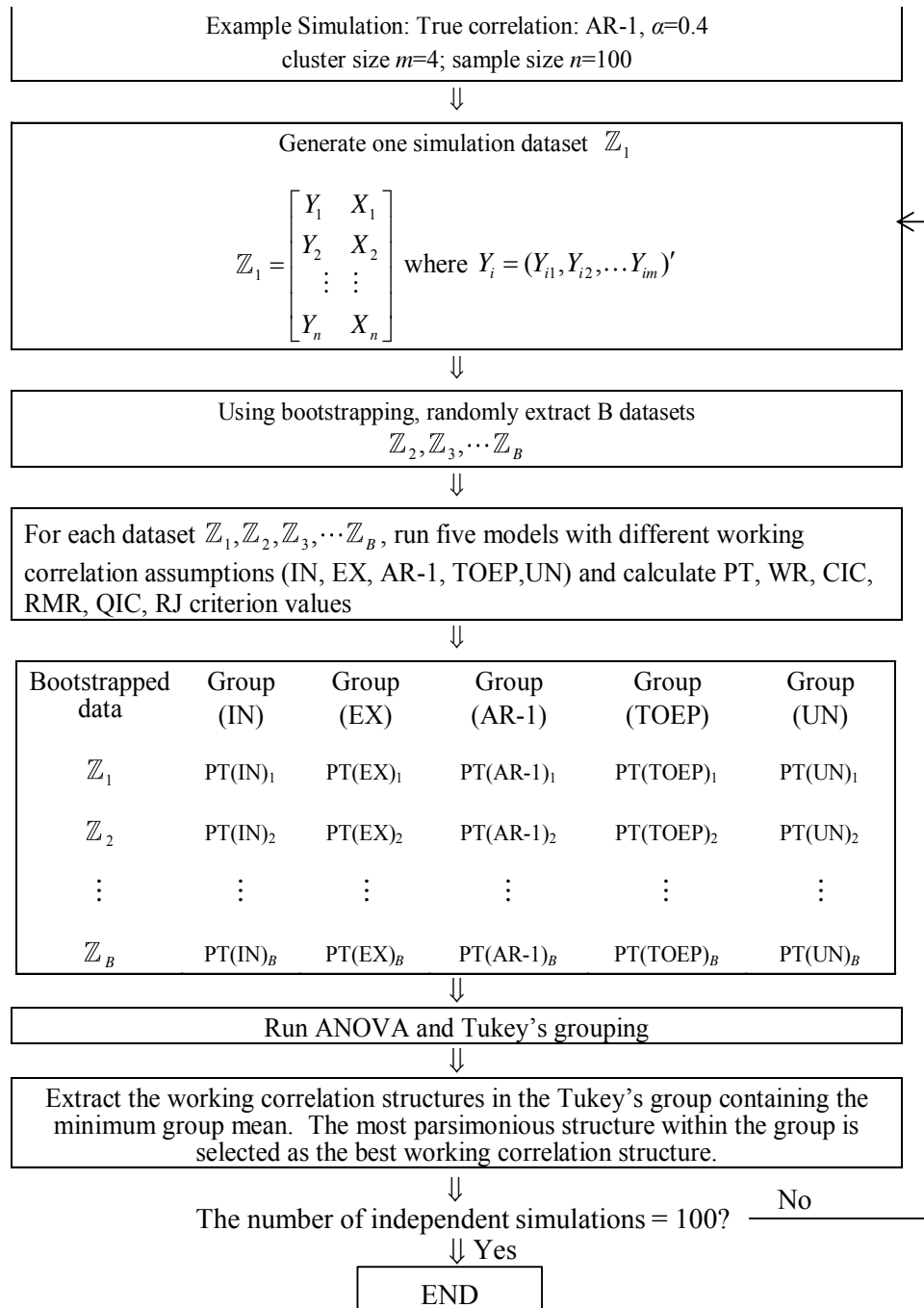


Figure 8.1 Flow chart of the bootstrap simulation study

Table 8.1 Simulation Study IV Design Parameters

Factor	Levels
Distribution (D)	Correlated binary responses $\text{logit}(\mu_{it}) = \beta_1 x_{1t} + \beta_2 x_{2t}$, where x_{1t} and x_{2t} are independently generated from the uniform distribution $U[0.5, 1]$. and $\beta_1 = \beta_2 = 0.3$
Response vector dimension (m)	$m = 4, 6$
True Correlation Structure $C(\alpha)$	Exchangeable: $\text{EX}(\alpha)$, $\alpha = 0.3, 0.7$ Autoregressive of order 1: $\text{AR-1}(\alpha)$, $\alpha = 0.3, 0.7$
Working Correlation structure $R(\alpha)$	Independence (IN), Exchangeable (EX), Autoregressive (AR-1), Toeplitz (TOEP), and unstructured (UN).
Sample Sizes (n)	$n = 100, 200$

The simulation results reveal that the second penalization method using the bootstrapped data is ineffective to improve the performance of PT, WR, RMR, CIC, QIC and RJ (Table 8.2 ~ Table 8.5).

Over-parameterized structures are still preferred by PT_b , WR_b . Even though CIC_b performs better than PT_b , WR_b , its overall correct identification rate is low even though the sample size is large. Note that we expect that the criterion value under R_T and the criterion value under R_{ov} will be “non-significantly” different. In other words, we expect that R_T and R_{ov} will be included in the same cluster. However, the proposed statistical test using the bootstrap sample tends to provide a significant difference between the two. For instance, $\text{PT}(\text{UN})$ is the minimum among candidates when the true correlation

structure is R_{AR-1} . Even though $PT(AR-1)$ is close to $PT(UN)$, the proposed statistical test concludes that $PT(UN)$ and $PT(AR-1)$ are significantly different. Thus, only R_{UN} is included in the cluster and R_{UN} is chosen as the best working correlation structure.

On the other hand, the independent working correlation structure is preferred by RMR_b , QIC_b and RJ_b . This is because these criterion measures under all five working correlation structures are not significantly different. Therefore, the most parsimonious structure, R_{IN} , is chosen as the best working correlation structure.

In fact, the statistical significance of the proposed statistical method is sensitive to the bootstrapped sample size (B) and the type I error rate. In particular, if we use a larger B , we are more likely to get significant results, capturing the difference in measures due to biases and over-fitting, rather than finding measures to be similar because they (asymptotically) estimate the same quantities. Different post-hoc pairwise comparison tests such as the Bonferroni and the Student Newman-Keuls test do not improve the performance of the proposed method using the bootstrap sample.

This simulation study was performed prior to the development of the work presented in chapter 7. The penalization for over-parameterization here is simply to choose the model with the smallest number of parameters among those with the “cluster”, without regard to the tradeoff of fit versus number of parameters, as was done in chapter 7.

Table 8.2 The correct identification rates (%) of the working correlation structure identified by PT_b , WR_b , RMR_b , CIC_b , QIC_b and RJ_b applying penalization method using bootstrapped data: $C(\alpha)$: AR-1 and $m=4$.

$C(\alpha)$	n	Criterion	IN	EX	AR	TP	UN
AR-1(0.3) $m=4$	100	PT_b	1	0	10	50	39
		WR_b	1	0	5	49	45
		RMR_b	96	0	4	0	0
		CIC_b	21	29	26	13	11
		QIC_b	100	0	0	0	0
		RJ_b	37	20	11	21	11
	200	PT_b	0	0	7	45	48
		WR_b	0	0	5	29	66
		RMR_b	62	10	28	0	0
		CIC_b	0	16	57	16	11
		QIC_b	98	1	1	0	0
		RJ_b	43	21	21	14	1
AR-1(0.7) $m=4$	100	PT_b	0	0	28	16	56
		WR_b	0	0	40	13	47
		RMR_b	2	12	67	5	14
		CIC_b	0	7	66	11	16
		QIC_b	84	10	3	2	1
		RJ_b	44	7	24	25	0
	200	PT_b	0	0	14	16	70
		WR_b	0	0	22	15	63
		RMR_b	0	0	77	11	12
		CIC_b	0	0	63	15	22
		QIC_b	77	14	8	1	0
		RJ_b	62	15	23	0	0

Table 8.3 The correct identification rates (%) of the working correlation structure identified by PT_b , WR_b , RMR_b , CIC_b , QIC_b and RJ_b applying penalization method using bootstrapped data: $C(\alpha)$: AR-1 and $m=6$.

$C(\alpha)$	n	Criterion	IN	EX	AR	TP	UN
AR-1(0.3) $m=6$	100	PT_b	0	0	0	0	100
		WR_b	0	0	0	0	100
		RMR_b	54	0	1	17	28
		CIC_b	3	3	8	25	61
		QIC_b	96	0	1	1	2
		RJ_b	44	25	17	14	0
	200	PT_b	0	0	0	0	100
		WR_b	0	0	0	0	100
		RMR_b	43	2	23	13	19
		CIC_b	0	1	21	16	62
		QIC_b	93	1	4	1	1
		RJ_b	50	24	19	7	0
AR-1(0.7) $m=6$	100	PT_b	1	0	11	6	82
		WR_b	1	0	13	4	82
		RMR_b	1	0	45	21	33
		CIC_b	2	2	39	17	40
		QIC_b	87	5	6	0	2
		RJ_b	19	9	8	64	0
	200	PT_b	0	0	3	2	95
		WR_b	0	0	3	2	95
		RMR_b	1	0	30	22	47
		CIC_b	0	0	30	29	41
		QIC_b	69	14	17	0	0
		RJ_b	34	11	35	19	1

Table 8.4 The correct identification rates (%) of the working correlation structure identified by PT_b , WR_b , RMR_b , CIC_b , QIC_b and RJ_b applying penalization method using bootstrapped data: $C(\alpha)$: EX and $m=4$.

$C(\alpha)$	n	Criterion	IN	EX	AR	TP	UN
EX(0.3) $m=4$	100	PT_b	0	0	0	17	33
		WR_b	0	0	0	16	34
		RMR_b	98	2	0	0	0
		CIC_b	3	29	0	14	4
		QIC_b	100	0	0	0	0
		RJ_b	54	17	22	7	0
	200	PT_b	0	1	0	23	76
		WR_b	0	0	0	20	80
		RMR_b	85	12	0	2	1
		CIC_b	1	64	0	17	18
		QIC_b	96	1	3	0	0
		RJ_b	59	18	21	2	0
EX(0.7) $m=4$	100	PT_b	0	4	0	19	27
		WR_b	0	4	0	25	21
		RMR_b	2	46	0	22	30
		CIC_b	0	30	2	9	9
		QIC_b	82	12	6	0	0
		RJ_b	65	15	10	9	1
	200	PT_b	0	5	0	37	58
		WR_b	0	9	0	49	42
		RMR_b	0	45	0	23	32
		CIC_b	0	58	0	21	21
		QIC_b	73	8	19	0	0
		RJ_b	68	19	12	1	0

Table 8.5 The correct identification rates (%) of the working correlation structure identified by PT_b , WR_b , RMR_b , CIC_b , QIC_b and RJ_b applying penalization method using bootstrapped data: $C(\alpha)$: EX and $m=6$.

$C(\alpha)$	n	Criterion	IN	EX	AR	TP	UN
EX(0.3) $m=6$	100	PT_b	0	0	0	1	49
		WR_b	0	0	0	1	49
		RMR_b	62	2	0	8	28
		CIC_b	4	15	0	8	23
		QIC_b	82	4	8	4	2
		RJ_b	47	31	19	3	0
	200	PT_b	0	0	0	0	100
		WR_b	0	0	0	0	100
		RMR_b	34	29	0	24	13
		CIC_b	0	24	0	24	52
		QIC_b	87	4	6	1	2
		RJ_b	43	26	29	2	0
EX(0.7) $m=6$	100	PT_b	22	0	0	1	27
		WR_b	22	0	0	1	27
		RMR_b	44	6	0	8	42
		CIC_b	4	26	2	4	14
		QIC_b	72	4	22	0	2
		RJ_b	50	12	1	35	2
	200	PT_b	38	0	0	0	62
		WR_b	38	0	0	2	60
		RMR_b	38	2	0	7	53
		CIC_b	0	60	2	10	28
		QIC_b	69	7	23	0	1
		RJ_b	67	24	5	4	0

8.3 Conclusion

To penalize the over-parameterized working correlation structure, the two different approaches have been proposed: (1) the cluster detection based on the spacings of the order statistics of the criterion measure (chapter 7), and (2) the statistical test using the bootstrapped data (chapter 8). These two approaches are developed based on the similar motivations in which the true correlation structure and the over-parameterized structure are expected to have relatively similar selection criterion values due to their asymptotic properties, whereas those that are misspecified and cannot capture the underlying correlation are biased and have larger values. Those candidate models with similar values are considered *clustered*. Simulation results show that the first approach can identify the *cluster* well, whereas the second approach does not do so even in large samples. Moreover, the second approach is sensitive to several factors, including the number of bootstrap samples taken. For penalizing the over-parameterized structure, the first approach is recommended, as it incorporates a tradeoff between fit and the number of parameters in the working correlation matrix more effectively than approach 2.

CHAPTER 9

CONCLUSIONS AND FUTURE PLANS

We close this dissertation with a summary of contributions and several directions for future work.

9.1 Conclusions

In this dissertation, a new selection criteria for working correlation structure in GEE was developed based on generalized eigenvalues. A set of generalized eigenvalues measures a disparity between the sandwich variance estimator under the hypothesized working correlation matrix and the model-based variance estimator under working independence assumption. A summary measure of the set of the generalized eigenvalues provides an indication of the disparity between the true correlation structure and the misspecified working correlation structure.

The proposed selection criteria use a bias-corrected sandwich variance estimator. The use of a bias-corrected sandwich variance estimator mitigates the negative bias of the sandwich variance estimator by Liang and Zeger (1986). Moreover, the use of a bias-corrected sandwich variance estimator improves the correct identification rates of the working correlation selection criteria. In this dissertation, four bias-corrected sandwich variance estimators were described. Their impacts on selecting the most appropriate working correlation structure were similar. Therefore, the simulation results using the Wong and Long (2010) bias-correction method are presented in this dissertation.

Our simulation experiments show that the new selection criteria can distinguish the true correlation matrix from the misspecified working correlation structure with high success rates as the sample size increases. However, like other selection criterion measures, they cannot distinguish the true underlying correlation matrix from the over-parameterized structures. Therefore, we developed a method to penalize over-parameterized structures based on our cluster detection method and an optimization

function that considers the trade-off between bias and variability. The simulation studies suggest that the proposed penalization method greatly improves the performances of the new selection criterion measures.

The GEE method is known to provide consistent regression parameter estimates regardless of the choice of the working correlation structure if $\sqrt{n}$ consistent estimates of nuisance parameters are used. In large samples, valid standard errors of the β estimates can be obtained from the sandwich variance estimator. This dissertation focuses on the statistical efficiency in small samples. The correct identification of the working correlation structure improves the statistical efficiency, as was shown in chapter 3. In that sense, our proposed working correlation selection criteria with the penalization for the over-parameterized structure would contribute to the improvement of the statistical efficiency in GEE.

9.2 Limitations and Future Directions

There are some limitations of this dissertation. A few suggestions for future research is also discussed.

The proposed working correlation selection criteria are one-dimensional summary measures of a set of generalized eigenvalues. One-dimensional summary measure loses some information on the disparity between two covariance matrices in the multi-dimensional space. Multivariate measures considering both direction and quantity of each generalized eigenvalue might provide more information on the disparity between the two covariance matrices.

The measures proposed are based upon the generalized eigenvalues of the comparison of the sandwich estimator of the variance of the regression parameters under a working correlation structure to the model-based estimator under independence. One might intuitively think that a comparison of the sandwich estimator to the model-based estimator both formed under the same working correlation would work to distinguish a

good working model choice from a poor one. If the right choice was made, then both should be measuring the same quantities and the generalized eigenvalues should all be close to 1. However, in small sample sizes, both estimators still try to fit the model to the data, and the sandwich estimator is not powerful enough to move far enough away from the model-based estimator to signal a poor choice of the working correlation.

The derivation of the QIC and CIC motivates the use of these particular generalized eigenvalues. But intuitively, we can appreciate that since the independence structure has no correlation parameters and since the estimates themselves are not highly influenced by the working correlation structure, the β elements of the variance matrix $\Sigma_{M(IN)}$ will tend to be pretty stable. Also, the higher the true correlations, the “larger” this matrix will be, relative to the ideal Σ_T . So, in effect, we expect independence to be the worst (but reasonably stable) choice, to which all others are compared.

In this dissertation, the marginal mean model and the variance function were correctly specified to evaluate the performance of the criterion measures on the working correlation structure selection. The effects of a misspecified or more complicated mean structure (thus increasing the size of p) and the variance function on the working correlation structure selection should be investigated.

Our simulation experiments used complete observations. Missing values are common issues in the longitudinal data analysis. The impact of the missing data on the working correlation structure selection should be investigated. Sparks (2009) proposed the model selection criteria in a GLM framework based on the Kullback-Leiber discrepancy that can be used in the presence of missing data. The criteria were developed by accounting for missing data using principles related to the expectation maximization (EM) algorithm and bootstrap methods. The approach taken by Sparks (2009) might be helpful to improve the performance of our proposed working correlation selection criteria in the presence of missing data.

Our simulation experiments considered the five working correlation structures: independence, exchangeable, AR-1, Toeplitz and unstructured matrices. They are commonly used in many statistical packages. In principle, many types of user-specified working correlation structure can be used. The performance of the proposed working correlation selection criteria should be evaluated when other types of working correlation structures are also considered.

In the modeling of the longitudinal data, the linear predictors, the link function, the variance function and the working correlation structure must be chosen. QIC is a widely used model selection criterion for the mean model and working correlation structure, even though it has some limitations when used for working correlation selection. Recently, several authors introduced other selection criteria for the covariate or working correlation structure. Wang and Hin (2010) proposed the extended QIC (EQIC) for the selection of the mean structure and variance function. Acion (2011) developed a new model selection criteria based on Kullback-Leibler symmetric discrepancy and demonstrated better performance than QIC. To our knowledge, there is no model selection criterion identifying all three: the mean structure, the variance function and working correlation structure, with high success rate. Combining our proposed working correlation selection method with the recently developed other selection criteria, the overall model selection strategies is planned for future research.

Finally, working to identify a better theoretical justification for the choice of an optimization function that penalizes for over-parameterization is also part of future research plans.

APPENDIX A

RELATIVE DIFFERENCE AND COVERAGE PROBABILITY OF
VARIANCE ESTIMATORS

Table A.1 Relative difference and coverage probability of each six variance estimator (MB, LZ, MD, KC, Pan, WL) when the true correlation structure for the binary responses is EX(0.3) and $m=4$.

		Relative difference (%)					Coverage probability (%)				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
$n=20$	MB	1.9	-14.0	-19.3	-20.1	-28.6	93.4	93.2	90.2	91.3	89.5
	LZ	-5.3	-20.2	-14.7	-24.9	-31.2	91.9	92.1	91.9	90.5	87.3
	MD	13.8	-6.7	0.7	-11.9	-18.4	94.4	93.6	93.4	92.3	90.4
	KC	3.7	-13.8	-7.4	-18.7	-25.2	93.3	92.7	92.7	91.7	89.0
	PAN	0.0	-16.1	-9.3	-20.7	-28.1	92.9	93.2	93.4	91.4	90.5
	WL	1.9	-14.0	-19.3	-20.1	-28.6	93.4	93.2	90.2	91.3	89.5
$n=40$	MB	8.3	-11.7	-4.1	-16.4	-23.7	94.0	93.5	93.9	92.6	90.9
	LZ	15.7	-2.2	-9.0	-5.4	-9.8	95.1	95.4	93.0	94.0	93.4
	MD	12.7	-5.7	2.5	-8.8	-12.4	94.2	94.0	93.8	93.4	92.8
	KC	23.5	1.9	11.2	-1.5	-5.1	95.3	94.7	95.1	94.2	93.1
	PAN	18.0	-2.0	6.8	-5.3	-8.8	94.9	94.5	94.3	93.7	92.9
	WL	14.7	-3.8	4.6	-6.9	-11.5	94.1	94.0	94.7	93.8	92.8
$n=50$	MB	19.2	-1.6	7.3	-4.7	-9.3	94.7	94.5	94.9	94.1	93.1
	LZ	21.7	2.3	-5.4	-0.5	-4.6	95.7	95.8	93.5	95.3	94.6
	MD	19.1	-0.7	7.8	-2.9	-6.7	94.7	95.5	94.3	94.6	93.8
	KC	28.2	5.5	15.1	3.3	-0.6	95.5	95.6	94.9	95.6	94.9
	PAN	23.6	2.3	11.4	0.1	-3.7	95.1	95.6	94.7	95.2	94.4
	WL	22.7	2.4	12.1	-0.2	-4.2	95.4	95.4	95.1	93.9	93.5

Table A.2 Relative difference and coverage probability of each six variance estimator (MB, LZ, MD, KC, Pan, WL) when the true correlation structure for the binary responses is EX(0.7) and $m=4$.

		Relative difference (%)					Coverage probability (%)				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
$n=20$	MB	142.1	7.4	-22.2	-7.5	-22.8	97.0	95.8	87.6	92.2	90.5
	LZ	127.4	-0.5	24.5	-9.7	-16.7	94.5	95.6	95.0	93.8	92.8
	MD	181.0	16.0	48.9	6.5	2.3	96.5	96.9	97.2	95.8	95.3
	KC	152.7	7.3	35.9	-2.8	-15.2	95.2	95.9	96.2	94.4	94.0
	PAN	137.9	4.4	32.0	-5.2	-10.1	96.3	95.1	95.5	93.3	91.3
	WL	142.1	7.4	-22.2	-7.5	-22.8	97.0	95.8	87.6	92.2	90.5
$n=40$	MB	165.2	9.7	40.4	0.4	36.8	97.3	96.3	96.4	94.5	92.9
	LZ	134.0	0.0	-31.4	-8.4	-18.9	95.4	95.7	85.3	94.6	92.9
	MD	128.3	-4.7	19.4	-9.7	-19.2	94.6	94.5	94.6	94.5	93.6
	KC	153.5	2.7	30.3	-2.3	-12.3	95.5	95.5	95.7	95.5	94.8
	PAN	140.5	-1.1	24.7	-6.2	-15.8	95.1	95.1	95.3	95.0	94.2
	WL	137.7	-0.6	25.3	-6.5	-15.6	93.9	94.4	94.2	93.4	91.8
$n=50$	MB	150.6	1.7	29.0	-4.1	-13.2	94.9	94.5	94.5	93.7	92.2
	LZ	140.3	3.6	-29.7	-3.0	-12.1	94.5	95.3	84.0	94.2	93.2
	MD	140.7	0.5	27.0	-4.8	-11.2	93.7	94.6	93.8	93.2	93.7
	KC	161.8	6.7	36.5	1.3	-5.3	94.7	95.5	94.7	94.4	94.1
	PAN	151.0	3.5	31.6	-1.8	-9.1	94.2	95.0	94.3	93.9	93.8
	WL	144.7	2.3	28.5	-2.5	-11.3	94.0	94.0	93.0	93.6	91.3

Table A.3 Relative difference and coverage probability of each six variance estimator (MB, LZ, MD, KC, Pan, WL) when the true correlation structure for the binary responses is EX(0.3) and $m=6$.

		Relative difference (%)					Coverage probability (%)				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
$n=20$	MB	15.4	-4.1	-16.0	-13.2	-33.6	95.5	96.1	91.5	92.8	84.5
	LZ	9.3	-10.3	1.9	-17.1	-33.6	94.5	92.9	92.6	90.8	84.1
	MD	28.8	2.7	18.0	-4.5	-22.1	96.3	94.7	94.7	93.0	88.0
	KC	18.6	-4.0	9.6	-11.1	-28.2	95.5	94.2	93.7	92.0	86.2
	PAN	14.7	-5.8	7.9	-12.4	-29.6	95.9	96.0	94.6	93.3	86.0
	WL	15.4	-4.1	-16.0	-13.2	-33.6	95.5	96.1	91.5	92.8	84.5
$n=40$	MB	22.4	-2.4	12.4	-9.0	-26.1	96.9	96.3	95.0	94.1	87.1
	LZ	16.7	-3.6	-17.2	-8.1	-17.9	95.0	95.0	90.3	94.2	92.3
	MD	15.3	-7.3	7.7	-10.6	-19.9	93.8	93.3	92.9	92.4	91.5
	KC	25.2	-0.9	15.7	-4.4	-14.1	95.0	94.4	94.0	93.3	92.4
	PAN	20.1	-4.1	11.6	-7.6	-17.1	94.6	93.8	93.5	92.9	92.2
	WL	18.2	-4.4	11.0	-8.3	-17.6	94.6	94.0	94.5	92.9	91.4
$n=50$	MB	22.0	-2.9	13.1	-6.7	-16.0	95.0	94.2	94.9	93.2	91.5
	LZ	17.9	-2.9	-17.6	-6.2	-14.2	95.6	95.2	88.9	94.7	92.1
	MD	17.8	-4.3	11.9	-7.5	-14.6	95.0	94.6	93.4	93.8	91.5
	KC	25.8	0.9	18.6	-2.4	-9.7	96.0	95.0	94.4	94.2	92.4
	PAN	21.7	-1.7	15.1	-4.9	-12.1	95.3	94.9	94.2	94.1	92.1
	WL	20.2	-2.9	12.9	-6.0	-13.5	95.6	94.9	93.6	94.3	92.0

Table A.4 Relative difference and coverage probability of each six variance estimator (MB, LZ, MD, KC, Pan, WL) when the true correlation structure for the binary responses is EX(0.7) and $m=6$.

		Relative difference (%)					Coverage probability (%)				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
$n=20$	MB	142.1	7.8	-36.6	-8.6	-35.6	95.3	96.8	80.8	92.6	85.8
	LZ	130.9	-1.0	44.1	2.7	-4.4	94.2	95.2	95.8	93.7	87.8
	MD	180.4	12.7	67.7	29.5	26.1	96.3	96.9	97.4	95.4	91.0
	KC	154.4	5.6	55.4	4.1	-2.5	95.4	96.0	96.9	94.6	89.5
	PAN	145.4	5.7	53.5	6.7	5.8	95.8	96.6	96.5	94.6	88.6
	WL	142.1	7.8	-36.6	-8.6	-35.6	95.3	96.8	80.8	92.6	85.8
$n=40$	MB	170.4	9.6	60.7	17.8	26.8	96.6	96.8	96.8	95.1	89.8
	LZ	155.9	6.4	-42.9	-5.0	-27.4	96.4	95.6	78.4	92.9	89.2
	MD	156.9	2.4	47.2	-2.0	-20.5	95.9	95.7	94.8	94.2	92.4
	KC	182.9	9.2	58.6	5.0	-14.1	96.6	96.5	95.8	95.3	93.6
	PAN	169.6	5.7	52.7	1.5	-19.3	96.2	96.1	95.4	94.6	93.3
	WL	166.0	7.2	52.5	1.5	-15.4	96.3	94.8	93.8	93.1	89.7
$n=50$	MB	179.0	9.1	56.0	3.5	-12.9	96.7	95.1	93.8	93.2	90.2
	LZ	133.8	-1.7	-47.6	-9.8	-27.4	94.6	94.1	75.5	93.2	91.2
	MD	140.9	-3.7	35.8	-8.9	-23.7	94.8	94.2	94.5	93.2	91.9
	KC	160.3	1.4	44.4	-3.8	-19.1	95.4	94.6	95.2	94.2	92.6
	PAN	150.4	-1.2	39.9	-6.4	-21.5	95.1	94.3	94.9	93.7	92.4
	WL	143.6	-3.2	37.7	-8.2	-24.8	94.7	93.7	94.5	93.2	90.0

Table A.5 Relative difference and coverage probability of each six variance estimator (MB, LZ, MD, KC, Pan, WL) when the true correlation structure for the binary responses is TOEP and $m=4$.

		Relative difference (%)					Coverage probability (%)				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
$n=20$	MB	34.0	10.4	7.0	-21.4	-27.5	95.1	94.2	92.3	91.3	89.2
	LZ	24.8	2.7	18.9	-22.6	-28.5	93.2	93.1	93.3	92.7	89.1
	MD	50.3	20.0	40.7	-7.0	-13.4	95.3	94.5	95.0	94.9	92.1
	KC	36.9	10.9	29.2	-14.9	-21.2	94.3	93.6	94.1	93.9	90.7
	PAN	31.2	7.4	25.4	-19.7	-26.3	95.0	93.8	94.1	92.9	91.0
	WL	34.0	10.4	7.0	-21.4	-27.5	95.1	94.2	92.3	91.3	89.2
$n=40$	MB	42.4	13.0	33.1	-14.4	-21.1	95.5	94.6	94.7	94.0	91.8
	LZ	39.2	15.2	11.4	-14.1	-17.3	95.0	95.4	91.4	93.7	92.9
	MD	36.1	11.1	30.0	-15.0	-18.9	94.9	94.3	93.7	93.5	92.3
	KC	49.3	19.9	41.2	-7.3	-11.5	95.1	95.2	94.8	95.4	93.7
	PAN	42.5	15.4	35.5	-11.1	-15.2	95.0	94.5	94.6	94.5	93.1
	WL	39.4	12.7	33.1	-12.9	-16.9	93.5	94.2	94.1	91.9	90.2
$n=50$	MB	45.0	15.3	36.7	-10.4	-14.5	94.3	94.3	94.3	92.6	90.7
	LZ	48.9	22.4	17.6	-6.3	-9.1	94.6	95.1	92.6	94.0	93.9
	MD	45.3	18.2	39.2	-6.7	-10.1	94.6	94.3	94.8	94.8	95.0
	KC	56.4	25.6	48.7	-0.1	-3.6	95.4	95.2	95.3	95.5	95.6
	PAN	50.7	21.9	43.9	-3.5	-6.9	95.2	94.7	95.1	95.1	95.3
	WL	48.8	21.7	43.1	-6.3	-9.6	94.6	94.6	94.6	93.3	93.5

Table A.6 Relative difference and coverage probability of each six variance estimator (MB, LZ, MD, KC, Pan, WL) when the true correlation structure for the binary responses is TOEP and $m=6$.

		Relative difference (%)					Coverage probability (%)				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
$n=20$	MB	68.4	33.3	16.1	-19.3	-38.4	95.7	96.1	90.3	92.3	85.5
	LZ	59.5	24.5	60.9	-11.7	-25.0	94.5	94.3	94.4	93.1	88.5
	MD	88.6	42.3	86.4	4.0	-7.6	95.5	95.7	95.9	95.3	90.8
	KC	73.4	33.1	73.1	-6.4	-25.4	95.2	94.9	95.3	94.3	89.5
	PAN	68.1	30.0	67.3	-9.7	-14.8	94.9	94.8	94.5	93.0	85.6
	WL	68.4	33.3	16.1	-19.3	-38.4	95.7	96.1	90.3	92.3	85.5
$n=40$	MB	80.2	34.6	74.6	-5.2	-6.0	95.6	95.2	94.6	93.5	86.6
	LZ	72.8	34.4	10.1	-15.1	-27.9	93.5	94.7	85.5	92.4	90.8
	MD	71.9	31.0	75.6	-12.6	-23.0	92.1	94.1	94.1	91.7	90.7
	KC	86.9	40.0	88.7	-5.9	-16.7	93.6	94.8	95.2	92.8	92.1
	PAN	79.2	35.4	82.0	-9.3	-20.5	92.6	94.5	94.6	92.4	91.3
	WL	75.3	35.0	78.7	-9.4	-21.1	92.8	93.3	93.5	92.3	90.5
$n=50$	MB	81.2	37.1	82.2	-7.6	-19.2	93.4	93.4	93.9	93.0	91.0
	LZ	81.9	42.4	17.8	-8.7	-19.8	94.3	95.1	85.6	93.5	91.3
	MD	81.2	37.7	82.9	-8.4	-17.5	93.6	93.6	92.7	93.3	91.1
	KC	93.7	45.2	93.8	-3.0	-12.2	94.0	94.2	93.5	93.6	93.0
	PAN	87.3	41.4	88.3	-5.8	-15.7	93.6	94.1	93.1	93.4	91.6
	WL	84.4	40.5	87.6	-5.5	-14.0	94.3	93.9	93.3	93.1	91.0

Table A.7 Relative difference and coverage probability of each six variance estimator (MB, LZ, MD, KC, Pan, WL) when the true correlation structure for the binary responses is UN and $m=4$.

		Relative difference (%)					Coverage probability (%)				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
$n=20$	MB	104.2	46.6	20.9	12.5	-25.2	95.2	95.4	93.7	91.7	90.3
	LZ	89.8	34.0	11.3	10.6	-28.5	92.8	93.5	92.5	90.8	89.8
	MD	130.6	56.5	31.0	31.1	-13.0	94.9	95.5	94.9	93.7	92.3
	KC	109.0	44.7	20.6	20.3	-23.7	93.8	94.8	94.0	92.6	91.0
	PAN	20.7	-1.2	15.4	-26.1	-32.3	95.0	93.8	94.1	92.9	91.0
	WL	104.2	46.6	20.9	12.5	-25.2	95.2	95.4	93.7	91.7	90.3
$n=40$	MB	31.0	3.9	22.4	-21.3	-27.4	95.5	94.6	94.7	94.0	91.8
	LZ	99.7	43.2	18.0	13.7	-21.3	94.9	95.7	94.9	94.1	92.1
	MD	95.6	37.1	13.8	14.6	-22.0	93.5	94.4	93.2	92.8	93.3
	KC	115.3	47.9	23.3	24.4	-14.4	94.0	95.6	95.1	94.1	94.2
	PAN	105.2	42.3	18.5	19.4	-18.2	93.7	95.3	94.4	93.8	93.5
	WL	38.2	11.7	31.9	-13.6	-17.7	93.5	94.2	94.1	91.9	90.2
$n=50$	MB	43.8	14.3	35.5	-11.2	-15.3	94.3	94.3	94.3	92.6	90.7
	LZ	108.3	47.5	21.0	18.5	-17.8	95.3	95.0	94.2	93.1	92.6
	MD	105.4	43.5	19.3	20.0	-15.4	94.5	93.9	94.5	93.3	93.9
	KC	121.7	52.5	27.2	28.1	-9.0	95.5	95.3	95.2	94.0	94.8
	PAN	113.4	47.9	23.2	23.9	-12.6	95.1	94.8	95.0	93.8	94.4
	WL	46.4	19.7	40.8	-7.8	-11.0	94.6	94.6	94.6	93.3	93.5

Table A.8 Relative difference and coverage probability of each six variance estimator (MB, LZ, MD, KC, Pan, WL) when the true correlation structure for the binary responses is UN and $m=6$.

		Relative difference (%)					Coverage probability (%)				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
$n=20$	MB	47.9	21.7	-3.4	-2.3	-41.6	95.6	95.8	93.5	92.8	84.4
	LZ	41.8	14.9	3.5	-5.3	-33.1	93.2	93.4	92.7	90.5	84.8
	MD	67.3	31.5	19.5	9.2	-18.1	94.9	94.7	94.2	91.8	88.3
	KC	53.9	22.9	11.2	1.7	-33.5	94.0	93.6	93.2	91.4	86.1
	PAN	51.5	17.2	50.8	-18.6	-23.2	94.9	94.8	94.5	93.0	85.6
	WL	47.9	21.7	-3.4	-2.3	-41.6	95.6	95.8	93.5	92.8	84.4
$n=40$	MB	47.9	21.7	-3.4	-2.3	-41.6	95.6	95.8	93.5	92.8	84.4
	LZ	41.8	14.9	3.5	-5.3	-33.1	93.2	93.4	92.7	90.5	84.8
	MD	67.3	31.5	19.5	9.2	-18.1	94.9	94.7	94.2	91.8	88.3
	KC	53.9	22.9	11.2	1.7	-33.5	94.0	93.6	93.2	91.4	86.1
	PAN	51.5	17.2	50.8	-18.6	-23.2	94.9	94.8	94.5	93.0	85.6
	WL	47.9	21.7	-3.4	-2.3	-41.6	95.6	95.8	93.5	92.8	84.4
$n=50$	MB	80.7	36.6	81.6	-7.9	-19.5	93.4	93.4	93.9	93.0	91.0
	LZ	74.3	42.6	11.5	19.6	-23.4	95.8	95.6	93.8	94.7	90.1
	MD	73.5	38.3	24.4	18.0	-21.7	95.0	94.5	94.9	94.6	91.7
	KC	85.3	45.8	31.6	24.7	-16.5	95.6	95.0	95.5	95.6	92.5
	PAN	79.2	42.0	27.9	21.3	-19.7	95.4	94.7	95.4	94.9	92.0
	WL	76.8	34.7	79.8	-9.4	-17.6	94.3	93.9	93.3	93.1	91.0

Table A.9 Monte Carlo variance of the variance estimators: Correlated binary responses,
 $C(\alpha)$: AR-1(0.3), $m=4$

		Monte Carlo variance of the variance estimators				
		IN	EX	AR-1	TOEP	UN
$n=20$	MB	0.066	0.071	0.080	0.081	0.088
	LZ	0.217	0.219	0.211	0.201	0.213
	MD	0.261	0.282	0.259	0.265	0.285
	KC	0.234	0.249	0.239	0.239	0.238
	PAN	0.077	0.079	0.085	0.080	0.087
	WL	0.089	0.090	0.093	0.090	0.098
$n=40$	MB	0.006	0.007	0.008	0.008	0.008
	LZ	0.024	0.024	0.024	0.023	0.023
	MD	0.030	0.029	0.029	0.029	0.029
	KC	0.027	0.026	0.027	0.026	0.026
	PAN	0.007	0.007	0.008	0.008	0.008
	WL	0.008	0.008	0.008	0.008	0.009
$n=50$	MB	0.003	0.003	0.004	0.004	0.004
	LZ	0.011	0.012	0.012	0.011	0.012
	MD	0.013	0.014	0.013	0.013	0.013
	KC	0.012	0.013	0.012	0.012	0.012
	PAN	0.004	0.004	0.004	0.004	0.004
	WL	0.004	0.004	0.004	0.004	0.004

Table A.10 Monte Carlo variance of the variance estimators: Correlated binary responses,
 $C(\alpha)$: AR-1(0.7), $m=4$

		Monte Carlo variance of the variance estimators				
		IN	EX	AR-1	TOEP	UN
$n=20$	MB	0.092	0.095	0.081	0.074	0.069
	LZ	0.210	0.185	0.136	0.131	0.158
	MD	0.253	0.254	0.192	0.188	0.221
	KC	0.231	0.222	0.165	0.155	0.187
	PAN	0.136	0.101	0.071	0.069	0.091
	WL	0.152	0.112	0.079	0.077	0.103
$n=40$	MB	0.007	0.009	0.008	0.006	0.007
	LZ	0.026	0.020	0.014	0.013	0.015
	MD	0.034	0.025	0.017	0.017	0.019
	KC	0.030	0.022	0.016	0.015	0.017
	PAN	0.013	0.009	0.006	0.006	0.009
	WL	0.014	0.009	0.007	0.006	0.010
$n=50$	MB	0.003	0.004	0.003	0.003	0.003
	LZ	0.012	0.010	0.007	0.006	0.007
	MD	0.015	0.011	0.008	0.007	0.009
	KC	0.013	0.011	0.007	0.007	0.007
	PAN	0.007	0.005	0.003	0.003	0.004
	WL	0.008	0.005	0.003	0.003	0.004

Table A.11 Monte Carlo variance of the variance estimators: Correlated binary responses,
 $C(\alpha)$: AR-1(0.3), $m=6$

		Monte Carlo variance of the variance estimators				
		IN	EX	AR-1	TOEP	UN
$n=20$	MB	0.015	0.016	0.020	0.020	0.022
	LZ	0.079	0.077	0.071	0.066	0.063
	MD	0.112	0.106	0.099	0.093	0.094
	KC	0.094	0.090	0.084	0.078	0.076
	PAN	0.020	0.021	0.020	0.020	0.024
	WL	0.023	0.023	0.022	0.022	0.026
$n=40$	MB	0.002	0.002	0.002	0.002	0.002
	LZ	0.010	0.010	0.009	0.009	0.009
	MD	0.012	0.012	0.011	0.011	0.010
	KC	0.011	0.011	0.010	0.010	0.010
	PAN	0.002	0.002	0.002	0.002	0.003
	WL	0.002	0.002	0.002	0.002	0.003
$n=50$	MB	0.001	0.001	0.001	0.001	0.001
	LZ	0.005	0.005	0.005	0.005	0.004
	MD	0.006	0.005	0.005	0.005	0.005
	KC	0.005	0.005	0.005	0.005	0.005
	PAN	0.001	0.001	0.001	0.001	0.001
	WL	0.001	0.001	0.001	0.001	0.001

Table A.12 Monte Carlo variance of the variance estimators: Correlated binary responses,
 $C(\alpha)$: AR-1(0.7), $m=6$

		Monte Carlo variance of the variance estimators				
		IN	EX	AR-1	TOEP	UN
$n=20$	MB	0.019	0.023	0.017	0.014	0.042
	LZ	0.097	0.072	0.034	0.037	0.086
	MD	0.146	0.099	0.047	0.045	0.112
	KC	0.119	0.084	0.039	0.044	0.087
	PAN	0.038	0.029	0.016	0.017	0.067
	WL	0.047	0.031	0.018	0.019	0.074
$n=40$	MB	0.002	0.003	0.002	0.002	0.002
	LZ	0.012	0.009	0.005	0.004	0.010
	MD	0.015	0.010	0.005	0.005	0.013
	KC	0.014	0.010	0.005	0.005	0.006
	PAN	0.004	0.003	0.002	0.001	0.002
	WL	0.005	0.003	0.002	0.002	0.003
$n=50$	MB	0.001	0.001	0.001	0.001	0.001
	LZ	0.006	0.004	0.002	0.002	0.002
	MD	0.007	0.005	0.003	0.002	0.003
	KC	0.007	0.005	0.002	0.002	0.003
	PAN	0.002	0.002	0.001	0.001	0.001
	WL	0.002	0.002	0.001	0.001	0.001

Table A.13 Monte Carlo variance of the variance estimators: Correlated binary responses,
 $C(\alpha)$: EX(0.3), $m=4$

		Monte Carlo variance of the variance estimators				
		IN	EX	AR-1	TOEP	UN
$n=20$	MB	0.073	0.084	0.113	0.089	0.099
	LZ	0.232	0.219	0.263	0.217	0.228
	MD	0.287	0.290	0.308	0.285	0.292
	KC	0.250	0.247	0.283	0.246	0.257
	PAN	0.087	0.086	0.100	0.088	0.103
	WL	0.104	0.098	0.115	0.100	0.113
$n=40$	MB	0.006	0.008	0.012	0.008	0.008
	LZ	0.025	0.024	0.028	0.023	0.023
	MD	0.031	0.028	0.035	0.028	0.029
	KC	0.028	0.026	0.031	0.025	0.026
	PAN	0.008	0.008	0.009	0.008	0.009
	WL	0.009	0.009	0.009	0.009	0.009
$n=50$	MB	0.003	0.004	0.006	0.004	0.004
	LZ	0.011	0.012	0.015	0.012	0.012
	MD	0.013	0.014	0.018	0.014	0.014
	KC	0.012	0.013	0.017	0.013	0.013
	PAN	0.004	0.004	0.004	0.004	0.004
	WL	0.004	0.004	0.005	0.004	0.005

Table A.14 Monte Carlo variance of the variance estimators: Correlated binary responses,
 $C(\alpha)$: EX(0.7), $m=4$

		Monte Carlo variance of the variance estimators				
		IN	EX	AR-1	TOEP	UN
$n=20$	MB	0.071	0.060	0.053	0.062	0.071
	LZ	0.220	0.132	0.243	0.134	0.124
	MD	0.262	0.193	0.340	0.196	0.180
	KC	0.249	0.159	0.287	0.160	0.138
	PAN	0.135	0.060	0.126	0.064	0.074
	WL	0.166	0.067	0.142	0.071	0.079
$n=40$	MB	0.008	0.007	0.005	0.008	0.008
	LZ	0.029	0.017	0.033	0.018	0.018
	MD	0.037	0.020	0.042	0.022	0.022
	KC	0.033	0.018	0.037	0.020	0.020
	PAN	0.016	0.007	0.015	0.008	0.013
	WL	0.018	0.007	0.016	0.008	0.014
$n=50$	MB	0.004	0.003	0.002	0.004	0.004
	LZ	0.014	0.008	0.017	0.008	0.012
	MD	0.018	0.009	0.020	0.009	0.013
	KC	0.016	0.009	0.018	0.009	0.009
	PAN	0.008	0.003	0.007	0.003	0.004
	WL	0.009	0.003	0.007	0.003	0.004

Table A.15 Monte Carlo variance of the variance estimators: Correlated binary responses,
 $C(\alpha)$: EX(0.3), $m=6$

		Monte Carlo variance of the variance estimators				
		IN	EX	AR-1	TOEP	UN
$n=20$	MB	0.018	0.023	0.043	0.024	0.030
	LZ	0.086	0.076	0.100	0.074	0.080
	MD	0.125	0.104	0.143	0.103	0.105
	KC	0.104	0.089	0.119	0.087	0.091
	PAN	0.026	0.025	0.032	0.028	0.041
	WL	0.031	0.027	0.035	0.031	0.046
$n=40$	MB	0.002	0.002	0.006	0.003	0.003
	LZ	0.011	0.009	0.012	0.009	0.009
	MD	0.013	0.011	0.015	0.010	0.010
	KC	0.012	0.010	0.013	0.010	0.010
	PAN	0.003	0.002	0.003	0.002	0.003
	WL	0.003	0.002	0.003	0.002	0.003
$n=50$	MB	0.001	0.001	0.002	0.001	0.001
	LZ	0.006	0.005	0.007	0.005	0.004
	MD	0.007	0.006	0.008	0.005	0.005
	KC	0.006	0.005	0.008	0.005	0.005
	PAN	0.001	0.001	0.002	0.001	0.001
	WL	0.001	0.001	0.002	0.001	0.001

Table A.16 Monte Carlo variance of the variance estimators: Correlated binary responses,
 $C(\alpha)$: EX(0.7), $m=6$

		Monte Carlo variance of the variance estimators				
		IN	EX	AR-1	TOEP	UN
$n=20$	MB	0.0223	0.0213	0.0129	0.0269	0.0379
	LZ	0.1015	0.0503	0.1182	0.0621	0.0850
	MD	0.1577	0.0678	0.1713	0.0832	0.0970
	KC	0.1262	0.0583	0.1418	0.0676	0.0826
	PAN	0.0535	0.0219	0.0608	0.0272	0.0844
	WL	0.0683	0.0236	0.0672	0.0320	0.0856
$n=40$	MB	0.0026	0.0018	0.0009	0.0020	0.0020
	LZ	0.0123	0.0046	0.0147	0.0055	0.0095
	MD	0.0153	0.0054	0.0176	0.0066	0.0120
	KC	0.0137	0.0050	0.0161	0.0060	0.0060
	PAN	0.0066	0.0020	0.0056	0.0025	0.0123
	WL	0.0075	0.0021	0.0058	0.0026	0.0045
$n=50$	MB	0.0011	0.0009	0.0004	0.0010	0.0011
	LZ	0.0059	0.0028	0.0075	0.0028	0.0026
	MD	0.0071	0.0031	0.0088	0.0032	0.0030
	KC	0.0065	0.0029	0.0081	0.0030	0.0028
	PAN	0.0031	0.0010	0.0029	0.0010	0.0013
	WL	0.0034	0.0010	0.0030	0.0010	0.0013

Table A.17 Monte Carlo variance of the variance estimators: Correlated binary responses,
 $C(\alpha)$: TOEP, $m=4$

		Monte Carlo variance of the variance estimators				
		IN	EX	AR-1	TOEP	UN
$n=20$	MB	0.070	0.078	0.123	0.113	0.099
	LZ	0.221	0.207	0.255	0.229	0.217
	MD	0.264	0.271	0.310	0.328	0.300
	KC	0.240	0.236	0.282	0.279	0.264
	PAN	0.093	0.088	0.098	0.110	0.109
	WL	0.108	0.099	0.111	0.124	0.124
$n=40$	MB	0.006	0.008	0.015	0.013	0.012
	LZ	0.025	0.024	0.030	0.030	0.028
	MD	0.031	0.029	0.037	0.038	0.036
	KC	0.028	0.026	0.033	0.034	0.031
	PAN	0.009	0.009	0.009	0.011	0.011
	WL	0.010	0.009	0.010	0.012	0.011
$n=50$	MB	0.003	0.004	0.007	0.006	0.006
	LZ	0.012	0.012	0.015	0.016	0.015
	MD	0.014	0.014	0.018	0.020	0.019
	KC	0.013	0.013	0.017	0.018	0.017
	PAN	0.004	0.004	0.004	0.005	0.005
	WL	0.005	0.005	0.005	0.006	0.005

Table A.18 Monte Carlo variance of the variance estimators: Correlated binary responses,
 $C(\alpha)$: TOEP, $m=6$

		Monte Carlo variance of the variance estimators				
		IN	EX	AR-1	TOEP	UN
$n=20$	MB	0.019	0.023	0.059	0.022	0.028
	LZ	0.086	0.072	0.109	0.061	0.076
	MD	0.123	0.098	0.152	0.091	0.101
	KC	0.105	0.084	0.130	0.074	0.077
	PAN	0.030	0.026	0.034	0.027	0.065
	WL	0.036	0.028	0.037	0.031	0.078
$n=40$	MB	0.002	0.002	0.007	0.002	0.003
	LZ	0.010	0.008	0.016	0.006	0.008
	MD	0.013	0.010	0.019	0.008	0.010
	KC	0.011	0.009	0.017	0.007	0.008
	PAN	0.003	0.003	0.004	0.002	0.004
	WL	0.003	0.003	0.004	0.002	0.004
$n=50$	MB	0.001	0.001	0.004	0.001	0.001
	LZ	0.005	0.004	0.007	0.003	0.005
	MD	0.006	0.005	0.009	0.004	0.006
	KC	0.005	0.005	0.008	0.003	0.004
	PAN	0.001	0.001	0.002	0.001	0.001
	WL	0.002	0.001	0.002	0.001	0.001

Table A.19 Monte Carlo variance of the variance estimators: Correlated binary responses,
 $C(\alpha)$: UN, $m=4$

		Monte Carlo variance of the variance estimators				
		IN	EX	AR-1	TOEP	UN
$n=20$	MB	0.068	0.079	0.072	0.067	0.068
	LZ	0.207	0.180	0.152	0.144	0.140
	MD	0.267	0.257	0.217	0.202	0.205
	KC	0.239	0.216	0.184	0.176	0.164
	PAN	0.093	0.088	0.098	0.110	0.109
	WL	0.108	0.099	0.111	0.124	0.124
$n=40$	MB	0.007	0.008	0.008	0.007	0.009
	LZ	0.025	0.023	0.018	0.018	0.020
	MD	0.032	0.027	0.022	0.022	0.025
	KC	0.028	0.025	0.020	0.020	0.022
	PAN	0.009	0.009	0.009	0.011	0.011
	WL	0.010	0.009	0.010	0.012	0.011
$n=50$	MB	0.003	0.004	0.004	0.003	0.005
	LZ	0.013	0.011	0.010	0.009	0.012
	MD	0.016	0.013	0.012	0.011	0.015
	KC	0.014	0.012	0.011	0.010	0.013
	PAN	0.004	0.004	0.004	0.005	0.005
	WL	0.005	0.005	0.005	0.006	0.005

Table A.20 Monte Carlo variance of the variance estimators: Correlated binary responses,
 $C(\alpha)$: UN, $m=6$

		Monte Carlo variance of the variance estimators				
		IN	EX	AR-1	TOEP	UN
$n=20$	MB	0.017	0.021	0.026	0.021	0.044
	LZ	0.087	0.074	0.077	0.061	0.082
	MD	0.127	0.102	0.109	0.084	0.110
	KC	0.105	0.087	0.091	0.072	0.079
	PAN	0.030	0.026	0.034	0.027	0.065
	WL	0.036	0.028	0.037	0.031	0.078
$n=40$	MB	0.002	0.002	0.003	0.002	0.004
	LZ	0.011	0.009	0.008	0.007	0.016
	MD	0.013	0.010	0.010	0.008	0.022
	KC	0.012	0.010	0.009	0.008	0.013
	PAN	0.003	0.003	0.004	0.002	0.004
	WL	0.003	0.003	0.004	0.002	0.004
$n=50$	MB	0.001	0.001	0.001	0.001	0.002
	LZ	0.006	0.005	0.005	0.004	0.005
	MD	0.006	0.005	0.005	0.004	0.006
	KC	0.006	0.005	0.005	0.004	0.005
	PAN	0.001	0.001	0.002	0.001	0.001
	WL	0.002	0.001	0.002	0.001	0.001

APPENDIX B

THE PERFORMANCE OF THE CRITERIA WITHOUT
PENALIZATION FOR OVER-PARAMETERIZED STRUCTURES

Table B.1 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated binary response; $C(\alpha)$: AR-1(0.3), $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	15	217	768	8	26	78	153	735
	WR	14	216	770	8	24	77	155	736
	RMR	112	233	655	38	25	42	88	807
	CIC	118	222	660	44	64	117	172	603
	QIC	304	250	446	171	115	129	191	394
	RJ	293	389	318	12	210	375	139	264
30	PT	3	168	829	2	11	33	93	861
	WR	2	168	830	1	12	29	95	863
	RMR	84	201	715	26	26	30	70	848
	CIC	80	187	733	34	40	89	149	688
	QIC	298	224	478	196	122	123	165	394
	RJ	217	467	316	2	222	368	155	253
50	PT	0	118	882	0	1	16	75	908
	WR	0	120	880	0	0	18	75	907
	RMR	71	173	756	28	9	43	50	870
	CIC	39	187	774	19	46	83	130	722
	QIC	288	246	466	206	139	136	148	371
	RJ	146	489	365	1	268	371	126	234
100	PT	0	58	942	0	0	7	40	953
	WR	0	56	944	0	0	8	41	951
	RMR	24	118	858	2	6	21	39	932
	CIC	4	106	890	0	23	106	136	735
	QIC	315	245	440	250	194	136	108	312
	RJ	36	570	394	0	287	367	135	211
200	PT	0	9	991	0	0	0	40	960
	WR	0	10	990	0	0	0	42	958
	RMR	5	52	943	2	0	15	45	938
	CIC	1	48	951	0	6	99	154	741
	QIC	282	253	465	247	209	131	104	309
	RJ	5	600	395	0	294	393	139	174

Table B.2 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated binary response; $C(\alpha)$: AR-1(0.6), $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	43	145	812	35	55	197	209	504
	WR	43	144	813	35	56	200	203	506
	RMR	39	184	777	15	24	83	150	728
	CIC	56	192	752	42	96	210	224	428
	QIC	304	263	433	240	169	139	204	248
	RJ	89	551	360	35	278	348	129	210
30	PT	3	81	916	2	27	90	149	732
	WR	3	76	921	2	28	92	148	730
	RMR	14	110	876	7	11	62	131	789
	CIC	4	150	846	2	59	132	226	581
	QIC	308	264	428	264	196	109	158	273
	RJ	6	590	404	2	309	357	112	220
50	PT	6	37	957	5	11	73	118	793
	WR	6	35	959	5	11	76	120	788
	RMR	14	83	903	7	10	68	136	779
	CIC	6	86	908	5	25	141	219	610
	QIC	292	258	450	244	194	114	167	281
	RJ	8	590	402	5	306	358	113	218
100	PT	0	5	995	0	3	28	78	891
	WR	0	5	995	0	3	25	74	898
	RMR	2	32	966	1	1	75	139	784
	CIC	0	35	965	0	6	133	164	697
	QIC	328	268	404	283	225	99	130	263
	RJ	0	586	414	0	340	338	125	197
200	PT	0	0	1000	0	0	26	62	912
	WR	0	0	1000	0	0	19	65	916
	RMR	0	5	995	0	0	71	129	800
	CIC	0	2	998	0	0	108	193	699
	QIC	314	254	432	287	228	130	117	238
	RJ	0	621	379	0	308	393	116	183

Table B.3 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated binary response; $C(\alpha)$: AR-1(0.3), $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	6	130	864	1	5	64	215	715
	WR	6	129	865	0	5	64	215	716
	RMR	94	245	661	4	8	14	38	936
	CIC	105	191	704	19	24	71	204	682
	QIC	293	234	473	150	101	134	249	366
	RJ	234	428	338	3	199	416	177	205
30	PT	0	45	955	0	1	19	77	903
	WR	0	46	954	0	1	20	76	903
	RMR	62	183	755	8	7	8	20	957
	CIC	52	148	800	11	12	44	109	824
	QIC	285	233	482	143	118	105	169	465
	RJ	144	484	372	0	202	385	158	255
50	PT	0	40	960	0	0	5	9	986
	WR	0	39	961	0	0	5	11	984
	RMR	41	171	788	4	4	3	6	983
	CIC	34	113	853	4	13	28	58	897
	QIC	312	219	469	193	108	116	135	448
	RJ	95	545	360	0	249	367	141	243
100	PT	0	5	995	0	0	0	2	998
	WR	0	4	996	0	0	0	2	998
	RMR	10	132	858	0	0	5	4	991
	CIC	7	55	938	0	7	32	47	914
	QIC	312	233	455	211	147	120	127	395
	RJ	20	561	419	0	265	384	137	214
200	PT	0	0	1000	0	0	0	3	997
	WR	0	0	1000	0	0	0	3	997
	RMR	0	73	927	0	0	2	3	995
	CIC	2	12	986	1	0	29	80	890
	QIC	326	243	431	243	176	120	123	338
	RJ	6	604	390	0	301	391	155	153

Table B.4 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated binary response; $C(\alpha)$: AR-1(0.6), $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	5	84	911	5	31	134	334	496
	WR	5	80	915	5	31	148	320	496
	RMR	21	118	861	2	9	51	102	836
	CIC	24	143	833	12	50	155	325	458
	QIC	285	253	462	226	160	135	243	236
	RJ	29	577	394	5	294	376	176	149
30	PT	0	22	978	1	19	81	242	657
	WR	0	22	978	1	19	91	227	662
	RMR	2	54	944	5	3	34	95	863
	CIC	5	69	926	2	32	132	249	585
	QIC	284	260	456	204	183	143	190	280
	RJ	1	638	361	1	304	361	154	180
50	PT	0	12	988	0	2	37	130	831
	WR	0	11	989	0	2	39	127	832
	RMR	4	47	949	1	1	27	65	906
	CIC	0	40	960	0	7	75	165	753
	QIC	274	244	482	205	171	131	183	310
	RJ	1	593	406	0	321	349	148	182
100	PT	0	1	999	0	0	7	26	967
	WR	0	1	999	0	0	6	25	969
	RMR	0	4	996	0	0	18	53	929
	CIC	0	4	996	0	0	38	107	855
	QIC	307	206	487	250	157	128	118	347
	RJ	0	602	398	0	301	345	141	213
200	PT	0	0	1000	0	0	5	11	984
	WR	0	0	1000	0	0	5	14	981
	RMR	0	0	1000	0	0	14	38	948
	CIC	0	0	1000	0	0	47	77	876
	QIC	302	272	426	258	231	113	111	287
	RJ	0	597	403	0	329	353	137	181

Table B.5 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated binary response; $C(\alpha)$: EX(0.3), $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	27	740	233	22	69	31	169	709
	WR	27	742	231	22	75	31	164	708
	RMR	175	557	268	44	55	30	77	794
	CIC	139	566	295	58	106	83	173	580
	QIC	309	345	346	189	89	173	156	393
	RJ	140	527	333	30	313	60	258	339
30	PT	10	815	175	2	55	9	113	821
	WR	10	814	176	2	51	10	116	821
	RMR	154	663	183	54	54	26	76	790
	CIC	91	678	231	33	105	63	144	655
	QIC	307	296	397	180	83	225	114	398
	RJ	62	572	366	4	339	29	253	375
50	PT	0	887	113	0	16	1	94	889
	WR	0	884	116	0	16	1	96	887
	RMR	116	729	155	36	43	17	69	835
	CIC	35	787	178	13	106	40	142	699
	QIC	329	301	370	218	72	255	96	359
	RJ	9	676	315	0	337	7	259	397
100	PT	0	968	32	0	11	0	71	918
	WR	0	967	33	0	10	0	71	919
	RMR	72	852	76	28	47	4	66	855
	CIC	8	894	98	2	106	11	150	731
	QIC	330	286	384	243	66	309	79	303
	RJ	0	748	252	0	331	0	278	391
200	PT	0	997	3	0	13	0	68	919
	WR	0	997	3	0	13	0	70	917
	RMR	17	963	20	6	53	0	65	876
	CIC	0	983	17	0	112	2	168	718
	QIC	320	284	396	252	59	358	68	263
	RJ	0	876	124	0	364	0	260	376

Table B.6 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated binary response; $C(\alpha)$: EX(0.6), $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	116	744	140	100	161	28	251	460
	WR	116	738	146	100	157	28	252	463
	RMR	102	746	152	38	92	20	151	699
	CIC	145	672	183	114	214	69	239	364
	QIC	381	375	244	311	162	162	166	199
	RJ	121	641	238	100	403	47	244	206
30	PT	81	831	88	72	122	14	172	620
	WR	81	824	95	72	120	14	172	622
	RMR	62	834	104	26	92	15	125	742
	CIC	88	763	149	76	190	57	197	480
	QIC	367	364	269	299	147	189	133	232
	RJ	81	705	214	72	394	14	225	295
50	PT	25	930	45	25	64	3	136	772
	WR	25	928	47	25	64	3	130	778
	RMR	32	917	51	18	84	3	121	774
	CIC	27	868	105	26	158	33	221	562
	QIC	348	349	303	277	122	242	101	258
	RJ	25	825	150	25	375	1	261	338
100	PT	2	990	8	2	28	0	83	887
	WR	2	990	8	2	25	0	77	896
	RMR	7	986	7	4	86	0	133	777
	CIC	2	959	39	2	142	10	185	661
	QIC	369	285	346	305	88	300	76	231
	RJ	2	932	66	2	376	0	234	388
200	PT	0	998	2	0	8	0	63	929
	WR	0	998	2	0	5	0	55	940
	RMR	0	998	2	0	77	0	116	807
	CIC	0	995	5	0	131	1	181	687
	QIC	380	217	403	336	50	369	64	181
	RJ	0	986	14	0	365	0	253	382

Table B.7 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated binary response; $C(\alpha)$: EX(0.3), $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	9	870	121	5	58	4	203	730
	WR	8	868	124	4	64	6	199	727
	RMR	102	680	218	22	26	15	45	892
	CIC	146	636	218	33	76	32	194	665
	QIC	319	356	325	158	91	161	164	426
	RJ	25	691	284	6	402	12	283	297
30	PT	1	945	54	1	19	0	100	880
	WR	1	948	51	1	20	0	99	880
	RMR	97	734	169	14	22	6	42	916
	CIC	76	765	159	8	51	26	117	798
	QIC	337	302	361	158	59	165	126	492
	RJ	7	745	248	1	350	2	280	367
50	PT	0	982	18	0	2	0	21	977
	WR	0	983	17	0	1	0	22	977
	RMR	53	845	102	5	14	4	19	958
	CIC	32	875	93	5	37	7	72	879
	QIC	304	295	401	159	63	242	73	463
	RJ	0	849	151	0	340	1	259	400
100	PT	0	999	1	0	3	0	9	988
	WR	0	999	1	0	3	0	11	986
	RMR	13	940	47	1	14	1	19	965
	CIC	2	973	25	0	41	2	69	888
	QIC	350	277	373	222	60	263	77	378
	RJ	0	931	69	0	316	0	256	428
200	PT	0	1000	0	0	0	0	5	995
	WR	0	1000	0	0	0	0	5	995
	RMR	1	988	11	0	6	0	14	980
	CIC	0	998	2	0	49	0	73	878
	QIC	370	264	366	279	61	298	70	292
	RJ	0	990	10	0	315	0	278	407

Table B.8 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated binary response; $C(\alpha)$: EX(0.6), $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	54	896	50	50	162	12	287	489
	WR	54	890	56	50	164	12	284	490
	RMR	58	872	70	14	64	14	118	790
	CIC	93	811	96	72	214	44	233	437
	QIC	379	431	190	275	171	133	188	233
	RJ	54	835	111	50	470	7	266	207
30	PT	26	958	16	22	127	2	272	577
	WR	26	956	18	22	125	1	273	579
	RMR	29	943	28	8	51	5	86	850
	CIC	37	897	66	25	189	21	262	503
	QIC	371	456	173	284	183	112	174	247
	RJ	26	922	52	22	453	0	277	248
50	PT	8	987	5	7	47	0	122	824
	WR	8	985	7	7	47	0	118	828
	RMR	11	985	4	3	31	0	61	905
	CIC	11	949	40	9	112	12	167	700
	QIC	360	409	231	254	135	178	140	293
	RJ	8	970	22	7	377	0	257	359
100	PT	0	999	1	0	6	0	24	970
	WR	0	999	1	0	6	0	24	970
	RMR	2	998	0	0	36	0	58	906
	CIC	0	998	2	0	90	1	135	774
	QIC	367	338	295	287	82	253	94	284
	RJ	0	996	4	0	379	0	235	386
200	PT	0	1000	0	0	3	0	8	989
	WR	0	1000	0	0	3	0	4	993
	RMR	0	1000	0	0	29	0	56	915
	CIC	0	1000	0	0	75	0	143	782
	QIC	374	242	384	317	55	338	57	233
	RJ	0	1000	0	0	354	0	255	391

Table B.9 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated binary response; $C(\alpha)$: TOEP, $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	18	945	37	15	277	11	132	565
	WR	16	946	38	13	278	11	133	565
	RMR	172	779	49	26	50	6	116	802
	CIC	126	804	70	59	266	27	153	495
	QIC	321	400	279	196	167	191	144	302
	RJ	77	598	325	15	524	5	211	245
30	PT	4	982	14	3	233	1	117	646
	WR	4	982	14	3	233	1	122	641
	RMR	158	813	29	17	33	0	105	845
	CIC	79	894	27	22	249	9	167	553
	QIC	310	394	296	178	170	231	146	275
	RJ	25	706	269	3	538	2	217	240
50	PT	1	997	2	1	143	0	64	792
	WR	1	997	2	1	143	0	64	792
	RMR	117	881	2	6	19	0	86	889
	CIC	32	964	4	7	171	1	170	651
	QIC	317	363	320	167	130	260	147	296
	RJ	4	805	191	1	492	1	219	287
100	PT	0	1000	0	0	61	0	65	874
	WR	0	1000	0	0	61	0	66	873
	RMR	48	952	0	2	3	0	77	918
	CIC	5	995	0	0	74	0	176	750
	QIC	333	329	338	191	80	297	155	277
	RJ	0	896	104	0	465	0	257	278
200	PT	0	1000	0	0	9	0	62	929
	WR	0	1000	0	0	9	0	62	929
	RMR	25	975	0	0	0	0	80	920
	CIC	0	1000	0	0	9	0	217	774
	QIC	326	319	355	161	54	321	173	291
	RJ	0	970	30	0	463	0	256	281

Table B.10 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated binary response; $C(\alpha)$: TOEP, $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	5	982	13	3	153	1	382	461
	WR	5	982	13	3	152	1	384	460
	RMR	100	778	122	3	29	6	125	837
	CIC	104	819	77	29	175	14	369	413
	QIC	331	397	272	202	158	159	230	251
	RJ	22	796	182	3	546	1	267	183
30	PT	2	998	0	2	106	0	327	565
	WR	2	998	0	2	106	0	323	569
	RMR	89	846	65	7	15	5	118	855
	CIC	44	922	34	8	125	3	345	519
	QIC	336	372	292	187	139	170	237	267
	RJ	4	850	146	2	523	0	265	210
50	PT	0	1000	0	0	44	0	250	706
	WR	0	1000	0	0	44	0	253	703
	RMR	40	926	34	1	9	1	78	911
	CIC	14	976	10	1	55	0	303	641
	QIC	358	320	322	207	94	213	213	273
	RJ	0	947	53	0	473	0	281	246
100	PT	0	1000	0	0	2	0	97	901
	WR	0	1000	0	0	2	0	99	899
	RMR	13	978	9	0	6	0	50	944
	CIC	0	1000	0	0	2	0	178	820
	QIC	346	276	378	203	74	259	151	313
	RJ	0	983	17	0	440	0	266	294
200	PT	0	1000	0	0	0	0	37	963
	WR	0	1000	0	0	0	0	40	960
	RMR	0	999	1	0	0	0	33	967
	CIC	0	1000	0	0	0	0	179	821
	QIC	374	249	377	238	66	264	127	305
	RJ	0	999	1	0	456	0	249	295

Table B.11 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated binary response; $C(\alpha)$: UN, $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	40	159	801	38	62	253	245	402
	WR	40	163	797	38	65	255	238	404
	RMR	32	96	872	5	8	75	104	808
	CIC	59	186	755	47	87	229	265	372
	QIC	323	267	410	260	196	135	238	171
	RJ	93	569	338	38	308	317	126	211
30	PT	8	91	901	7	30	195	225	543
	WR	8	99	893	7	30	211	209	543
	RMR	15	64	921	2	5	38	76	879
	CIC	14	124	862	10	49	200	253	488
	QIC	300	240	460	241	170	130	247	212
	RJ	34	598	368	7	283	366	117	227
50	PT	4	36	960	4	8	175	176	637
	WR	4	37	959	4	8	186	165	637
	RMR	4	29	967	0	1	24	39	936
	CIC	4	67	929	4	15	174	214	593
	QIC	295	255	450	249	192	133	226	200
	RJ	5	582	413	4	300	309	131	256
100	PT	0	3	997	0	1	107	104	788
	WR	0	3	997	0	1	113	98	788
	RMR	0	4	996	0	0	5	9	986
	CIC	0	15	985	0	3	95	129	773
	QIC	300	254	446	234	195	99	228	244
	RJ	0	595	405	0	282	305	75	338
200	PT	0	0	1000	0	0	39	32	929
	WR	0	0	1000	0	0	43	28	929
	RMR	0	0	1000	0	0	0	0	1000
	CIC	0	0	1000	0	0	27	45	928
	QIC	298	250	452	249	190	92	189	280
	RJ	0	579	421	0	303	256	90	351

Table B.12 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated binary response; $C(\alpha)$: UN, $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	10	302	688	5	39	68	467	421
	WR	9	288	703	5	35	79	459	422
	RMR	95	471	434	10	21	17	74	878
	CIC	71	400	529	25	102	108	355	410
	QIC	311	271	418	215	142	214	210	219
	RJ	51	576	373	5	396	147	272	180
30	PT	3	258	739	2	10	48	417	523
	WR	3	235	762	2	9	59	407	523
	RMR	75	521	404	3	16	5	61	915
	CIC	31	381	588	12	58	84	334	512
	QIC	302	266	432	212	149	220	207	212
	RJ	19	575	406	2	385	91	274	248
50	PT	0	168	832	0	1	14	338	647
	WR	0	145	855	0	1	16	336	647
	RMR	44	529	427	1	7	2	13	977
	CIC	7	355	638	1	34	45	279	641
	QIC	314	282	404	237	149	215	133	266
	RJ	1	652	347	0	386	46	267	301
100	PT	0	105	895	0	0	5	174	821
	WR	0	89	911	0	0	6	173	821
	RMR	10	616	374	0	1	0	3	996
	CIC	1	320	679	0	6	16	163	815
	QIC	328	242	430	241	138	264	67	290
	RJ	0	726	274	0	370	9	241	380
200	PT	0	40	960	0	0	0	66	934
	WR	0	31	969	0	0	1	65	934
	RMR	2	709	289	0	0	0	0	1000
	CIC	0	270	730	0	1	0	65	934
	QIC	324	261	415	213	130	250	53	354
	RJ	0	840	160	0	380	1	181	438

Table B.13 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: AR-1(0.3), $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	8	242	750	1	6	35	207	751
	WR	8	243	749	1	8	35	214	742
	RMR	24	287	689	6	20	56	168	750
	CIC	124	267	609	50	64	107	193	586
	QIC	124	267	609	50	64	107	193	586
	RJ	347	374	279	15	246	344	166	229
30	PT	0	162	838	0	0	20	132	848
	WR	0	151	849	0	0	15	148	837
	RMR	8	264	728	2	18	45	90	845
	CIC	68	229	703	24	58	115	177	626
	QIC	68	229	703	24	58	115	177	626
	RJ	232	410	358	0	301	321	164	214
50	PT	0	144	856	0	0	12	122	866
	WR	0	141	859	0	1	12	132	855
	RMR	1	202	797	0	13	41	92	854
	CIC	40	189	771	9	53	118	167	653
	QIC	40	189	771	9	53	118	167	653
	RJ	205	424	371	0	255	348	149	248
100	PT	0	51	949	0	0	6	90	904
	WR	0	49	951	0	0	7	106	887
	RMR	0	118	882	0	8	28	78	886
	CIC	17	127	856	9	37	128	177	649
	QIC	17	127	856	9	37	128	177	649
	RJ	118	486	396	0	295	355	143	207
200	PT	0	4	996	0	0	5	63	932
	WR	0	4	996	0	0	6	71	923
	RMR	0	49	951	0	1	35	59	905
	CIC	0	59	941	0	21	123	176	680
	QIC	0	59	941	0	21	123	176	680
	RJ	27	523	450	0	332	358	142	168

Table B.14 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: AR-1(0.6), $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	0	117	883	0	15	117	457	411
	WR	0	114	886	0	17	123	463	397
	RMR	0	175	825	0	17	128	400	455
	CIC	19	186	795	10	68	194	318	410
	QIC	19	186	795	10	68	194	318	410
	RJ	166	440	394	0	306	338	209	147
30	PT	0	35	965	0	0	100	475	425
	WR	0	32	968	0	1	114	474	411
	RMR	0	70	930	0	0	102	421	477
	CIC	3	132	865	0	41	204	350	405
	QIC	3	132	865	0	41	204	350	405
	RJ	37	546	417	0	374	353	177	96
50	PT	0	21	979	0	0	93	490	417
	WR	0	20	980	0	1	102	484	413
	RMR	0	43	957	0	0	81	429	490
	CIC	0	89	911	0	28	181	369	422
	QIC	0	89	911	0	28	181	369	422
	RJ	22	570	408	0	330	386	192	92
100	PT	0	0	1000	0	0	83	481	436
	WR	0	1	999	0	0	87	497	416
	RMR	0	8	992	0	0	65	391	544
	CIC	0	28	972	0	7	199	368	426
	QIC	0	28	972	0	7	199	368	426
	RJ	0	580	420	0	342	382	197	79
200	PT	0	1	999	0	0	83	455	462
	WR	0	1	999	0	0	96	460	444
	RMR	0	1	999	0	0	65	365	570
	CIC	0	6	994	0	1	218	328	453
	QIC	0	6	994	0	1	218	328	453
	RJ	0	562	438	0	357	375	188	80

Table B.15 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: AR-1(0.3), $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	1	149	850	0	1	26	233	740
	WR	1	141	858	0	1	24	237	738
	RMR	16	256	728	3	10	48	192	747
	CIC	101	231	668	18	45	94	195	648
	QIC	101	231	668	18	45	94	195	648
	RJ	337	361	302	1	251	378	206	164
30	PT	0	51	949	0	0	9	91	900
	WR	0	50	950	0	0	7	99	894
	RMR	2	175	823	0	2	22	58	918
	CIC	42	148	810	6	19	69	138	768
	QIC	42	148	810	6	19	69	138	768
	RJ	236	400	364	0	250	377	183	190
50	PT	0	48	952	0	0	5	63	932
	WR	0	48	952	0	0	5	74	921
	RMR	1	153	846	0	0	23	38	939
	CIC	25	152	823	7	24	60	132	777
	QIC	25	152	823	7	24	60	132	777
	RJ	157	442	401	0	271	344	175	210
100	PT	0	5	995	0	0	1	18	981
	WR	0	5	995	0	0	1	19	980
	RMR	1	50	949	0	1	2	16	981
	CIC	4	68	928	2	9	62	103	824
	QIC	4	68	928	2	9	62	103	824
	RJ	88	482	430	0	317	341	160	182
200	PT	0	0	1000	0	0	0	14	986
	WR	0	0	1000	0	0	0	15	985
	RMR	0	8	992	0	0	3	19	978
	CIC	0	15	985	0	3	58	114	825
	QIC	0	15	985	0	3	58	114	825
	RJ	21	573	406	0	354	346	152	148

Table B.16 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: AR-1(0.6), $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	0	28	972	0	2	110	544	344
	WR	0	28	972	0	2	111	543	344
	RMR	2	68	930	1	4	128	509	358
	QIC	12	102	886	4	31	186	429	350
	RJ	84	546	370	0	302	349	183	166
30	PT	0	1	999	0	0	68	534	398
	WR	0	1	999	0	0	68	548	384
	RMR	0	5	995	0	0	68	493	439
	CIC	0	43	957	0	13	197	404	386
	QIC	0	43	957	0	13	197	404	386
	RJ	16	548	436	0	346	385	195	74
50	PT	0	0	1000	0	0	50	523	427
	WR	0	0	1000	0	0	66	519	415
	RMR	0	3	997	0	0	45	470	485
	CIC	0	31	969	0	3	179	412	406
	QIC	0	31	969	0	3	179	412	406
	RJ	3	579	418	0	363	365	205	67
100	PT	0	0	1000	0	0	42	506	452
	WR	0	0	1000	0	0	54	523	423
	RMR	0	0	1000	0	0	47	413	540
	CIC	0	1	999	0	0	191	366	443
	QIC	0	1	999	0	0	191	366	443
	RJ	0	573	427	0	374	367	206	53
200	PT	0	0	1000	0	0	31	483	486
	WR	0	0	1000	0	0	36	509	455
	RMR	0	0	1000	0	0	34	347	619
	CIC	0	0	1000	0	0	169	379	452
	QIC	0	0	1000	0	0	169	379	452
	RJ	0	587	413	0	387	373	198	42

Table B.17 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: EX(0.3), $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	2	784	214	0	43	5	169	783
	WR	2	788	210	0	51	6	182	761
	RMR	8	715	277	4	32	25	126	813
	CIC	118	609	273	42	113	64	169	612
	QIC	118	609	273	42	113	64	169	612
	RJ	205	478	317	3	388	56	265	288
30	PT	0	902	98	0	20	1	107	872
	WR	0	903	97	0	28	1	122	849
	RMR	1	842	157	0	8	6	48	938
	CIC	53	766	181	16	135	45	151	653
	QIC	53	766	181	16	135	45	151	653
	RJ	81	586	333	0	401	12	256	331
50	PT	0	915	85	0	5	0	101	894
	WR	0	914	86	0	14	0	110	876
	RMR	0	842	158	0	6	5	46	943
	CIC	55	763	182	18	122	42	168	650
	QIC	55	763	182	18	122	42	168	650
	RJ	63	627	310	0	418	7	256	319
100	PT	0	973	27	0	6	0	44	950
	WR	0	974	26	0	7	0	62	931
	RMR	0	934	66	0	1	2	26	971
	CIC	7	882	111	1	111	27	188	673
	QIC	7	882	111	1	111	27	188	673
	RJ	5	742	253	0	388	0	294	318
200	PT	0	999	1	0	3	0	30	967
	WR	0	999	1	0	7	0	46	947
	RMR	0	987	13	0	3	0	16	981
	CIC	0	968	32	0	105	6	173	716
	QIC	0	968	32	0	105	6	173	716
	RJ	0	868	132	0	382	0	268	350

Table B.18 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: EX(0.6), $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	0	918	82	0	269	5	378	348
	WR	0	917	83	0	279	8	373	340
	RMR	0	929	71	0	232	12	344	412
	CIC	10	784	206	2	247	104	255	392
	QIC	10	784	206	2	247	104	255	392
	RJ	31	747	222	0	536	7	263	194
30	PT	0	980	20	0	243	1	333	423
	WR	0	980	20	0	254	1	345	400
	RMR	0	976	24	0	163	0	293	544
	CIC	2	908	90	0	251	31	285	433
	QIC	2	908	90	0	251	31	285	433
	RJ	2	903	95	0	578	0	280	142
50	PT	0	992	8	0	223	0	341	436
	WR	0	989	11	0	235	0	355	410
	RMR	0	992	8	0	150	0	290	560
	CIC	1	935	64	1	247	22	280	450
	QIC	1	935	64	1	247	22	280	450
	RJ	0	937	63	0	578	0	291	131
100	PT	0	999	1	0	181	0	331	488
	WR	0	999	1	0	206	0	336	458
	RMR	0	1000	0	0	93	0	246	661
	CIC	0	985	15	0	250	4	280	466
	QIC	0	985	15	0	250	4	280	466
	RJ	0	989	11	0	560	0	284	156
200	PT	0	1000	0	0	189	0	304	507
	WR	0	1000	0	0	209	0	311	480
	RMR	0	1000	0	0	81	0	202	717
	CIC	0	1000	0	0	243	0	255	502
	QIC	0	1000	0	0	243	0	255	502
	RJ	0	999	1	0	575	0	266	159

Table B.19 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: EX(0.3), $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	0	912	88	0	39	1	224	736
	WR	0	913	87	0	45	1	228	726
	RMR	1	871	128	0	31	8	187	774
	CIC	78	719	203	12	95	35	189	669
	QIC	78	719	203	12	95	35	189	669
	RJ	78	640	282	0	446	10	312	232
30	PT	0	978	22	0	13	0	74	913
	WR	0	981	19	0	16	0	92	892
	RMR	0	940	60	0	3	0	35	962
	CIC	26	852	122	1	79	21	129	770
	QIC	26	852	122	1	79	21	129	770
	RJ	11	768	221	0	460	1	297	242
50	PT	0	988	12	0	8	0	66	926
	WR	0	989	11	0	10	0	80	910
	RMR	0	962	38	0	3	0	25	972
	CIC	16	891	93	3	88	14	124	771
	QIC	16	891	93	3	88	14	124	771
	RJ	1	824	175	0	449	0	282	269
100	PT	0	1000	0	0	2	0	8	990
	WR	0	999	1	0	3	0	15	982
	RMR	0	991	9	0	0	0	1	999
	CIC	0	970	30	0	56	4	82	858
	QIC	0	970	30	0	56	4	82	858
	RJ	0	919	81	0	403	0	285	312
200	PT	0	1000	0	0	0	0	5	995
	WR	0	1000	0	0	0	0	10	990
	RMR	0	1000	0	0	0	0	0	1000
	CIC	0	999	1	0	58	0	101	841
	QIC	0	999	1	0	58	0	101	841
	RJ	0	988	12	0	393	0	275	332

Table B.20 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: EX(0.6), $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	0	992	8	0	247	1	411	341
	WR	0	988	12	0	255	1	410	334
	RMR	1	994	5	0	224	0	413	363
	CIC	7	895	98	4	268	38	335	355
	QIC	7	895	98	4	268	38	335	355
	RJ	1	951	48	0	598	0	267	135
30	PT	0	1000	0	0	233	0	352	415
	WR	0	999	1	0	256	0	340	404
	RMR	0	1000	0	0	167	0	353	480
	CIC	0	967	33	0	263	18	303	416
	QIC	0	967	33	0	263	18	303	416
	RJ	0	995	5	0	654	0	260	86
50	PT	0	1000	0	0	244	0	316	440
	WR	0	1000	0	0	253	0	319	428
	RMR	0	1000	0	0	165	0	310	525
	CIC	0	990	10	0	228	1	314	457
	QIC	0	990	10	0	228	1	314	457
	RJ	0	999	1	0	645	0	270	85
100	PT	0	1000	0	0	191	0	242	567
	WR	0	1000	0	0	211	0	252	537
	RMR	0	1000	0	0	107	0	184	709
	CIC	0	998	2	0	201	0	237	562
	QIC	0	998	2	0	201	0	237	562
	RJ	0	1000	0	0	647	0	255	98
200	PT	0	1000	0	0	160	0	194	646
	WR	0	1000	0	0	178	0	211	611
	RMR	0	1000	0	0	69	0	117	814
	CIC	0	1000	0	0	209	0	241	550
	QIC	0	1000	0	0	209	0	241	550
	RJ	0	1000	0	0	672	0	230	98

Table B.21 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: TOEP, $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	2	944	54	0	99	1	176	724
	WR	2	943	55	0	99	1	192	708
	RMR	11	893	96	2	118	16	149	715
	CIC	112	768	120	20	128	18	250	584
	QIC	112	768	120	20	128	18	250	584
	RJ	164	551	285	5	444	7	302	242
30	PT	0	998	2	0	76	0	109	815
	WR	0	998	2	0	73	0	117	810
	RMR	1	981	18	0	97	3	81	819
	CIC	54	910	36	6	101	6	227	660
	QIC	54	910	36	6	101	6	227	660
	RJ	51	711	238	0	483	0	242	275
50	PT	0	999	1	0	48	0	61	891
	WR	0	999	1	0	45	0	80	875
	RMR	0	993	7	0	64	0	49	887
	CIC	32	948	20	2	79	2	231	686
	QIC	32	948	20	2	79	2	231	686
	RJ	38	736	226	0	496	0	241	263
100	PT	0	1000	0	0	12	0	24	964
	WR	0	1000	0	0	11	0	31	958
	RMR	0	1000	0	0	21	0	44	935
	CIC	5	994	1	0	26	0	215	759
	QIC	5	994	1	0	26	0	215	759
	RJ	1	837	162	0	474	0	246	280
200	PT	0	1000	0	0	0	0	11	989
	WR	0	1000	0	0	0	0	20	980
	RMR	0	1000	0	0	2	0	47	951
	CIC	0	1000	0	0	3	0	227	770
	QIC	0	1000	0	0	3	0	227	770
	RJ	0	933	67	0	499	0	235	266

Table B.22 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: TOEP, $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	0	996	4	0	24	0	591	385
	WR	0	996	4	0	23	0	594	383
	RMR	3	978	19	0	37	0	586	377
	CIC	55	879	66	1	73	6	525	395
	QIC	55	879	66	1	73	6	525	395
	RJ	38	754	208	1	516	0	296	187
30	PT	0	1000	0	0	4	0	598	398
	WR	0	1000	0	0	4	0	608	388
	RMR	0	999	1	0	5	0	569	426
	CIC	13	981	6	0	14	0	587	399
	QIC	13	981	6	0	14	0	587	399
	RJ	2	905	93	0	541	0	346	113
50	PT	0	1000	0	0	3	0	585	412
	WR	0	1000	0	0	3	0	594	403
	RMR	0	999	1	0	9	0	546	445
	CIC	13	982	5	0	13	0	546	441
	QIC	13	982	5	0	13	0	546	441
	RJ	3	918	79	0	569	0	321	110
100	PT	0	1000	0	0	0	0	562	438
	WR	0	1000	0	0	0	0	574	426
	RMR	0	1000	0	0	1	0	499	500
	CIC	0	1000	0	0	0	0	545	455
	QIC	0	1000	0	0	0	0	545	455
	RJ	0	972	28	0	558	0	358	84
200	PT	0	1000	0	0	0	0	587	413
	WR	0	1000	0	0	0	0	607	393
	RMR	0	1000	0	0	0	0	489	511
	CIC	0	1000	0	0	0	0	538	462
	QIC	0	1000	0	0	0	0	538	462
	RJ	0	999	1	0	546	0	378	76

Table B.23 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: UN, $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	2	944	54	0	99	1	176	724
	WR	2	943	55	0	99	1	192	708
	RMR	11	893	96	2	118	16	149	715
	CIC	112	768	120	20	128	18	250	584
	QIC	112	768	120	20	128	18	250	584
	RJ	164	551	285	5	444	7	302	242
30	PT	0	998	2	0	76	0	109	815
	WR	0	998	2	0	73	0	117	810
	RMR	1	981	18	0	97	3	81	819
	CIC	54	910	36	6	101	6	227	660
	QIC	54	910	36	6	101	6	227	660
	RJ	51	711	238	0	483	0	242	275
50	PT	0	32	968	0	0	57	98	845
	WR	0	29	971	0	0	53	90	857
	RMR	0	51	949	0	0	55	137	808
	CIC	1	86	913	0	11	83	127	779
	QIC	1	86	913	0	11	83	127	779
	RJ	24	530	446	0	292	307	104	297
100	PT	0	9	991	0	0	23	41	936
	WR	0	9	991	0	0	22	34	944
	RMR	0	14	986	0	0	25	66	909
	CIC	0	30	970	0	0	38	63	899
	QIC	0	30	970	0	0	38	63	899
	RJ	2	550	448	0	302	261	97	340
200	PT	0	0	1000	0	0	3	13	984
	WR	0	0	1000	0	0	2	8	990
	RMR	0	0	1000	0	0	3	16	981
	CIC	0	8	992	0	0	14	8	978
	QIC	0	8	992	0	0	14	8	978
	RJ	0	542	458	0	306	267	87	340

Table B.24 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: UN, $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	0	996	4	0	24	0	591	385
	WR	0	996	4	0	23	0	594	383
	RMR	3	978	19	0	37	0	586	377
	CIC	55	879	66	1	73	6	525	395
	QIC	55	879	66	1	73	6	525	395
	RJ	38	754	208	1	516	0	296	187
30	PT	0	1000	0	0	4	0	598	398
	WR	0	1000	0	0	4	0	608	388
	RMR	0	999	1	0	5	0	569	426
	CIC	13	981	6	0	14	0	587	399
	QIC	13	981	6	0	14	0	587	399
	RJ	2	905	93	0	541	0	346	113
50	PT	0	164	836	0	0	8	167	825
	WR	0	149	851	0	0	6	160	834
	RMR	0	317	683	0	2	14	180	804
	CIC	6	314	680	1	17	38	150	794
	QIC	6	314	680	1	17	38	150	794
	RJ	5	640	355	0	401	47	249	303
100	PT	0	73	927	0	0	1	86	913
	WR	0	63	937	0	0	1	79	920
	RMR	0	238	762	0	0	3	102	895
	CIC	0	213	787	0	1	8	97	894
	QIC	0	213	787	0	1	8	97	894
	RJ	0	722	278	0	379	13	243	365
200	PT	0	29	971	0	0	0	28	972
	WR	0	23	977	0	0	0	27	973
	RMR	0	139	861	0	0	0	49	951
	CIC	0	147	853	0	0	7	40	953
	QIC	0	147	853	0	0	7	40	953
	RJ	0	815	185	0	362	0	251	387

Table B.25 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: AR-1(0.3), $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	14	243	743	2	17	73	249	659
	WR	13	243	744	2	21	66	261	650
	RMR	146	247	607	53	60	108	165	614
	CIC	155	246	599	63	69	112	202	554
	QIC	293	261	446	164	125	94	163	454
	RJ	310	403	287	9	227	387	162	215
30	PT	0	237	763	0	28	59	213	700
	WR	0	229	771	0	27	56	224	693
	RMR	140	259	601	51	60	84	122	683
	CIC	133	215	652	66	54	87	187	606
	QIC	303	276	421	200	140	114	127	419
	RJ	278	396	326	2	247	345	168	238
50	PT	2	165	833	2	9	46	199	744
	WR	2	160	838	2	9	48	218	723
	RMR	116	243	641	48	63	80	106	703
	CIC	77	221	702	43	62	107	162	626
	QIC	322	229	449	235	133	142	119	371
	RJ	179	437	384	0	271	323	148	258
100	PT	0	68	932	0	1	40	187	772
	WR	0	62	938	0	2	40	205	753
	RMR	66	183	751	28	53	100	102	717
	CIC	31	134	835	14	46	120	166	654
	QIC	330	266	404	262	182	116	115	325
	RJ	91	526	383	0	286	367	123	224
200	PT	0	20	980	0	2	42	186	770
	WR	0	17	983	0	1	40	205	754
	RMR	40	140	820	15	46	101	99	739
	CIC	10	86	904	6	33	144	187	630
	QIC	315	253	432	265	203	135	110	287
	RJ	23	518	459	0	334	358	139	169

Table B.26 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: AR-1(0.6), $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	1	156	843	0	29	161	395	415
	WR	1	143	856	0	26	165	400	409
	RMR	87	239	674	52	77	169	248	454
	CIC	49	188	763	33	68	196	267	436
	QIC	316	256	428	216	141	125	169	349
	RJ	112	484	404	0	325	315	185	175
30	PT	0	112	888	0	18	166	404	412
	WR	0	111	889	0	14	177	411	398
	RMR	45	219	736	28	81	143	276	472
	CIC	20	179	801	7	74	159	314	446
	QIC	319	251	430	247	169	106	120	358
	RJ	24	527	449	0	358	336	177	129
50	PT	0	67	933	0	9	157	400	434
	WR	0	59	941	0	8	179	408	405
	RMR	30	200	770	17	74	147	229	533
	CIC	3	139	858	3	52	185	284	476
	QIC	307	250	443	241	175	120	114	350
	RJ	11	580	409	0	377	330	188	105
100	PT	0	7	993	0	0	149	415	436
	WR	0	7	993	0	0	162	428	410
	RMR	4	109	887	3	32	138	243	584
	CIC	0	67	933	0	16	200	302	482
	QIC	309	274	417	269	206	125	93	307
	RJ	0	550	450	0	388	316	205	91
200	PT	0	0	1000	0	0	152	426	422
	WR	0	0	1000	0	0	156	450	394
	RMR	2	38	960	0	11	130	224	635
	CIC	0	11	989	0	0	187	291	522
	QIC	345	234	421	317	194	126	89	274
	RJ	0	581	419	0	394	305	187	114

Table B.27 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: AR-1(0.3), $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	3	153	844	0	8	52	262	678
	WR	3	147	850	0	7	51	268	674
	RMR	109	258	633	15	42	86	177	680
	CIC	134	189	677	26	32	94	191	657
	QIC	318	240	442	158	95	119	179	449
	RJ	308	391	301	2	263	403	165	167
30	PT	5	113	882	0	3	21	200	776
	WR	5	105	890	0	3	19	212	766
	RMR	98	234	668	20	23	63	105	789
	CIC	116	149	735	20	22	70	147	741
	QIC	304	241	455	158	114	121	147	460
	RJ	238	427	335	1	228	390	188	193
50	PT	0	63	937	0	1	19	113	867
	WR	0	56	944	0	0	17	126	857
	RMR	75	221	704	13	27	48	65	847
	CIC	54	126	820	12	19	69	124	776
	QIC	324	233	443	187	138	110	125	440
	RJ	137	444	419	0	261	364	160	215
100	PT	0	11	989	0	0	10	79	911
	WR	0	10	990	0	0	9	88	903
	RMR	39	156	805	4	10	23	60	903
	CIC	21	86	893	5	19	68	105	803
	QIC	343	258	399	252	172	100	86	390
	RJ	40	540	420	0	300	364	161	175
200	PT	0	0	1000	0	0	10	61	929
	WR	0	0	1000	0	0	9	72	919
	RMR	11	113	876	0	6	22	43	929
	CIC	4	35	961	1	9	51	103	836
	QIC	345	258	397	263	178	112	94	353
	RJ	4	558	438	0	322	360	131	187

Table B.28 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: AR-1(0.6), $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	0	49	951	0	3	130	495	372
	WR	0	43	957	0	3	135	500	362
	RMR	52	196	752	17	55	141	377	410
	CIC	29	138	833	14	38	175	345	428
	QIC	342	250	408	199	146	121	156	378
	RJ	45	546	409	1	300	333	198	168
30	PT	0	24	976	0	1	115	481	403
	WR	0	24	976	0	0	131	494	375
	RMR	37	161	802	15	34	133	336	482
	CIC	16	110	874	6	26	172	365	431
	QIC	332	249	419	218	146	112	141	383
	RJ	17	557	426	0	349	320	208	123
50	PT	0	8	992	0	0	98	487	415
	WR	0	6	994	0	0	119	501	380
	RMR	16	126	858	4	35	82	340	539
	CIC	7	60	933	1	20	153	344	482
	QIC	339	240	421	254	139	106	109	392
	RJ	0	557	443	0	385	303	218	94
100	PT	0	0	1000	0	0	98	470	432
	WR	0	0	1000	0	0	114	494	392
	RMR	1	38	961	0	6	92	269	633
	CIC	0	26	974	0	3	143	353	501
	QIC	340	244	416	268	166	85	90	391
	RJ	0	591	409	0	402	317	215	66
200	PT	0	0	1000	0	0	79	484	437
	WR	0	0	1000	0	0	86	526	388
	RMR	0	5	995	0	0	70	225	705
	CIC	0	4	996	0	0	156	300	544
	QIC	348	244	408	288	178	96	86	352
	RJ	0	555	445	0	407	307	227	59

Table B.29 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: EX(0.3), $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	12	732	256	0	94	40	207	659
	WR	13	735	252	0	97	38	223	642
	RMR	195	467	338	62	73	90	118	657
	CIC	166	520	314	56	96	84	163	601
	QIC	317	304	379	156	89	194	128	433
	RJ	166	468	366	6	389	63	254	288
30	PT	3	816	181	1	96	14	190	699
	WR	1	820	179	1	98	16	205	680
	RMR	199	527	274	62	60	59	99	720
	CIC	152	616	232	58	97	78	140	627
	QIC	304	302	394	179	85	236	91	409
	RJ	88	544	368	0	390	24	263	323
50	PT	1	879	120	0	63	7	191	739
	WR	1	887	112	0	73	8	217	702
	RMR	159	584	257	53	56	55	87	749
	CIC	102	689	209	35	107	58	168	632
	QIC	333	291	376	209	72	253	83	383
	RJ	37	602	361	0	360	15	288	337
100	PT	0	946	54	0	64	2	170	764
	WR	0	955	45	0	74	1	188	737
	RMR	106	703	191	53	71	62	85	729
	CIC	41	832	127	19	131	30	162	658
	QIC	317	291	392	227	63	280	88	342
	RJ	5	722	273	0	381	2	255	362
200	PT	0	991	9	0	49	0	167	784
	WR	0	989	11	0	56	0	183	761
	RMR	38	853	109	15	69	31	97	788
	CIC	6	948	46	1	118	17	179	685
	QIC	350	268	382	292	67	320	75	246
	RJ	0	851	149	0	361	0	269	370

Table B.30 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: EX(0.6), $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	0	853	147	0	226	37	311	426
	WR	0	868	132	0	231	34	328	407
	RMR	99	672	229	57	179	83	219	462
	CIC	75	653	272	37	198	113	207	445
	QIC	314	258	428	207	71	285	103	334
	RJ	8	675	317	0	483	25	288	204
30	PT	0	908	92	0	263	22	318	397
	WR	0	911	89	0	285	25	321	369
	RMR	86	749	165	47	181	51	223	498
	CIC	38	781	181	23	229	95	219	434
	QIC	363	232	405	267	63	280	75	315
	RJ	3	790	207	0	543	4	273	180
50	PT	0	958	42	0	262	7	327	404
	WR	0	953	47	0	285	8	342	365
	RMR	58	846	96	34	155	27	225	559
	CIC	9	873	118	3	214	46	239	498
	QIC	396	202	402	290	48	300	53	309
	RJ	0	891	109	0	517	0	282	201
100	PT	0	995	5	0	266	0	290	444
	WR	0	995	5	0	279	0	315	406
	RMR	5	966	29	3	154	4	179	660
	CIC	1	957	42	0	223	13	218	546
	QIC	366	201	433	293	51	358	52	246
	RJ	0	970	30	0	530	0	289	181
200	PT	0	1000	0	0	255	0	302	443
	WR	0	1000	0	0	278	0	315	407
	RMR	0	997	3	0	126	0	185	689
	CIC	0	993	7	0	213	3	225	559
	QIC	400	220	380	359	55	341	50	195
	RJ	0	995	5	0	528	0	281	191

Table B.31 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: EX(0.3), $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	9	850	141	0	62	11	274	653
	WR	7	854	139	0	71	11	282	636
	RMR	131	587	282	24	67	49	153	707
	CIC	154	620	226	34	77	58	189	642
	QIC	354	319	327	162	82	154	134	468
	RJ	52	625	323	3	424	13	321	239
30	PT	2	915	83	0	61	2	228	709
	WR	1	924	75	0	68	4	248	680
	RMR	117	634	249	18	53	35	110	784
	CIC	123	696	181	27	89	39	152	693
	QIC	360	293	347	183	58	153	87	519
	RJ	6	707	287	0	411	3	317	269
50	PT	0	978	22	0	48	0	162	790
	WR	0	980	20	0	57	1	185	757
	RMR	80	737	183	12	34	17	74	863
	CIC	61	824	115	18	68	14	144	756
	QIC	336	298	366	170	64	214	87	465
	RJ	0	812	188	0	421	0	311	268
100	PT	0	995	5	0	36	0	108	856
	WR	0	997	3	0	43	0	131	826
	RMR	21	879	100	6	20	0	33	941
	CIC	18	928	54	3	62	5	105	825
	QIC	355	239	406	231	41	268	58	402
	RJ	0	917	83	0	362	0	276	362
200	PT	0	1000	0	0	19	0	59	922
	WR	0	1000	0	0	27	0	81	892
	RMR	4	961	35	1	17	1	27	954
	CIC	4	991	5	1	52	0	83	864
	QIC	353	277	370	259	64	286	58	333
	RJ	0	984	16	0	375	0	254	371

Table B.32 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: EX(0.6), $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	0	923	77	0	221	24	391	364
	WR	0	916	84	0	230	34	379	357
	RMR	64	832	104	28	167	26	360	419
	CIC	60	766	174	26	170	81	312	411
	QIC	379	232	389	226	68	267	107	332
	RJ	0	875	125	0	556	1	305	138
30	PT	0	956	44	0	241	17	379	363
	WR	0	951	49	0	267	18	379	336
	RMR	35	913	52	20	194	10	326	450
	CIC	21	850	129	10	232	59	293	406
	QIC	385	231	384	239	51	243	94	373
	RJ	0	961	39	0	586	0	315	99
50	PT	0	997	3	0	261	3	324	412
	WR	0	994	6	0	286	4	333	377
	RMR	14	978	8	9	157	0	291	543
	CIC	11	945	44	6	217	14	283	480
	QIC	362	197	441	231	32	301	78	358
	RJ	0	996	4	0	616	0	283	101
100	PT	0	1000	0	0	239	0	294	467
	WR	0	999	1	0	267	0	301	432
	RMR	4	996	0	2	144	0	198	656
	CIC	0	992	8	0	188	1	240	571
	QIC	384	209	407	283	36	313	43	325
	RJ	0	1000	0	0	564	0	289	147
200	PT	0	1000	0	0	209	0	268	523
	WR	0	1000	0	0	246	0	289	465
	RMR	0	1000	0	0	99	0	144	757
	CIC	0	1000	0	0	177	0	203	620
	QIC	387	183	430	321	37	378	42	222
	RJ	0	1000	0	0	594	0	269	137

Table B.33 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: TOEP, $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	6	938	56	0	91	6	276	627
	WR	5	941	54	0	93	4	290	613
	RMR	188	660	152	55	120	49	175	601
	CIC	196	668	136	58	115	33	218	576
	QIC	301	338	361	128	69	229	167	407
	RJ	106	560	334	4	446	5	252	293
30	PT	1	966	33	0	71	4	250	675
	WR	2	970	28	0	74	2	263	661
	RMR	198	709	93	48	111	40	164	637
	CIC	147	789	64	30	99	16	243	612
	QIC	337	345	318	173	73	236	161	357
	RJ	44	622	334	1	483	0	285	231
50	PT	0	995	5	0	44	0	225	731
	WR	0	995	5	0	42	0	241	717
	RMR	157	805	38	34	96	28	166	676
	CIC	119	858	23	23	73	2	242	660
	QIC	338	339	323	171	69	263	153	344
	RJ	16	720	264	0	493	0	240	267
100	PT	0	1000	0	0	9	0	176	815
	WR	0	1000	0	0	9	0	196	795
	RMR	90	901	9	11	41	3	151	794
	CIC	43	957	0	2	26	0	223	749
	QIC	339	306	355	194	59	310	145	292
	RJ	0	867	133	0	459	0	244	297
200	PT	0	1000	0	0	0	0	198	802
	WR	0	1000	0	0	1	0	209	790
	RMR	41	959	0	2	13	0	182	803
	CIC	3	997	0	0	2	0	223	775
	QIC	343	272	385	178	47	338	170	267
	RJ	0	949	51	0	489	0	249	262

Table B.34 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: TOEP, $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	7	965	28	0	35	0	586	379
	WR	5	968	27	0	33	0	583	384
	RMR	128	706	166	33	86	30	429	422
	CIC	137	732	131	15	81	18	473	413
	QIC	363	322	315	180	87	159	230	344
	RJ	23	733	244	1	495	0	301	203
30	PT	0	992	8	0	15	0	568	417
	WR	0	993	7	0	12	0	583	405
	RMR	103	766	131	26	85	16	393	480
	CIC	112	823	65	13	60	12	448	467
	QIC	389	294	317	182	70	165	178	405
	RJ	4	838	158	0	500	0	329	171
50	PT	0	1000	0	0	1	0	602	397
	WR	0	1000	0	0	1	0	629	370
	RMR	54	853	93	5	40	20	425	510
	CIC	53	921	26	1	24	0	522	453
	QIC	376	278	346	185	73	187	143	412
	RJ	0	920	80	0	507	0	394	99
100	PT	0	1000	0	0	0	0	581	419
	WR	0	1000	0	0	0	0	619	381
	RMR	20	950	30	1	7	1	372	619
	CIC	8	992	0	0	3	0	484	513
	QIC	404	246	350	219	69	228	123	361
	RJ	0	985	15	0	522	0	374	104
200	PT	0	1000	0	0	0	0	564	436
	WR	0	1000	0	0	0	0	595	405
	RMR	2	994	4	0	1	0	288	711
	CIC	1	999	0	0	0	0	442	558
	QIC	398	222	380	247	59	251	102	341
	RJ	0	997	3	0	552	0	346	102

Table B.35 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: UN, $m=4$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	0	166	834	0	25	109	184	682
	WR	0	161	839	0	23	108	178	691
	RMR	72	214	714	26	70	137	226	541
	CIC	48	193	759	21	53	113	212	601
	QIC	330	245	425	223	150	91	188	348
	RJ	100	504	396	0	279	295	149	277
30	PT	0	87	913	0	6	101	155	738
	WR	0	85	915	0	9	102	135	754
	RMR	39	133	828	16	38	120	230	596
	CIC	16	135	849	4	29	116	206	645
	QIC	331	256	413	227	172	93	170	338
	RJ	43	520	437	0	288	315	153	244
50	PT	0	55	945	0	2	68	113	817
	WR	0	56	944	0	3	68	101	828
	RMR	18	110	872	5	21	108	195	671
	CIC	3	93	904	1	17	110	130	742
	QIC	336	233	431	248	159	88	169	336
	RJ	8	536	456	0	303	302	119	276
100	PT	0	6	994	0	0	24	45	931
	WR	0	6	994	0	0	23	38	939
	RMR	4	44	952	1	4	40	134	821
	CIC	1	36	963	0	2	33	80	885
	QIC	325	258	417	245	174	83	154	344
	RJ	0	597	403	0	281	317	127	275
200	PT	0	0	1000	0	0	11	13	976
	WR	0	1	999	0	0	11	10	979
	RMR	1	9	990	0	1	11	50	938
	CIC	0	8	992	0	0	14	26	960
	QIC	344	250	406	267	183	89	150	311
	RJ	0	542	458	0	288	300	94	318

Table B.36 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: UN, $m=6$

n	Criterion	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
20	PT	2	310	688	0	17	66	275	642
	WR	2	287	711	0	14	61	274	651
	RMR	125	407	468	32	69	82	223	594
	CIC	114	310	576	19	54	105	224	598
	QIC	336	252	412	181	114	163	146	396
	RJ	69	518	413	0	346	145	244	265
30	PT	0	292	708	0	7	34	255	704
	WR	0	265	735	0	6	44	244	706
	RMR	116	456	428	32	71	63	176	658
	CIC	57	394	549	11	60	70	197	662
	QIC	373	243	384	224	92	173	115	396
	RJ	23	560	417	0	378	81	281	260
50	PT	0	228	772	0	1	21	169	809
	WR	0	207	793	0	1	19	172	808
	RMR	60	496	444	14	52	34	117	783
	CIC	31	385	584	6	35	52	129	778
	QIC	320	259	421	205	127	188	78	402
	RJ	4	645	351	0	441	47	241	271
100	PT	0	142	858	0	0	6	105	889
	WR	0	119	881	0	0	5	101	894
	RMR	30	506	464	5	22	20	87	866
	CIC	8	316	676	0	8	20	92	880
	QIC	329	275	396	227	130	204	57	382
	RJ	0	701	299	0	419	10	264	307
200	PT	0	69	931	0	0	0	47	953
	WR	0	44	956	0	0	0	45	955
	RMR	0	622	378	0	8	0	47	945
	CIC	0	311	689	0	0	3	52	945
	QIC	353	248	399	244	138	211	37	370
	RJ	0	825	175	0	439	1	228	332

APPENDIX C

THE PERFORMANCE OF THE CRITERIA WHEN THE SCALE
PARAMETER IS FIXED

Table C.1 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications when the scale parameter is fixed:
Correlated Gaussian response; $C(\alpha)$: AR-1 and $n=50$

$C(\alpha)$	Criteri on	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
AR-1(0.3) $m=4$	PT	49	202	749	22	71	111	204	592
	WR	53	203	744	27	76	112	204	581
	RMR	97	253	650	45	88	111	168	588
	CIC	42	200	758	23	66	104	193	614
	QIC	199	245	556	142	117	151	116	474
	RJ	271	389	340	236	249	209	119	187
AR-1(0.6) $m=4$	PT	0	71	929	0	25	220	332	423
	WR	0	70	930	0	28	235	325	412
	RMR	4	175	821	1	81	190	292	436
	CIC	0	89	911	0	38	198	335	429
	QIC	112	242	646	102	185	233	191	289
	RJ	56	506	438	45	342	222	189	202
AR-1(0.3) $m=6$	PT	34	127	839	9	27	82	138	744
	WR	37	128	835	9	31	81	146	733
	RMR	96	199	705	18	49	88	135	710
	CIC	34	120	846	9	17	74	130	770
	QIC	155	227	618	76	95	102	79	648
	RJ	206	427	367	153	253	201	167	226
AR-1(0.6) $m=6$	PT	0	9	991	0	1	186	423	390
	WR	0	13	987	0	3	199	426	372
	RMR	4	65	931	1	33	191	362	413
	CIC	0	19	981	0	8	186	402	404
	QIC	90	202	708	84	154	232	255	275
	RJ	15	517	468	9	343	231	223	194

Table C.2 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications when the scale parameter is fixed:
Correlated Gaussian response; $C(\alpha)$: EX and $n=50$

$C(\alpha)$	Criteri on	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
EX(0.3) $m=4$	PT	43	738	219	19	120	66	147	648
	WR	45	731	224	29	135	74	145	617
	RMR	66	674	260	28	129	95	124	624
	CIC	39	751	210	14	114	67	129	676
	QIC	161	598	241	109	155	123	99	514
	RJ	101	596	303	101	264	251	173	211
EX(0.6) $m=4$	PT	0	891	109	0	280	42	267	411
	WR	0	884	116	0	281	52	271	396
	RMR	3	824	173	2	257	81	234	426
	CIC	1	887	112	0	278	48	253	421
	QIC	118	643	239	100	282	175	173	270
	RJ	31	605	364	26	368	266	167	173
EX(0.3) $m=6$	PT	13	872	115	0	82	18	144	756
	WR	20	854	126	3	93	23	160	721
	RMR	23	832	145	6	93	29	145	727
	CIC	10	877	113	0	70	9	132	789
	QIC	105	641	254	71	114	120	89	606
	RJ	16	756	228	16	285	195	230	274
EX(0.6) $m=6$	PT	0	963	37	0	260	14	299	427
	WR	0	947	53	0	271	25	285	419
	RMR	0	911	89	0	274	39	276	411
	CIC	0	952	48	0	278	19	279	424
	QIC	91	637	272	80	257	213	189	261
	RJ	1	709	290	1	418	202	222	157

Table C.3 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications when the scale parameter is fixed: Correlated Gaussian response; $C(\alpha)$: TOEP and $n=50$

$C(\alpha)$	Criteri on	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
TOEP $m=4$	PT	33	933	34	4	60	1	274	661
	WR	44	915	41	6	61	3	307	623
	RMR	75	880	45	14	133	10	203	640
	CIC	33	942	25	8	68	2	234	688
	QIC	192	699	109	83	191	55	169	502
	RJ	76	616	308	73	366	262	142	157
TOEP $m=6$	PT	7	988	5	0	6	0	571	423
	WR	16	976	8	0	7	0	593	400
	RMR	15	954	31	0	50	1	516	433
	CIC	7	986	7	0	13	0	552	435
	QIC	100	726	174	33	175	84	378	330
	RJ	5	838	157	4	414	87	339	156

Table C.4 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ from 1000 independent replications when the scale parameter is fixed: Correlated Gaussian response; $C(\alpha)$: UN and $n=50$

$C(\alpha)$	Criteri on	Wc1={IN, EX, AR-1}			Wc2={IN, EX, AR-1, TOEP, UN}				
		IN	EX	AR-1	IN	EX	AR-1	TOEP	UN
UN $m=4$	PT	0	73	927	0	4	62	92	842
	WR	1	80	919	0	5	71	83	841
	RMR	5	128	867	0	31	84	176	709
	CIC	1	77	922	0	5	63	123	809
	QIC	115	230	655	72	96	95	76	661
	RJ	70	503	427	54	336	223	202	185
UN $m=6$	PT	2	223	775	0	14	38	106	842
	WR	4	206	790	0	16	41	93	850
	RMR	8	373	619	1	44	59	131	765
	CIC	1	263	736	0	21	39	117	823
	QIC	116	342	542	68	68	72	72	720
	RJ	22	527	451	18	299	286	201	196

APPENDIX D THE EFFECT OF THE PENALIZATION FOR THE
OVER-PARAMETERIZED STRUCTURES ON THE WORKING
CORRELATION SELECTION

Table D.1 Frequencies of the working correlation structure identified by the six criteria with considering penalization for the over-penalized structures Correlated Gaussian response; $C(\alpha)$: AR-1(0.3), $m=4$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	1	6	35	207	751	15	89	377	102	417
	WR	1	8	35	214	742	9	90	379	109	413
	RMR	6	20	56	168	750	38	71	320	83	488
	CIC	50	64	107	193	586	12	94	383	94	417
	QIC	50	64	107	193	586	131	107	277	102	383
	RJ	15	246	344	166	229	44	423	517	2	14
30	PT	0	0	20	132	848	3	87	437	59	414
	WR	0	0	15	148	837	2	93	441	66	398
	RMR	2	18	45	90	845	24	84	376	59	457
	CIC	24	58	115	177	626	3	87	440	61	409
	QIC	24	58	115	177	626	83	105	373	100	339
	RJ	0	301	321	164	214	12	452	535	0	1
50	PT	0	0	12	122	866	0	92	635	26	247
	WR	0	1	12	132	855	0	91	651	24	234
	RMR	0	13	41	92	854	10	97	524	23	346
	CIC	9	53	118	167	653	0	96	634	18	252
	QIC	9	53	118	167	653	53	107	524	46	270
	RJ	0	255	348	149	248	0	466	533	0	1
100	PT	0	0	6	90	904	0	44	868	5	83
	WR	0	0	7	106	887	0	42	883	5	70
	RMR	0	8	28	78	886	2	70	713	4	211
	CIC	9	37	128	177	649	0	49	858	5	88
	QIC	9	37	128	177	649	19	101	718	17	145
	RJ	0	295	355	143	207	0	472	528	0	0
200	PT	0	0	5	63	932	0	4	988	0	8
	WR	0	0	6	71	923	0	4	989	0	7
	RMR	0	1	35	59	905	0	38	906	0	56
	CIC	0	21	123	176	680	0	9	983	0	8
	QIC	0	21	123	176	680	4	51	898	3	44
	RJ	0	332	358	142	168	0	490	510	0	0

Table D.2 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: AR-1(0.6), $m=4$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	1	6	35	207	751	0	103	796	25	76
	WR	1	8	35	214	742	0	100	801	34	65
	RMR	6	20	56	168	750	12	133	712	28	115
	CIC	50	64	107	193	586	0	113	798	23	66
	QIC	50	64	107	193	586	135	124	585	35	121
	RJ	15	246	344	166	229	2	477	485	0	36
30	PT	0	0	20	132	848	0	47	928	2	23
	WR	0	0	15	148	837	0	44	934	6	16
	RMR	2	18	45	90	845	1	84	853	8	54
	CIC	24	58	115	177	626	0	56	922	4	18
	QIC	24	58	115	177	626	54	128	749	20	49
	RJ	0	301	321	164	214	0	469	511	0	20
50	PT	0	0	12	122	866	0	21	975	0	4
	WR	0	1	12	132	855	0	20	976	0	4
	RMR	0	13	41	92	854	0	41	941	1	17
	CIC	9	53	118	167	653	0	31	966	0	3
	QIC	9	53	118	167	653	15	82	878	1	24
	RJ	0	255	348	149	248	0	497	501	0	2
100	PT	0	0	6	90	904	0	0	1000	0	0
	WR	0	0	7	106	887	0	1	999	0	0
	RMR	0	8	28	78	886	0	8	992	0	0
	CIC	9	37	128	177	649	0	1	999	0	0
	QIC	9	37	128	177	649	0	28	971	0	1
	RJ	0	295	355	143	207	0	494	506	0	0
200	PT	0	0	5	63	932	0	1	999	0	0
	WR	0	0	6	71	923	0	1	999	0	0
	RMR	0	1	35	59	905	0	1	999	0	0
	CIC	0	21	123	176	680	0	1	999	0	0
	QIC	0	21	123	176	680	0	6	994	0	0
	RJ	0	332	358	142	168	0	491	509	0	0

Table D.3 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: AR-1(0.3), $m=6$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	1	26	233	740	12	25	256	109	598
	WR	0	1	24	237	738	4	18	245	152	581
	RMR	3	10	48	192	747	29	38	228	80	625
	CIC	18	45	94	195	648	6	15	243	147	589
	QIC	18	45	94	195	648	121	53	230	96	500
	RJ	1	251	378	206	164	14	436	531	1	18
30	PT	0	0	9	91	900	1	13	315	63	608
	WR	0	0	7	99	894	0	12	292	114	582
	RMR	0	2	22	58	918	5	17	224	56	698
	CIC	6	19	69	138	768	0	14	284	99	603
	QIC	6	19	69	138	768	54	52	303	82	509
	RJ	0	250	377	183	190	3	464	528	4	1
50	PT	0	0	5	63	932	0	12	448	14	526
	WR	0	0	5	74	921	0	12	468	36	484
	RMR	0	0	23	38	939	0	14	306	12	668
	CIC	7	24	60	132	777	0	15	430	31	524
	QIC	7	24	60	132	777	22	66	423	34	455
	RJ	0	271	344	175	210	0	490	510	0	0
100	PT	0	0	1	18	981	0	3	794	1	202
	WR	0	0	1	19	980	0	3	826	1	170
	RMR	0	1	2	16	981	0	12	566	3	419
	CIC	2	9	62	103	824	0	4	783	1	212
	QIC	2	9	62	103	824	5	42	658	16	279
	RJ	0	317	341	160	182	0	477	523	0	0
200	PT	0	0	0	14	986	0	0	983	0	17
	WR	0	0	0	15	985	0	0	991	0	9
	RMR	0	0	3	19	978	0	4	848	0	148
	CIC	0	3	58	114	825	0	0	977	0	23
	QIC	0	3	58	114	825	1	13	907	1	78
	RJ	0	354	346	152	148	0	509	491	0	0

Table D.4 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: AR-1(0.6), $m=6$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	2	110	544	344	0	17	839	50	94
	WR	0	2	111	543	344	0	14	779	140	67
	RMR	1	4	128	509	358	4	41	768	45	142
	CIC	4	31	186	429	350	0	22	790	114	74
	QIC	4	31	186	429	350	147	57	616	28	152
	RJ	0	302	349	183	166	0	428	467	7	98
30	PT	0	0	68	534	398	0	8	945	11	36
	WR	0	0	68	548	384	0	5	908	69	18
	RMR	0	0	68	493	439	1	19	895	12	73
	CIC	0	13	197	404	386	0	7	917	60	16
	QIC	0	13	197	404	386	130	48	729	10	83
	RJ	0	346	385	195	74	0	446	503	6	45
50	PT	0	0	50	523	427	0	0	998	0	2
	WR	0	0	66	519	415	0	0	979	21	0
	RMR	0	0	45	470	485	0	3	979	1	17
	CIC	0	3	179	412	406	0	0	986	14	0
	QIC	0	3	179	412	406	53	30	893	1	23
	RJ	0	363	365	205	67	0	504	478	6	12
100	PT	0	0	42	506	452	0	0	1000	0	0
	WR	0	0	54	523	423	0	0	1000	0	0
	RMR	0	0	47	413	540	0	0	1000	0	0
	CIC	0	0	191	366	443	0	0	1000	0	0
	QIC	0	0	191	366	443	8	1	990	0	1
	RJ	0	374	367	206	53	0	480	518	2	0
200	PT	0	0	31	483	486	0	0	1000	0	0
	WR	0	0	36	509	455	0	0	1000	0	0
	RMR	0	0	34	347	619	0	0	1000	0	0
	CIC	0	0	169	379	452	0	0	1000	0	0
	QIC	0	0	169	379	452	0	0	1000	0	0
	RJ	0	387	373	198	42	0	501	499	0	0

Table D.5 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: EX(0.3), $m=4$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	43	5	169	783	37	326	141	78	418
	WR	0	51	6	182	761	29	332	141	89	409
	RMR	4	32	25	126	813	184	130	81	35	570
	CIC	42	113	64	169	612	47	316	140	70	427
	QIC	42	113	64	169	612	323	111	197	80	289
	RJ	3	388	56	265	288	33	868	88	4	7
30	PT	0	20	1	107	872	11	450	110	42	387
	WR	0	28	1	122	849	7	466	115	45	367
	RMR	0	8	6	48	938	247	146	76	29	502
	CIC	16	135	45	151	653	18	439	109	34	400
	QIC	16	135	45	151	653	322	105	240	49	284
	RJ	0	401	12	256	331	5	950	43	0	2
50	PT	0	5	0	101	894	1	640	87	10	262
	WR	0	14	0	110	876	0	652	92	10	246
	RMR	0	6	5	46	943	293	181	67	17	442
	CIC	18	122	42	168	650	3	604	85	10	298
	QIC	18	122	42	168	650	370	109	272	35	214
	RJ	0	418	7	256	319	0	985	15	0	0
100	PT	0	6	0	44	950	0	887	29	1	83
	WR	0	7	0	62	931	0	895	31	2	72
	RMR	0	1	2	26	971	499	184	44	7	266
	CIC	1	111	27	188	673	0	863	30	0	107
	QIC	1	111	27	188	673	397	108	315	10	170
	RJ	0	388	0	294	318	0	1000	0	0	0
200	PT	0	3	0	30	967	0	990	3	0	7
	WR	0	7	0	46	947	0	992	3	0	5
	RMR	0	3	0	16	981	655	219	14	2	110
	CIC	0	105	6	173	716	0	986	1	0	13
	QIC	0	105	6	173	716	440	121	345	5	89
	RJ	0	382	0	268	350	0	1000	0	0	0

Table D.6 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: EX(0.6), $m=4$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	269	5	378	348	100	495	264	24	117
	WR	0	279	8	373	340	100	477	273	46	104
	RMR	0	232	12	344	412	168	328	75	31	398
	CIC	2	247	104	255	392	116	489	248	29	118
	QIC	2	247	104	255	392	388	208	268	49	87
	RJ	0	536	7	263	194	102	783	115	0	0
30	PT	0	243	1	333	423	72	608	219	8	93
	WR	0	254	1	345	400	72	600	226	25	77
	RMR	0	163	0	293	544	172	469	62	10	287
	CIC	0	251	31	285	433	76	619	198	11	96
	QIC	0	251	31	285	433	393	185	276	47	99
	RJ	0	578	0	280	142	72	846	81	1	0
50	PT	0	223	0	341	436	25	767	177	2	29
	WR	0	235	0	355	410	25	754	181	14	26
	RMR	0	150	0	290	560	221	580	42	3	154
	CIC	1	247	22	280	450	25	773	166	4	32
	QIC	1	247	22	280	450	395	173	304	28	100
	RJ	0	578	0	291	131	25	915	60	0	0
100	PT	0	181	0	331	488	2	912	82	0	4
	WR	0	206	0	336	458	2	912	82	1	3
	RMR	0	93	0	246	661	207	750	7	0	36
	CIC	0	250	4	280	466	2	919	75	0	4
	QIC	0	250	4	280	466	462	120	335	9	74
	RJ	0	560	0	284	156	2	962	36	0	0
200	PT	0	189	0	304	507	0	975	25	0	0
	WR	0	209	0	311	480	0	974	26	0	0
	RMR	0	81	0	202	717	224	774	2	0	0
	CIC	0	243	0	255	502	0	980	20	0	0
	QIC	0	243	0	255	502	449	107	391	3	50
	RJ	0	575	0	266	159	0	986	14	0	0

Table D.7 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: EX(0.3), $m=6$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	39	1	224	736	7	229	70	88	606
	WR	0	45	1	228	726	6	216	71	113	594
	RMR	0	31	8	187	774	50	96	22	29	803
	CIC	12	95	35	189	669	16	214	67	99	604
	QIC	12	95	35	189	669	275	105	185	98	337
	RJ	0	446	10	312	232	7	950	38	3	2
30	PT	0	13	0	74	913	2	235	47	41	675
	WR	0	16	0	92	892	2	223	46	87	642
	RMR	0	3	0	35	962	53	123	18	18	788
	CIC	1	79	21	129	770	3	216	43	50	688
	QIC	1	79	21	129	770	268	85	200	58	389
	RJ	0	460	1	297	242	1	980	17	1	1
50	PT	0	8	0	66	926	0	425	14	8	553
	WR	0	10	0	80	910	0	459	15	25	501
	RMR	0	3	0	25	972	41	218	22	13	706
	CIC	3	88	14	124	771	0	373	16	15	596
	QIC	3	88	14	124	771	288	100	246	35	331
	RJ	0	449	0	282	269	0	998	2	0	0
100	PT	0	2	0	8	990	0	827	3	0	170
	WR	0	3	0	15	982	0	866	3	1	130
	RMR	0	0	0	1	999	21	439	13	6	521
	CIC	0	56	4	82	858	0	728	2	0	270
	QIC	0	56	4	82	858	336	105	282	17	260
	RJ	0	403	0	285	312	0	1000	0	0	0
200	PT	0	0	0	5	995	0	989	0	0	11
	WR	0	0	0	10	990	0	992	0	0	8
	RMR	0	0	0	0	1000	15	764	4	2	215
	CIC	0	58	0	101	841	0	965	0	0	35
	QIC	0	58	0	101	841	430	126	293	7	144
	RJ	0	393	0	275	332	0	1000	0	0	0

Table D.8 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: EX(0.6), $m=6$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	247	1	411	341	53	355	256	100	236
	WR	0	255	1	410	334	53	350	267	107	223
	RMR	0	224	0	413	363	63	263	25	24	625
	CIC	4	268	38	335	355	63	384	225	112	216
	QIC	4	268	38	335	355	361	193	252	80	114
	RJ	0	598	0	267	135	51	842	99	8	0
30	PT	0	233	0	352	415	22	434	313	38	193
	WR	0	256	0	340	404	22	396	312	103	167
	RMR	0	167	0	353	480	40	409	16	10	525
	CIC	0	263	18	303	416	25	451	277	62	185
	QIC	0	263	18	303	416	371	215	250	58	106
	RJ	0	654	0	260	86	22	855	123	0	0
50	PT	0	244	0	316	440	7	586	307	8	92
	WR	0	253	0	319	428	7	571	311	54	57
	RMR	0	165	0	310	525	18	740	2	4	236
	CIC	0	228	1	314	457	8	606	278	30	78
	QIC	0	228	1	314	457	349	180	289	66	116
	RJ	0	645	0	270	85	7	872	121	0	0
100	PT	0	191	0	242	567	0	797	202	0	1
	WR	0	211	0	252	537	0	789	203	8	0
	RMR	0	107	0	184	709	6	977	0	0	17
	CIC	0	201	0	237	562	0	815	179	1	5
	QIC	0	201	0	237	562	371	156	333	47	93
	RJ	0	647	0	255	98	0	922	78	0	0
200	PT	0	160	0	194	646	0	913	87	0	0
	WR	0	178	0	211	611	0	912	88	0	0
	RMR	0	69	0	117	814	0	1000	0	0	0
	CIC	0	209	0	241	550	0	915	85	0	0
	QIC	0	209	0	241	550	410	128	383	24	55
	RJ	0	672	0	230	98	0	967	33	0	0

Table D.9 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: TOEP, $m=4$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	99	1	176	724	23	427	17	327	206
	WR	0	99	1	192	708	22	351	16	433	178
	RMR	2	118	16	149	715	55	388	36	189	332
	CIC	20	128	18	250	584	30	414	23	329	204
	QIC	20	128	18	250	584	154	314	29	270	233
	RJ	5	444	7	302	242	51	890	10	34	15
30	PT	0	76	0	109	815	13	457	2	379	149
	WR	0	73	0	117	810	13	361	2	498	126
	RMR	0	97	3	81	819	32	470	11	225	262
	CIC	6	101	6	227	660	18	453	4	389	136
	QIC	6	101	6	227	660	141	365	19	283	192
	RJ	0	483	0	242	275	42	932	0	14	12
50	PT	0	48	0	61	891	2	508	0	460	30
	WR	0	45	0	80	875	1	358	0	624	17
	RMR	0	64	0	49	887	11	573	2	285	129
	CIC	2	79	2	231	686	9	499	0	462	30
	QIC	2	79	2	231	686	95	500	10	312	83
	RJ	0	496	0	241	263	35	961	0	2	2
100	PT	0	12	0	24	964	1	510	0	489	0
	WR	0	11	0	31	958	1	284	0	715	0
	RMR	0	21	0	44	935	2	661	0	312	25
	CIC	0	26	0	215	759	2	466	0	532	0
	QIC	0	26	0	215	759	46	595	1	349	9
	RJ	0	474	0	246	280	6	992	0	1	1
200	PT	0	0	0	11	989	0	544	0	456	0
	WR	0	0	0	20	980	0	157	0	843	0
	RMR	0	2	0	47	951	0	768	0	232	0
	CIC	0	3	0	227	770	0	410	0	590	0
	QIC	0	3	0	227	770	18	713	0	269	0
	RJ	0	499	0	235	266	0	1000	0	0	0

Table D.10 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: TOEP, $m=6$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	24	0	591	385	4	184	0	672	140
	WR	0	23	0	594	383	0	182	0	700	118
	RMR	0	37	0	586	377	20	349	5	433	193
	CIC	1	73	6	525	395	2	301	1	593	103
	QIC	1	73	6	525	395	212	298	13	317	160
	RJ	1	516	0	296	187	3	887	0	17	93
30	PT	0	4	0	598	398	2	127	0	820	51
	WR	0	4	0	608	388	1	123	0	833	43
	RMR	0	5	0	569	426	6	341	1	536	116
	CIC	0	14	0	587	399	0	217	0	746	37
	QIC	0	14	0	587	399	177	360	8	360	95
	RJ	0	541	0	346	113	3	909	0	8	80
50	PT	0	3	0	585	412	0	65	0	917	18
	WR	0	3	0	594	403	0	69	0	919	12
	RMR	0	9	0	546	445	1	303	0	650	46
	CIC	0	13	0	546	441	0	113	0	874	13
	QIC	0	13	0	546	441	112	433	1	419	35
	RJ	0	569	0	321	110	0	961	0	5	34
100	PT	0	0	0	562	438	0	5	0	995	0
	WR	0	0	0	574	426	0	4	0	996	0
	RMR	0	1	0	499	500	0	238	0	762	0
	CIC	0	0	0	545	455	0	20	0	980	0
	QIC	0	0	0	545	455	68	605	0	323	4
	RJ	0	558	0	358	84	0	997	0	1	2
200	PT	0	0	0	587	413	0	0	0	1000	0
	WR	0	0	0	607	393	0	0	0	1000	0
	RMR	0	0	0	489	511	0	190	0	810	0
	CIC	0	0	0	538	462	0	1	0	999	0
	QIC	0	0	0	538	462	24	764	0	212	0
	RJ	0	546	0	378	76	0	1000	0	0	0

Table D.11 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: UN, $m=4$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	99	1	176	724	1	76	515	7	401
	WR	0	99	1	192	708	0	82	529	13	376
	RMR	2	118	16	149	715	11	96	510	22	361
	CIC	20	128	18	250	584	0	96	575	11	318
	QIC	20	128	18	250	584	115	124	413	23	325
	RJ	5	444	7	302	242	1	505	418	0	76
30	PT	0	76	0	109	815	0	34	503	3	460
	WR	0	73	0	117	810	0	37	537	4	422
	RMR	0	97	3	81	819	3	43	511	8	435
	CIC	6	101	6	227	660	0	44	606	5	345
	QIC	6	101	6	227	660	80	76	495	9	340
	RJ	0	483	0	242	275	2	498	442	0	58
50	PT	0	0	57	98	845	0	16	486	0	498
	WR	0	0	53	90	857	0	18	530	0	452
	RMR	0	0	55	137	808	0	30	513	6	451
	CIC	0	11	83	127	779	0	25	632	0	343
	QIC	0	11	83	127	779	60	57	538	4	341
	RJ	0	292	307	104	297	0	501	460	0	39
100	PT	0	0	23	41	936	0	12	454	1	533
	WR	0	0	22	34	944	0	12	511	1	476
	RMR	0	0	25	66	909	0	15	506	0	479
	CIC	0	0	38	63	899	0	17	626	0	357
	QIC	0	0	38	63	899	41	45	545	1	368
	RJ	0	302	261	97	340	0	494	480	0	26
200	PT	0	0	3	13	984	0	0	371	0	629
	WR	0	0	2	8	990	0	0	451	0	549
	RMR	0	0	3	16	981	0	0	416	0	584
	CIC	0	0	14	8	978	0	1	627	0	372
	QIC	0	0	14	8	978	13	8	575	0	404
	RJ	0	306	267	87	340	0	542	454	0	4

Table D.12 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Gaussian response; $C(\alpha)$: UN, $m=6$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	24	0	591	385	0	103	340	55	502
	WR	0	23	0	594	383	0	79	318	121	482
	RMR	0	37	0	586	377	10	174	290	44	482
	CIC	1	73	6	525	395	0	125	312	106	457
	QIC	1	73	6	525	395	176	105	244	33	442
	RJ	1	516	0	296	187	0	716	187	11	86
30	PT	0	4	0	598	398	0	58	323	28	591
	WR	0	4	0	608	388	0	41	291	89	579
	RMR	0	5	0	569	426	3	116	295	27	559
	CIC	0	14	0	587	399	0	63	324	96	517
	QIC	0	14	0	587	399	140	94	250	32	484
	RJ	0	541	0	346	113	0	828	109	6	57
50	PT	0	0	8	167	825	0	68	266	14	652
	WR	0	0	6	160	834	0	39	256	74	631
	RMR	0	2	14	180	804	1	114	249	13	623
	CIC	1	17	38	150	794	0	74	282	79	565
	QIC	1	17	38	150	794	117	91	242	16	534
	RJ	0	401	47	249	303	0	853	101	0	46
100	PT	0	0	1	86	913	0	29	243	8	720
	WR	0	0	1	79	920	0	15	231	55	699
	RMR	0	0	3	102	895	1	69	237	3	690
	CIC	0	1	8	97	894	0	42	282	49	627
	QIC	0	1	8	97	894	89	80	239	7	585
	RJ	0	379	13	243	365	0	903	66	0	31
200	PT	0	0	0	28	972	0	7	145	0	848
	WR	0	0	0	27	973	0	1	143	19	837
	RMR	0	0	0	49	951	0	39	140	0	821
	CIC	0	0	7	40	953	0	16	174	22	788
	QIC	0	0	7	40	953	44	43	187	1	725
	RJ	0	362	0	251	387	0	969	9	0	22

Table D.13 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: AR-1(0.3), $m=4$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	2	17	73	249	659	31	87	318	135	429
	WR	2	21	66	261	650	13	96	345	136	410
	RMR	53	60	108	165	614	185	37	199	116	463
	CIC	63	69	112	202	554	28	79	319	130	444
	QIC	164	125	94	163	454	348	57	121	113	361
	RJ	9	227	387	162	215	39	398	547	0	16
30	PT	0	28	59	213	700	10	112	405	67	406
	WR	0	27	56	224	693	4	112	424	75	385
	RMR	51	60	84	122	683	170	49	223	71	487
	CIC	66	54	87	187	606	14	108	386	80	412
	QIC	200	140	114	127	419	370	70	171	76	313
	RJ	2	247	345	168	238	13	459	527	0	1
50	PT	2	9	46	199	744	4	88	551	38	319
	WR	2	9	48	218	723	4	97	572	36	291
	RMR	48	63	80	106	703	126	61	310	38	465
	CIC	43	62	107	162	626	4	85	518	36	357
	QIC	235	133	142	119	371	408	81	217	53	241
	RJ	0	271	323	148	258	1	475	523	1	0
100	PT	0	1	40	187	772	0	57	802	3	138
	WR	0	2	40	205	753	0	54	816	3	127
	RMR	28	53	100	102	717	98	68	437	10	387
	CIC	14	46	120	166	654	0	64	752	6	178
	QIC	262	182	116	115	325	439	113	229	26	193
	RJ	0	286	367	123	224	0	453	547	0	0
200	PT	0	2	42	186	770	0	18	953	0	29
	WR	0	1	40	205	754	0	15	962	0	23
	RMR	15	46	101	99	739	48	76	610	4	262
	CIC	6	33	144	187	630	0	27	922	0	51
	QIC	265	203	135	110	287	434	159	264	12	131
	RJ	0	334	358	139	169	0	488	512	0	0

Table D.14 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: AR-1(0.6), $m=4$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	29	161	395	415	7	128	727	26	112
	WR	0	26	165	400	409	0	124	743	38	95
	RMR	52	77	169	248	454	205	90	377	70	258
	CIC	33	68	196	267	436	11	146	658	40	145
	QIC	216	141	125	169	349	374	115	217	67	227
	RJ	0	325	315	185	175	1	474	481	1	43
30	PT	0	18	166	404	412	3	101	824	14	58
	WR	0	14	177	411	398	0	100	836	19	45
	RMR	28	81	143	276	472	154	100	479	37	230
	CIC	7	74	159	314	446	3	118	765	20	94
	QIC	247	169	106	120	358	396	125	214	43	222
	RJ	0	358	336	177	129	0	508	480	0	12
50	PT	0	9	157	400	434	0	66	921	1	12
	WR	0	8	179	408	405	0	58	934	4	4
	RMR	17	74	147	229	533	86	129	573	14	198
	CIC	3	52	185	284	476	1	92	871	6	30
	QIC	241	175	120	114	350	365	140	251	32	212
	RJ	0	377	330	188	105	0	524	475	0	1
100	PT	0	0	149	415	436	0	7	993	0	0
	WR	0	0	162	428	410	0	7	993	0	0
	RMR	3	32	138	243	584	24	84	816	9	67
	CIC	0	16	200	302	482	0	32	968	0	0
	QIC	269	206	125	93	307	387	181	278	16	138
	RJ	0	388	316	205	91	0	525	475	0	0
200	PT	0	0	152	426	422	0	0	1000	0	0
	WR	0	0	156	450	394	0	0	1000	0	0
	RMR	0	11	130	224	635	5	35	942	0	18
	CIC	0	0	187	291	522	0	3	997	0	0
	QIC	317	194	126	89	274	440	184	290	7	79
	RJ	0	394	305	187	114	0	564	436	0	0

Table D.15 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: AR-1(0.3), $m=6$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	8	52	262	678	17	36	268	120	559
	WR	0	7	51	268	674	3	38	256	166	537
	RMR	15	42	86	177	680	122	22	163	101	592
	CIC	26	32	94	191	657	11	33	253	137	566
	QIC	158	95	119	179	449	365	41	110	94	390
	RJ	2	263	403	165	167	14	406	556	3	21
30	PT	0	3	21	200	776	9	20	269	85	617
	WR	0	3	19	212	766	0	22	271	122	585
	RMR	20	23	63	105	789	69	25	155	66	685
	CIC	20	22	70	147	741	7	17	242	94	640
	QIC	158	114	121	147	460	337	59	130	81	393
	RJ	1	228	390	188	193	1	428	564	2	5
50	PT	0	1	19	113	867	1	15	392	30	562
	WR	0	0	17	126	857	0	16	438	37	509
	RMR	13	27	48	65	847	47	33	176	28	716
	CIC	12	19	69	124	776	0	17	327	32	624
	QIC	187	138	110	125	440	361	68	163	58	350
	RJ	0	261	364	160	215	1	458	540	1	0
100	PT	0	0	10	79	911	0	6	628	5	361
	WR	0	0	9	88	903	0	6	684	7	303
	RMR	4	10	23	60	903	16	22	274	16	672
	CIC	5	19	68	105	803	0	6	554	8	432
	QIC	252	172	100	86	390	425	98	187	26	264
	RJ	0	300	364	161	175	0	474	526	0	0
200	PT	0	0	10	61	929	0	0	923	0	77
	WR	0	0	9	72	919	0	0	946	0	54
	RMR	0	6	22	43	929	7	23	420	3	547
	CIC	1	9	51	103	836	0	0	854	0	146
	QIC	263	178	112	94	353	431	116	214	19	220
	RJ	0	322	360	131	187	0	466	534	0	0

Table D.16 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: AR-1(0.6), $m=6$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	3	130	495	372	0	28	741	64	167
	WR	0	3	135	500	362	0	27	697	143	133
	RMR	17	55	141	377	410	117	75	424	82	302
	CIC	14	38	175	345	428	5	66	599	122	208
	QIC	199	146	121	156	378	347	102	191	58	302
	RJ	1	300	333	198	168	0	445	453	6	96
30	PT	0	1	115	481	403	0	21	880	16	83
	WR	0	0	131	494	375	0	13	835	89	63
	RMR	15	34	133	336	482	74	59	528	41	298
	CIC	6	26	172	365	431	1	42	732	75	150
	QIC	218	146	112	141	383	350	122	190	42	296
	RJ	0	349	320	208	123	0	497	439	7	57
50	PT	0	0	98	487	415	0	8	955	2	35
	WR	0	0	119	501	380	0	2	960	23	15
	RMR	4	35	82	340	539	45	57	627	24	247
	CIC	1	20	153	344	482	0	23	875	25	77
	QIC	254	139	106	109	392	366	120	212	28	274
	RJ	0	385	303	218	94	0	536	452	1	11
100	PT	0	0	98	470	432	0	0	1000	0	0
	WR	0	0	114	494	392	0	0	999	1	0
	RMR	0	6	92	269	633	10	20	827	8	135
	CIC	0	3	143	353	501	0	5	986	3	6
	QIC	268	166	85	90	391	389	126	227	25	233
	RJ	0	402	317	215	66	0	533	464	0	3
200	PT	0	0	79	484	437	0	0	1000	0	0
	WR	0	0	86	526	388	0	0	1000	0	0
	RMR	0	0	70	225	705	0	4	982	1	13
	CIC	0	0	156	300	544	0	0	1000	0	0
	QIC	288	178	96	86	352	387	162	270	7	174
	RJ	0	407	307	227	59	0	573	427	0	0

Table D.17 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: EX(0.3), $m=4$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	94	40	207	659	32	370	119	75	404
	WR	0	97	38	223	642	13	400	129	78	380
	RMR	62	73	90	118	657	207	118	127	56	492
	CIC	56	96	84	163	601	36	324	133	71	436
	QIC	156	89	194	128	433	349	77	173	76	325
	RJ	6	389	63	254	288	12	890	83	3	12
30	PT	1	96	14	190	699	12	493	96	46	353
	WR	1	98	16	205	680	6	523	102	50	319
	RMR	62	60	59	99	720	201	142	113	39	505
	CIC	58	97	78	140	627	15	437	105	39	404
	QIC	179	85	236	91	409	358	84	233	45	280
	RJ	0	390	24	263	323	3	959	32	0	6
50	PT	0	63	7	191	739	3	646	90	24	237
	WR	0	73	8	217	702	1	688	86	22	203
	RMR	53	56	55	87	749	211	156	137	27	469
	CIC	35	107	58	168	632	10	562	94	22	312
	QIC	209	72	253	83	383	374	97	250	27	252
	RJ	0	360	15	288	337	0	988	12	0	0
100	PT	0	64	2	170	764	0	877	45	2	76
	WR	0	74	1	188	737	0	903	39	1	57
	RMR	53	71	62	85	729	316	224	125	8	327
	CIC	19	131	30	162	658	2	801	69	2	126
	QIC	227	63	280	88	342	383	106	300	18	193
	RJ	0	381	2	255	362	0	1000	0	0	0
200	PT	0	49	0	167	784	0	988	9	0	3
	WR	0	56	0	183	761	0	987	11	0	2
	RMR	15	69	31	97	788	369	323	88	3	217
	CIC	1	118	17	179	685	0	961	17	0	22
	QIC	292	67	320	75	246	443	125	318	8	106
	RJ	0	361	0	269	370	0	1000	0	0	0

Table D.18 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: EX(0.6), $m=4$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	226	37	311	426	5	749	128	13	105
	WR	0	231	34	328	407	2	770	117	31	80
	RMR	57	179	83	219	462	259	321	152	36	232
	CIC	37	198	113	207	445	20	665	160	16	139
	QIC	207	71	285	103	334	347	95	308	40	210
	RJ	0	483	25	288	204	1	956	15	2	26
30	PT	0	263	22	318	397	0	862	89	4	45
	WR	0	285	25	321	369	0	870	84	10	36
	RMR	47	181	51	223	498	221	447	121	19	192
	CIC	23	229	95	219	434	9	804	98	7	82
	QIC	267	63	280	75	315	404	99	299	22	176
	RJ	0	543	4	273	180	2	989	1	0	8
50	PT	0	262	7	327	404	0	949	42	0	9
	WR	0	285	8	342	365	0	947	47	2	4
	RMR	34	155	27	225	559	176	593	83	7	141
	CIC	3	214	46	239	498	2	899	65	1	33
	QIC	290	48	300	53	309	410	96	333	10	151
	RJ	0	517	0	282	201	0	1000	0	0	0
100	PT	0	266	0	290	444	0	995	5	0	0
	WR	0	279	0	315	406	0	995	5	0	0
	RMR	3	154	4	179	660	148	774	28	0	50
	CIC	0	223	13	218	546	0	987	12	0	1
	QIC	293	51	358	52	246	416	112	374	9	89
	RJ	0	530	0	289	181	0	1000	0	0	0
200	PT	0	255	0	302	443	0	1000	0	0	0
	WR	0	278	0	315	407	0	1000	0	0	0
	RMR	0	126	0	185	689	76	913	3	0	8
	CIC	0	213	3	225	559	0	999	1	0	0
	QIC	359	55	341	50	195	474	100	361	1	64
	RJ	0	528	0	281	191	0	1000	0	0	0

Table D.19 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: EX(0.3), $m=6$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	62	11	274	653	9	334	51	94	512
	WR	0	71	11	282	636	3	321	47	143	486
	RMR	24	67	49	153	707	133	125	79	55	608
	CIC	34	77	58	189	642	14	257	70	107	552
	QIC	162	82	154	134	468	326	83	149	59	383
	RJ	3	424	13	321	239	3	962	7	4	24
30	PT	0	61	2	228	709	2	465	28	43	462
	WR	0	68	4	248	680	0	475	31	75	419
	RMR	18	53	35	110	784	87	160	64	48	641
	CIC	27	89	39	152	693	5	340	49	55	551
	QIC	183	58	153	87	519	302	70	170	36	422
	RJ	0	411	3	317	269	1	992	2	0	5
50	PT	0	48	0	162	790	0	590	11	12	387
	WR	0	57	1	185	757	0	630	11	24	335
	RMR	12	34	17	74	863	44	226	56	17	657
	CIC	18	68	14	144	756	2	437	20	24	517
	QIC	170	64	214	87	465	298	102	222	20	358
	RJ	0	421	0	311	268	0	1000	0	0	0
100	PT	0	36	0	108	856	0	860	4	1	135
	WR	0	43	0	131	826	0	899	2	1	98
	RMR	6	20	0	33	941	21	385	24	3	567
	CIC	3	62	5	105	825	0	719	3	0	278
	QIC	231	41	268	58	402	322	98	305	9	266
	RJ	0	362	0	276	362	0	1000	0	0	0
200	PT	0	19	0	59	922	0	989	0	0	11
	WR	0	27	0	81	892	0	996	0	0	4
	RMR	1	17	1	27	954	12	607	18	0	363
	CIC	1	52	0	83	864	0	927	0	0	73
	QIC	259	64	286	58	333	376	154	302	5	163
	RJ	0	375	0	254	371	0	1000	0	0	0

Table D.20 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: EX(0.6), $m=6$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	221	24	391	364	3	765	64	36	132
	WR	0	230	34	379	357	3	701	67	120	109
	RMR	28	167	26	360	419	119	543	66	28	244
	CIC	26	170	81	312	411	17	680	57	94	152
	QIC	226	68	267	107	332	356	117	283	28	216
	RJ	0	556	1	305	138	0	952	0	2	46
30	PT	0	241	17	379	363	0	896	40	6	58
	WR	0	267	18	379	336	0	851	44	68	37
	RMR	20	194	10	326	450	80	668	34	21	197
	CIC	10	232	59	293	406	6	811	34	54	95
	QIC	239	51	243	94	373	339	106	295	28	232
	RJ	0	586	0	315	99	0	984	0	0	16
50	PT	0	261	3	324	412	0	990	3	0	7
	WR	0	286	4	333	377	0	970	5	23	2
	RMR	9	157	0	291	543	38	831	5	5	121
	CIC	6	217	14	283	480	1	954	3	12	30
	QIC	231	32	301	78	358	325	83	336	26	230
	RJ	0	616	0	283	101	0	997	0	0	3
100	PT	0	239	0	294	467	0	1000	0	0	0
	WR	0	267	0	301	432	0	999	1	0	0
	RMR	2	144	0	198	656	6	970	0	1	23
	CIC	0	188	1	240	571	0	996	0	1	3
	QIC	283	36	313	43	325	373	101	347	5	174
	RJ	0	564	0	289	147	0	1000	0	0	0
200	PT	0	209	0	268	523	0	1000	0	0	0
	WR	0	246	0	289	465	0	1000	0	0	0
	RMR	0	99	0	144	757	0	1000	0	0	0
	CIC	0	177	0	203	620	0	1000	0	0	0
	QIC	321	37	378	42	222	400	103	398	1	98
	RJ	0	594	0	269	137	0	1000	0	0	0

Table D.21 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: TOEP, $m=4$

n	Criteri on	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	91	6	276	627	42	341	21	351	245
	WR	0	93	4	290	613	32	313	17	423	215
	RMR	55	120	49	175	601	274	140	38	149	399
	CIC	58	115	33	218	576	53	339	32	301	275
	QIC	128	69	229	167	407	332	70	176	154	268
	RJ	4	446	5	252	293	32	911	7	32	18
30	PT	0	71	4	250	675	20	408	7	366	199
	WR	0	74	2	263	661	13	365	3	454	165
	RMR	48	111	40	164	637	271	191	39	157	342
	CIC	30	99	16	243	612	28	425	13	301	233
	QIC	173	73	236	161	357	357	88	184	166	205
	RJ	1	483	0	285	231	24	955	1	15	5
50	PT	0	44	0	225	731	8	446	3	467	76
	WR	0	42	0	241	717	6	385	2	553	54
	RMR	34	96	28	166	676	308	206	26	191	269
	CIC	23	73	2	242	660	17	489	4	378	112
	QIC	171	69	263	153	344	376	95	187	166	176
	RJ	0	493	0	240	267	15	978	0	6	1
100	PT	0	9	0	176	815	0	497	0	493	10
	WR	0	9	0	196	795	0	389	0	604	7
	RMR	11	41	3	151	794	523	210	5	149	113
	CIC	2	26	0	223	749	0	634	0	345	21
	QIC	194	59	310	145	292	397	97	245	143	118
	RJ	0	459	0	244	297	4	996	0	0	0
200	PT	0	0	0	198	802	0	529	0	471	0
	WR	0	1	0	209	790	0	395	0	605	0
	RMR	2	13	0	182	803	782	119	0	74	25
	CIC	0	2	0	223	775	0	761	0	237	2
	QIC	178	47	338	170	267	390	86	289	181	54
	RJ	0	489	0	249	262	0	1000	0	0	0

Table D.22 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: TOEP, $m=6$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	35	0	586	379	23	250	1	538	188
	WR	0	33	0	583	384	12	217	1	619	151
	RMR	33	86	30	429	422	243	176	50	200	331
	CIC	15	81	18	473	413	34	321	12	407	226
	QIC	180	87	159	230	344	364	78	144	141	273
	RJ	1	495	0	301	203	4	876	0	13	107
30	PT	0	15	0	568	417	4	197	0	678	121
	WR	0	12	0	583	405	2	134	0	779	85
	RMR	26	85	16	393	480	208	237	37	199	319
	CIC	13	60	12	448	467	12	289	3	535	161
	QIC	182	70	165	178	405	356	76	144	120	304
	RJ	0	500	0	329	171	1	924	0	4	71
50	PT	0	1	0	602	397	0	141	0	823	36
	WR	0	1	0	629	370	0	67	0	911	22
	RMR	5	40	20	425	510	165	251	30	261	293
	CIC	1	24	0	522	453	1	207	1	710	81
	QIC	185	73	187	143	412	355	62	172	106	305
	RJ	0	507	0	394	99	0	964	0	1	35
100	PT	0	0	0	581	419	0	38	0	960	2
	WR	0	0	0	619	381	0	10	0	989	1
	RMR	1	7	1	372	619	66	467	7	280	180
	CIC	0	3	0	484	513	0	60	0	933	7
	QIC	219	69	228	123	361	389	78	228	107	198
	RJ	0	522	0	374	104	0	995	0	1	4
200	PT	0	0	0	564	436	0	4	0	996	0
	WR	0	0	0	595	405	0	1	0	999	0
	RMR	0	1	0	288	711	10	696	1	242	51
	CIC	0	0	0	442	558	0	6	0	994	0
	QIC	247	59	251	102	341	392	78	269	89	172
	RJ	0	552	0	346	102	0	1000	0	0	0

Table D.23 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: UN, $m=4$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	25	109	184	682	4	85	456	17	438
	WR	0	23	108	178	691	2	86	458	22	432
	RMR	26	70	137	226	541	177	72	328	63	360
	CIC	21	53	113	212	601	8	108	482	28	374
	QIC	223	150	91	188	348	409	87	167	91	246
	RJ	0	279	295	149	277	3	481	441	0	75
30	PT	0	6	101	155	738	1	40	523	8	428
	WR	0	9	102	135	754	0	40	529	18	413
	RMR	16	38	120	230	596	119	52	414	50	365
	CIC	4	29	116	206	645	3	47	564	23	363
	QIC	227	172	93	170	338	404	100	176	65	255
	RJ	0	288	315	153	244	0	461	494	0	45
50	PT	0	2	68	113	817	0	18	507	0	475
	WR	0	3	68	101	828	0	19	521	2	458
	RMR	5	21	108	195	671	82	39	454	19	406
	CIC	1	17	110	130	742	0	17	590	5	388
	QIC	248	159	88	169	336	402	114	181	66	237
	RJ	0	303	302	119	276	0	507	469	0	24
100	PT	0	0	24	45	931	0	3	416	0	581
	WR	0	0	23	38	939	0	3	435	0	562
	RMR	1	4	40	134	821	26	15	511	5	443
	CIC	0	2	33	80	885	0	4	582	0	414
	QIC	245	174	83	154	344	394	125	201	65	215
	RJ	0	281	317	127	275	0	506	490	0	4
200	PT	0	0	11	13	976	0	0	316	0	684
	WR	0	0	11	10	979	0	0	349	0	651
	RMR	0	1	11	50	938	2	2	495	0	501
	CIC	0	0	14	26	960	0	1	571	0	428
	QIC	267	183	89	150	311	410	137	205	51	197
	RJ	0	288	300	94	318	0	557	443	0	0

Table D.24 Frequencies of the working correlation structure identified by PT, WR, RMR, CIC, QIC and RJ with considering penalization for the over-penalized structures from 1000 independent replications:
Correlated Poisson response; $C(\alpha)$: UN, $m=6$

n	Criterion	Without penalization					With penalization				
		IN	EX	AR-1	TOEP	UN	IN	EX	AR-1	TOEP	UN
20	PT	0	17	66	275	642	5	108	299	48	540
	WR	0	14	61	274	651	1	94	289	85	531
	RMR	32	69	82	223	594	146	97	181	47	529
	CIC	19	54	105	224	598	13	115	269	65	538
	QIC	181	114	163	146	396	346	79	197	51	327
	RJ	0	346	145	244	265	0	711	198	9	82
30	PT	0	7	34	255	704	3	76	298	30	593
	WR	0	6	44	244	706	0	65	289	71	575
	RMR	32	71	63	176	658	157	74	161	28	580
	CIC	11	60	70	197	662	10	95	264	41	590
	QIC	224	92	173	115	396	386	79	178	26	331
	RJ	0	378	81	281	260	0	809	128	8	55
50	PT	0	1	21	169	809	0	49	245	8	698
	WR	0	1	19	172	808	0	41	233	43	683
	RMR	14	52	34	117	783	97	76	105	19	703
	CIC	6	35	52	129	778	0	70	215	20	695
	QIC	205	127	188	78	402	369	88	205	21	317
	RJ	0	441	47	241	271	0	909	53	0	38
100	PT	0	0	6	105	889	0	8	166	4	822
	WR	0	0	5	101	894	0	3	181	14	802
	RMR	5	22	20	87	866	58	44	77	5	816
	CIC	0	8	20	92	880	0	32	146	6	816
	QIC	227	130	204	57	382	388	109	204	11	288
	RJ	0	419	10	264	307	0	973	11	0	16
200	PT	0	0	0	47	953	0	3	76	2	919
	WR	0	0	0	45	955	0	0	86	3	911
	RMR	0	8	0	47	945	31	14	27	4	924
	CIC	0	0	3	52	945	0	13	64	6	917
	QIC	244	138	211	37	370	414	99	205	7	275
	RJ	0	439	1	228	332	0	998	1	0	1

REFERENCES

- Acion, L. (2011). Criteria for generalized linear model selection based on Kullback's symmetric divergence. Ph.D. Dissertation, University of Iowa.
- Albert, Paul S. and Lisa M. McShane (1995). A generalized estimating equations approach for spatially correlated binary data: applications to the analysis of neuroimaging data. *Biometrics* **51**:627-638.
- Barnett, A. G., Koper, N., Dobson, A. J., Schmiedel, F., and Manseau, M. (2010). Using information criteria to select the correct variance-covariance structure for longitudinal data in ecology. *Methods in Ecology and Evolution* **1**, 15-25.
- Bock, R. D. (1975). *Multivariate Statistical Methods in Behavioral Research*. New York: McGraw-Hill.
- Crowder, M. (1995). On the use of a working correlation matrix in using generalised linear models for repeated measures. *Biometrika* **82**(2), 407.
- Demattei, C. and Molinari, N. (2007). P-value calculations for multiple temporal cluster detection. *C.R.Acad.Sci.Paris, Ser. I.* **344**, 697-701.
- Fitzmaurice, G. M. (1995). A Caveat concerning independence estimating equations with multivariate binary data. *Biometrics* **51**:309-317.
- Fitzmaurice, G. M., Laird, N. M., and Ware, J. H. (2004). *Applied longitudinal analysis*. NJ: John Wiley & Sons.
- Gentle J. E. (2007). *Matrix Algebra*. New York: Wiley.
- Hardin, J. and Hilbe, J. (2003). *Generalized Estimating Equations*. Florida: Chapman and Hall.
- Heagerty, P. J. (1999). Marginally specified logistic-normal models for longitudinal binary data. *Biometrics*, **55**(3) 688-698.
- Hin, L. Y., Carey, V. J., and Wang, Y. G. (2007). Criteria for working-correlation-structure selection in GEE. *The American Statistician* **61**(4), 360-364.
- Hin, L. Y. and Wang, Y. G. (2009). Working-correlation-structure identification in generalized estimating equations. *Statistics in Medicine* **28**(4), 642-658.
- Johnson, R. A. and Wichern, D. W. (2007). *Applied Multivariate Statistical Analysis*, 6th ed. New Jersey: Prentice Hall.
- Kauermann, G. and Carroll, R. J. (2001). A note on the efficiency of sandwich covariance matrix estimation. *Journal of the American Statistical Association* **96**(456), 1387-1397.
- Liang, K. Y. and Zeger, S. L. (1986). Longitudinal data analysis using generalized linear models. *Biometrika* **73**(1), 13.

- Liang, K. Y., Zeger, S. L., and Qaqish, B. (1992). Multivariate regression analyses for categorical data (with discussion). *Journal of the Royal Statistical Society, Series B* **54**, 3-40.
- Lipsitz, S. R., Laird, N. M., and Harrington, D. P. (1990). Maximum likelihood regression methods for paired binary data. *Statistics in Medicine* **9**, 1517-1525.
- Lu, B., Preisser, J. S., Qaqish, B. F., Suchindran, C., Bangdiwala, S. I., and Wolfson, M. (2007). A comparison of two bias-corrected covariance estimators for generalized estimating equations. *Biometrics* **63**, 935-941.
- Mancl, L. A. and DeRouen, T. A. (2001). A covariance estimator for GEE with improved small-sample properties. *Biometrics* **57**, 126-134.
- McDonald, B. W. (1993). Estimating logistic regression parameters for bivariate binary data. *Journal of the Royal Statistical Society, Series B* **55**:391-397.
- Molinari, N., Bonaldi, C., and Daurès, J. P. (2001). Multiple temporal cluster detection. *Biometrics* **57**, 577-583.
- Paik, M. C. (1998). Repeated measurement analysis for nonnormal data in small samples. *Communications in Statistics: Simulations* **17**, 1155-1171.
- Pan, W. (2001). Akaike's information criterion in generalized estimating equations. *Biometrics* **57**(1), 120-125.
- Pan, W. (2001). On the robust variance estimator in generalized estimating equation. *Biometrika* **88**(3), 901-906.
- Prentice, R. L. (1988). Correlated binary regression with covariates specific to each binary observation. *Biometrics* **44**:1033-1048.
- Qaqish, B.F. (2003). A family of multivariate binary distributions for simulating correlated binary variables with specified marginal means and correlations. *Biometrika*, **90**, 455-463.
- Qu, Y., Piedmonte, M. R., and Williams, G. W. (1994). Small sample validity of latent variable models for correlated binary data. *Communications in Statistics: Simulations* **23**, 243-269.
- Rotnitzky, A. and Jewell, N. P. (1990). Hypothesis testing of regression parameters in semiparametric generalized linear models for cluster correlated data. *Biometrika* **77**(3), 485-497.
- Seber, G.A. F. (1984). *Multivariate Observations*. New York: Wiley.
- Seber, G.A.F. (2007). *A Matrix Handbook for Statisticians*. New York: Wiley.
- Schott, J. R. (2005). *Matrix Analysis for Statistics*. New York: Wiley.
- Sherman, M. and le Cessie, S. (1997). A comparison between bootstrap methods and generalized estimating equations for correlated outcomes in generalized linear model. *Communications in Statistics: Simulation* **26**, 901-925.

- Shults, J. and Chaganty, N. R. (1998). Analysis of serially correlated data using quasi-least squares. *Biometrics* **54**(4), 1622-1630.
- Shults, J., Sun, W., Tu, X., Kim, H., Amsterdam, J., Hilbe, J. M., and Ten-Have, T. (2009). A comparison of several approaches for choosing between working correlation structures in generalized estimating equation analysis of longitudinal binary data. *Statistics in Medicine* **28**(18), 2338-2355.
- Sparks, J. D. (2009). Model selection criteria in the presence of missing data based on the Kullback-Leibler discrepancy. Ph.D. Dissertation, University of Iowa.
- Thisted, R.A. (1988). *Elements of Statistical Computing*. London: Chapman and Hall.
- Wang, Y. G. and Hin, L. (2010). Modeling strategies in longitudinal data analysis: Covariate, variance function and correlation structure selection. *Computational Statistics & Data Analysis* **54**, 3359-3370.
- Wang, Y. G. and Carey, V. (2003). Working correlation structure misspecification, estimation and covariate design: implications for GEE performance. *Biometrika* **90**, 29-41.
- Wang, Y. G. and Lin, X. (2005). Effects of variance-function misspecification in analysis of longitudinal data. *Biometrics* **61**, 413-421.
- Wilks, S.S. (1932). Certain generalizations in the analysis of variance. *Biometrika* **24**, 471-494.
- Wong, M. and Long, Q. (2010). Modified robust variance estimator for generalized estimating equations with improved small-sample performance. *Statistics in Medicine* **30**(11), 1278-1291.
- Yahav, I. and Shmueli, G. (2011). On generating multivariate Poisson data in management science applications. *Applied Stochastic Models in Business and Industry* doi: 10.1002/asmb.901.
- Zhao, L. P. and Prentice, R. L. (1990). Correlated binary regression using a quadratic exponential model. *Biometrika* **77**:642-648.
- Zhao, L. P., Prentice, R. L., and Self, S. G. (1992). Multivariate mean parameter estimation by using a partially exponential model. *Journal of the Royal Statistical Society, Series B* **54**:805-811.